

Mensuration

2-D: Under 2 dimensions we will study

- Triangles
- Quadrilateral
- Polygons
- Circle

3-D: Under 3 dimensions we will study

- Cube
- Cuboids
- Box
- Cylinder
- Prism
- Cone
- Pyramid
- Frustum
- Sphere
- Hemisphere
- Tetrahedral

Note: For 2D figures, area and perimeter are calculated and for 3D figures, volume and surface area are calculated.

TWO DIMENSIONAL FIGURES

TRIANGLE

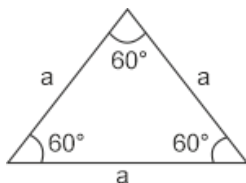
For a triangle with height, h, and base, b;

$$\text{Area} = \frac{1}{2} \times b \times h$$

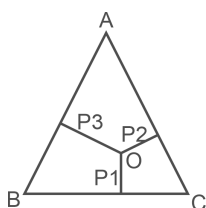
Perimeter = Sum of all sides

Special cases:

1) Equilateral Triangle - All sides are equal and each angle is 60° .



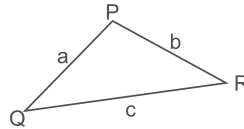
- Area = $\frac{\sqrt{3}}{4} a^2$
- Height = $\frac{\sqrt{3}}{2} a$
- Perimeter = $3a$
- Inradius(r) = $\frac{a}{2\sqrt{3}}$
- Circumradius(R) = $\frac{a}{\sqrt{3}}$



If P1, P2 and P3 are perpendicular to a side from a point inside the triangle, then

- Height of triangle = $P1 + P2 + P3$

2) Scalene Triangle - All sides will be unequal lengths.

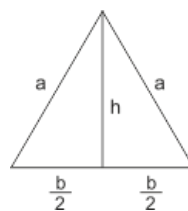


Area = $\sqrt{s(s-a)(s-b)(s-c)}$,
This formula is called Heron's formula.

Where, Semi-perimeter,
$$s = \frac{a+b+c}{2}$$

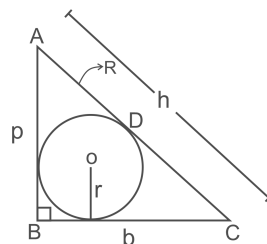
- Perimeter = $a + b + c$

3) Isosceles Triangle :- two sides and two angles are equal. Altitude bisects the base.



- Area = $\frac{b}{4} \sqrt{4a^2 - b^2}$
- Height = $\sqrt{a^2 - \left(\frac{b}{2}\right)^2}$
 $= \frac{1}{2} \sqrt{4a^2 - b^2}$
- Perimeter = $a + a + b$
 $= 2a + b$

4) Right angled Triangle :- One of the angles is 90° .



Here, p = perpendicular, b = base and h = hypotenuse

- Area = $\frac{1}{2} \times b \times p$
- Perimeter = $p + b + h$
- Pythagoras Theorem:
$$h^2 = p^2 + b^2$$
- Inradius(r) = $\frac{p+b-h}{2}$
- Circumradius(R) = $\frac{h}{2}$

Note: Common Pythagoras triplets:

(1, 1, $\sqrt{2}$); (1, 2, $\sqrt{5}$); (3, 4, 5); (5, 12, 13);
(7, 24, 25); (20, 21, 29); (9, 40, 41);
(8, 15, 17) (12, 35, 37) (60, 11, 61)
(65, 72, 97) (96, 110, 146).

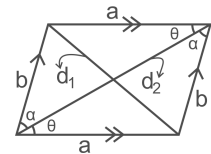
Try remembering them.

QUADRILATERAL

A figure enclosed by four sides is called a quadrilateral. A quadrilateral has four angles and sum of these angles is equal to 360° .

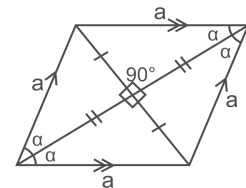
Special Cases:

1) Parallelogram - It is a quadrilateral with opposite sides parallel and equal.



- Area = base $\times$ height
- Perimeter = $2(a + b)$
- $d_1^2 + d_2^2 = 2(a^2 + b^2)$
- In a parallelogram, opposite sides are equal, opposite angles are equal and diagonals bisect each other.

2) Rhombus - It is a parallelogram with all four sides equal. The opposite angles in a rhombus are equal.

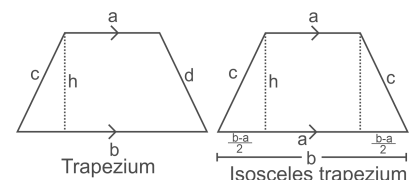


Here, a = side; d_1 and d_2 are diagonals.

- Area = $\frac{1}{2} \times d_1 \times d_2$
- Side (a) = $\frac{1}{2} \sqrt{d_1^2 + d_2^2}$
- Perimeter = $4a$
- $4a^2 = d_1^2 + d_2^2$

Diagonals bisect each other at right angles.

3) Trapezium - It is a quadrilateral with one pair of opposite sides parallel.



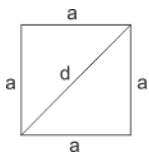
Here, a and b are parallel sides and h is the height or perpendicular distance between a and b.

- Area = $\frac{1}{2} \times \text{height} \times (\text{sum of parallel sides})$
$$= \frac{1}{2} \times h \times (a + b)$$

4) Rectangle – It is a parallelogram with equal opposite sides and each angle is 90°

- Area = Length $\times$ Breadth = $L \times B$
- Perimeter = $2(L + B)$
- Diagonal (d) = $\sqrt{L^2 + B^2}$
- Area of the walls of a rectangular room
 $= 2 \times (L + B) \times H$

5) Square – It is a parallelogram with all four sides equal and each angle is equal to 90° .



- Area = $(side)^2 = a^2 = \frac{1}{2}d^2$
- Perimeter = $4 \times side = 4a$
- Diagonal (d) = $a\sqrt{2}$

NOTE: Important points about Quadrilaterals

1. The diagonals of a parallelogram bisect each other.
2. Diagonal of a parallelogram divides it into two triangles of equal area.
3. The diagonals of a rectangle are of equal lengths and bisect each other.
4. The diagonals of a square are equal and bisect each other at right angles.
5. A rhombus has unequal diagonals and they bisect each other at right angles.
6. A parallelogram and a rectangle have equal areas if they are on the same base and between the same parallel lines.

REGULAR POLYGON

In a regular polygon all sides and all interior angles are equal. A polygon is called a pentagon, hexagon, heptagon, octagon, nonagon and a decagon as they have 5, 6, 7, 8, 9, 10 sides, respectively.

If each side of a regular polygon of 'n' sides is equal to 'a' then:

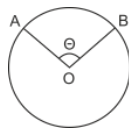
- Area of regular pentagon
 $= \frac{6.9}{4} \times a^2$
- Area of regular hexagon
 $= 6 \times \frac{\sqrt{3}}{4} a^2$
- Each exterior angle = $\frac{360^\circ}{n}$
- Each interior angle

= 180° – Exterior angle

- Number of diagonals
 $= \frac{n(n-3)}{2}$
- Sum of all interior angles
 $= (n-2) \times 180$
- Sum of all exterior angle = 360°

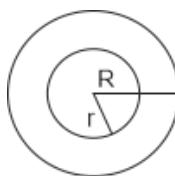
CIRCLE

It is a plane figure enclosed by a line on which every point is equally distant from a fixed point (centre) inside the circle.



- Area = πr^2
- Circumference (perimeter)
 $= 2\pi r$
- Diameter = $2r$
- Length of Arc (AB)
 $= 2\pi r \times \frac{\theta}{360^\circ}$
- Area of sector AOB
 $= \pi r^2 \times \frac{\theta}{360^\circ}$
- Length of Arc (AB) = *angle* $\times$ *radius*, (where angle in radian)

CIRCULAR RING



Here, R = radius of bigger ring,
 r = radius of smaller ring

- Area = $\pi(R^2 - r^2)$
- Difference in circumference of both the rings = $2\pi R - 2\pi r$

Short Cut methods/Tricks

1. If the length and breadth of a rectangle are increased by a% and b%, the area of the rectangle will be increased by

$$(a + b + \frac{ab}{100})\%$$

2. If any of the two sides of the rectangle is decreased then use '-ve' values for that side.

3. All the sides of any two dimensional figure changed by a%, then its area will

$$\text{change by } (2a + \frac{a^2}{100})\%$$

Whenever there is a decrease, use negative value for 'a'

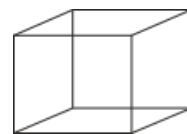
4. If all the sides of any two dimensional

figure has changed (increased or decreased) by a% then its perimeter also changes by a%. In the case of a circle such changes take place because of the change in radius (or diameter).

5. If the area of a square is 'a' square unit. Then the area of the circle formed with the same perimeter is given by square units.
6. Area of a square inscribed in a circle of radius 'r' is equal to $2r^2$.
7. The area of the largest triangle inscribed in a semicircle of radius r is equal to r^2 .
8. If a pathway of width x is made inside or outside a rectangular plot of length L and breadth B, then area of the pathway is
 - (i) $2x(L + B + 2x)$, if path is made outside the plot
 - (ii) $2x(L + B - 2x)$, if a path is made inside the plot.
9. If two paths, each of width x are made parallel to length (L) and breadth (B) of the rectangular plot in the middle of the plot crossing each other, then,
 Area of the path = $x(L + B - x)$

THREE DIMENSIONAL FIGURES

CUBE



All sides are equal. It has six faces and 12 edges.

- Volume = a^3
- Total surface area = $6a^2$
- Diagonal = $a\sqrt{3}$
- Sum of all edges = $12a$

Here, a = length of the side

CUBOID

A rectangular body having 3D rectangular shape, is called a cuboid.

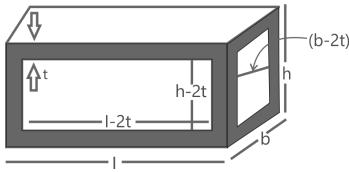


- Volume = $l \times b \times h$
- Total surface area =
 $2(lb + bh + lh)$
- Diagonal = $\sqrt{l^2 + b^2 + h^2}$

BOX

A box has its shape like a cube or cuboid. The amount that a box can hold or

contain is called the capacity of the box.
Capacity means internal volume.



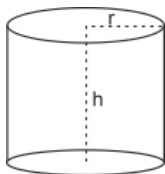
- Surface area of an open box
 $= 2(\text{length} + \text{breadth}) \times \text{height} + \text{length} \times \text{breadth}$
 $= 2(l + b) \times h + l \times b$
- Capacity of box
 $= (l - 2t)(b - 2t)(h - 2t);$
 where, t = thickness of box
- Volume of the material of the box = External volume – Internal volume(or capacity) =
 $lbh - (l - 2t)(b - 2t)(h - 2t)$

ROOM

A rectangular room has four walls (surfaces) and opposite walls have equal area.

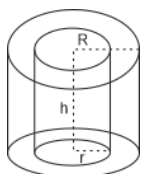
- Total Area of walls
 $= 2(l + b) \times h$
- Total volume of the room
 $= l \times b \times h$
- Area of floor or roof = $l \times b$

CYLINDER



- Volume of cylinder = area of base $\times$ height = $\pi r^2 h$
- Curved surface area = Perimeter of base $\times$ height = $2\pi r h$
- Total surface Area = curved surface area + area of both the circles
 $= 2\pi r h + 2\pi r^2 = 2\pi(r + h)r$
- When the rectangular sheet is folded along its length, then the length becomes the circumference of the base of the cylinder and breadth becomes the height of the cylinder.

HOLLOW CYLINDER



- Volume of hollow cylinder
 $= \pi(R^2 - r^2)h$
- Curved surface area
 $= 2\pi(R + r)h$
- Total surface area
 $= 2\pi(R + r)h + 2\pi(R^2 - r^2)$
 $= 2\pi(R + r)\{h + R - r\}$

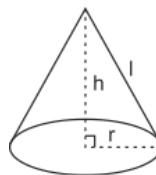
Where, R = External radius of cylinder, r = internal radius of cylinder, h = height

PRISM

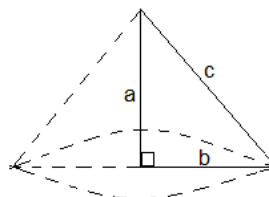
- Volume of prism
 $= \text{area of base} \times \text{height}$
- Lateral surface area = Perimeter of base $\times$ height
- Total surface Area = Lateral surface area + area of base and top surface

CONE

A solid and round body with a round base and pointed peak.



- Volume = $\frac{1}{3} \times \text{base area} \times \text{height} = \frac{1}{3} \pi r^2 h$
- Slant height (l) = $\sqrt{r^2 + h^2}$
- Curved surface area = $\pi r l$
 $= \pi r \sqrt{r^2 + h^2}$
- Total surface area = $\pi r l + \pi r^2$
 $= \pi r(l + r)$
- Cone formed by rotating right angled triangle about its height:



- Volume of cone so formed
 $= \frac{1}{3} \pi b^2 a$
- Similarly, Cone formed by rotating right angled triangle about its base:
 Volume of cone so formed
 $= \frac{1}{3} \pi a^2 b$

- Similarly, Cone formed by rotating right angled triangle about its hypotenuse :
 Volume of cone so formed
 $= \frac{1}{3} \pi r^2 c$, (where r is the altitude on hypotenuse and $r = \frac{a \times b}{c}$)

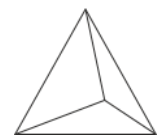
Note: If the base is not round, it will be called a pyramid. A pyramid can have various shapes of the base example: square, rectangular, triangular etc.

PYRAMID

- Pyramid means a structure with regular polygon as its base and sloping sides that meet in a point at the top.
- In Pyramid, with n sided regular polygon at its base, total number of vertices = $n + 1$
- Volume = $\frac{1}{3} \times \text{base area} \times \text{height}$
- Slant height (l) = $\sqrt{r^2 + h^2}$
- Lateral surface area
 $= \frac{\text{Perimeter} \times \text{slant height}}{2}$
- Total Surface area = Lateral surface area + Area of base

TETRAHEDRON

It is a 3D figure made by joining four equilateral triangles.



- Volume (V)
 $= \frac{1}{3} \times \text{base area} \times \text{height}$
 $\Rightarrow V = \frac{1}{3} \times \frac{\sqrt{3}}{4}$

$$a^2 \times \left(\frac{\sqrt{6}}{12} a + \frac{\sqrt{6}}{4} a \right)$$

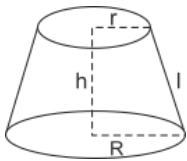
$$\Rightarrow V = \frac{\sqrt{2}}{12} a^3 \text{ (Remember this formula)}$$

- Lateral surface area = $3 \times \frac{\sqrt{3}}{4} a^2$
- Total surface area
 $= 4 \times \frac{\sqrt{3}}{4} a^2$
- Height of Tetrahedron = $\frac{\sqrt{6}}{3} a$

FRUSTUM OF CONE

If a cone is cut by a plane parallel to its base, so as to divide the cone into two

parts: upper part and lower part, then the lower part is called frustum.



- Slant height (l)

$$= \sqrt{h^2 + (R - r)^2}$$
- Curved Surface Area

$$= \pi(R + r)l$$
- Total surface area

$$= \pi(R + r)l + \pi R^2 + \pi r^2$$

$$= \pi\{(R + r)l + R^2 + r^2\}$$
- Volume $= \frac{1}{3}\pi h(r^2 + R^2 + rR)$

SPHERE

- Volume of sphere $= \frac{4}{3}\pi r^3$
- Curved Surface area

$$= \text{Total surface area} = 4\pi r^2$$

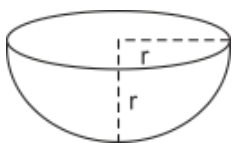
HOLLOW SPHERE OR SPHERICAL SHELL:

- Volume of hollow sphere

$$= \frac{4}{3}\pi(R^3 - r^3)$$
- Internal surface area $= 4\pi r^2$
- External surface area $= 4\pi R^2$

Here R = external radius and
 r = internal radius

HEMISPHERE



- Volume of the hemisphere

$$= \frac{2}{3}\pi r^3$$
 - Total surface area $= 3\pi r^2$
 - Curved surface area $= 2\pi r^2$
- Where, r = radius

IMP. UNIT CONVERSION :

- $1m^3 = 1000$ litres and
- $1 \text{ litre} = 1000 \text{ cm}^3$
- $1 \text{ meter} = 10 \text{ decimeter}$
 $= 100 \text{ cm} = 1000 \text{ millimeter}$
- $1 \text{ meter} = 10^{-1} \text{ decameter}$
 $= 10^{-2} \text{ hectometer}$
 $= 10^{-3} \text{ kilometer}$

Variety Questions

Q.1. The base of a right prism is an equilateral triangle with each side measuring 4 cm. If the lateral surface area is 120 cm^2 , find the volume (in cm^3) of the prism.

SSC CHSL Tier II (10/01/2024)

- (a) $30\sqrt{3}$ (b) $40\sqrt{3}$ (c) $10\sqrt{3}$ (d) $20\sqrt{3}$

Q.2. Some ice pieces, spherical in shape, of diameter 6 cm are dropped in a cylindrical container containing some juice and are fully submerged. If the diameter of the container is 18 cm and level of juice rises by 40 cm, then how many ice pieces are dropped in the container?

SSC CPO 05/10/2023 (3rd Shift)

- (a) 90 (b) 80 (c) 85 (d) 95

Q.3. Some medicine in liquid form is prepared in a hemispherical container of diameter 36 cm. When the container is full of medicine, the medicine is transferred to small cylindrical bottles of diameter 6 cm and height 6 cm. How many bottles are required to empty the container?

SSC CPO 04/10/2023 (2nd Shift)

- (a) 70 (b) 75 (c) 72 (d) 76

Q.4. The length and the breadth of the floor of a rectangular hall are 126 feet and 90 feet, respectively. What will be the area (in square feet) of each of the largest identical square tiles that can be used to tile this floor in a way that no part of the floor remains uncovered?

SSC CPO 03/10/2023 (3rd Shift)

- (a) 196 (b) 256 (c) 324 (d) 484

Q.5. The radii of the ends of a frustum of a solid right-circular cone 45 cm high are 28 cm and 7 cm. If this frustum is melted and reconstructed into a solid right circular cylinder whose radius of base and height are in the ratio 3 : 5, find the curved surface area (in cm^2) of this cylinder. [Use $\pi = \frac{22}{7}$.]

SSC CPO 03/10/2023 (1st Shift)

- (a) 4610 (b) 4620 (c) 4580 (d) 4640

Q.6. A square and a rhombus have the same base and the rhombus is inclined at 45° , then what will be the ratio of the area of the square to the area of the rhombus?

SSC MTS 14/09/2023 (3rd Shift)

- (a) $\sqrt{3} : 1$ (b) $1 : \sqrt{3}$ (c) $\sqrt{2} : \sqrt{3}$ (d) $\sqrt{2} : 1$

Q.7. A closed wooden box measures externally 10 cm long, 8 cm broad and 6

cm high. If the thickness of the wood is 0.5 cm, then the volume of wood required is

SSC MTS 14/09/2023 (3rd Shift)

- (a) 165 cm^3 (b) 300 cm^3
 (c) 230 cm^3 (d) 150 cm^3

Q.8. The weight of a cube varies directly as the product of its volume and its density. The ratio of densities of the materials of the first cube and second cube is 27:16. If the weight of the first cube is 4 times the weight of the second cube, then what is the ratio of the edges of the first cube and second cube?

SSC MTS 14/09/2023 (2nd Shift)

- (a) 2 : 3 (b) 3 : 2 (c) 3 : 4 (d) 4 : 3

Q.9. The weight of a circular disc varies directly as the product of the square of the radius and its thickness. Two identical/similar discs have their thickness in the ratio of 16 : 9. What is the ratio of their radii if the weight of the first is four times that of the second?

SSC MTS 12/09/2023 (3rd Shift)

- (a) 3 : 2 (b) 2 : 3 (c) 1 : 2 (d) 2 : 1

Q.10. A path of uniform width 2.5 m runs around the outside of a rectangular field of dimensions 45 m $\times$ 35 m. What is the area of the path (in m^2)?

SSC MTS 08/09/2023 (2nd Shift)

- (a) 500 (b) 425 (c) 475 (d) 525

Q.11. A hemispherical dome of a building needs to be painted. If the circumference of the base of the dome is 154 cm, then find the cost of painting it if the cost of painting is ₹4 per 100 cm^2 (use $\pi = \frac{22}{7}$).

SSC CHSL 10/08/2023 (4th Shift)

- (a) ₹150.92 (b) ₹ 150.66
 (c) ₹ 105.29 (d) ₹ 105.66

Q.12. The area of a sector of a circle is 616 cm^2 with a central angle of 10° . The radius of the circle is _____. (use $\pi = \frac{22}{7}$)

SSC CHSL 08/08/2023 (2nd Shift)

- (a) 84 cm (b) 21 cm (c) 48 cm (d) 28 cm

Q.13. Find the volume of the largest right circular cone that can be cut out from a cube whose edge is $3\frac{1}{2} \text{ cm}$, correct to

two places of decimals (use $\pi = \frac{22}{7}$).

SSC CHSL 07/08/2023 (4th Shift)

- (a) 13.21 cm^3 (b) 21.31 cm^3
 (c) 11.23 cm^3 (d) 12.13 cm^3

Q.14. The sum of the radius of the base and the height of a cylinder is 42 m. If the total surface area of the cylinder is 6336 m^2 , find the curved surface area of the

cylinder correct to two places of decimals (use $\pi = \frac{22}{7}$)

SSC CHSL 02/08/2023 (4th Shift)

- (a) 2157.43 m^2 (b) 2571.43 m^2
(c) 2715.43 m^2 (d) 2517.43 m^2

Q.15. Three cubes with sides in the ratio of 3 : 4 : 5 are melted to form a single cube whose diagonal is $18\sqrt{3}$ cm. The sides of the three cubes are:

SSC CGL 20/07/2023 (1st shift)

- (a) 21 cm, 28 cm and 35 cm
(b) 9 cm, 12 cm and 15 cm
(c) 18 cm, 24 cm and 30 cm
(d) 12 cm, 16 cm and 20 cm

Q.16. The perimeter of equilateral triangle is $3 \times \left(\sqrt[4]{\frac{16}{3}}\right)$ units. Determine the area of the triangle.

Matriculation Level 30/06/2023 (Shift - 4)

- (a) 4 unit^2 (b) 1 unit^2 (c) 3 unit^2 (d) 2 unit^2

Q.17. A spherical ball of diameter 8 cm is cut into two equal parts. The curved area of one such part has to be painted with green colour, while the other part has to be painted with red colour. The bases of both the hemispheres are to be painted with blue colour. The cost of painting with blue is ₹2/cm², while the cost of painting the curved area is ₹3/cm². What will be the cost (in ₹) of painting the hemispheres? Take $\pi = 3.14$

Matriculation Level 28/06/2023 (Shift - 4)

- (a) ₹451.92 (b) ₹492.92
(c) ₹803.84 (d) ₹401.92

Q.18. A cube of side 5 cm, has been assembled using 125 cubes each of 1 cm edge. One cube is removed from the middle of each of the faces of the cube. What is the surface area (in cm²) of the remaining solid?

SSC CHSL Tier II (26/06/2023)

- (a) 155 (b) 144 (c) 150 (d) 174

Q.19. A prism and a pyramid have the same base and the same height. Find the ratio of the volumes of the prism and the pyramid.

SSC CGL Tier II (07/03/2023)

- (a) 2 : 3 (b) 3 : 1 (c) 1 : 3 (d) 3 : 2

Q.20. How many metres of 2 m wide cloth will be required to make a conical tent with the diameter of the base as 14 m and slant height as 9 m to ignore wastage?

SSC CGL 12/12/2022 (1st Shift)

- (a) 66 m (b) 88 m (c) 99 m (d) 77 m

Q.21. The minute hand of a clock is 20 cm long. Find the area on the face of the

clock swept by the minute hand between 8 a.m. and 8:45 a.m.

SSC CGL 12/12/2022 (1st Shift)

- (a) $\frac{6600}{7} \text{ cm}^2$ (b) $\frac{6600}{9} \text{ cm}^2$
(c) $\frac{6600}{18} \text{ cm}^2$ (d) $\frac{6600}{14} \text{ cm}^2$

Q.22. The external diameter of an iron pipe is 20 cm and its length is 12 cm. If the thickness of the pipe is 1 cm, find the surface area of the pipe (take $\pi = \frac{22}{7}$)

correct to two places of decimal.

SSC CGL 06/12/2022 (2nd Shift)

- (a) $1,662.67 \text{ cm}^2$ (b) $1,552.57 \text{ cm}^2$
(c) $1,442.48 \text{ cm}^2$ (d) $1,772.76 \text{ cm}^2$

Q.23. If the length of certain rectangle is decreased by 4 cm and breadth is increased by 2 cm, it would result in a square of the same area. What is the perimeter of the original rectangle?

SSC CGL 06/12/2022 (1st Shift)

- (a) 15 cm (b) 24 cm (c) 20 cm (d) 10 cm

Q.24. Two identical circles each of radius 30 cm intersect each other such that the circumference of each one passes through the center of the other. What is the area of the intersecting region?

SSC CGL 03/12/2022 (4th Shift)

- (a) $400\pi - 250\sqrt{3} \text{ cm}^2$
(b) $300\pi - 150\sqrt{3} \text{ cm}^2$
(c) $500\pi - 350\sqrt{3} \text{ cm}^2$
(d) $600\pi - 450\sqrt{3} \text{ cm}^2$

Q.25. The area of a trapezium shaped field is 1785 square feet. The distance between the two parallel sides is 35 feet and one of the parallel sides is 42 feet long. What is the length of the other parallel side?

SSC CPO 11/11/2022 (Evening)

- (a) 60 ft (b) 35 ft (c) 40 ft (d) 65 ft

Q.26. If the sum of the diagonals of a rhombus is L and the perimeter is 4P, find the area of the rhombus?

SSC CPO 11/11/2022 (Morning)

- (a) $\frac{1}{4}(L^2 - P^2)$ (b) $\frac{1}{4}(L^2 - 4P^2)$
(c) $\frac{1}{2}(L^2 - 4P^2)$ (d) $\frac{1}{4}(L^2 + 3P^2)$

Q.27. Water is flowing through a cylindrical canal with an internal diameter of 7 m at the speed of 18 km/h. Find the volume of water flowing through the canal in 30 minutes. (take $\pi = \frac{22}{7}$)

SSC CPO 09/11/2022 (Afternoon)

- (a) $3,76,500 \text{ m}^3$ (b) $3,56,500 \text{ m}^3$
(c) $3,66,500 \text{ m}^3$ (d) $3,46,500 \text{ m}^3$

Q.28. ABCDEF is a regular hexagon. Side of the hexagon is 36 cm. What is the area of the triangle ABC?

SSC CGL Tier II (08/08/2022)

- (a) $324\sqrt{3} \text{ cm}^2$ (b) $360\sqrt{3} \text{ cm}^2$
(c) $240\sqrt{3} \text{ cm}^2$ (d) $192\sqrt{3} \text{ cm}^2$

Q.29. The frustum of a right circular cone has the radius of the base as 5 cm, radius of the top as 3 cm, and height as 6 cm. What is its volume?

SSC CHSL 27/05/2022 (Afternoon)

- (a) $98\pi \text{ cm}^3$ (b) $100\pi \text{ cm}^3$
(c) $96\pi \text{ cm}^3$ (d) $90\pi \text{ cm}^3$

Q.30. Three sides of a triangle are $\sqrt{a^2 + b^2}$, $\sqrt{(2a)^2 + b^2}$ and $\sqrt{a^2 + (2b)^2}$ units. What is the area (in unit squares) of a triangle?

SSC CGL Tier II (03/02/2022)

- (a) $\frac{5}{2} ab$ (b) $3ab$ (c) $4ab$ (d) $\frac{3}{2} ab$

Q.31. A square park has been divided into two rectangles of equal area. If the perimeter of each of these rectangles is 39 m, then what will be the perimeter of the square park?

SSC CHSL 09/08/2021 (Afternoon)

- (a) 104 m (b) 39 m (c) 78 m (d) 52 m

Q.32. The circumference of a circle is 'aπ' units and the area of the circle is 'bπ' square units. If a:b is equal to 4:5, then the radius of the circle is:

SSC CHSL 06/08/2021 (Evening)

- (a) 3 cm (b) 2.5 cm (c) 5 cm (d) 2 cm

Q.33. The area of a triangular field whose sides are 65m, 72m and 97m is equal to the area of a rectangular park whose sides are in the ratio of 5 : 13. What is the perimeter (in m) of the rectangular park?

SSC CHSL 19/04/2021 (Morning)

- (a) 108 (b) 180 (c) 216 (d) 144

Q.34. A bicycle wheel has a radius of 42 cm. It makes 40 revolutions in 25 seconds. What is its speed (in kmph, up to one decimal place)?

SSC CHSL 16/04/2021 (Morning)

- (a) 3.5 (b) 11.6 (c) 15.2 (d) 9.5

Q.35. A wire is bent to form a square of area 169 cm^2 . If the same wire is bent to form a circle, then what is its area in cm^2 (to the nearest whole number)?

SSC CHSL 13/04/2021 (Afternoon)

- (a) 215 (b) 227 (c) 532 (d) 531

Q.36. A sector of radius 10.5 cm with the central angle 120° is folded to form a cone by joining the two bounding radii of the sector. What is the volume (in cm^3) of

the cone so formed ?

SSC CGL Tier II (13/09/2019)

- (a) $\frac{343\sqrt{2}}{6} \pi$ (b) $\frac{343\sqrt{3}}{6} \pi$
(c) $\frac{343\sqrt{3}}{12} \pi$ (d) $\frac{343\sqrt{2}}{12} \pi$

Q.37. The base of a right prism is a triangle with sides 20 cm, 21 cm and 29 cm. If its volume is 7560cm^3 , then its lateral surface area (in cm^2) is :

SSC CGL Tier II (12/09/2019)

- (a) 2484 (b) 2556 (c) 2520 (d) 2448

Q.38. If the radius of a sphere is increased by 4 cm, its surface area is increased by $464\pi\text{cm}^2$. What is the volume (in cm^3) of the original sphere ?

SSC CGL Tier II (11/09/2019)

- (a) $\frac{15625}{6} \pi$ (b) $\frac{35937}{8} \pi$
(c) $\frac{11979}{2} \pi$ (d) $\frac{15625}{8} \pi$

Q.39. The base of right prism is a trapezium whose parallel sides are 11 cm and 15cm and the distance between them is 9 cm. If the volume of the prism is 1731.6cm^3 , then the height (in cm) of the prism will be :

SSC CGL Tier II (11/09/2019)

- (a) 15.6 (b) 15.2 (c) 14.8 (d) 14.2

Q.40. A ribbon of equal width and length l is wrapped around the curved surface of a right circular cylinder to cover it completely. If the base circumference of this cylinder is c , then how many times the ribbon was wrapped around the cylinder ?

SSC MTS 05/08/2019 (Evening)

- (a) $\frac{l}{4c}$ (b) $\frac{l}{c}$ (c) $\frac{l}{2c}$ (d) $\frac{2l}{c}$

Q.41. The volume of a right circular cone is equal to that of a sphere, whose radius is half the radius of the base of the cone. What is the ratio of the radius of the base to the height of the cone ?

SSC CHSL 11/07/2019 (Afternoon)

- (a) 1 : 4 (b) 1 : 2 (c) 4 : 1 (d) 2 : 1

Q.42. ABCD is a rhombus with each side being equal to 8 cm. If $BD = 10$ cm, $AC = 2\sqrt{x}$ cm, what is the value of $\sqrt{x + 10}$?

SSC CHSL 11/07/2019 (Afternoon)

- (a) $2\sqrt{3}$ (b) $3\sqrt{2}$ (c) 7 (d) 5

Q.43. The two diagonals of a rhombus are respectively, 14 cm and 48 cm. The perimeter of the rhombus is equal to :

SSC CHSL 11/07/2019 (Morning)

- (a) 120cm (b) 160cm (c) 80cm (d) 100cm

Q.44. The area of a sector of a circle with central angle 60° is A. The circumference of the circle is C. Then A is equal to :

SSC CHSL 10/07/2019 (Evening)

- (a) $\frac{c^2}{6\pi}$ (b) $\frac{c^2}{18\pi}$ (c) $\frac{c^2}{24\pi}$ (d) $\frac{c^2}{4\pi}$

Q.45. Equilateral triangles are drawn on the hypotenuse and one of the perpendicular sides of a right-angled isosceles triangle. Their areas are H and

A respectively. $\frac{A}{H}$ is equal to :

SSC CHSL 10/07/2019 (Afternoon)

- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{1}{\sqrt{2}}$ (d) $\frac{1}{\sqrt{2}}$

Q.46. Twelve sticks, each of length one unit, are used to form an equilateral triangle. The area of the triangle is :

SSC CHSL 10/07/2019 (Morning)

- (a) $3\sqrt{3}$ (b) $2\sqrt{3}$ (c) $4\sqrt{3}$ (d) $8\sqrt{3}$

Q.47. If the length of a rectangle is decreased by 11% and the breadth is increased by 11%, its area will undergo :

SSC CHSL 09/07/2019 (Afternoon)

- (a) 13.13% increase (b) 1.21% increase
(c) 1.21% decrease (d) 13.13% decrease

Q.48. If each side of a rectangle is increased by 22%, then its area will increase by :

SSC CHSL 08/07/2019 (Morning)

- (a) 44% (b) 50% (c) 46.65% (d) 48.84%

Q.49. In triangle ABC, the length of BC is less than twice the length of AB by 2 cm. The length of AC exceeds the length of AB by 10 cm. The perimeter is 32cm. The length (in cm) of the smallest side of the triangle is :

SSC CHSL 05/07/2019 (Afternoon)

- (a) 4 (b) 10 (c) 8 (d) 6

Q.50. The parallel sides of a trapezium are 20 cm and 10 cm and its non-parallel sides are equal to each other. If its area is 180cm^2 , then what is the length (in cm) of each non parallel side?

SSC CHSL 04/07/2019 (Morning)

- (a) 11 (b) 13 (c) 12 (d) 15

Q.51. The length of a rectangular park is 20m more than its breadth. If the cost of fencing the park at Rs. 53 per metre is Rs. 21,200, then what is the area (in square metres) of the park ?

SSC CHSL 03/07/2019 (Evening)

- (a) 9504 (b) 8925 (c) 9240 (d) 9900

Q.52. A wire is in the shape of a rectangle whose sides are in the ratio 7 : 4. It was initially in the shape of a circle of radius , very nearly equal to 31.5 cm.

The length of smaller side of the rectangle is : (Take $\pi = \frac{22}{7}$)

SSC CHSL 03/07/2019 (Afternoon)

- (a) 44 cm (b) 36 cm (c) 40 cm (d) 32cm

Q.53. A circle circumscribes a rectangle whose sides are in the ratio 4 : 3. If the perimeter of the rectangle is 56 cm, then what is the area (in square cm) of the circle ?

SSC CHSL 03/07/2019 (Morning)

- (a) 70π (b) 96π (c) 90π (d) 100π

Q.54. The radius of the base of a cylinder is 7 cm and its curved surface area is 440 square cm. Its volume (in cubic cm)

will be: (Take $\pi = \frac{22}{7}$)

SSC CHSL 02/07/2019 (Evening)

- (a) 1760 (b) 1430 (c) 1540 (d) 1650

Q.55. The curved surface area and volume of a cylindrical pole are 132 square metres and 528 cubic metres, respectively. What is the height (in m) of the pole ? (Take $\pi = \frac{22}{7}$)

SSC CHSL 02/07/2019 (Afternoon)

- (a) $2\frac{1}{2}$ (b) $3\frac{5}{8}$ (c) $3\frac{1}{2}$ (d) $2\frac{5}{8}$

Q.56. The area of a field in the shape of a hexagon is $2400\sqrt{3}$ square meters. What will be the cost(Rs.) of fencing it at Rs. 18.50 per meter?

SSC CHSL 01/07/2019 (Evening)

- (a) 4440 (b) 5920 (c) 5550 (d) 5180

Q.57. Six cubes, each of edge 2 cm, are joined end to end. What is the total surface area of the resulting cuboid

SSC CGL 13/06/2019 (Morning)

- (a) 96cm^2 (b) 144cm^2
(c) 104cm^2 (d) 128cm^2

Q.58. The radius of a sphere is reduced by 40%. By what percent, will its volume decrease?

SSC CGL 10/06/2019 (Afternoon)

- (a) 60% (b) 64% (c) 72.5% (d) 78.4%

Q.59. The area of a field in the shape of a triangle with each side x meter is equal to the area of another triangular field having sides 50m, 70m and 80m. The value of x is closest to :

SSC CGL 07/06/2019 (Afternoon)

- (a) 65.5 (b) 63.2 (c) 62.4 (d) 61.8

Q.60. A sector is cut out from a circle of diameter 42 cm. If the angle of the sector is 150° , then its area (in square cm) is : (Take $\pi = \frac{22}{7}$)

SSC CGL 07/06/2019 (Morning)

(a) 564 (b) 574 (c) 580.6 (d) 577.5

Q.61. The radii of three concentric circles are in the ratio of 4:5:7. What is the ratio of the area between the two inner circles to that between the two outer circles ?

SSC CGL 06/06/2019 (Evening)

(a) 4 : 7 (b) 5 : 9 (c) 4 : 5 (d) 3 : 8

Q.62. The length of the metallic pipe is 7.56 m. Its external and internal radii are 2.5 cm and 1.5 cm respectively. If 1 cubic cm of the metal weigh 7.5 g, then the weight(kg) of the pipe is

SSC CGL 06/06/2019 (Afternoon)

(a) 72.82 (b) 70.14 (c) 71.28 (d) 69.68

Q.63. The volume of a metallic cylindrical pipe is 7480 cm^3 . If its length is 1.4 m and its external radius is 9 cm, then its thickness (given $\pi = \frac{22}{7}$) is:

SSC CGL 06/06/2019 (Morning)

(a) 1 cm (b) 0.8 cm (c) 0.9 cm (d) 1.2cm

Q.64. The areas of the three adjacent faces of a cuboid are 32 cm^2 , 24 cm^2 and 48 cm^2 . What is the volume (in cm^3) of the cuboid ?

SSC CGL 04/06/2019 (Evening)

(a) 192 (b) 256 (c) 288 (d) 128

Q.65. The area of a triangle is 15 sq cm and the radius of its incircle is 3 cm. Its perimeter is equal to :

SSC CGL 04/06/2019 (Evening)

(a) 12 cm (b) 20 cm (c) 5 cm (d) 10 cm

Q.66. How much iron sheet (in m^2) will be needed to construct a rectangular tank measuring $10 \text{ m} \times 8 \text{ m} \times 6 \text{ m}$, if a circular opening of radius one meter is to be left at the top of the tank? (correct to one decimal place)

SSC CGL 04/06/2019 (Afternoon)

(a) 371.6 (b) 370.4 (c) 372.9 (d) 370.8

Q.67. A solid cube of volume 13824 cm^3 is cut into 8 cubes of equal volumes. The ratio of the surface area of the original cube to the sum of the surface areas of three of the smaller cubes is :

SSC CGL 04/06/2019 (Morning)

(a) 2 : 3 (b) 4 : 3 (c) 8 : 3 (d) 2 : 1

Q.68. The unequal side of an isosceles triangle is 2cm. The medians drawn to the equal sides are perpendicular. The area of the triangle is :

SSC CPO 16/03/2019 (Evening)

(a) 2 cm^2 (b) 3 cm^2 (c) 5 cm^2 (d) 1 cm^2

Q.69. What will be total cost of polishing curved surface of a wooden cylinder at rate of Rs 20 per m^2 , if its diameter is 40

cm and height is 7m ?

SSC CPO 16/03/2019 (Evening)

(a) 176 rs (b) 184 rs (c) 175 rs (d) 186 rs

Q.70. A steel vessel has a base of length 60 cm and breadth 30 cm. Water is poured in the vessel. A cubical steel box having an edge of 30 cm is immersed completely in the vessel. By how much will the water rise?

SSC CPO 16/03/2019 (Evening)

(a) 12 cm (b) 15 cm (c) 10 cm (d) 9 cm

Q.71. Original breadth of a rectangular box is 20cm. The box was then remade in such a way that its length increased by 30% but the breadth decreased by 20% and the area increased by 100 cm^2 . What is the new area(in cm^2) of the box ?

SSC CPO 16/03/2019 (Evening)

(a) 2500 (b) 2200 (c) 2600 (d) 2400

Q.72. 12 buckets of water fill a tank when the capacity of each bucket is 13.5 liters. How many buckets will be needed to fill the same tank, if the capacity of each bucket in 9 liters ?

SSC CPO 16/03/2019 (Evening)

(a) 15 (b) 16 (c) 18 (d) 17

Q.73. Find the cost of carpeting a room which is 11m long and 6m broad by a carpet which is 60cm broad at the rate of rs 112.50 per meter.

SSC CPO 16/03/2019 (Afternoon)

(a) 12,375 (b) 13,280
(c) 11,695 (d) 12,040

Q.74. A field is $119 \text{ m} \times 18 \text{ m}$ in dimension. A tank $17 \text{ m} \times 6 \text{ m} \times 3 \text{ m}$ is dug out in the middle and the soil removed is evenly spread over the remaining part of the field. The increase in the level on the remaining part of the field is: (in cm)

SSC CPO 16/03/2019 (Afternoon)

(a) 14 cm (b) 13 cm (c) 15 cm (d) 12 cm

Q.75. A swimming pool is 40 m in length, 30 m in breadth and 2.2 m in depth. The cost of cementing its floor and the four sides at Rs. $25/\text{m}^2$ is:

SSC CPO 16/03/2019 (Afternoon)

(a) Rs.43,980 (b) Rs.37540
(c) Rs.34260 (d) Rs.37700

Q.76. The base of an isosceles triangle is 6 cm and its perimeter is 16 cm. Its area is ?

SSC CPO 16/03/2019 (Morning)

(a) 9 cm^2 (b) 11 cm^2 (c) 10 cm^2 (d) 12 cm^2

Q.77. Find the inner surface area of all walls of a rectangular room with length 7 m breadth 5 m and height 3.5 m .

SSC CPO 16/03/2019 (Morning)

(a) 84 m^2 (b) 168 m^2 (c) 126 m^2 (d) 42 m^2

Q.78. A square cardboard with side 3 m is folded through one of its diagonal to make a triangle, the height(m) of the triangle is :

SSC CPO 16/03/2019 (Morning)

(a) $\frac{3}{\sqrt{2}}$ (b) $2\sqrt{3}$ (c) $3\sqrt{2}$ (d) $\frac{2}{\sqrt{3}}$

Q.79. The surface area of a cube is 1176 cm^2 Its volume(in cm^3) is:

SSC CPO 16/03/2019 (Morning)

(a) 3486 (b) 3964 (c) 3206 (d) 2744

Q.80. The liquid in a container is sufficient to paint an area of 11.28 m^2 . How many boxes of dimension $30 \text{ cm} \times 25 \text{ cm} \times 12 \text{ cm}$ can be painted with the liquid in this container.

SSC CPO 16/03/2019 (Morning)

(a) 40 (b) 24 (c) 32 (d) 12

Q.81. If the area of a regular hexagon is $108\sqrt{3} \text{ cm}^2$, its perimeter (in cm) is:

SSC CPO 15/03/2019 (Evening)

(a) $36\sqrt{2}$ (b) $42\sqrt{3}$ (c) $28\sqrt{3}$ (d) 24

Q.82. A square piece of cardboard with side 12 cm has a small square of 2 cm cut out from each of the corners. The resulting flaps are turned up to make a box 2 cm deep. The volume (in cm^3) of the box is:

SSC CPO 15/03/2019 (Evening)

(a) 128 (b) 94 (c) 102 (d) 11

Q.83. The area (in cm^2) of a right angled triangle having base 24 cm and hypotenuse 25 cm is :

SSC CPO 15/03/2019 (Evening)

(a) 72 (b) 108 (c) 92 (d) 84

Q.84. The length of the longest pole that can be placed in a room 16 m long, 8 m wide and 11 m high is:

SSC CPO 15/03/2019 (Evening)

(a) 21 m (b) 20 m (c) 18 m (d) 19 m

Q.85. If the height of an equilateral triangle is $10\sqrt{3} \text{ cm}$, the area (in cm^2) is:

SSC CPO 15/03/2019 (Morning)

(a) $124\sqrt{3}$ (b) $75\sqrt{3}$ (c) $80\sqrt{3}$ (d) $100\sqrt{3}$

Q.86. A tall rectangular vessel is half filled with water. The base dimension of the vessel is $62 \text{ cm} \times 45 \text{ cm}$. A heavy metal cube of edge 15 cm is dropped into the vessel. The rise in level of the vessel is:

SSC CPO 15/03/2019 (Morning)

(a) 1.21 cm (b) 1.15 cm
(c) 1.07 cm (d) 1 cm

Q.87. The dimensions of a swimming pool are $66\text{m} \times 35\text{m} \times 3\text{m}$. How many hours will it take to fill the pool by a pipe of diameter 35 cm with water flowing at speed 8 m/s ?

SSC CPO 15/03/2019 (Morning)

(a) 2.75 h (b) 3.5 h (c) 2.5 h (d) 3.2 h

Q.88. Which of the following has the maximum number of vertices ?

SSC CPO 14/03/2019 (Evening)

- (a) Cuboid
(b) Triangular Prism
(c) Hexagonal pyramid
(d) Tetrahedron

Q.89. A diagonal of a quadrilateral is 40 cm. The length of the perpendicular to the opposite ends is 7.5 cm and 8.6 cm. What is the area (in cm^2) of quadrilateral ?

SSC CPO 14/03/2019 (Evening)

(a) 434 (b) 322 (c) 368 (d) 28

Q.90. The area of each square of a chessboard having 64 squares is 4 cm^2 . If there is a border on all the sides of the chessboard of 2 cm, then the perimeter of the chessboard is:

SSC CPO 14/03/2019 (Morning)

(a) 128cm (b) 80cm (c) 70cm (d) 256cm

Q.91. The sides of a triangle are 24cm, 45cm and 51cm. At each of its vertices, circles of radius 10.5cm are drawn. What is the area of the triangle, excluding the portion covered by sectors of the circles?

SSC CPO 13/03/2019 (Evening)

(a) 244.75cm^2 (b) 366.75cm^2
(c) 464.75cm^2 (d) 327.75cm^2

Q.92. The radius of a cylinder is increased by 150 % and its height is decreased by 20 %. What is the percentage increase in its volume?

SSC CPO 12/03/2019 (Morning)

(a) 400% (b) 600% (c) 500% (d) 80%

Practice Questions

SSC CHSL 2023 Tier - 2

Q.93. The diameter of a sphere is increased by 30%. What is the percentage increase in its volume ?

SSC CHSL Tier II (02/11/2023)

(a) 69% (b) 119.7%
(c) 41.7% (d) 90.5%

Q.94. The diameter of a hemispherical bowl is 21 cm. The volume of the bowl is equal to: (use $\pi = \frac{22}{7}$)

SSC CHSL Tier II (02/11/2023)

(a) $2,624.5\text{ cm}^3$ (b) $2,425.5\text{ cm}^3$

(c) $2,725.5\text{ cm}^3$ (d) $2,524.5\text{ cm}^3$

Q.95. The radius and height of a right circular cone is increased by 25%. Find the increase in percentage of the volume of the cone (rounded off to 2 decimal places).

SSC CHSL Tier II (02/11/2023)

(a) 94.31% (b) 95.31%
(c) 94.51% (d) 93.51%

Q.96. A hollow cube is made of paper to have a volume of 512 cubic units. How much paper in square units will be required to make the cube?

SSC CHSL Tier II (10/01/2024)

(a) 328 (b) 348 (c) 288 (d) 384

Q.97. A right rectangular box's measurement are 1.6 m long, 90 cm wide, 60 cm in height. Soap cakes of measurement 6 cm $\times$ 5 cm $\times$ 40 mm are to be packed in the box, so that no space is left. How many cakes can be packed in the box ?

SSC CHSL Tier II (10/01/2024)

(a) 6000 (b) 5600 (c) 6500 (d) 7200

SSC CGL 2023 Tier - 2

Q.98. The ratio of the length, width and height of a closed cuboid is given as 6 : 3 : 2. The total surface area of this cuboid is given as 1800 cm^2 . Find the volume (in cm^3) of this cuboid.

SSC CGL Tier II (26/10/2023)

(a) 4650 (b) 4800 (c) 4500 (d) 4200

Q.99. A solid cube, whose each edge is of length 48 cm, is melted. Identical solid cubes, each of volume 64 cm^3 , are made out of this molten cube, without any wastage. How many such small cubes are obtained ?

SSC CGL Tier II (26/10/2023)

(a) 1728 (b) 1718 (c) 1748 (d) 1738

SSC CPO 2023 Tier - 1

Q.100. The radii of two cylinders are in the ratio of 4 : 5 and their heights are in the ratio of 5 : 2. The ratio of their volume is:

SSC CPO 03/10/2023 (1st Shift)

(a) 9 : 7 (b) 9 : 4 (c) 8 : 5 (d) 2 : 1

Q.101. The cost of painting the total surface area of a 30 m high solid right circular cylinder at the rate of ₹25 per m^2 is ₹18,425. What is the volume (in m^3) of

this cylinder [use $\pi = \frac{22}{7}$]?

SSC CPO 03/10/2023 (2nd Shift)

(a) 1210 (b) 1155 (c) 1145 (d) 1122

Q.102. What will be the difference between the total surface area and the curved surface area of a hemisphere having 4 cm diameter ?

SSC CPO 03/10/2023 (2nd Shift)

(a) $4.4\pi\text{ cm}^2$ (b) $5\pi\text{ cm}^2$
(c) $4\pi\text{ cm}^2$ (d) $8\pi\text{ cm}^2$

Q.103. A conical shape vessel has a radius of 21 cm and has a slant height of 25 cm. If the curved part of the vessel is to be painted white, find the cost (in ₹) of painting at the rate of ₹1.5 per cm^2 .

SSC CPO 03/10/2023 (2nd Shift)

(a) 2475 (b) 825 (c) 1650 (d) 1250

Q.104. The volume of a right circular cone having a base diameter of 14 cm is $196\pi\text{ cm}^3$. Find the perpendicular height of this cone

SSC CPO 03/10/2023 (2nd Shift)

(a) 8 cm (b) 14 cm (c) 10 cm (d) 12 cm

Q.105. A solid sphere made of wax of radius 12 cm is melted and cast into solid hemispheres of radius 4 cm each. Find the number of such solid hemispheres.

SSC CPO 03/10/2023 (3rd Shift)

(a) 14 (b) 54 (c) 28 (d) 27

Q.106. Find the area of a rhombus if the perimeter of the rhombus is 52 cm, and one of its diagonals is 10 cm long.

SSC CPO 03/10/2023 (3rd Shift)

(a) 164 cm^2 (b) 144 cm^2
(c) 160 cm^2 (d) 120 cm^2

Q.107. The curved surface area (CSA) and the total surface area (TSA) of a hemisphere whose radius is 7 cm are:

SSC CPO 03/10/2023 (3rd Shift)

(a) CSA - 308 cm^2 ; TSA - 462 cm^2
(b) CSA - 320 cm^2 ; TSA - 480 cm^2
(c) CSA - 412 cm^2 ; TSA - 544 cm^2
(d) CSA - 350 cm^2 ; TSA - 500 cm^2

Q.108. If the radius of a hemispherical balloon increases from 4 cm to 7 cm as air is pumped into it, find the ratio of the surface area of the new balloon to its original.

SSC CPO 04/10/2023 (1st Shift)

(a) 20:49 (b) 16:21 (c) 49:16 (d) 21:12

Q.109. The total surface area of a square-based right pyramid is 1536 m^2 , of which 37.5% is the area of the base of the pyramid. What is the volume (in m^3) of this pyramid ?

SSC CPO 04/10/2023 (1st Shift)

(a) 3108 (b) 3048 (c) 3072 (d) 3144

Q.110. The area of a rectangular field is 480 m^2 . If the length is 20% more than the breadth, then the length of the

rectangular field is:

SSC CPO 04/10/2023 (2nd Shift)

(a) 24 m (b) 20 m (c) 27 m (d) 30 m

Q.111. If the diameter of a sphere is reduced to its half, then the volume would be :

SSC CPO 04/10/2023 (2nd Shift)

(a) reduced by $\frac{1}{8}$ of the former volume

(b) increased by $\frac{1}{4}$ of the former volume

(c) reduced by $\frac{1}{4}$ of the former volume

(d) increased by $\frac{1}{8}$ of the former volume

Q.112. The heights of two cones are in the ratio 7 : 5 and their diameters are in the ratio 10 : 21. What is the ratio of their

volumes ? (Where $\pi = \frac{22}{7}$)

SSC CPO 04/10/2023 (2nd Shift)

(a) 17 : 21 (b) 14 : 19

(c) 20 : 63 (d) 26 : 47

Q.113. If the base radius and the height of a right circular cone are increased by 10%, then the percentage increase in the volume is:

SSC CPO 04/10/2023 (3rd Shift)

(a) 33.1% (b) 20.2% (c) 22.1% (d) 30.2 %

Q.114. The diameter of a hemisphere is equal to the diagonal of a rectangle of length 4 cm and breadth 3 cm. Find the total surface area (in cm^2) of the hemisphere.

SSC CPO 04/10/2023 (3rd Shift)

(a) 25π (b) $\frac{50\pi}{4}$ (c) $\frac{25\pi}{4}$ (d) $\frac{75\pi}{4}$

Q.115. A cylindrical metallic rod of diameter 2 cm and length 45 cm is melted and converted into wire of uniform thickness and length 5 m. The diameter of the wire is:

SSC CPO 04/10/2023 (3rd Shift)

(a) 5 mm (b) 3 mm (c) 2 mm (d) 6 mm

Q.116. The height and the radius of the base of a right circular cone are in the ratio of 12 : 5. If its volume is 314 cm^3 , then what is the slant height of the cone ? (Use $\pi = 3.14$)

SSC CPO 05/10/2023 (1st Shift)

(a) 11 cm (b) 14 cm (c) 12 cm (d) 13 cm

Q.117. A right square pyramid having lateral surface area is 624 cm^2 . If the length of the diagonal of the square is $24\sqrt{2}$, then the volume of the pyramid is:

SSC CPO 05/10/2023 (1st Shift)

(a) 780 cm^3 (b) 1150 cm^3

(c) 1083 cm^3 (d) 960 cm^3

Q.118. A cylinder has some water in it at a height of 16 cm. If a sphere of radius 9 cm is put into it, then find the rise in the height of the water if the radius of the cylinder is 12 cm.

SSC CPO 05/10/2023 (1st Shift)

(a) 1.6 cm (b) 8.75 cm

(c) 8 cm (d) 6.75 cm

Q.119. A sphere and another solid hemisphere have the same surface area. The ratio of their volumes is:

SSC CPO 05/10/2023 (2nd Shift)

(a) $2\sqrt{3} : 8$ (b) $3\sqrt{3} : 8$

(c) $\sqrt{3} : 4$ (d) $3\sqrt{3} : 4$

Q.120. What will be the area of a plot of quadrilateral shape, one of whose diagonals is 20 m and lengths of the perpendiculars from the opposite vertices on it are 12 m and 18 m, respectively ?

SSC CPO 05/10/2023 (2nd Shift)

(a) 250 m^2 (b) 400 m^2

(c) 200 m^2 (d) 300 m^2

Q.121. Two metallic balls P and Q are such that the diameter of P is four times the diameter of Q. What is the ratio between the volumes of P and Q?

SSC CPO 05/10/2023 (2nd Shift)

(a) 8 : 1 (b) 16 : 1 (c) 64 : 1 (d) 32 : 1

Q.122. If the slant height of a cone is 29 cm and its height is 20 cm, find the ratio between the magnitudes of total surface area and the volume.

SSC CPO 05/10/2023 (3rd Shift)

(a) 3 : 7 (b) 3 : 14 (c) 5 : 14 (d) 7 : 15

Q.123. The radius of a metallic spherical ball is 3 cm. If the metallic ball is melted and recast into x number of hemispheres of radius equal to half the radius of the metallic spherical ball, then find the value of x.

SSC CPO 05/10/2023 (3rd Shift)

(a) 14 (b) 16 (c) 13 (d) 15

Q.124. A hollow cylindrical tube made of plastic is 3 cm thick. If the external diameter is 20 cm and length of the tube is 49 cm, find the volume of the plastic.

SSC CPO 05/10/2023 (3rd Shift)

(a) 7644 cm^3 (b) 7238 cm^3

(c) 7854 cm^3 (d) 7546 cm^3

Q.125. The number of balls, each one is 4 cm in the diameter that can be made from a solid metallic sphere of diameter 32 cm is:

SSC CPO 05/10/2023 (3rd Shift)

(a) 512 (b) 256 (c) 1024 (d) 64

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Q.126. The diameter of a metallic sphere is 6 cm. It is melted and drawn into a wire having a diameter of the cross section as 0.4 cm. Find the length of the wire (in meters).

SSC MTS 01/09/2023 (1st Shift)

(a) 18 (b) 9 (c) 12 (d) 10

Q.127. The area of a trapezium shaped field is 1720 m^2 , the distance between its parallel sides is 40 m and the length of one of the parallel sides is 48 m. What is the length (in m) of the other parallel side?

SSC MTS 01/09/2023 (2nd Shift)

(a) 56 (b) 38 (c) 42 (d) 34

Q.128. From a cuboid of dimension of $8 \times 6 \times 6$ units, small cubes of dimension $0.5 \times 0.5 \times 0.5$ unit are cut. How many small cubes can be cut from the big cuboid ?

SSC MTS 01/09/2023 (3rd Shift)

(a) 576 (b) 1152 (c) 2304 (d) 1552

Q.129. The radius and the height of a right circular cone are in the ratio 3 : 4. If its volume is 301.44 m^3 , find the slant height (use $\pi = 3.14$).

SSC MTS 04/09/2023 (1st Shift)

(a) 5 cm (b) 20 cm (c) 10 cm (d) 15 cm

Q.130. Water is pouring into a cuboidal reservoir at the rate of 50 litres per minute. The volume of the reservoir is 120 m^3 . How many hours will it take to fill the reservoir ?

SSC MTS 04/09/2023 (3rd Shift)

(a) 35 (b) 45 (c) 40 (d) 30

Q.131. Find the number of solid lead balls each 2 cm in diameter that can be made from solid sphere made up of lead of diameter 14 cm.

SSC MTS 05/09/2023 (1st Shift)

(a) 433 (b) 343 (c) 430 (d) 340

Q.132. What is the total surface area (in cm^2) of a cone with a base radius of 28 cm and height of 45 cm? (use $\pi = \frac{22}{7}$)

SSC MTS 05/09/2023 (1st Shift)

(a) 8168 (b) 6288 (c) 7128 (d) 9216

Q.133. A right circular cone is 9.8 cm high with the radius of its base 4 cm. It is melted and recast into a right circular cone with the radius of its base 3.5 cm.

Find its height (use $\pi = \frac{22}{7}$).

SSC MTS 05/09/2023 (2nd Shift)

(a) 21.8 cm (b) 28.1 cm

(c) 12.8 cm (d) 18.2 cm

Q.134. Using the middle points of sides of a square with an area of 36 square units, another square is made. What is the area of a new square made in square units?

SSC MTS 05/09/2023 (2nd Shift)

(a) 24 (b) 27 (c) 18 (d) 21

Q.135. The length of the median of an equilateral triangle is $6\sqrt{3}$ cm. The area of the triangle is :

SSC MTS 05/09/2023 (3rd Shift)

(a) $36\sqrt{3}$ cm² (b) 36 cm²
(c) 72 cm² (d) $72\sqrt{3}$ cm²

Q.136. What is the length of the longest pole that can be placed in an indoor stadium 24 m long, 18 m wide and 16 m high ?

SSC MTS 06/09/2023 (1st Shift)

(a) 38 m (b) 34 m (c) 42 m (d) 36 m

Q.137. Metallic spheres of radii 6 cm, 8 cm and 10 cm, respectively, are melted to form a single solid sphere. Find the radius of the resulting sphere (in cm).

SSC MTS 06/09/2023 (2nd Shift)

(a) 18 (b) 12 (c) 24 (d) 16

Q.138. The areas of three adjacent faces of a cuboid are p , q and r unit², respectively. What is its volume ?

SSC MTS 06/09/2023 (3rd Shift)

(a) $p^3 q^3 r^3$ unit³ (b) $\sqrt{pqr}$ unit³
(c) pqr unit³ (d) $\sqrt[3]{pqr}$ unit³

Q.139. The radius of a cone is 10.5 cm and the area of the curved surface is 396 cm². Find its slant height. Take $\pi = \frac{22}{7}$

SSC MTS 08/09/2023 (1st Shift)

(a) 9 cm (b) 11.5 cm (c) 12 cm (d) 8.5 cm

Q.140. Find the volume of a cuboid whose dimensions are 12.2 cm, 16 cm and 8.5 cm.

SSC MTS 08/09/2023 (1st Shift)

(a) 3072 cm³ (b) 1659.2 cm³
(c) 832 cm³ (d) 1664 cm³

Q.141. A cylinder has a total surface area of 64π square units, where the height of the cylinder is equal to its radius. If the height is doubled and the radius is halved, then the total surface area of the new cylinder is _____ square units.

SSC MTS 08/09/2023 (3rd Shift)

(a) 54π (b) 40π (c) 64π (d) 48π

Q.142. Find the area of a triangle whose base is 7 cm and corresponding height is 82 cm.

SSC MTS 08/09/2023 (3rd Shift)

(a) 191 cm² (b) 574 cm²
(c) 143 cm² (d) 287 cm²

Q.143. A hollow metal sphere of internal and external radii 5 cm and 6 cm, respectively, is melted into a solid cone of base radius 5.2 cm. What is the height (in cm, rounded off to 1 decimal place) of the cone ?

SSC MTS 11/09/2023 (1st Shift)

(a) 11.8 (b) 12.8 (c) 13.5 (d) 14.5

Q.144. Find the area (in cm²) of the rhombus, lengths of whose diagonals are, respectively, 7.5 cm and 5.6 cm.

SSC MTS 11/09/2023 (3rd Shift)

(a) 21 (b) 28 (c) 27 (d) 30

Q.145. Some solid metallic right circular cones, each with diameter of the base 8 cm and height 6 cm are melted to form a solid sphere of radius 12 cm. Find the number of right circular cones which are melted.

SSC MTS 11/09/2023 (3rd Shift)

(a) 60 (b) 66 (c) 72 (d) 70

Q.146. Find the diameter of the sphere whose volume is $381\frac{6}{7}$ m³.

SSC MTS 12/09/2023 (1st Shift)

(a) 8 m (b) 7 m (c) 9 m (d) 18 m

Q.147. If a hollow sphere of an external and internal diameters of 6 cm and 4 cm, respectively, is melted into a cone of base diameter 8 cm, then the height of the cone is:

SSC MTS 12/09/2023 (1st Shift)

(a) $4\frac{3}{4}$ cm (b) $3\frac{1}{4}$ cm
(c) $5\frac{1}{4}$ cm (d) $2\frac{3}{4}$ cm

Q.148. A conical tent is set to accommodate 15 persons. Each person must have 5 square metres of space on the ground and 30 cubic metres of air to breathe. What is the height of the cone?

SSC MTS 12/09/2023 (2nd Shift)

(a) 15 m (b) 22 m (c) 13 m (d) 18 m

Q.149. A steel sphere of radius 1 cm is melted and cast into a wire of diameter 4 mm, then the length of the wire is:

SSC MTS 12/09/2023 (3rd Shift)

(a) $31\frac{1}{3}$ cm (b) $35\frac{1}{7}$ cm
(c) $37\frac{1}{7}$ cm (d) $33\frac{1}{3}$ cm

Q.150. The length of an edge of a cube is 3.5 cm. Find its total surface area.

SSC MTS 13/09/2023 (1st Shift)

(a) 73.5 cm² (b) 36 cm²
(c) 27 cm² (d) 154 cm²

Q.151. If the diameter of the base of a cone is 70 cm and its curved surface

area is 4070 cm², then what will be its

volume (in cm³)? (use $\pi = \frac{22}{7}$)

SSC MTS 13/09/2023 (1st Shift)

(a) 1180 (b) 16,300 (c) 15,400 (d) 17,600

Q.152. The longest diagonal of a cube is $6\sqrt{3}$ cm. Find the total surface area of the cube.

SSC MTS 13/09/2023 (3rd Shift)

(a) 648 cm² (b) 216 cm²
(c) 36 cm² (d) 324 cm²

Q.153. A solid metallic cuboid of length 11 cm, breadth 8 cm and height 6 cm is melted and reshaped to form a cone of height 14 cm. What is the radius of the cone thus formed ?

SSC MTS 13/09/2023 (3rd Shift)

(a) 6 cm (b) 9 cm (c) 7 cm (d) 8 cm

Q.154. The length, breadth and height of a cuboid are in the ratio of 6 : 4 : 3. If the total surface area of the cuboid is 972 cm², what is the length (in cm) of the diagonal of the cuboid ?

SSC MTS 14/09/2023 (1st Shift)

(a) $5\sqrt{65}$ (b) $3\sqrt{61}$ (c) $3\sqrt{65}$ (d) $5\sqrt{61}$

Q.155. A triangle has its sides 5 cm, 12 cm and 13 cm. What will be the area of the triangle formed by joining the mid point of the three sides ?

SSC MTS 14/09/2023 (2nd Shift)

(a) $7\frac{1}{2}$ cm² (b) $19\frac{1}{2}$ cm²
(c) $13\frac{1}{2}$ cm² (d) 15 cm²

Q.156. The height of a right circular cone is 20 cm, and its base diameter is 30 cm. What is the curved surface area and the total surface area of the right circular cone, respectively ?

SSC MTS 14/09/2023 (2nd Shift)

(a) 335π cm² and 525π cm²
(b) 390 cm² and 475π cm²
(c) 450 cm² and 625π cm²
(d) 375π cm² and 600π cm²

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Q.157. If a right circular cone of height 24 cm has the circumference of its base 42π cm, then the volume of the cone is (use $\pi = \frac{22}{7}$) :

SSC CHSL 02/08/2023 (1st Shift)

(a) 15211 cm³ (b) 11088 cm³
(c) 12034 cm³ (d) 21011 cm³

Q.158. Aman wants to get his spherical overhead water tank having a diameter of 7 m, painted. If the cost of painting is ₹13 per square metre, find the cost of

painting Aman's spherical overhead water tank. (Use $\pi = \frac{22}{7}$)

SSC CHSL 02/08/2023 (2nd Shift)

- (a) ₹1,872 (b) ₹1,984
(c) ₹2,002 (d) ₹1,926

Q.159. If the height and slant height of a cone are 21 cm and 29 cm, respectively.

Find the volume. (use $\pi = \frac{22}{7}$)

SSC CHSL 02/08/2023 (3rd Shift)

- (a) 8800 cm³ (b) 8780 cm³
(c) 8440 cm³ (d) 8080 cm³

Q.160. A circle with centre O has a chord AB that is 20 cm in length. If the radius of the circle is 12 cm, then the area of triangle AOB is:

SSC CHSL 02/08/2023 (3rd Shift)

- (a) $20\sqrt{15}$ cm² (b) $22\sqrt{11}$ cm²
(c) $20\sqrt{11}$ cm² (d) $22\sqrt{15}$ cm²

Q.161. The length of each side of an equilateral triangle is 22 cm. Find the area (in cm²) of this triangle.

SSC CHSL 02/08/2023 (4th Shift)

- (a) 121 (b) $242\sqrt{3}$ (c) $121\sqrt{3}$ (d) 242

Q.162. The perimeter of a parallelogram is 48 cm. If the height of the parallelogram is 6 cm and the length of the adjacent side is 8 cm. find its area.

SSC CHSL 03/08/2023 (1st Shift)

- (a) 90 cm² (b) 80 cm²
(c) 84 cm² (d) 96 cm²

Q.163. In an isosceles triangle, if the unequal side is 8 cm and equal side is 5 cm, then the area of the triangle is:

SSC CHSL 03/08/2023 (1st Shift)

- (a) 12 cm² (b) 25 cm²
(c) 6 cm² (d) 11 cm²

Q.164. Find the volume of the largest possible sphere circumscribed by a cube of edge 7 cm.

SSC CHSL 03/08/2023 (2nd Shift)

- (a) 179.67 cm³ (b) 169.67 cm³
(c) 189.67 cm³ (d) 199.67 cm³

Q.165. The length, breadth and height of a room are 7m, 5m and 3m, respectively. Find the cost of whitewashing the walls of the room and the ceiling at the rate of ₹16 per m².

SSC CHSL 03/08/2023 (3rd Shift)

- (a) ₹1,712 (b) ₹1,742
(c) ₹1,698 (d) ₹1,684

Q.166. The difference between the total surface area and the lateral surface area of a cube of side 12 cm is:

SSC CHSL 03/08/2023 (4th Shift)

- (a) 292 cm² (b) 290 cm²
(c) 288 cm² (d) 286 cm²

Q.167. If a map of a city was drawn on a scale of 0.2 cm to a km, what area would represent a country if it has an area of 22500 km²?

SSC CHSL 04/08/2023 (2nd Shift)

- (a) 2500 cm² (b) 2250 cm²
(c) 5625 cm² (d) 900 cm²

Q.168. The sides of the three cubes of metal are 15 cm, 18 cm and 25 cm, respectively. Find the side (in cm) of the new cube formed by melting these cubes together

SSC CHSL 04/08/2023 (2nd Shift)

- (a) $9\sqrt[3]{388}$ (b) $6\sqrt[3]{388}$ (c) $7\sqrt[3]{388}$ (d) $4\sqrt[3]{388}$

Q.169. A shopkeeper usually makes one big besan laddoo (spherical in shape) with a radius of 9 cm. With the same laddoo, How many besan laddoos of radius 3 cm can be made? (Use $\pi = \frac{22}{7}$)

SSC CHSL 04/08/2023 (3rd Shift)

- (a) 9 (b) 18 (c) 36 (d) 27

Q.170. How many litres of water can a hemispherical tank of radius 2.1 m contain? (Take $\pi = \frac{22}{7}$)

SSC CHSL 04/08/2023 (4th Shift)

- (a) 18984 (b) 19324 (c) 18404 (d) 19404

Q.171. The area of a triangle is half the area of a square. The perimeter of the square is 172 cm. What is the area of the triangle (in cm²) ?

SSC CHSL 04/08/2023 (4th Shift)

- (a) 924.5 (b) 1984 (c) 1849 (d) 992

Q.172. The radius of a circle is 21 cm. What will be the area (in cm²) of a sector of the circle that subtends a 22.5° angle at the centre? [Use $\pi = \frac{22}{7}$]

SSC CHSL 07/08/2023 (2nd Shift)

- (a) 86.6205 (b) 86.0625
(c) 86.6250 (d) 86.6025

Q.173. A solid rectangular block of metal of dimensions 88 cm × 21 cm × 21 cm, is melted to form a solid sphere. What is its radius (in cm) ? Take ($\pi = \frac{22}{7}$)

SSC CHSL 07/08/2023 (2nd Shift)

- (a) 28 (b) 35 (c) 21 (d) 14

Q.174. A sphere of diameter 18 cm has been melted to make cylinder of radius 3 cm and height 2 cm. The total number of cylinder made out will be:

SSC CHSL 07/08/2023 (3rd Shift)

- (a) 40 (b) 100 (c) 54 (d) 45

Q.175. A cuboidal water tank is 12 m long, 9 m wide and 7.5 m deep. How many litres of water can it hold?

SSC CHSL 08/08/2023 (1st Shift)

- (a) 840000 (b) 805000
(c) 780000 (d) 810000

Q.176. How many solid spheres are made if a metallic cone of radius 12 cm and height 24 cm is melted into sphere of radius 2 cm each ?

SSC CHSL 08/08/2023 (3rd Shift)

- (a) 105 (b) 101 (c) 108 (d) 103

Q.177. What is the area of a triangular flower bed of a garden whose perimeter is 20 metres and two sides are 5 metres and 6 metres.

SSC CHSL 08/08/2023 (4th Shift)

- (a) $10\sqrt{2}$ m² (b) $2\sqrt{10}$ m²
(c) $5\sqrt{2}$ m² (d) $4\sqrt{5}$ m²

Q.178. A solid cuboid is melted to form a number of cubes. If the length, breadth and height of a cuboid are 20 cm, 16 cm and 8 cm, and the edge of each cube is 4 cm, then the number of cubes is:

SSC CHSL 09/08/2023 (2nd Shift)

- (a) 50 (b) 44 (c) 40 (d) 48

Q.179. The length, breadth of a cuboidal box are 25 cm and 15 cm, respectively. If the volume of the box is 1875 cm³, then the height of the box (in cm) is:

SSC CHSL 09/08/2023 (3rd Shift)

- (a) 4.0 (b) 5.0 (c) 5.5 (d) 4.5

Q.180. The diameter of a sphere is 14 cm, then the volume of this sphere is (use $\pi = \frac{22}{7}$):

SSC CHSL 09/08/2023 (4th Shift)

- (a) $1437\frac{1}{3}$ cm³ (b) $1683\frac{1}{3}$ cm³
(c) $1521\frac{2}{3}$ cm³ (d) $2125\frac{1}{3}$ cm³

Q.181. The perimeter of a sector of a circle of radius 7.4 cm is 20.6 cm. The area of this sector is:

SSC CHSL 10/08/2023 (3rd Shift)

- (a) 42.92 cm² (b) 5.8 cm²
(c) 10.73 cm² (d) 21.46 cm²

Q.182. The total surface area of a solid hemisphere of diameter 14 cm is (use $\pi = \frac{22}{7}$):

SSC CHSL 11/08/2023 (1st Shift)

- (a) 522 cm² (b) 462 cm²
(c) 428 cm² (d) 584 cm²

Q.183. The perimeter of a minor sector of a circle of radius 5 cm is 15.5 cm. Find the area of the minor sector (use $\pi = \frac{22}{7}$).

SSC CHSL 11/08/2023 (1st Shift)

- (a) 17.53 cm² (b) 13.75 cm²
(c) 13.57 cm² (d) 17.35 cm²

Q.184. A copper wire is bent in the form of a square and it encloses an area of 30.25 cm². If the same wire is bent to form a circle, then find the area of the circle. (use $\pi = \frac{22}{7}$).

SSC CHSL 11/08/2023 (2nd Shift)

- (a) 38.50 cm² (b) 42.25 cm²
(c) 35 cm² (d) 30.25 cm²

Q.185. A metallic sphere of radius 21 cm is melted and then recast into smaller cones each with a radius of 7 cm, and a height of 3 cm. Find the number of cones obtained.

SSC CHSL 11/08/2023 (4th Shift)

- (a) 225 (b) 325 (c) 522 (d) 252

Q.186. If the slant height and radius of a right circular cone are 28 cm and 21 cm, respectively, then the total surface area of the right circular cone (in cm²) is:

(Taken $\pi = \frac{22}{7}$)

SSC CHSL 14/08/2023 (1st Shift)

- (a) 3234 (b) 3342 (c) 3243 (d) 3324

Q.187. The sides of a triangle are of length 8 cm, 15 cm and 17 cm. Find the area of the triangle.

SSC CHSL 14/08/2023 (1st Shift)

- (a) 65 cm² (b) 75 cm²
(c) 60 cm² (d) 70 cm²

Q.188. The total surface area of a cylinder whose radius is 6cm and height is 8 cm is (use $\pi = \frac{22}{7}$).

SSC CHSL 14/08/2023 (2nd Shift)

- (a) 528 cm² (b) 575 cm²
(c) 658 cm² (d) 625 cm²

Q.189. Find the surface area of a sphere whose radius is 3.5 cm (use $\pi = \frac{22}{7}$).

SSC CHSL 14/08/2023 (3rd Shift)

- (a) 154 cm² (b) 152 cm²
(c) 146 cm² (d) 160 cm²

Q.190. The curved surface area of a right circular cone of diameter 42 cm is 990 cm². What is the slant height (in cm) of the cone ? (use $\pi = \frac{22}{7}$).

SSC CHSL 14/08/2023 (4th Shift)

- (a) 10 (b) 15 (c) 24 (d) 12

Q.191. A cylindrical pillar is 84 cm in diameter and 5 m in height. Find the cost of painting the curved surface of the pillar at the rate of ₹15 per m².

(use $\pi = \frac{22}{7}$).

SSC CHSL 17/08/2023 (1st Shift)

- (a) ₹198 (b) ₹184 (c) ₹208 (d) ₹216

Q.192. A hemispherical bowl is made of 1 cm thick steel. The inside radius of the bowl is 4 cm. The volume of the steel (in cm³) used in making the bowl is:

SSC CHSL 17/08/2023 (2nd Shift)

- (a) $40\frac{2}{3}\pi$ (b) $27\frac{1}{3}\pi$
(c) $41\frac{1}{3}\pi$ (d) $35\frac{1}{3}\pi$

Q.193. A rod of 5 cm radius and 52 cm length is converted into a wire of 100 cm length of uniform thickness. The radius of the wire (in cm) is:

SSC CHSL 17/08/2023 (4th Shift)

- (a) $\sqrt{13}$ (b) $\sqrt{15}$ (c) $\sqrt{23}$ (d) $\sqrt{17}$

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Q.194. The volume of a sphere of radius

4.2 cm is: (use $\pi = \frac{22}{7}$)

SSC CGL 14/07/2023 (1st shift)

- (a) 278.234 cm³ (b) 312.725 cm³
(c) 297.824 cm³ (d) 310.464 cm³

Q.195. If the total surface area of a cube is 24 sq. units, then what is the volume of the cube?

SSC CGL 14/07/2023 (2nd shift)

- (a) 8 cu.units (b) 16 cu.units
(c) 10 cu.units (d) 4 cu.units

Q.196. Ranu carries water to school in a cylindrical flask with diameter 12 cm and height 21 cm. Determine the amount of water that she can carry in the flask.

(Use $\pi = \frac{22}{7}$)

SSC CGL 14/07/2023 (4th shift)

- (a) 2372 cm³ (b) 2370 cm³
(c) 2376 cm³ (d) 2374 cm³

Q.197. The volume of a cone with height equal to radius, and slant height 5 cm is:

SSC CGL 17/07/2023 (1st shift)

- (a) $\frac{125\pi}{12\sqrt{3}}$ cm³ (b) $\frac{125\pi}{6\sqrt{3}}$ cm³
(c) $\frac{125\pi}{12\sqrt{2}}$ cm³ (d) $\frac{125\pi}{6\sqrt{2}}$ cm³

Q.198. If the area of a circle is 616 cm² and a chord XY = 10 cm, then find the perpendicular distance from the center of the circle to the chord XY.

SSC CGL 17/07/2023 (1st shift)

- (a) $\sqrt{171}$ cm (b) $\sqrt{161}$ cm
(c) $\sqrt{117}$ cm (d) $\sqrt{181}$ cm

Q.199. The area of two triangles is in the ratio 5 : 3 and their heights are in the

ratio 5 : 7. Find the ratio of their bases.

SSC CGL 17/07/2023 (1st shift)

- (a) 3 : 5 (b) 7 : 3 (c) 7 : 5 (d) 2 : 3

Q.200. If the side of an equilateral triangle is increased by 34%, then by what percentage will its area increase?

SSC CGL 17/07/2023 (2nd shift)

- (a) 70.65% (b) 79.56%
(c) 68.25% (d) 75.15%

Q.201. An arc of length 23.1 cm subtends an 18° angle at the centre. What is the area of the circle ?

[Use $\pi = \frac{22}{7}$]

SSC CGL 17/07/2023 (2nd shift)

- (a) 16978.50 cm² (b) 16988.50 cm²
(c) 16878.50 cm² (d) 16798.50 cm²

Q.202. What is the volume of a cube if the perimeter of one face of the cube is 40 cm ?

SSC CGL 17/07/2023 (3rd shift)

- (a) 1000 cm³ (b) 820 cm³
(c) 600 cm³ (d) 800 cm³

Q.203. Find the area of the sector of a circle with radius 5 cm and angle 60° (rounded off to one decimal).

SSC CGL 17/07/2023 (3rd shift)

- (a) 12.8 cm² (b) 14.1 cm²
(c) 13.1 cm² (d) 15.1 cm²

Q.204. A sector of a circle of radius 10 cm is formed at 60° angle at the centre. What will be its area (take $\pi = 3.14$)?

SSC CGL 17/07/2023 (4th shift)

- (a) 52.33 cm² (b) 75.28 cm²
(c) 60.67 cm² (d) 55.00 cm²

Q.205. A spherical ball of lead, 3 cm in diameter, is melted and recast into three spherical balls. The diameters of two of these balls are $\frac{3}{2}$ cm and 2cm, respectively. Find the diameter of the third ball.

SSC CGL 17/07/2023 (4th shift)

- (a) 2.1 cm (b) 3.3 cm (c) 3 cm (d) 2.5 cm

Q.206. The area of the sector of a circle of radius 12 cm is 32 π cm². Find the length of the corresponding arc of the sector.

SSC CGL 17/07/2023 (4th shift)

- (a) $\frac{16}{3}\pi$ cm (b) $\frac{13}{3}\pi$ cm
(c) $\frac{10}{3}\pi$ cm (d) $\frac{8}{3}\pi$ cm

Q.207. The ratio of the length of each equal side and the third side of an isosceles triangle is 3 : 5. If the area of the triangle is $30\sqrt{11}$ cm², then the length of the third side (in cm) is:

SSC CGL 18/07/2023 (2nd shift)

(a) $10\sqrt{6}$ (b) $5\sqrt{6}$ (c) $13\sqrt{6}$ (d) $11\sqrt{6}$

Q.208. A conical tent with radius 6 units and height 8 units is to be made by canvas. How much canvas is needed to make the tent? (Rounded off of two places of decimals)

SSC CGL 18/07/2023 (3rd shift)

(a) 188.57 units (b) 155.87 units
(c) 166.57 units (d) 177.55 units

Q.209. Area of a sector is 1848 m^2 and the central angle of the sector is 270° . Find the radius of the circle.

(Take $\pi = \frac{22}{7}$)

SSC CGL 18/07/2023 (4th shift)

(a) 784 m (b) 22 m (c) 27 m (d) 28 m

Q.210. A conical tent has a radius of 7 m and a vertical height of 24 m. How many full bags of rice can be emptied in it if the space occupied by the rice in each bag is 2 m^3 ?

SSC CGL 19/07/2023 (1st shift)

(a) 566 (b) 616 (c) 716 (d) 650

Q.211. Find the length of the arc of the sector of a circle of diameter 7 cm with a central angle of 108° . [Use $\pi = \frac{22}{7}$]

SSC CGL 19/07/2023 (1st shift)

(a) 6.6 cm (b) 5.6 cm
(c) 13.2 cm (d) 11.2 cm

Q.212. The lateral area of a cylinder is 3168 cm^2 and its height is 48 cm. Find the volume.

SSC CGL 19/07/2023 (2nd shift)

(a) 5244 cm^3 (b) 5544 cm^3
(c) 5644 cm^3 (d) 16632 cm^3

Q.213. The wheel of a lorry has radius 182 cm. The number of revolutions (approximately) per minute the lorry wheel will make is ____ (if the speed of the lorry is 66 km/h).

SSC CGL 19/07/2023 (2nd shift)

(a) 100 (b) 96 (c) 1144 (d) 66

Q.214. To have a surface area of 9π square units of a ball, what should be the diameter (in units) of the ball?

SSC CGL 19/07/2023 (3rd shift)

(a) 2.5 (b) 2 (c) 1.5 (d) 3

Q.215. If the surface area of a cube is 5046 cm^2 , then the volume of the cube is:

SSC CGL 19/07/2023 (4th shift)

(a) 26189 cm^3 (b) 25145 cm^3 (c) 22812 cm^3 (d) 24389 cm^3

Q.216. A solid cone with curved surface area twice its base area has slant height of $6\sqrt{3}$ cm. Its height is :

SSC CGL 20/07/2023 (2nd shift)

(a) $6\sqrt{2}$ cm (b) 9 cm (c) 6 cm (d) $3\sqrt{6}$ cm

Q.217. Find the total surface area (in cm^2) of a cuboid having dimensions 5 cm, 7 cm and 11 cm.

SSC CGL 20/07/2023 (2nd shift)

(a) 385 (b) 334 (c) 343 (d) 167

Q.218. The slant height of a cone is 10 cm, and the radius is 6 cm. What is the total surface area of the cone? (Take $\pi = 3.14$)

SSC CGL 20/07/2023 (3rd shift)

(a) 292.4 cm^2 (b) 312.2 cm^2
(c) 305.4 cm^2 (d) 301.44 cm^2

Q.219. The width of a rectangle is 2 m less than its length. If the perimeter of the rectangle is 68 m, then what is the length (in metres) of the rectangle?

SSC CGL 20/07/2023 (4th shift)

(a) 20 (b) 18 (c) 16 (d) 17

Q.220. The length, breadth and height of a cuboid are in the ratio 1 : 2 : 3. The length, breadth and height of the cuboid are increased by 200%, 300% and 300%, respectively. Then compared to the original volume, the increase in the volume of the cuboid will be:

SSC CGL 20/07/2023 (4th shift)

(a) 47 times (b) 26 times
(c) 17 times (d) 15 times

Q.221. The volume of a cuboid is twice that of a cube. If the dimensions of the cuboid are (8 m × 8 m × 16 m), the total surface area of the cube is:

SSC CGL 21/07/2023 (1st shift)

(a) 316 m^2 (b) 288 m^2
(c) 324 m^2 (d) 384 m^2

Q.222. The perimeter of an isosceles triangle is 100 cm. If the base is 36 cm, then find its semi perimeter (in centimetres).

SSC CGL 21/07/2023 (2nd shift)

(a) 64 (b) 60 (c) 45 (d) 50

Q.223. Three solid metallic spheres of radii 1 cm, 6 cm and 8 cm, respectively, are melted and recast into a single solid sphere. The radius of the new sphere so formed is:

SSC CGL 21/07/2023 (3rd shift)

(a) 9.0 cm (b) 5.9 cm (c) 7.7 cm (d) 8.5 cm

Q.224. The area of the sector of a circle of radius 28 cm is 112 cm^2 . Find the length of the corresponding arc of the sector.

SSC CGL 21/07/2023 (3rd shift)

(a) 4 cm (b) 8 cm (c) 6 cm (d) 5 cm

Q.225. If the volume of one brick is

0.0014 m^3 , then how many bricks will be required to construct a wall of length 14 m, breadth 0.125 m and height 5 m?

SSC CGL 21/07/2023 (3rd shift)

(a) 5250 (b) 3250 (c) 6250 (d) 4250

Q.226. An arc on a circle, whose length is 19.25 cm, subtends an 18° angle at the centre. What is the area of the circle?

[Use $\pi = \frac{22}{7}$]

SSC CGL 21/07/2023 (4th shift)

(a) 11796.625 cm^2 (b) 11786.625 cm^2
(c) 11780.625 cm^2 (d) 11790.625 cm^2

Q.227. If the inner radius of a hemispherical bowl is 5 cm and its thickness is 0.25 cm, then find the volume of the material required in making the bowl. (use $\pi = \frac{22}{7}$) (Rounded up to two places of decimals).

SSC CGL 21/07/2023 (4th shift)

(a) 34 cm^3 (b) 44 cm^3
(c) 45.34 cm^3 (d) 41.28 cm^3

Q.228. A one metre pipe is made with inner diameter equal to outer radius. How much material (in cubic units) is required to make the pipe, if it can hold $\frac{88}{7}$ cubic metres water in it?

SSC CGL 24/07/2023 (1st shift)

(a) 37.7 (b) 35.5 (c) 33.3 (d) 36.6

Q.229. The ratio of the length to width of a certain rectangle is 3 : 2 and the area is 150 cm^2 . The perimeter of the rectangle (in cm) is :

SSC CGL 24/07/2023 (1st shift)

(a) 20 (b) 30 (c) 50 (d) 40

Q.230. The length of each side of a triangle is 12 cm. What is the length of the circumradius of the triangle?

SSC CGL 24/07/2023 (1st shift)

(a) $8\sqrt{3}$ cm (b) $2\sqrt{3}$ cm
(c) $6\sqrt{3}$ cm (d) $4\sqrt{3}$ cm

Q.231. The length, breadth, and height of a room are 10 m, 8 m and 6 m respectively. Find the cost of white washing the walls of the room and the ceiling at the rate of ₹7.50 per m^2 .

SSC CGL 24/07/2023 (2nd shift)

(a) ₹2,220 (b) ₹1,850
(c) ₹2,150 (d) ₹2,000

Q.232. If the diameter of a sphere is 7 cm, then find the volume of the sphere.

(use $\pi = \frac{22}{7}$) (Rounded up to two places of decimal).

SSC CGL 24/07/2023 (3rd shift)

(a) 184 cm^3 (b) 201.34 cm^3
(c) 128.87 cm^3 (d) 179.67 cm^3

Q.233. The side of an equilateral triangle is 12 cm. What is the area (in cm^2 , rounded off to 2 decimal places) of the triangle? Given: $\sqrt{3} = 1.732$

SSC CGL 24/07/2023 (4th shift)

(a) 68.07 (b) 63.89 (c) 62.35 (d) 65.23

Q.234. A ball is to be made with inner radius of 2 units and outside radius of 3 units. How much material is required to make the ball?

SSC CGL 24/07/2023 (4th shift)

(a) $\frac{19}{3}\pi$ (b) 19π (c) $\frac{76}{3}\pi$ (d) π

Q.235. A metallic cube has each of its side of length 12 cm. It is melted and recast into three small cubes. Out of these cubes, two have their sides as 6 cm and 8 cm, respectively. The length of each side of the third cube is:

SSC CGL 25/07/2023 (1st shift)

(a) 7 cm (b) 9 cm (c) 10 cm (d) 5 cm

Q.236. The field is 150 m long and 80 m wide. How many times (rounded off to 1 decimal place) is its perimeter to the length of its diagonal?

SSC CGL 25/07/2023 (1st shift)

(a) 2.3 (b) 2.5 (c) 2.7 (d) 3.1

Q.237. The length, breadth and height of a hall are 10 m, 20 m and 15m respectively. Find the cost of whitewashing the walls of the inside of the hall and ceiling at the rate of ₹10.20/ m^2

SSC CGL 25/07/2023 (2nd shift)

(a) ₹13,394 (b) ₹11,220

(c) ₹15,320 (d) ₹16,542

Q.238. What is the area of the sector of a circle of radius 8 cm and formed by an arc of length 12 cm?

SSC CGL 25/07/2023 (3rd shift)

(a) 45 cm^2 (b) 47 cm^2

(c) 48 cm^2 (d) 84 cm^2

Q.239. The curved surface area of a solid circular cylinder of height 12 cm is 2640 cm^2 . What is the volume (in cm^3) of the cylinder? (Take $\pi = \frac{22}{7}$)

SSC CGL 25/07/2023 (3rd shift)

(a) 46200 (b) 37900

(c) 42000 (d) 55200

Q.240. The length of the longest diagonal of a cube is $7\sqrt{3}\text{ cm}$. Find its volume (in cm^3).

SSC CGL 25/07/2023 (4th shift)

(a) $343\sqrt{3}$ (b) $49\sqrt{3}$ (c) 334 (d) 343

Q.241. A cuboid with sides 4, 6 and 8 units is covered with paper. The paper is removed and a square is made from it.

What is the side (in units) of the square? SSC CGL 25/07/2023 (4th shift)

(a) 12.24 (b) 16.62 (c) 14.42 (d) 12.62

Q.242. If the slant height of a cone is 60 cm and the radius of its base is 21 cm, then find its curved surface area.

(use $\pi = \frac{22}{7}$)

SSC CGL 26/07/2023 (1st shift)

(a) 3880 cm^2 (b) 3820 cm^2

(c) 3960 cm^2 (d) 4020 cm^2

Q.243. What is the surface area (in cm^2) of a spherical sculpture whose radius is 35 cm? (Take $\pi = \frac{22}{7}$)

SSC CGL 26/07/2023 (2nd shift)

(a) 16540 (b) 15700

(c) 15400 (d) 14500

Q.244. The area of a sector of a circle is 110 cm^2 and the central angle of the sector is 56° , what is the radius of the circle? (Take $\pi = \frac{22}{7}$)

SSC CGL 26/07/2023 (2nd shift)

(a) 35 cm (b) 20 cm (c) 25 cm (d) 15 cm

Q.245. A petrol tank is of length 50 m, breadth 25 m, and depth 10 m. How many litres of petrol can be accommodated in the petrol tank when it is completely filled?

SSC CGL 26/07/2023 (2nd shift)

(a) 1,25,00,000 (b) 12,50,000

(c) 1,25,000 (d) 12500

Q.246. Find the area of an equilateral triangle whose sides are 16 cm each.

SSC CGL 26/07/2023 (3rd shift)

(a) $60\sqrt{3}\text{ cm}^2$ (b) $62\sqrt{3}\text{ cm}^2$

(c) $64\sqrt{3}\text{ cm}^2$ (d) $66\sqrt{3}\text{ cm}^2$

Q.247. If the side of a cubical box is 12 cm, then find its total surface area.

SSC CGL 26/07/2023 (3rd shift)

(a) 952 cm^2 (b) 864 cm^2

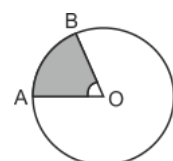
(c) 664 cm^2 (d) 792 cm^2

Q.248. If the volume of a cube is 4096 cm^3 , then what is the length of the side of the cube?

SSC CGL 26/07/2023 (4th shift)

(a) 12 cm (b) 18 cm (c) 16 cm (d) 14 cm

Q.249. In the following figure, the length of arc AB is equal to five times of the radius r of the circle. Find the area of sector AOB?



SSC CGL 27/07/2023 (1st shift)

(a) $2.5r^2$ (b) $\frac{1}{2}r^2$ (c) $3r^2$ (d) $2r^2$

Q.250. A hemispherical bowl has a radius of 7 cm. It is to be painted from inside as well as outside. Find the cost of painting it at the rate of Rs.40 per 20 cm^2 ? (Take the thickness of the bowl as negligible).

SSC CGL 27/07/2023 (1st shift)

(a) Rs.1,232 (b) Rs.1,125

(c) Rs.1,432 (d) Rs.1,550

Q.251. The length of a hall is 21 m and the width is 14 m. The sum of the areas of the floor and the flat roof is equal to the sum of the areas of the four walls. Find the height of the hall.

SSC CGL 27/07/2023 (2nd shift)

(a) 8.4 m (b) 8.2 m (c) 8.6 m (d) 8.8 m

Q.252. A circular arc whose radius is 4 cm makes an angle 45° at the centre. Find the perimeter of the sector formed.

(Take $\pi = \frac{22}{7}$)

SSC CGL 27/07/2023 (3rd shift)

(a) $\frac{78}{7}\text{ cm}$ (b) $\frac{72}{7}\text{ cm}$

(c) $\frac{74}{7}\text{ cm}$ (d) $\frac{76}{7}\text{ cm}$

Q.253. The curved surface area of a right circular cylinder of height 56 cm is 1408 cm^2 . Find the diameter of the base of the cylinder.

SSC CGL 27/07/2023 (3rd shift)

(a) 8m (b) 0.04m (c) 0.08m (d) 0.008m

SSC Selection Post (Phase - XI)

Q.254. If the volume of a cube is given as 12167 cm^3 , then the surface area of the cube will be:

Graduate Level 27/06/2023 (Shift - 4)

(a) 3475 cm^2 (b) 4574 cm^2

(c) 3174 cm^2 (d) 5413 cm^2

Q.255. What is the side of a cube (in metres) in which an iron rod of maximum length 10 metres can be put?

Graduate Level 28/06/2023 (Shift - 3)

(a) 5.77 (b) 7.57 (c) 5.57 (d) 7.75

Q.256. The volume of a cylinder is 1650 m^3 , whereas the surface area of its base is $78\frac{4}{7}\text{ m}^2$. Find the height of the cylinder.

Graduate Level 30/06/2023 (Shift - 1)

(a) 7.5 m (b) 14 m (c) 21 m (d) 2.1 m

Q.257. Melting a hemisphere of radius 7 cm, four small identical spheres are made. What is the radius of these spheres?

Higher Secondary 27/06/2023 (Shift - 3)
(a) 3 cm (b) 3.5 cm (c) 4 cm (d) 4.5 cm

Q.258. What is the volume (in cm^3) of a hemisphere having a diameter of 21 cm?

(Take $\pi = \frac{22}{7}$)

Higher Secondary 28/06/2023 (Shift - 2)
(a) 2625.5 (b) 2255.5
(c) 1925.5 (d) 2425.5

Q.259. The area of the triangle whose sides are 3 cm, 5 cm, and 6 cm is:

Higher Secondary 30/06/2023 (Shift - 2)

(a) $\sqrt{14} \text{ cm}^2$ (b) $2\sqrt{14} \text{ cm}^2$
(c) $4\sqrt{14} \text{ cm}^2$ (d) $3\sqrt{14} \text{ cm}^2$

Q.260. The diagonal (in cm) of a cuboid with dimensions $9\text{cm} \times 12\text{cm} \times 20\text{cm}$ is:

Higher Secondary 30/06/2023 (Shift - 2)

(a) 9 (b) 12 (c) 25 (d) 20

Q.261. Find the length of a diagonal of a cuboid whose dimensions are 12 m, 10 m and 8 m. (up to one decimal point)

Matriculation Level 27/06/2023 (Shift - 1)

(a) 18m (b) 16.5m (c) 17m (d) 17.54m

Q.262. The internal and external diameters of a hollow hemispherical bowl are 6 cm and 10 cm, respectively. If it is melted and recast into a solid cylinder of diameter 14 cm, what is the height of the cylinder in cm?

Matriculation Level 27/06/2023 (Shift - 1)

(a) 1.1 (b) 1.22 (c) 1.33 (d) 1

Q.263. The curved surface area of a hemisphere is 693 cm^2 . Its radius will be

_____. (Take $\pi = \frac{22}{7}$)

Matriculation Level 27/06/2023 (Shift - 2)
(a) 10.5 cm (b) 9 cm (c) 10 cm (d) 9.5 cm

Q.264. The perimeter of an isosceles right - angled triangle having an area of 200 cm^2 is:

Matriculation Level 28/06/2023 (Shift - 1)

(a) 68.3 cm (b) 78.2 cm
(c) 70.6 cm (d) 58.6 cm

Q.265. A solid sphere has a surface area of 616 cm^2 . This sphere is now cut into two hemispheres. What is the total surface area of one of the hemispheres?

Matriculation Level 28/06/2023 (Shift - 1)

(a) 440 cm^2 (b) 462 cm^2
(c) 452 cm^2 (d) 390 cm^2

Q.266. A solid iron sphere of radius 21 cm is melted down into a cone of height 84 cm. What will be the radius of the cone?

Matriculation Level 28/06/2023 (Shift - 4)

(a) 24 cm (b) 28 cm (c) 32 cm (d) 21 cm

Q.267. The curved surface area of a cylindrical pillar is 528 m^2 and its volume is 1848 m^3 . The ratio between its radius and height is _____.

Matriculation Level 30/06/2023 (Shift - 3)

(a) 9 : 10 (b) 7 : 12 (c) 7 : 3 (d) 9 : 5

Q.268. The radius of the base of a solid right circular cone is 20 cm and its height is 21 cm. What is its total surface area (in cm^2)?

Matriculation Level 30/06/2023 (Shift - 3)

(a) 760π (b) 980π (c) 890π (d) 680π

Q.269. To pack a set of books, Surbhi made a carton that was 48 inches long, 27 inches wide and 30 inches tall. What will be the total surface area (in square feet) of this carton, once it is closed from all sides? [Use 1 foot = 12 inches]

Matriculation Level 30/06/2023 (Shift - 4)

(a) 49.25 (b) 48.25 (c) 48.75 (d) 49.75

SSC CHSL 2022 Tier - 2

Q.270. The length of a rectangular plot is thrice its breadth. If the area of the rectangular plot is 2700 m^2 , then what is the breadth of the rectangular plot?

SSC CHSL Tier II (26/06/2023)

(a) 20 m (b) 12 m (c) 30 m (d) 28 m

SSC MTS 2022 Tier - 1

Q.271. If the surface area of two spheres is in the ratio 81 : 25, then what is the ratio of their radius.

SSC MTS 02/05/2023 (Morning)

(a) 5 : 8 (b) 7 : 9 (c) 5 : 7 (d) 9 : 5

Q.272. When a wire is bent in the form of a square, then the area enclosed by it is 5929 cm^2 . If wire is bent into the form of a circle, then what will be the area enclosed by the wire?

SSC MTS 02/05/2023 (Afternoon)

(a) 7546 cm^2 (b) 8542 cm^2
(c) 7632 cm^2 (d) 6954 cm^2

Q.273. What will be the area of the semi-circle of perimeter 3.6 meters?

SSC MTS 02/05/2023 (Afternoon)

(a) 1.54 m^2 (b) 1.76 m^2
(c) 1.10 m^2 (d) 0.77 m^2

Q.274. What will be the length (in cm) of the longest diagonal of the cuboid having length 13 cm, width 11 cm and height 20 cm?

SSC MTS 02/05/2023 (Evening)

(a) 23.45 (b) 26.27 (c) 34.65 (d) 22.65

Q.275. The ratio of the outer and the inner circumference of a circular path is 17 : 15. If path is 40 metres wide, then

what is the radius of the inner circle?

SSC MTS 02/05/2023 (Evening)

(a) 280 metres (b) 340 metres
(c) 360 metres (d) 300 metres

Q.276. The radius of a right circular cylinder is thrice of its height. If the height of the cylinder is 3.5 cm. then what is the volume of cylinder?

SSC MTS 02/05/2023 (Evening)

(a) 1124.25 cm^3 (b) 1324.75 cm^3
(c) 1468.25 cm^3 (d) 1212.75 cm^3

Q.277. If the area of a square is 529 cm^2 , then what is the length of its diagonal?

SSC MTS 03/05/2023 (Morning)

(a) 23 cm (b) $26\sqrt{3}$ (c) $23\sqrt{2}$ (d) 46 cm

Q.278. The length of two parallel sides of a trapezium are 15 metre and 20 metre. If its height is 22 metre, then what is the area of the trapezium?

SSC MTS 03/05/2023 (Morning)

(a) 295 metre^2 (b) 385 metre^2
(c) 425 metre^2 (d) 315 metre^2

Q.279. The volume of a hemisphere is $486\pi \text{ cm}^3$. Find the radius.

SSC MTS 03/05/2023 (Afternoon)

(a) 7 cm (b) 9 cm (c) 4 cm (d) 8 cm

Q.280. A farmer's land is in the shape of a trapezium which has its parallel sides measuring 6.32 yards and 7.68 yards and the distance between the parallel sides is 5.50 yards. The cost of ploughing the land is Rs. 1200 per square yard. What amount have to be spent in order to plough the entire land?

SSC MTS 03/05/2023 (Afternoon)

(a) Rs. 36600 (b) Rs. 32500
(c) Rs. 55400 (d) Rs. 46200

Q.281. The length of a cuboid is six times of its breadth and the height of a cuboid is four times of its breadth. If the breadth of the cuboid is 3 cm, then what is the total surface area of the cuboid?

SSC MTS 03/05/2023 (Evening)

(a) 484 cm^2 (b) 612 cm^2
(c) 576 cm^2 (d) 676 cm^2

Q.282. The ratio of radii of a cylinder to a cone is 3 : 1. If their heights are equal, What is the ratio of their volumes?

SSC MTS 03/05/2023 (Evening)

(a) 1 : 3 (b) 27 : 1 (c) 9 : 1 (d) 1 : 9

Q.283. Find the area of triangle having base 34 cm and corresponding height as 22 cm.

SSC MTS 04/05/2023 (Morning)

(a) 453 cm^2 (b) 374 cm^2
(c) 234 cm^2 (d) 378 cm^2

Q.284. The ratio of the sides of a triangle is 11 : 11 : 4. If the area of triangle is $2\sqrt{117} \text{ cm}^2$, then what is the length of the equal sides?

SSC MTS 04/05/2023 (Morning)

(a) 3 cm (b) 13 cm (c) 11 cm (d) 9 cm

Q.285. The two adjacent sides of a parallelogram are 12 cm and 5 cm respectively. If one of the diagonals is 13 cm long, then what is the area of the parallelogram?

SSC MTS 04/05/2023 (Afternoon)

(a) 60 cm^2 (b) 30 cm^2
(c) 75 cm^2 (d) 25 cm^2

Q.286. What is the area of a circle with diameter 12 mm?

SSC MTS 04/05/2023 (Afternoon)

(a) $12\pi \text{ mm}^2$ (b) $24\pi \text{ mm}^2$
(c) $36\pi \text{ mm}^2$ (d) $48\pi \text{ mm}^2$

Q.287. A solid cylinder having radius of base as 21 cm and height as 16 cm is bisected from its height to get two identical cylinders. What will be the percentage increase in the total surface area?

SSC MTS 04/05/2023 (Evening)

(a) 56.76 percent (b) 42.52 percent
(c) 62.34 percent (d) 48.88 percent

Q.288. What will be the curved surface area of cylinder having radius 20 cm and height 34 cm? (use $\pi = 3.14$)

SSC MTS 04/05/2023 (Evening)

(a) 3245.7 cm^2 (b) 4328.98 cm^2
(c) 4270.4 cm^2 (d) 4320.6 cm^2

Q.289. A cuboid whose sides are 8 cm, 27 cm and 64 cm is melted to form a new cube. What is the respective ratio of the total surface area of the cuboid and cube?

SSC MTS 08/05/2023 (Morning)

(a) 307: 216 (b) 291: 203
(c) 349: 248 (d) 329: 237

Q.290. A wire is in the form of square with side 44 cm. It is bent to form a circle. What is the radius of the circle?

SSC MTS 08/05/2023 (Morning)

(a) 28 cm (b) 21 cm (c) 16 cm (d) 32 cm

Q.291. Find $\frac{1}{21}$ th of the volume of a hemisphere whose radius is 4.2 m

SSC MTS 08/05/2023 (Afternoon)

(a) 6.6 m^3 (b) 7.3 m^3 (c) 8.2 m^3 (d) 7.1 m^3

Q.292. Two cubes each of edge 9 cm are joined to form a single cuboid. What is the total surface area of the cuboid formed?

SSC MTS 08/05/2023 (Afternoon)

(a) 775 cm^2 (b) 810 cm^2

(c) 990 cm^2 (d) 1024 cm^2

Q.293. What is the area of a triangle whose base is 120 cm and the corresponding height is 35 cm?

SSC MTS 08/05/2023 (Evening)

(a) 2120 cm^2 (b) 2100 cm^2
(c) 2010 cm^2 (d) 2630 cm^2

Q.294. The radius of the base of a conical tent is 8 m and its height is 15 m, what is the cost of the material needed to make it if it costs Rs.54 per $\pi \text{ m}^2$?

SSC MTS 08/05/2023 (Evening)

(a) Rs.6454 (b) Rs.7344
(c) Rs.8678 (d) Rs.7454

Q.295. The Largest possible sphere is carved out of a cube of side 6 cm. What is the volume of the sphere?

SSC MTS 09/05/2023 (Morning)

(a) $\frac{792}{7} \text{ cm}^3$ (b) $\frac{144}{7} \text{ cm}^3$
(c) $\frac{133}{3} \text{ cm}^3$ (d) $\frac{444}{7} \text{ cm}^3$

Q.296. Find the area of the rhombus whose diagonals are 16 cm and 20 cm.

SSC MTS 09/05/2023 (Morning)

(a) 110 cm^2 (b) 150 cm^2
(c) 160 cm^2 (d) 120 cm^2

Q.297. Two cubes of iron whose edges are 7 cm and 11 cm are melted and formed into a single cube. Find the volume (in cm^3) of the new cube formed.

SSC MTS 09/05/2023 (Afternoon)

(a) 1881 (b) 1674 (c) 1444 (d) 1221

Q.298. A rectangle has sides of 45 cm and 55 cm. What is the area (in cm^2) of this rectangle?

SSC MTS 09/05/2023 (Afternoon)

(a) 2675 (b) 2475 (c) 2550 (d) 2350

Q.299. Length of a rectangle is 58 meters, while its breadth is 30 meters. What is the total cost of covering it with grass bed if cost is Rs. 20 per m^2 ?

SSC MTS 09/05/2023 (Evening)

(a) Rs. 34800 (b) Rs. 32400
(c) Rs. 28500 (d) Rs. 17400

Q.300. If the volume of a cube is 5832 cm^3 , then what is the lateral surface area of this cube?

SSC MTS 09/05/2023 (Evening)

(a) 1024 cm^2 (b) 384 cm^2
(c) 576 cm^2 (d) 1296 cm^2

Q.301. Find the total surface area of cuboid having length 23 cm, breadth 18 cm and height 35 cm.

SSC MTS 09/05/2023 (Evening)

(a) 6541 cm^2 (b) 2341 cm^2
(c) 3698 cm^2 (d) 5436 cm^2

Q.302. Find the weight of a cylindrical iron container with base radius 12.6 cm and height is 4 m, where the weight of 1 cm^3 of iron is 2 grams?

SSC MTS 10/05/2023 (Morning)

(a) 299.284 kg (b) 199.584 kg
(c) 399.168 kg (d) 499.482 kg

Q.303. Sides of a triangle are 20 cm, 48 cm and 52 cm. What is the area of the triangle?

SSC MTS 10/05/2023 (Afternoon)

(a) 320 cm^2 (b) 480 cm^2
(c) 560 cm^2 (d) 245 cm^2

Q.304. Find the area of the Kite whose diagonals are 22 cm and 6 cm.

SSC MTS 10/05/2023 (Afternoon)

(a) 66 cm^2 (b) 100 cm^2
(c) 88 cm^2 (d) 50 cm^2

Q.305. Find the area of the triangle whose height is 56 cm and the corresponding base is $\frac{1}{4}$ th of the height.

SSC MTS 10/05/2023 (Evening)

(a) 544 cm^2 (b) 440 cm^2
(c) 392 cm^2 (d) 480 cm^2

Q.306. There is a rectangular garden of 150 metres $\times$ 25 metres. A path of width 10 metres is built around the garden. What is the area of the path?

SSC MTS 10/05/2023 (Evening)

(a) 3700 metre^2 (b) 3550 metre^2
(c) 3900 metre^2 (d) 3220 metre^2

Q.307. The outer radius of a spherical shell is 13 cm and the thickness of the shell is 3 cm. What is the volume of the metal used for the shell? ($\pi = \frac{22}{7}$)

SSC MTS 11/05/2023 (Morning)

(a) 1197 cm^3 (b) 2197 cm^3
(c) 3024 cm^3 (d) 5016 cm^3

Q.308. What is the area of a square whose diagonal is $5\sqrt{2} \text{ cm}$?

SSC MTS 11/05/2023 (Morning)

(a) 64 cm^2 (b) 25 cm^2
(c) 100 cm^2 (d) 36 cm^2

Q.309. What will be the approximately volume of sphere with radius 65 cm?

SSC MTS 11/05/2023 (Afternoon)

(a) $4.5 \times 10^6 \text{ cm}^3$ (b) $6 \times 10^6 \text{ cm}^3$
(c) $1.15 \times 10^6 \text{ cm}^3$ (d) $2 \times 10^6 \text{ cm}^3$

Q.310. What is the volume of a cone whose base diameter is 12 cm and the height is 21.7 cm?

SSC MTS 11/05/2023 (Afternoon)

(a) 626.3 cm^3 (b) 548.6 cm^3
(c) 818.4 cm^3 (d) 334.5 cm^3

Q.311. A picture is 40 cm wide and 1.6 m long. The ratio of its width to its perimeter is :

SSC MTS 11/05/2023 (Evening)

- (a) 4 : 1 (b) 10 : 1 (c) 1 : 10 (d) 1 : 4

Q.312. The lateral area is thrice as big as the base area of a cone. If the diameter of the cone is 1 cm, then what is the total surface area of the cone ?

SSC MTS 11/05/2023 (Evening)

- (a) π cm² (b) 3π cm² (c) 2π cm² (d) 4π cm²

Q.313. The circumference of a circle is 220 cm. What is the area of that circle ?

SSC MTS 11/05/2023 (Evening)

- (a) 3850 cm² (b) 4550 cm²
(c) 2850 cm² (d) 4650 cm²

Q.314. If the radius of a sphere is $\frac{3}{4}$ of the radius of a hemisphere, then what will be the ratio of the volumes of sphere and hemisphere ?

SSC MTS 12/05/2023 (Morning)

- (a) 9:16 (b) 51 : 64 (c) 27 : 32 (d) 18:64

Q.315. The area of the base of a rectangular tank is 720 m² and the volume of water contained in it is 28800 m³. What is the depth of the water ?

SSC MTS 12/05/2023 (Morning)

- (a) 30 m (b) 40 m (c) 50 m (d) 60 m

Q.316. The radius of a wheel is 21 cm. What is the distance (in cm) travelled by the wheel in 20 revolutions ?

SSC MTS 12/05/2023 (Afternoon)

- (a) 1440 (b) 1500 (c) 2640 (d) 2100

Q.317. Height of a cone is 12 cm and radius of its base is 14 cm. What is the volume of the cone?

SSC MTS 12/05/2023 (Afternoon)

- (a) 2464 cm³ (b) 2692 cm³
(c) 2248 cm³ (d) 2872 cm³

Q.318. Sides of a triangle are 6 cm, 8 cm and 10 cm. What is the area of the triangle ?

SSC MTS 12/05/2023 (Afternoon)

- (a) 20 cm² (b) 28 cm²
(c) 24 cm² (d) 16 cm²

Q.319. If the ratio of the volumes of two cylinders is 6: 5 and the ratio of their heights is 5 : 24, then what is the ratio of their radii ?

SSC MTS 12/05/2023 (Evening)

- (a) 12 : 7 (b) 5 : 6 (c) 1 : 2 (d) 12 : 5

Q.320. What is the area of the lateral surface of a right circular cylinder, If the circumference of the base is 22 cm and its height is four times of its radius?

SSC MTS 12/05/2023 (Evening)

- (a) 388 cm² (b) 308 cm²
(c) 408 cm² (d) 288 cm²

Q.321. A sphere of radius 18 cm is melted and then a solid cylinder of base radius 4 cm is made from it. What is the height of the solid cylinder?

SSC MTS 15/05/2023 (Morning)

- (a) 196 cm (b) 486 cm
(c) 224 cm (d) 384 cm

Q.322. A metallic solid sphere has radius 28 cm. If it is melted to form small spheres of radius 3.5 cm, then how many small spheres will be obtained ?

SSC MTS 15/05/2023 (Morning)

- (a) 512 (b) 624 (c) 496 (d) 424

Q.323. The radius of a hemisphere is 5 cm. Find the volume. (in cm³)

SSC MTS 15/05/2023 (Afternoon)

- (a) 330.4 (b) 117.9 (c) 150.6 (d) 261.9

Q.324. If the heights of two cylinders are in the ratio of 7 : 9 and their diameters are in the ratio of 6 : 7, then what is the ratio of the volumes of the cylinders ?

SSC MTS 15/05/2023 (Afternoon)

- (a) 6 : 7 (b) 7 : 9 (c) 5 : 9 (d) 4 : 7

Q.325. If the curved surface area of a solid cylinder is two-fifth of its total surface area, then what is the ratio of its diameter to its height ?

SSC MTS 15/05/2023 (Evening)

- (a) 2 : 5 (b) 1 : 2 (c) 2 : 3 (d) 3 : 1

Q.326. If the circumference of a circle is 13.2 cm, then what is its radius?

SSC MTS 15/05/2023 (Evening)

- (a) 1.7 cm (b) 6 cm (c) 21 cm (d) 2.1 cm

Q.327. Difference between the circumferences of two circles is 4.4 cm and radius of the smaller circle is 1.4 cm. The radius of the larger circle is:

SSC MTS 16/05/2023 (Morning)

- (a) 4.1 cm (b) 0.7 cm
(c) 2.1 cm (d) 3.5 cm

Q.328. The height of a right circular cone is 24 cm. If the diameter of its base is 36 cm, then what will be the curved surface area of the cone?

SSC MTS 16/05/2023 (Morning)

- (a) 1444.6 cm² (b) 2400.9 cm²
(c) 1697.14 cm² (d) 2144.2 cm²

Q.329. Nine times the area of a circle is same as the three times the area of a square. What is the ratio of the diameter of the circle and diagonal of the square?

SSC MTS 16/05/2023 (Afternoon)

- (a) $\sqrt{2} : \sqrt{3}\pi$ (b) $2 : \sqrt{3}\pi$
(c) $2 : 3\pi$ (d) $\sqrt{5} : \sqrt{7}\pi$

Q.330. How many litre of water can a hemispherical bowl of diameter 21 m hold? (1m³ - 1000 litre)

SSC MTS 16/05/2023 (Afternoon)

- (a) 2100000 litre (b) 2425500 litre
(c) 2322500 litre (d) 2250500 litre

Q.331. A solid cylinder having radius of base as 28 cm and height as 24 cm is bisected from its height to get two identical cylinders. What will be the percentage increase in the total surface area ?

SSC MTS 17/05/2023 (Morning)

- (a) 61.72 percent (b) 41.92 percent
(c) 53.85 percent (d) 48.64 percent

Q.332. The height and curved surface area of the cylinder are 15 cm and 330 cm². Find the diameter of the cylinder.

SSC MTS 17/05/2023 (Morning)

- (a) 2 cm (b) 5 cm (c) 7 cm (d) 6 cm

Q.333. The length, breadth, height of a room are 6 m 40 cm, 4 m 80 cm and 3 m 20 cm respectively. Find the longest tape which can measure the dimensions of the room exactly.

SSC MTS 17/05/2023 (Morning)

- (a) 1 m 80 cm (b) 1 m 20 cm
(c) 1 m 60 cm (d) 1 m 40 cm

Q.334. A solid hemisphere of radius 8 cm is melted into N identical cone whose radius is 4 cm and height is 2 cm. Then, what is the value of N ?

SSC MTS 17/05/2023 (Afternoon)

- (a) 26 (b) 10 (c) 32 (d) 25

Q.335. A cuboid is of dimensions 80 cm × 64 cm × 50 cm How many small cubes with side 8 cm can be placed in the given cuboid ?

SSC MTS 17/05/2023 (Afternoon)

- (a) 480 (b) 700 (c) 1000 (d) 200

Q.336. If the diameter of a circle is 14 cm, find the area of the sector of the circle whose central angle measure 60 degree.

SSC MTS 17/05/2023 (Afternoon)

- (a) 18.12 cm² (b) 25. 66 cm²
(c) 38.12 cm² (d) 21.66 cm²

Q.337. The ratio of the outer and the inner circumferences of a circular path is 11 : 7. If path is 20 metres wide, then what is the radius of the inner circle ?

SSC MTS 17/05/2023 (Evening)

- (a) 20 metres (b) 55 metres
(c) 65 metres (d) 35 metres

Q.338. The longest side of the obtuse triangle is 7 cm and the other two sides of the triangle are 4 cm and 5 cm. Find the area of the triangle.

SSC MTS 18/05/2023 (Morning)

- (a) $1\sqrt{3} \text{ cm}^2$ (b) $6\sqrt{3} \text{ cm}^2$
(c) $3\sqrt{2} \text{ cm}^2$ (d) $4\sqrt{6} \text{ cm}^2$

Q.339. The circumference of base of a cylinder is 28π cm. If the height of cylinder is 17 cm, then what will be the curved surface area of cylinder?

SSC MTS 18/05/2023 (Afternoon)

- (a) $372\pi \text{ cm}^2$ (b) $436\pi \text{ cm}^2$
(c) $524\pi \text{ cm}^2$ (d) $476\pi \text{ cm}^2$

Q.340. Find the curved surface area of cone of radius 12 cm and slant height 14 cm. (Use $\pi = 3.14$)

SSC MTS 18/05/2023 (Afternoon)

- (a) 4325.5 cm^2 (b) 527.52 cm^2
(c) 5436.6 cm^2 (d) 501.54 cm^2

Q.341. How many solid spherical balls each of 33 cm radius can be made out of a solid spherical ball of radius 66 cm?

SSC MTS 18/05/2023 (Evening)

- (a) 8 (b) 4 (c) 7 (d) 3

Q.342. If the diagonal of a square is 18.4 cm, then the area of that square is: (in cm^2)

SSC MTS 18/05/2023 (Evening)

- (a) 84.64 (b) 338.56 (c) 324.42 (d) 169.28

Q.343. If the volume of a cylinder is 5000 m^3 and its radius is 10 meter, then what is the approximate curved surface area of the cylinder? ($\pi = 3.14$)

SSC MTS 19/05/2023 (Morning)

- (a) 954 m^2 (b) 1024 m^2
(c) 1000 m^2 (d) 968 m^2

Q.344. A cone and a cylinder have same height and the radius of the cone is twice of the radius of the cylinder. What is the ratio of the volume of the cone to that of the cylinder?

SSC MTS 19/05/2023 (Afternoon)

- (a) 2 : 5 (b) 4 : 5 (c) 3 : 2 (d) 4 : 3

Q.345. The difference between the length and breadth of a rectangle is 40 metres. If its perimeter is 96 metres, then what will be the area of rectangle?

SSC MTS 19/05/2023 (Evening)

- (a) 254 metre^2 (b) 110 metre^2
(c) 176 metre^2 (d) 152 metre^2

Q.346. The area of a rectangular floor is 600 square feet. If the length of the floor is 30 feet, then what is the breadth of the floor?

SSC MTS 19/05/2023 (Evening)

- (a) 20 feet (b) 60 feet
(c) 30 feet (d) 10 feet

Q.347. A farmer's land is in the shape of a trapezium which has its parallel sides measuring 7.58 yards and 6.42 yards and the distance between the parallel sides is

6.50 yards. The cost of ploughing the land is Rs. 2000 per square yard. What amount have to be spent in order to plough the entire land?

SSC MTS 13/06/2023 (Morning)

- (a) Rs. 85000 (b) Rs. 91000
(c) Rs. 72000 (d) Rs. 102000

Q.348. The ratio of the volume of two cylinders is 13 : 1 and the ratio of their heights is 13 : 9. If the area of the base of the second cylinder is 154 cm^2 , then what will be the radius of the first cylinder?

SSC MTS 13/06/2023 (Afternoon)

- (a) 42 cm (b) 21 cm (c) 28 cm (d) 14 cm

Q.349. What will be the curved surface area of cone of radius 5 cm and slant height 30 cm? (use $\pi = 3.14$)

SSC MTS 13/06/2023 (Afternoon)

- (a) 234 m^2 (b) 715 m^2
(c) 543 m^2 (d) 471 m^2

Q.350. The length of two parallel sides of a trapezium are 25 metre and 30 metre. If its height is 20 metre, then what is the area (in metre^2) of the trapezium?

SSC MTS 13/06/2023 (Evening)

- (a) 420 (b) 350 (c) 590 (d) 550

Q.351. If the radius of a cylinder is decreased by 12 percent, then by how much percent its height must be increased so that the volume of the cylinder remains same?

SSC MTS 14/06/2023 (Morning)

- (a) 29.13 percent (b) 21.78 percent
(c) 42.56 percent (d) 34.27 percent

Q.352. If the ratio of the volumes of two cylinders is 7 : 5 and the ratio of their heights is 5 : 7, then what is the ratio of their radii?

SSC MTS 14/06/2023 (Morning)

- (a) 7 : 5 (b) 3 : 4 (c) 5 : 9 (d) 4 : 5

Q.353. The radius of a right circular cylinder is 7 cm. Its height is thrice its radius. What is the curved surface area of the cylinder?

SSC MTS 14/06/2023 (Afternoon)

- (a) 724 cm^2 (b) 666 cm^2
(c) 924 cm^2 (d) 264 cm^2

Q.354. The ratio of the sides of a triangle is 3 : 3 : 4. If the area of triangle is $32\sqrt{5} \text{ cm}^2$, then what is the length of the equal sides?

SSC MTS 14/06/2023 (Evening)

- (a) 12 cm (b) 16 cm (c) 10 cm (d) 15 cm

Q.355. A square park has sides of length 18 cm. A 1 cm wide path along the edges is inside it. What is the area of the path?

SSC MTS 14/06/2023 (Evening)

- (a) 92 cm^2 (b) 68 cm^2
(c) 180 cm^2 (d) 76 cm^2

Q.356. Find out the area of triangle of base 7 cm and corresponding height 12 cm.

SSC MTS 15/06/2023 (Morning)

- (a) 42 cm^2 (b) 54 cm^2
(c) 43 cm^2 (d) 44 cm^2

Q.357. Two cubes each of edge 14 cm are joined to form a single cuboid. What is the volume of the cuboid formed?

SSC MTS 15/06/2023 (Afternoon)

- (a) 5022 cm^3 (b) 5244 cm^3
(c) 5666 cm^3 (d) 5488 cm^3

Q.358. The length of a cuboid is 4 cm. If the breadth of the cuboid is four times of its length and height of the cuboid is twice of its length, then what is the lateral surface area of the cuboid?

SSC MTS 15/06/2023 (Afternoon)

- (a) 380 cm^2 (b) 440 cm^2
(c) 260 cm^2 (d) 320 cm^2

Q.359. One of the diagonals of a rhombus is 70 percent of the other diagonal. What is the ratio of the area of rhombus to the square of the length of the larger diagonal?

SSC MTS 15/06/2023 (Evening)

- (a) 5 : 17 (b) 7 : 20 (c) 6 : 19 (d) 20 : 7

Q.360. If the radius of a cylinder is decreased by 16 percent, then by how much percent its height must be increased, so that the volume of the cylinder remains the same?

SSC MTS 15/06/2023 (Evening)

- (a) 32.96% (b) 41.72%
(c) 45.28% (d) 36.43%

Q.361. The ratio of the volume of two cylinders is 27 : 25 and the ratio of their heights is 3 : 4. If the area of the base of the second cylinder is 3850 cm^2 , then what will be the radius of the first cylinder?

SSC MTS 16/06/2023 (Morning)

- (a) 42 cm (b) 56 cm (c) 63 cm (d) 34 cm

Q.362. The sum of the length and breadth of a cuboid is 16 cm. If the height of the cuboid is $\frac{1}{4}$ of the sum of its length and breadth, then what is the lateral surface area of the cuboid?

SSC MTS 16/06/2023 (Afternoon)

- (a) 128 cm^2 (b) 196 cm^2
(c) 96 cm^2 (d) 156 cm^2

Q.363. The radius of two spheres is in the ratio of 1 : 5. What is the ratio of the volume of the two spheres?

SSC MTS 16/06/2023 (Afternoon)

(a) 8 : 27 (b) 1 : 25 (c) 1 : 125 (d) 1 : 64

Q.364. The length of a cuboid is 10 cm. If the breadth of the cuboid is half of its length and height of the cuboid is twice of its length, then what is the lateral surface area of the cuboid?

SSC MTS 16/06/2023 (Evening)

(a) 600 cm² (b) 500 cm²
(c) 720 cm² (d) 560 cm²

Q.365. The length, breadth and height of a cuboid are 4 cm, 6 cm and 10 cm respectively. Find the total surface area.

SSC MTS 19/06/2023 (Morning)

(a) 248 cm² (b) 244 cm²
(c) 144 cm² (d) 222 cm²

Q.366. There is a rectangular garden of 220 m × 70 m. A path 4 m wide has been made around the garden. What is the area of the path?

SSC MTS 19/06/2023 (Afternoon)

(a) 1836 metre² (b) 2384 metre²
(c) 2162 metre² (d) 2472 metre²

Q.367. The curved surface area of a cylinder is 968 cm². If the height of the cylinder is 11 cm, then what will be the diameter of its base?

SSC MTS 19/06/2023 (Evening)

(a) 26 cm (b) 22 cm (c) 28 cm (d) 24 cm

Q.368. The length of a cuboid is half of its breadth and the height of cuboid is thrice of its breadth. If the breadth of the cuboid is 12 cm, then what is the total surface area of cuboid?

SSC MTS 20/06/2023 (Morning)

(a) 1680 cm² (b) 1320 cm²
(c) 1440 cm² (d) 1560 cm²

Q.369. The perimeter of the top of a rectangular table is 56 meter and its area is 192 meter². What is the length of its diagonal?

SSC MTS 20/06/2023 (Morning)

(a) 22 metre (b) 20 metre
(c) 16 metre (d) 18 metre

Q.370. Find the surface area of a spherical ball that has a radius of 1.1 cm.

SSC MTS 20/06/2023 (Afternoon)

(a) 16 cm² (b) 12.1 cm²
(c) 11.2 cm² (d) 15.2 cm²

Q.371. If the area of non-spherical portion of a solid hemisphere is 1560 m², then the total surface area of the hemisphere is:

SSC MTS 20/06/2023 (Afternoon)

(a) 6240 m² (b) 4680 m²
(c) 5140 m² (d) 3120 m²

Q.372. The sum of the length and breadth of a cuboid is 40 cm. If the height of the cuboid is $\frac{1}{5}$ of the sum of its length and breadth, then what is the lateral surface area of the cuboid?

SSC MTS 20/06/2023 (Evening)

(a) 580 cm² (b) 320 cm²
(c) 640 cm² (d) 720 cm²

SSC CHSL 2022 Tier - 1

Q.373. Find the area of a triangle whose length of each side is 13 inches, 15 inches and 14 inches.

SSC CHSL 09/03/2023 (1st Shift)

(a) 84 square inches
(b) 81.2 square inches
(c) 42.1 square inches
(d) 84.2 square inches

Q.374. The circumference of base of a right circular cone is 88 cm. If the height of the cone is 28 cm, then what is the curved surface area of the cone?

SSC CHSL 09/03/2023 (2nd Shift)

(a) $670\sqrt{3}$ cm² (b) $616\sqrt{5}$ cm²
(c) $627\sqrt{3}$ cm² (d) $561\sqrt{5}$ cm²

Q.375. If the ratio of the volumes of two cylinders is 6 : 1 and the ratio of their heights is 1 : 6, then what is the ratio of their radii?

SSC CHSL 10/03/2023 (1st Shift)

(a) 9 : 1 (b) 3 : 1 (c) 8 : 1 (d) 6 : 1

Q.376. If the radius of the sphere is decreased by 10 percent, then volume of the sphere will be decreased by what percentage?

SSC CHSL 10/03/2023 (1st Shift)

(a) 32.5 percent (b) 30.6 percent
(c) 25.6 percent (d) 27.1 percent

Q.377. Three circles of radius 7 cm are placed in such a way that each circle touches the other two. What will be the area of the portion enclosed by these three circles?

SSC CHSL 10/03/2023 (3rd Shift)

(a) $49\sqrt{3} - 77$ cm² (b) $40\sqrt{3} - 66$ cm²
(c) $50\sqrt{3} - 66$ cm² (d) $55\sqrt{3} - 77$ cm²

Q.378. There is a rectangular garden of 240 metres × 80 metres. A path of width 4 metres is build outside the garden along its four sides. What is the area of the path?

SSC CHSL 10/03/2023 (4th Shift)

(a) 2826 metre² (b) 2542 metre²
(c) 2916 metre² (d) 2624 metre²

Q.379. The sum of the length, breadth, and height of a cuboid box is 20 cm and

the total surface area of cuboid is 256 sq . cm. What is the maximum length (in approximate) of a stick that can be placed inside the cuboid box?

SSC CHSL 13/03/2023 (2nd Shift)

(a) 32 cm (b) 24 cm (c) 16 cm (d) 12 cm

Q.380. X, Y and Z are three equilateral triangles. The sum of the areas of X and Y is equal to the area of Z. If the side lengths of X and Y are 6 cm and 8 cm respectively, then what is the side length of Z?

SSC CHSL 13/03/2023 (2nd Shift)

(a) 10.5 cm (b) 10 cm (c) 9.5 cm (d) 9 cm

Q.381. An incircle with center O is drawn inside an equilateral triangle ABC and the length of the side is 12 cm. Find the area (in sq .cm) of the triangle which is not in the incircle.

SSC CHSL 13/03/2023 (2nd Shift)

(a) 21.48 (b) 30.64 (c) 37.71 (d) 24.64

Q.382. The length, width and height of a cuboidal box are 18 cm, 12 cm and 10 cm, respectively. What is the total surface area of the box?

SSC CHSL 13/03/2023 (3rd Shift)

(a) 1000 cm² (b) 1048 cm²
(c) 1032 cm² (d) 1120 cm²

Q.383. What is the area of a rhombus whose diagonals are of 8 cm and 13 cm?

SSC CHSL 13/03/2023 (4th Shift)

(a) 52 cm² (b) 96 cm²
(c) 44 cm² (d) 84 cm²

Q.384. If the radius of a sphere is thrice than the radius of a hemisphere, then what will be the respective ratio of their volumes?

SSC CHSL 14/03/2023 (1st Shift)

(a) 27 : 1 (b) 16 : 1 (c) 54 : 1 (d) 18 : 1

Q.385. The ratio of the volume of the first and second cylinder is 32 : 9 and the ratio of their heights is 8 : 9. If the area of the base of the second cylinder is 616 cm², then what will be the radius of the first cylinder?

SSC CHSL 14/03/2023 (2nd Shift)

(a) 24 cm (b) 20 cm (c) 28 cm (d) 36 cm

Q.386. Half of the perimeter of a rectangle is 30 cm. If the length of the rectangle is 6 cm more than its breadth, then what is the area of the rectangle?

SSC CHSL 14/03/2023 (4th Shift)

(a) 216 cm² (b) 250 cm²
(c) 185 cm² (d) 195 cm²

Q.387. 24 equal solid hemispheres are melted to form a right circular cylinder of radius 12 cm and height 24 cm. Find the

radius of each solid hemisphere.

SSC CHSL 15/03/2023 (1st Shift)

(a) 6 cm (b) 8 cm (c) 3 cm (d) 4 cm

Q.388. The three sides of a triangle are 7 cm, 9 cm and 8 cm. What is the area of the triangle ?

SSC CHSL 15/03/2023 (1st Shift)

(a) $12\sqrt{5}$ Sq. cm (b) $10\sqrt{3}$ Sq. cm
(c) $12\sqrt{3}$ Sq. cm (d) $2\sqrt{5}$ Sq. cm

Q.389. If the square of product of three sides of the triangle is 1024 units and the area of its circumcircle is 16π , what is the area of the given triangle ?

SSC CHSL 15/03/2023 (1st Shift)

(a) 2 sq. units (b) 3 sq. units
(c) 2.5 sq. units (d) 4 sq. units

Q.390. The breadth of a rectangular floor is $\frac{3}{5}$ of its length. If the area of the floor is 60 metre^2 , then what is the difference between the length and breadth of the floor ?

SSC CHSL 15/03/2023 (2nd Shift)

(a) 6 m (b) 7.5 m (c) 5 m (d) 4 m

Q.391. How many bricks each measuring $25 \text{ cm} \times 12.5 \text{ cm} \times 16 \text{ cm}$ will be needed to construct a wall 8 m long, 5 m high and 22.5 cm thick ?

SSC CHSL 15/03/2023 (3rd Shift)

(a) 1800 (b) 3600 (c) 1200 (d) 2400

Q.392. A solid cylinder having radius of base as 7 cm and height as 5 cm is bisected from its height to get two identical cylinders. What will be the percentage increase in the total surface area?

SSC CHSL 16/03/2023 (2nd Shift)

(a) 58.33 percent (b) 64.56 percent
(c) 69.44 percent (d) 52.56 percent

Q.393. If the volume of a cube is 1728 cm^3 , then what is the total surface area of the cube?

SSC CHSL 16/03/2023 (3rd Shift)

(a) 804 cm^2 (b) 729 cm^2
(c) 864 cm^2 (d) 926 cm^2

Q.394. If in triangle ABC, MN is parallel to BC, and M and N are points on AB and AC respectively. The area of quadrilateral MBCN = 130 cm^2 . If AN:NC = 4:5, then the area of Triangle MAN is:

SSC CHSL 17/03/2023 (1st Shift)

(a) 40 cm^2 (b) 65 cm^2
(c) 32 cm^2 (d) 45 cm^2

Q.395. The circumference of the two circles is 264 m and 308 m respectively. What is the difference between the area of the larger circle and the smaller circle?

SSC CHSL 17/03/2023 (1st Shift)

(a) 2002 m^2 (b) 2010 m^2
(c) 1890 m^2 (d) 1990 m^2

Q.396. Find the area of the circle whose circumference is 22 cm.

SSC CHSL 17/03/2023 (1st Shift)

(a) 38.5 cm^2 (b) 42 cm^2
(c) 30 cm^2 (d) 77 cm^2

Q.397. Parallel sides of a trapezium are 21 cm and 14 cm and its area is 875 cm^2 . What is the distance between parallel sides?

SSC CHSL 17/03/2023 (3rd Shift)

(a) 45 cm (b) 50 cm (c) 60 cm (d) 65 cm

Q.398. If the heights of two cylinders are in the ratio of 2 : 3 and their radius are in the ratio of 6 : 5, then what is the ratio of the volumes of the cylinders ?

SSC CHSL 17/03/2023 (4th Shift)

(a) 24 : 29 (b) 21 : 31
(c) 24 : 25 (d) 28 : 31

Q.399. The sum of the length and breadth of a cuboid is 4 cm. Length is three times of breadth. If the height of the cuboid is half of the sum of its length and breadth, then what is the total surface area of the cuboid ?

SSC CHSL 20/03/2023 (2nd Shift)

(a) 64 cm^2 (b) 22 cm^2
(c) 32 cm^2 (d) 30 cm^2

Q.400. The area of a square is 1156 cm^2 . What is the respective ratio between the length and the breadth of a rectangle whose length is twice the side of the square and whose breadth is 14 cm less than the side of the square?

SSC CHSL 20/03/2023 (4th Shift)

(a) 16 : 7 (b) 19 : 5 (c) 17 : 5 (d) 21 : 19

Q.401. In triangle ABC, AB=20 cm, BC=16 cm and AC = 12 cm and the radius of circle inscribed in it is 4 cm. Find the area of triangle ABC.

SSC CHSL 21/03/2023 (1st Shift)

(a) 48 cm^2 (b) 72 cm^2
(c) 96 cm^2 (d) 120 cm^2

Q.402. A trapezium plate having two parallel sides of length 15 cm and 9 cm. and distance between them is 6 cm. Copper plating is to be done on the plate at a rate of Rs.3 per square cm. What will be the total cost of copper plating on the upper side of the plate?

SSC CHSL 21/03/2023 (1st Shift)

(a) Rs.432 (b) Rs.216 (c) Rs.72 (d) Rs.108

Q.403. $\triangle ABC$ is similar to $\triangle PQR$, AB = 4 cm, BC = 8 cm and AC = 10 cm. If QR = 16 cm, then what is the area of $\triangle PQR$?

SSC CHSL 21/03/2023 (2nd Shift)

(a) $2\sqrt{231} \text{ cm}^2$ (b) $4\sqrt{231} \text{ cm}^2$
(c) $4\sqrt{462} \text{ cm}^2$ (d) $8\sqrt{462} \text{ cm}^2$

Q.404. Three metallic cubes whose sides are 6 cm, 8 cm and 10 cm respectively are melted and converted into a single cube. If there is no loss of metal in this process, then what is the length of the side of the new cube?

SSC CHSL 21/03/2023 (3rd Shift)

(a) 16 cm (b) 14 cm (c) 12 cm (d) 20 cm

SSC CGL 2022 Tier - 2

Q.405. What is the volume of the largest sphere that can be carved out of a wooden cube of sides 21 cm? ($\pi = \frac{22}{7}$)

SSC CGL Tier II (02/03/2023)

(a) 3851 cm^3 (b) 6858 cm^3
(c) 4851 cm^3 (d) 5821 cm^3

Q.406. A rectangular park is 120 m long and 104 m wide. A 1 - m wide path runs along the boundary of the park, remaining completely inside the park area. Thus, the outside edges of the path run along the boundary wall of the park. The inside edges of the path are marked with a white line of negligible thickness. If it costs ₹2.50 to mark each metre with the white line, then how much would it cost (in ₹) to fully mark the inside edges of the path ?

SSC CGL Tier II (02/03/2023)

(a) 1090 (b) 1080 (c) 1120 (d) 1100

Q.407. A hemispherical bowl of internal radius 18 cm is full of liquid. This liquid is to be filled in cylindrical bottles each of radius 3 cm and height 6 cm. How many bottles are required to empty the bowl?

SSC CGL Tier II (03/03/2023)

(a) 72 (b) 70 (c) 68 (d) 66

Q.408. What is the total surface area of a pyramid whose base is a square with side 8 cm and height of the pyramid is 3 cm ?

SSC CGL Tier II (03/03/2023)

(a) 169 cm^2 (b) 121 cm^2
(c) 144 cm^2 (d) 184 cm^2

Q.409. The breadth 'b' of a room is twice its height and half of its length. Find the length of the longest diagonal of the room.

SSC CGL Tier II (03/03/2023)

(a) $\frac{\sqrt{20}b}{2}$ (b) $\frac{b}{2}$ (c) $\frac{\sqrt{21}b}{2}$ (d) $\frac{\sqrt{19}b}{2}$

Q.410. A pyramid has an equilateral triangle as its base, of which each side is

8 cm. Its slant edge is 24 cm. The whole surface area of the pyramid (in cm^2) is:

SSC CGL Tier II (06/03/2023)

- (a) $(16\sqrt{3} + 24\sqrt{35})$ (b) $(12\sqrt{3} + 24\sqrt{35})$
(c) $(24\sqrt{3} + 36\sqrt{35})$ (d) $(16\sqrt{3} + 48\sqrt{35})$

Q.411. If the curved surface area of a cylinder is $126\pi cm^2$ and its height is 14 cm, what is the volume of the cylinder.
SSC CGL Tier II (06/03/2023)

- (a) $283\frac{1}{2}\pi cm^3$ (b) $137\frac{1}{2}\pi cm^3$
(c) $128\frac{1}{2}\pi cm^3$ (d) $125\frac{1}{2}\pi cm^3$

Q.412. The difference between the length of two parallel sides of a trapezium is 12 cm. The perpendicular distance between these two parallel sides is 60 cm. If the area of the trapezium is $1380 cm^2$, then find the length of each of the parallel sides (in cm).
SSC CGL Tier II (06/03/2023)

- (a) 27, 15 (b) 31, 19 (c) 29, 17 (d) 24, 12

SSC CGL 2022 Tier - 1

Q.413. The length and breadth of a rectangle are increased by 8% and 5%, respectively. By how much percentage will the area of the rectangle increase?
SSC CGL 01/12/2022 (1st Shift)

- (a) 13.4% (b) 15.4% (c) 12.4% (d) 16.4%

Q.414. How many spherical lead shots each of diameter 8.4 cm can be obtained from a rectangular solid of lead with dimension 88 cm, 63 cm, 42 cm.

(take $\pi = \frac{22}{7}$) ?

SSC CGL 01/12/2022 (1st Shift)
(a) 920 (b) 750 (c) 650 (d) 860

Q.415. A solid metallic sphere of radius 13 cm is melted and recast into a cone having diameter of the base as 13 cm. What is the height (cm) of the cone ?
SSC CGL 01/12/2022 (3rd Shift)

- (a) 246 (b) 152 (c) 174 (d) 208

Q.416. The area of a triangle is $480 cm^2$ and the ratio of its sides is 10 : 24 : 26. What is the perimeter of the triangle ?
SSC CGL 01/12/2022 (4th Shift)

- (a) 120 cm (b) 30 cm
(c) 150 cm (d) 60 cm

Q.417. Which of the following two will provide more volume ?
I) A cuboid of edges 6 cm, 7 cm and 8 cm
II) A cube of edge 7 cm
SSC CGL 02/12/2022 (1st Shift)

- (a) Cube will have more volume
(b) Both will have equal volume

- (c) Cuboid will have more volume
(d) Cannot be determined

Q.418. A hemispherical bowl made of iron has an inner diameter of 84 cm. Find the cost of tin plating it on the inside at the rate of ₹21 per $100 cm^2$ (take $\pi = \frac{22}{7}$)

) correct to two places of decimal.

SSC CGL 02/12/2022 (1st Shift)

- (a) ₹2,328.48 (b) ₹2,425.48
(c) ₹2,425.60 (d) ₹2,355.48

Q.419. If the side of an equilateral triangle is 24 cm, then what is its area?

SSC CGL 02/12/2022 (2nd Shift)

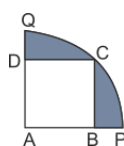
- (a) $169\sqrt{3} cm^2$ (b) $125\sqrt{3} cm^2$
(c) $256\sqrt{3} cm^2$ (d) $144\sqrt{3} cm^2$

Q.420. The radius of a right circular cylinder is five times of its height. If the height of the cylinder is 3.5 cm, then what is the volume of the cylinder?

SSC CGL 02/12/2022 (3rd Shift)

- (a) $3368.75 cm^3$ (b) $3872.75 cm^3$
(c) $3146.75 cm^3$ (d) $3524.25 cm^3$

Q.421. In the given figure, a square ABCD is inscribed in a quadrant APCQ. If AB = 16 cm, find the area of the shaded region (take $\pi = 3.14$) correct to two places of decimal.



SSC CGL 02/12/2022 (4th Shift)

- (a) $155.98 cm^2$ (b) $179.68 cm^2$
(c) $163.85 cm^2$ (d) $145.92 cm^2$

Q.422. If the volume of a sphere is $24,416.64 cm^3$, find its surface area (take $\pi = 3.14$) correct to two places of decimal.
SSC CGL 03/12/2022 (1st Shift)

- (a) $3069.55 cm^2$ (b) $4069.44 cm^2$
(c) $5069.66 cm^2$ (d) $6069.67 cm^2$

Q.423. The area of a sector of a circle is $66 cm^2$ and the angle of the sector is 60° . Find the radius of the circle.

SSC CGL 03/12/2022 (1st Shift)

- (a) $5\sqrt{15} cm$ (b) $6\sqrt{14} cm$
(c) $7\sqrt{19} cm$ (d) $3\sqrt{14} cm$

Q.424. The radius of a right circular cylinder is thrice its height. If the height of the cylinder is 2.1 cm, then what is the volume of the cylinder ?

SSC CGL 03/12/2022 (3rd Shift)

- (a) $224.65 cm^3$ (b) $194.72 cm^3$
(c) $324.86 cm^3$ (d) $261.95 cm^3$

Q.425. A solid copper sphere of radius 9 cm is hammered and moulded into a wire of radius 2 cm. What is the length of this

wire ?

SSC CGL 03/12/2022 (3rd Shift)

- (a) 224 cm (b) 183 cm
(c) 198 cm (d) 243 cm

Q.426. The length of the side of a cube is $5.6 cm$. What is the volume of the largest sphere that can be taken out of the cube ?

SSC CGL 05/12/2022 (1st Shift)

- (a) $91.98 cm^3$ (b) $99.96 cm^3$
(c) $96.98 cm^3$ (d) $90.69 cm^3$

Q.427. A hollow sphere has an outer radius of 4 cm and inner radius of 1 cm. What is the volume of this hollow sphere?

SSC CGL 05/12/2022 (3rd Shift)

- (a) $84\pi cm^3$ (b) $68\pi cm^3$
(c) $48\pi cm^3$ (d) $92\pi cm^3$

Q.428. Three cubes of equal volume are joined end to end. Find the surface area of the resulting cuboid if the diagonal of the cube is $6\sqrt{3} cm$.

SSC CGL 05/12/2022 (4th Shift)

- (a) $509 cm^2$ (b) $504 cm^2$
(c) $516 cm^2$ (d) $512 cm^2$

Q.429. A solid hemisphere has radius 21 cm. It is melted to form a cylinder such that the ratio of its curved surface area to total surface area is 2 : 5. What is the radius (in cm) of its base (take $\pi = \frac{22}{7}$) ?

SSC CGL 06/12/2022 (1st Shift)

- (a) 23 (b) 21 (c) 17 (d) 19

Q.430. Calculate the area of a sector of a circle with radius 10 metres and angle of 90 degrees at the centre.

SSC CGL 06/12/2022 (3rd Shift)

- (a) $79.3 m^2$ (b) $79.5 m^2$
(c) $78.5 m^2$ (d) $75 m^2$

Q.431. The ratio of the outer and the inner circumference of a circular path is 5 : 4. If the path is 50 metres wide, then what is the radius of the inner circle?

SSC CGL 06/12/2022 (3rd Shift)

- (a) 250 metres (b) 300 metres
(c) 200 metres (d) 210 metres

Q.432. The diagonal of the square is $8\sqrt{2} cm$. Find the diagonal of another square whose area is triple that of the first square.

SSC CGL 06/12/2022 (4th Shift)

- (a) $8\sqrt{5} cm$ (b) $8\sqrt{3} cm$
(c) $8\sqrt{2} cm$ (d) $8\sqrt{6} cm$

Q.433. Steel is used to make a hemispherical bowl that is 0.37 cm thick. The bowl's inner radius is 6 cm. Find the

bowl's outer curved surface area

(take $\pi = \frac{22}{7}$).

SSC CGL 07/12/2022 (1st Shift)

- (a) 532 cm² (b) 255.0548 cm²
(c) 484 cm² (d) 523.4107 cm²

Q.434. The length and breadth of a cuboid is increased by 20% and 25%, respectively, while its height is reduced by 30%. What must be the total percentage increase/decrease in the volume of the cuboid ?

SSC CGL 07/12/2022 (3rd Shift)

- (a) Increase by 5% (b) Increase by 3%
(c) Decrease by 2% (d) Decrease by 4%

Q.435. What is the area of the curved surface of a right circular cone with height 8 m and slant height 10 m?

SSC CGL 08/12/2022 (1st Shift)

- (a) 60 m² (b) 60 π m² (c) 40 m² (d) 40 π m²

Q.436. An iron rod with diameter 4 cm and length 12 cm is drawn into a wire of length 12 m of uniform thickness. Determine the thickness of the wire.

SSC CGL 08/12/2022 (2nd Shift)

- (a) 0.6 cm (b) 0.7 cm
(c) 0.3 cm (d) 0.4 cm

Q.437. A solid cone of height 42 cm with diameter of its base 42 cm is cut out from a wooden solid sphere of radius 24 cm. Find the percentage of wood wasted correct to two places of decimal.

SSC CGL 09/12/2022 (1st Shift)

- (a) 75.56% (b) 56.65%
(c) 66.50% (d) 67.50%

Q.438. The length of the side of a cube is 2.8 cm. What is the volume of the largest sphere that can be taken out of the cube ?

SSC CGL 09/12/2022 (2nd Shift)

- (a) 11.50 cm³ (b) 1.15 cm³
(c) 11.55 cm³ (d) 115 cm³

Q.439. The curved surface area of a right circular cone of base radius 21 cm is 594 sq cm. What is the slant height of the cone ?

SSC CGL 09/12/2022 (3rd Shift)

- (a) 15 cm (b) 11 cm (c) 9 cm (d) 6 cm

Q.440. The number of solid spheres, each of diameter 6 cm, that could be moulded to form a solid metal cylinder of height 90 cm and diameter 4 cm is:

SSC CGL 09/12/2022 (4th Shift)

- (a) 10 (b) 14 (c) 16 (d) 13

Q.441. The radius of a right circular cylinder is four times of its height. If the height of the cylinder is 14 cm, then what is the volume of the cylinder?

SSC CGL 12/12/2022 (2nd Shift)

- (a) 15468 cm³ (b) 14262 cm³
(c) 137984 cm³ (d) 11296 cm³

Q.442. The radii of the ends of a frustum of a cone 7 cm high are 5 cm and 3 cm. Find its volume correct to one decimal place. (Use $\pi = \frac{22}{7}$)

SSC CGL 12/12/2022 (2nd Shift)

- (a) 345.6 cm³ (b) 359.3 cm³
(c) 379.3 cm³ (d) 369.3 cm³

Q.443. Area of the floor of a cubical room is 64 m². The length of the longest rod that can be kept in the room is:

SSC CGL 12/12/2022 (3rd Shift)

- (a) $16\sqrt{3}$ m (b) $4\sqrt{3}$ m
(c) $12\sqrt{3}$ m (d) $8\sqrt{3}$ m

Q.444. A conical vessel, whose internal radius is 20 cm and height is 27 cm, is full of water. If this water is poured into a cylindrical vessel with internal radius 15 cm, what will be the height to which the water rises in it ?

SSC CGL 12/12/2022 (4th Shift)

- (a) 16 cm (b) 10 cm
(c) 12 cm (d) 14 cm

Q.445. If the numerical value of twice the curved surface area of a right circular cylinder is equal to the numerical value of its volume, then what is the numerical value of the radius of the base of the cylinder ?

SSC CGL 13/12/2022 (1st Shift)

- (a) 2 (b) 3 (c) 5 (d) 4

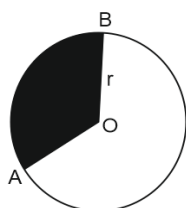
SSC CPO 2022 Tier - 1

Q.446. A field is in the shape of a rectangle of length 90 m and breadth 75 m. In one corner of the field, a pit, which is 18 m long, 15 m broad and 6 m deep, has been dug out. The Earth taken out of it is evenly spread over the remaining part of the field. Find the rise in the level of the field ?

SSC CPO 09/11/2022 (Morning)

- (a) 27 cm (b) 25 cm (c) 28 cm (d) 24 cm

Q.447. In the given figure, the length of arc AB is equal to twice the length of radius r of the circle. Find the area of sector OAB in terms of the radius r.



SSC CPO 09/11/2022 (Morning)

- (a) 3r (b) 2r (c) πr^2 (d) r^2

Q.448. Determine the total surface area of a hemisphere closed at bottom.

Radius of the hemisphere is $\sqrt{\frac{25}{\pi}}$ unit.

SSC CPO 09/11/2022 (Morning)

- (a) 75 unit² (b) 70 unit²
(c) 60 unit² (d) 50 unit²

Q.449. Find the perimeter of major sector of a circle of radius 12 metres, whose minor sector subtends an angle of 75° at the centre.

SSC CPO 09/11/2022 (Morning)

- (a) $24 + 5\pi$ metres (b) $24 + 19\pi$ metres
(c) $24 - 5\pi$ metres (d) $24 - 19\pi$ metres

Q.450. A frustum of a right circular cone, whose height is 18 cm, large base radius is 25 cm, and small base radius is 20 cm, is melted to form a small sphere of diameter 2 cm. How many spheres will be formed ?

SSC CPO 09/11/2022 (Afternoon)

- (a) 6287 (b) 6862 (c) 4575 (d) 5857

Q.451. The radius of a hollow sphere is 21 cm. It is cut into two equal halves by a plane passing through its centre. What is 75% of the curved surface area (in cm², rounded off to the nearest tens) of the hemisphere ? (Take $\pi = \frac{22}{7}$)

SSC CPO 09/11/2022 (Evening)

- (a) 4160 (b) 3470 (c) 2080 (d) 2770

Q.452. What is the total surface area of a solid right circular cylinder of radius 7 cm and height 8 cm? ($\pi = \frac{22}{7}$)

SSC CPO 09/11/2022 (Evening)

- (a) 560 cm² (b) 660 cm²
(c) 850 cm² (d) 760 cm²

Q.453. The total surface area of a solid metallic hemisphere is 462 cm². This is melted and moulded into a right circular cone. If the radius of the base of the cone is the same as that of the hemisphere, then its height is: (use $\pi = \frac{22}{7}$)

SSC CPO 09/11/2022 (Evening)

- (a) 14 cm (b) 7 cm (c) 21 cm (d) 28 cm

Q.454. As shown in the given figure, inside the large semicircle, three semicircles (with equal radii) are drawn so that their diameters all sit on the large semicircle's diameter. What is the ratio between the unshaded and shaded areas?



SSC CPO 09/11/2022 (Evening)

- (a) 4 : 3 (b) 1 : 2 (c) 2 : 1 (d) 3 : 4

Q.455. What is the length of the longest pole that can fit itself in a hall 60 feet long, 30 feet broad and 20 feet high?

SSC CPO 10/11/2022 (Morning)

(a) 50 ft. (b) 70 ft. (c) 30 ft. (d) 20 ft.

Q.456. The volume of a wall, six times as high as its breadth and half as long as its height, is 23958 cm^3 . What is the breadth of the wall?

SSC CPO 10/11/2022 (Morning)

(a) 21 cm (b) 15 cm (c) 11 cm (d) 18 cm

Q.457. The curved surface area and circumference at the base of a solid right circular cylinder are 2200 cm^2 and 110 cm, respectively. Find the height of the cylinder.

SSC CPO 10/11/2022 (Morning)

(a) 24 cm (b) 22 cm (c) 20 cm (d) 18 cm

Q.458. The length of a cold storage is 5 times its breadth, and its height is 8 m. The area of its four walls (including the doors) is 720 m^2 . Find the total surface area of cold storage.

SSC CPO 10/11/2022 (Afternoon)

(a) 1290.6 m^2 (b) 1275.3 m^2

(c) 1265.25 m^2 (d) 1282.5 m^2

Q.459. A cylinder of height 8 cm and radius 6 cm is melted and converted into 3 cones of the same radius and height that of cylinder. Determine the total curved surface area of cones.

SSC CPO 10/11/2022 (Evening)

(a) $180\pi \text{ cm}^2$ (b) $60\pi \text{ cm}^2$

(c) $144\pi \text{ cm}^2$ (d) $120\pi \text{ cm}^2$

Q.460. The heights of two right circular cones are in the ratio 1 : 5 and the perimeter of their bases are in the ratio 5: 3. Find the ratio of their volumes.

SSC CPO 10/11/2022 (Evening)

(a) 8 : 11 (b) 7 : 6 (c) 5 : 9 (d) 3 : 4

Q.461. The area of the sector of a circle is 128 cm^2 . If the length of the arc of that sector is 64 cm, then find the radius of the circle.

SSC CPO 10/11/2022 (Evening)

(a) 4 cm (b) 8 cm (c) 2 cm (d) 16 cm

Q.462. A cone-shaped storage tank's height and radius are 9 and 7 feet, respectively. Determine how much liquid the tank can contain. (Take $\pi = 3.14$)

SSC CPO 11/11/2022 (Morning)

(a) 471 cubic feet

(b) 450 cubic feet

(c) 435.32 cubic feet

(d) 461.58 cubic feet

Q.463. A hall of 10 m length, 5 m width and 6 m height is to be painted (one side

only). Determine the total surface area to be painted. (neglect the area of windows and doors) & (floor area is NOT to be painted)

SSC CPO 11/11/2022 (Morning)

(a) 300 m^2

(b) 280 m^2

(c) 330 m^2

(d) 230 m^2

Q.464. A hemispherical bowl of internal radius 6 cm contains a liquid. This liquid is to be filled into cylindrical shaped small bottles of diameter 2 cm and height 4 cm. How many bottles will be needed to empty the bowl?

SSC CPO 11/11/2022 (Afternoon)

(a) 32

(b) 37

(c) 38

(d) 36

Q.465. The total surface area of a hemisphere is 462 cm^2 . The diameter of this hemisphere is:

SSC CPO 11/11/2022 (Evening)

(a) 28 cm (b) 21 cm (c) 7 cm (d) 14 cm

Q.466. The difference between the semi-perimeter and the sides of $\triangle PQR$ are 18 cm, 17 cm and 25 cm, respectively. Find the area of the triangle

SSC CPO 11/11/2022 (Evening)

(a) $330\sqrt{510} \text{ cm}^2$ (b) $230\sqrt{510} \text{ cm}^2$

(c) $30\sqrt{510} \text{ cm}^2$ (d) $130\sqrt{510} \text{ cm}^2$

SSC CGL 2021 Tier - 2

Q.467. The sum of the curved surface area and the total surface area of a solid cylinder is 2068 cm^2 . If the radius of its base is 7 cm, then what is the volume of this cylinder? (use $\pi = \frac{22}{7}$)

SSC CGL Tier II (08/08/2022)

(a) 2060 cm^3 (b) 2480 cm^3

(c) 3080 cm^3 (d) 2760 cm^3

Q.468. The slant height of a cone is 20 cm. If area of its base is 616 cm^2 , then what is the curved surface area of this cone?

SSC CGL Tier II (08/08/2022)

(a) 960 cm^2 (b) 440 cm^2

(c) 1760 cm^2 (d) 880 cm^2

Q.469. The height of a cylinder is 6 cm more than the radius of its base. If its radius is 14 cm, then what will be volume of this cylinder? (use $\pi = \frac{22}{7}$)

SSC CGL Tier II (08/08/2022)

(a) 13560 cm^3 (b) 14340 cm^3

(c) 10440 cm^3 (d) 12320 cm^3

Q.470. Radius of a large solid sphere is 14 cm. It is melted to form 8 equal solid spheres. What is the sum of total surface

areas of all the 8 small solid spheres?

(use $\pi = \frac{22}{7}$)

SSC CGL Tier II (08/08/2022)

(a) 3648 cm^2 (b) 4928 cm^2

(c) 4244 cm^2 (d) 4158 cm^2

Q.471. ABCD is a quadrilateral. The length of diagonal AC is 24 cm. The sum of perpendiculars drawn from vertex B and D on the diagonal AC is 10 cm. What is the area of the quadrilateral?

SSC CGL Tier II (08/08/2022)

(a) 240 cm^2 (b) 180 cm^2

(c) 120 cm^2 (d) 90 cm^2

Q.472. The base of a prism is a right angle triangle whose sides are 9 cm, 12 cm and 15 cm. Volume of this prism is 648 cm^3 . What will be the height of prism?

SSC CGL Tier II (08/08/2022)

(a) 14 cm (b) 12 cm (c) 9 cm (d) 16 cm

Q.473. The difference between the areas of two concentric circles is 264 cm^2 . What is the difference between the square of their radius? (use $\pi = \frac{22}{7}$)

SSC CGL Tier II (08/08/2022)

(a) 70 cm^2 (b) 140 cm^2

(c) 84 cm^2 (d) 64 cm^2

Q.474. Volume of a cone whose radius of a base and height are r and h respectively, is 400 cm^3 . What will be the volume of a cone whose radius of base and height are 2r cm and h cm respectively?

SSC CGL Tier II (08/08/2022)

(a) 100 cm^3 (b) 1600 cm^3

(c) 1200 cm^3 (d) 800 cm^3

SSC Selection Post (Phase - X)

Q.475. A conical vessel has radius 4 cm and its curved surface area is $20\pi \text{ cm}^2$. The volume of conical vessel is:

Graduate Level 02/08/2022 (Shift - 3)

(a) $18\pi \text{ cm}^3$ (b) $12\pi \text{ cm}^3$

(c) $26\pi \text{ cm}^3$ (d) $16\pi \text{ cm}^3$

Q.476. A student wants to know the area of a rhombus, one of whose diagonals is 6 cm. He measures another diagonal as 8 cm. What is the area of the rhombus?

Graduate Level 05/08/2022 (Shift - 3)

(a) 25 cm^2 (b) 20 cm^2

(c) 24 cm^2 (d) 48 cm^2

Q.477. The volumes of two cubes are in the ratio of 8 : 27. If the volume of the larger cube is 513 cm^3 more than that of

the smaller cube, find the length of each side of the smaller cube.

Higher Secondary 01/08/2022 (Shift - 3)

(a) 27 cm (b) 7 cm (c) 6 cm (d) 16 cm

Q.478. A chord of a circle of radius 21 cm makes a right angle at the center. Find the area of major segment.

(take $\pi = \frac{22}{7}$) closest to nearest integer.

Higher Secondary 03/08/2022 (Shift - 4)

(a) 1360 cm^2 (b) 1436 cm^2
(c) 1260 cm^2 (d) 1160 cm^2

SSC MTS 2021 Tier - 1

Q.479. A goat is tied with a 10.5 m-long rope to one corner of a plot which is in the shape of an equilateral triangle each side of which is 16 m long. Find the area that the goat can graze on.

SSC MTS 05/07/2022 (Morning)

(a) 62.25 m^2 (b) 53.5 m^2
(c) 57.75 m^2 (d) 51.75 m^2

Q.480. If the height and the radius of the base of a cone are both doubled, the volume of the cone becomes _____ of its previous volume.

SSC MTS 05/07/2022 (Morning)

(a) 4 times (b) 3 times
(c) 6 times (d) 8 times

Q.481. A smaller circle touches a bigger circle internally and also passes through the centre 'O' of the bigger circle. If the area of the smaller circle is 192 cm^2 , the area of the bigger circle (in cm^2) is :

SSC MTS 05/07/2022 (Evening)

(a) 768 (b) 384 (c) 1024 (d) 720

Q.482. The perimeter of a square is equal to the perimeter of a rectangle, and the latter has a width of 12 cm. The area of the square is 400 cm^2 . Find the area of the rectangle.

SSC MTS 06/07/2022 (Morning)

(a) 412 cm^2 (b) 394 cm^2
(c) 324 cm^2 (d) 336 cm^2

Q.483. If the total surface area of a sphere of radius r cm is equal to the total surface area of a cube of side s cm, which of the following is true ?

SSC MTS 06/07/2022 (Morning)

(a) $r > s$ (b) $r \geq s$ (c) $r = s$ (d) $r < s$

Q.484. The volume of a wall whose height is six times its breadth and length is half its height, is 23958 cm^3 . What is the breadth of the wall ?

SSC MTS 06/07/2022 (Afternoon)

(a) 21 cm (b) 15 cm (c) 11 cm (d) 18 cm

Q.485. Find the area of a trapezium whose parallel sides are of length 15 cm and 20 cm, respectively, and whose distance between the parallel sides is 12 cm.

SSC MTS 06/07/2022 (Evening)

(a) 280 cm^2 (b) 210 cm^2
(c) 360 cm^2 (d) 120 cm^2

Q.486. A goldsmith uses 110 cm^3 of gold to make a wire 1 mm in diameter. He will use this wire to make gold chains of equal lengths. If each chain requires 56 cm length of wire, then how many gold chains can he make? [use $\pi = \frac{22}{7}$]

SSC MTS 07/07/2022 (Morning)

(a) 320 (b) 250 (c) 160 (d) 210

Q.487. The measures of the outer dimensions of a closed metal box are given as $15 \text{ cm} \times 18 \text{ cm} \times 8 \text{ cm}$. The thickness of the metal is 1.5 cm. What will be the volume of the metal required to make the box ?

SSC MTS 07/07/2022 (Evening)

(a) 1680 cm^3 (b) 950 cm^3
(c) 1440 cm^3 (d) 1260 cm^3

Q.488. The length of a side of an equilateral triangle is 120 cm less than its perimeter. What is the area of the equilateral triangle?

SSC MTS 07/07/2022 (Evening)

(a) $1800\sqrt{3} \text{ cm}^2$ (b) $900\sqrt{3} \text{ cm}^2$
(c) $450\sqrt{3} \text{ cm}^2$ (d) $1200\sqrt{3} \text{ cm}^2$

Q.489. The volume of a cuboid is 3300 cm^3 . The sum of the length and the breadth of the cuboid is 37 cm. If the difference between the length and the breadth of the cuboid is 13 cm, then what is the height of the cuboid?

SSC MTS 08/07/2022 (Morning)

(a) 18 cm (b) 11 cm (c) 14 cm (d) 16 cm

Q.490. The length and the breadth of a rectangular cardboard are 40 cm and 35 cm, respectively. If the biggest possible square is cut out from this cardboard, what is the area of the remaining part of the cardboard ?

SSC MTS 08/07/2022 (Afternoon)

(a) 175 cm^2 (b) 275 cm^2
(c) 425 cm^2 (d) 325 cm^2

Q.491. The diagonal of a rectangle is 12 cm long and it is twice as long as one of the sides of the rectangle. Find area of the rectangle ?

SSC MTS 08/07/2022 (Evening)

(a) $25\sqrt{3} \text{ cm}^2$ (b) $30\sqrt{3} \text{ cm}^2$
(c) $36\sqrt{3} \text{ cm}^2$ (d) $42\sqrt{3} \text{ cm}^2$

Q.492. A solid sphere whose radius is 21 cm is melted and recast into a regular pyramid of square base. If the height of the pyramid is 66 cm, then what is the length of each side of the base of the pyramid ? [use $\pi = \frac{22}{7}$].

SSC MTS 08/07/2022 (Evening)

(a) 68 cm (b) 85 cm (c) 42 cm (d) 48 cm

Q.493. the total surface area of a closed cube is 128 cm^2 more than the area of a square. If the length of each side of the cube is 8 cm, then what is the length of each side of the square?

SSC MTS 11/07/2022 (Morning)

(a) 16 cm (b) 12 cm (c) 18 cm (d) 9 cm

Q.494. In a camp, there are tents of the same shape and size. Each tent is cylindrical up to a height of 4 m and conical above it. The diameters of the bases of the cylinder and the cone are both 10.5 m and the slant height of the conical part is 10 m. If a total of 3861 m^2 canvas is used in making all the tents, then how many tents are there in the

camp ? [use $\pi = \frac{22}{7}$]

SSC MTS 11/07/2022 (Evening)

(a) 11 (b) 7 (c) 19 (d) 13

Q.495. A rectangular LCD has an area of 480 cm^2 and a perimeter of 92 cm. If the size of the LCD is defined in term of length of the diagonal, then what is the size of the LCD?

SSC MTS 12/07/2022 (Evening)

(a) 34 cm (b) 39 cm (c) 31 cm (d) 35 cm

Q.496. If the circumference of a circle is equal to the perimeter of a square and the radius of the given circle is positive, then which of the following options is correct ?

SSC MTS 12/07/2022 (Evening)

(a) Area of the circle > Area of the square
(b) Area of the circle $\geq$ Area of the square
(c) Area of the circle < Area of the square
(d) Area of the circle = Area of the square

Q.497. If the length of one side of a rectangle is twice that of the other side, and its perimeter is 54 cm, then what is the length of the longer side of this rectangle ?

SSC MTS 13/07/2022 (Evening)

(a) 14 cm (b) 9 cm (c) 27 cm (d) 18 cm

Q.498. A person uses a rectangular

metallic sheet of 45 m × 30 m for making an open box. From each corner of the sheet he cut offs a square. If the length of one side of each of the squares is 10 m, then what will be the volume of the box (in m³) ?

SSC MTS 13/07/2022 (Evening)

(a) 5000 (b) 4500 (c) 7000 (d) 2500

Q.499. The total surface area of a closed hemisphere is 16632 cm². What is the sum of the radius and the diameter of this hemisphere? [use $\pi = \frac{22}{7}$]

SSC MTS 14/07/2022 (Afternoon)

(a) 172 cm (b) 168 cm
(c) 126 cm (d) 154 cm

Q.500. The length of the longest diagonal that can be placed in a closed cube is $50\sqrt{3}$ cm. What is the total surface area of this cube ?

SSC MTS 14/07/2022 (Evening)

(a) 15000 cm² (b) 10000 cm²
(c) 12000 cm² (d) 6000 cm²

Q.501. The larger diagonal of a rhombus is 150% of its smaller diagonal, and its area is 432 cm². Find the length (in cm) of the side of the rhombus ?

SSC MTS 14/07/2022 (Evening)

(a) $8\sqrt{13}$ (b) $4\sqrt{13}$ (c) $6\sqrt{13}$ (d) $2\sqrt{13}$

Q.502. The surface area of a sphere is 2464 cm². Find the radius of the sphere.

[use $\pi = \frac{22}{7}$]

SSC MTS 15/07/2022 (Afternoon)

(a) 14 cm (b) 28 cm (c) 42 cm (d) 40 cm

Q.503. The length of each side of a rhombus is 10 cm. If the length of one of its diagonals is 16 cm, then what is the area of the rhombus?

SSC MTS 15/07/2022 (Evening)

(a) 108 cm² (b) 112 cm²
(c) 128 cm² (d) 96 cm²

Q.504. Two cubes, each having a total surface area of 150cm^2 , are joined together to form a cuboid. Find the total surface area of the resulting cuboid.

SSC MTS 18/07/2022 (Afternoon)

(a) 225 (b) 250 (c) 275 (d) 300

Q.505. The circumference of a circle is equal to the perimeter of an equilateral triangle. If the radius of the circle is 84 cm, then what will be the length of each side of the equilateral triangle ?

[use $\pi = \frac{22}{7}$]

SSC MTS 19/07/2022 (Afternoon)

(a) 192 cm (b) 176 cm
(c) 172 cm (d) 184 cm

Q.506. The height of a cylinder is 3 cm more than the radius of its base. If the height is 10 cm, then what is the volume of this cylinder ? [use $\pi = \frac{22}{7}$]

SSC MTS 20/07/2022 (Morning)

(a) 1540 cm^3 (b) 1860 cm^3
(c) 2160 cm^3 (d) 1240 cm^3

Q.507. What is the radius of a circular field whose area is equal to the sum of the area of three smaller circular fields of radii 8 m, 9m and 12m respectively?

SSC MTS 20/07/2022 (Morning)

(a) 20 m (b) 21 m (c) 29 m (d) 17 m

Q.508. A toy is formed by placing a solid circular cone on a solid hemisphere and the radius of the common base of the cone and the hemisphere is 15 cm, If the volume of the toy is $2850\pi\text{cm}^3$, then find the total surface area of that toy.

SSC MTS 20/07/2022 (Morning)

(a) $815\pi\text{ cm}^2$ (b) $755\pi\text{ cm}^2$
(c) $705\pi\text{ cm}^2$ (d) $685\pi\text{ cm}^2$

Q.509. A large solid cube is melted and cast into 'N' small solid spheres, each of radius 3 cm, and 'N+2' small solid cuboids, Each of dimensions 4 cm × 4cm × 6.5cm. If the length of each side of the large solid cube is 12 cm, then find the value of 'N' [use $\pi = \frac{22}{7}$]

SSC MTS 20/07/2022 (Afternoon)

(a) 8 (b) 5 (c) 7 (d) 6

Q.510. The height of a trapezium is 68 cm and the sum of its parallel sides is 75 cm. If the area of the trapezium is $\frac{6}{17}$ times of the area of a square, then the length of the diagonal of the square is:

(take $\sqrt{2} = 1.41$)

SSC MTS 21/07/2022 (Morning)

(a) 127.39 cm (b) 183.49 cm
(c) 119.85 cm (d) 102.39 cm

Q.511. The length of a cuboidal room is 56 m, its breadth is 34 m and its height is 8 m. Find the length of the longest ladder that can be placed in the room.

SSC MTS 21/07/2022 (Afternoon)

(a) 66 m (b) 58 m (c) 42 m (d) 33 m

Q.512. Find the total surface area of a closed hemisphere whose diameter is 7 cm. [Use $\pi = \frac{22}{7}$]

SSC MTS 21/07/2022 (Evening)

(a) 77 cm^2 (b) 154 cm^2
(c) 110 cm^2 (d) 115.5 cm^2

Q.513. The ratio of the length to the width of a rectangle is 3:2. If the length of this rectangle is increased by 25% and its width is kept constant, then the area of the rectangle increases by 24 m^2 . What is the width of this rectangle?

SSC MTS 22/07/2022 (Morning)

(a) 12 m (b) 10 m (c) 8 m (d) 15 m

Q.514. If the ratio of an external angle and an internal angle of a regular polygon is 1 : 5, then what is the number of sides in the polygon ?

SSC MTS 22/07/2022 (Afternoon)

(a) 24 (b) 22 (c) 12 (d) 30

Q.515. The length, the breadth, and the height of a cuboidal room are given as 80 m, 70 m, and 80 m, respectively. What will be the total cost of painting the four walls of this room, at the rate of 50 paise per square meter ?

SSC MTS 25/07/2022 (Morning)

(a) Rs 1,20,000 (b) Rs 28000
(c) Rs 48000 (d) Rs 12000

Q.516. The area of a square is 1296 cm^2 and the radius of a circle is $\frac{7}{6}$ of the length of a side of the square. What is the ratio of the perimeter of the square and the circumference of the circle ?

(use $\pi = \frac{22}{7}$)

SSC MTS 25/07/2022 (Morning)

(a) 13:11 (b) 8:11 (c) 6:11 (d) 3:7

Q.517. The length, the breadths, and the volumes of two cuboids are in the ratios of 4 : 5, 3 : 4 and 2 : 3 respectively. What is the ratio of their heights ?

SSC MTS 25/07/2022 (Afternoon)

(a) 3 : 5 (b) 10 : 9 (c) 8 : 15 (d) 2 : 3

Q.518. The breadth of a rectangle is $\frac{4}{5}$ of the radius of a circle. The radius of the circle is $\frac{1}{5}$ of the side of a square, whose area is 625 cm^2 . What is the area of the rectangle if the length of rectangle is 20 cm ?

SSC MTS 25/07/2022 (Afternoon)

(a) 150 cm^2 (b) 60 cm^2
(c) 100 cm^2 (d) 80 cm^2

Q.519. The diameter of a circle is $\frac{7}{4}$ times of the base of a triangle, and the height of the triangle is 14 cm. If the area of the triangle is 56cm^2 , then what is the circumference (in m) of the circle - (use $\pi = \frac{22}{7}$)

SSC MTS 26/07/2022 (Morning)

- (a) 0.44 m (b) 2.46 m
(c) 0.48 m (d) 1.74 m

Q.520. A square of side p is taken. A rectangle is cut out from this square such that the length of one side of the rectangle is equal to half of the length of one side of the square and the length of another side of the rectangle is equal to $\frac{1}{3}$ of the length of the first side of the rectangle. What is the area of the portion of the square that remained after the rectangle is cut out?

SSC MTS 26/07/2022 (Afternoon)

- (a) $\frac{7}{8}p^2$ (b) $\frac{3}{4}p^2$ (c) $\frac{11}{12}p^2$ (d) $\frac{15}{16}p^2$

Q.521. The length of a rope, by which a calf is tied, is decreased from 9 m to 5 m. Find the decrease in the area that can be grazed by the calf. [Use $\pi = \frac{22}{7}$]

SSC MTS 26/07/2022 (Evening)

- (a) $112m^2$ (b) $154m^2$
(c) $122m^2$ (d) $176m^2$

SSC CHSL 2021 Tier - 1

Q.522. A pillar 11 m in radius is 21 m high. How much material (in m^3) was used to construct it? (Use $\pi = \frac{22}{7}$)

SSC CHSL 24/05/2022 (Morning)

- (a) 7996 (b) 7886 (c) 7989 (d) 7986

Q.523. A sphere passes through the eight corners of a cube of side 14 cm. Find the volume (in cm^3) of the sphere.
($\pi = \frac{22}{7}$)

SSC CHSL 25/05/2022 (Morning)

- (a) $4312\sqrt{3}$ (b) $4012\sqrt{3}$
(c) $4312\sqrt{2}$ (d) $4012\sqrt{2}$

Q.524. A pond is 50 m long, 35 m wide and 2.5 m deep. Find the capacity of the pond.

SSC CHSL 25/05/2022 (Afternoon)

- (a) 4165 (b) 4375 (c) 4165 (d) 3850

Q.525. A rectangular water tank is 1.5 m high, 3 m long and 3.5 m wide. How much water (in litres) can it hold?

SSC CHSL 25/05/2022 (Evening)

- (a) 13520 (b) 14510 (c) 15750 (d) 11780

Q.526. How many $25\text{ cm} \times 11.25\text{ cm} \times 6\text{ cm}$ bricks will be required to construct an $8\text{ m} \times 6\text{ m} \times 22.5\text{ cm}$ wall? (ignoring other material used)

SSC CHSL 26/05/2022 (Morning)

- (a) 7020 (b) 6400 (c) 5800 (d) 6800

Q.527. What is the volume of the largest right circular cone that can be cut out from a cube whose edge is 10 cm?

SSC CHSL 26/05/2022 (Morning)

- (a) $\frac{250\pi}{3}$ (b) $\frac{351\pi}{3}$ (c) $\frac{145\pi}{3}$ (d) $\frac{150\pi}{3}$

Q.528. 15 identical solid spherical balls of radius 5 cm are melted to form a single sphere. In this process, 35% of the mass is wasted. What is the radius (in cm) of the single sphere that is now formed? [Use $\pi = \frac{22}{7}$, and give your answer correct to two decimal places.]

SSC CHSL 26/05/2022 (Evening)

- (a) 12.68 (b) 16.68 (c) 10.68 (d) 34.68

Q.529. As air is pumped into a spherical balloon, its radius increases from 7 cm to 14 cm. Find the ratio of volume of the balloon's in both circumstances.

SSC CHSL 27/05/2022 (Morning)

- (a) 2 : 5 (b) 1 : 9 (c) 1 : 8 (d) 3 : 7

Q.530. A solid right-circular cylinder, whose radius of the base is 15 cm and height is 12 cm, is melted and moulded into the solid right - circular cone, whose radius of the base is 24 cm. What will be the height of this cone?

SSC CHSL 27/05/2022 (Evening)

- (a) 14.0625 cm (b) 14.0675 cm
(c) 14.6025 cm (d) 14.0525 cm

Q.531. Find the total surface area (cm^2) of a container in cylindrical shape whose diameter is 14 cm and height is 25 cm.

[Use $\pi = \frac{22}{7}$]

SSC CHSL 30/05/2022 (Afternoon)

- (a) 1452 (b) 1392 (c) 1408 (d) 1424

Q.532. A solid toy is in the shape which is a combination of a cylinder, cone and a hemispherical bowl. The cylinder contributes to 50% of the total volume of the toy, the cone contributes to 20% of the volume. Find the ratio of the contribution (in terms of volume) of the cone, cylinder and hemisphere.

SSC CHSL 30/05/2022 (Afternoon)

- (a) 2 : 3 : 5 (b) 5 : 2 : 3
(c) 2 : 5 : 3 (d) 4 : 5 : 3

Q.533. What is the volume (cm^3) of a right circular cylinder whose curved surface area is $12\pi\text{ cm}^2$ and the circumference of its base is $4\pi\text{ cm}$?

SSC CHSL 30/05/2022 (Evening)

- (a) 48π (b) 36π (c) 12π (d) 24π

Q.534. The sum of the length, breadth and depth of a cuboid is 23 cm, and its diagonal is $5\sqrt{7}\text{ cm}$. Its total surface area (cm^2) is:

SSC CHSL 31/05/2022 (Morning)

- (a) 144 (b) 354 (c) 188 (d) 111

Q.535. A cone-shaped temple top has slant height of 10 m and radius of base is 11 m. Find the total cost for painting its curved surface area at the rate of $\text{₹}7/m^2$.

SSC CHSL 31/05/2022 (Afternoon)

- (a) ₹2440 (b) ₹2040
(c) ₹2420 (d) ₹2240

Q.536. The area of a triangle is 96 cm^2 and the length of one of its sides is 24 cm. What is the length of the perpendicular drawn to this side of the triangle from the opposite vertex?

SSC CHSL 01/06/2022 (Morning)

- (a) 16 cm (b) 8 cm (c) 4 cm (d) 12 cm

Q.537. Find the volume (in cm^3) of the cuboid, if its length is twice more than its breadth and its breadth is 8 cm, which is twice its height.

SSC CHSL 01/06/2022 (Afternoon)

- (a) 768 (b) 769 (c) 767 (d) 766

Q.538. If the volume of a cube is 64 cm^3 , then find its total surface area (cm^2).

SSC CHSL 01/06/2022 (Evening)

- (a) 96 (b) 82 (c) 88 (d) 64

Q.539. If the length of a cuboid is 20 m, breadth is 5 m and height is 4 m, then find the length of diagonal of the cuboid.

SSC CHSL 01/06/2022 (Evening)

- (a) 12 m (b) 21 m (c) 18 m (d) 24 m

Q.540. The radius of a hemispherical bowl is given as 7 cm. What is the capacity (cm^3) of the bowl? [$\pi = \frac{22}{7}$]

SSC CHSL 02/06/2022 (Afternoon)

- (a) $718\frac{2}{3}$ (b) $812\frac{1}{3}$ (c) $618\frac{1}{3}$ (d) $518\frac{2}{3}$

Q.541. The length of each side of a regular hexagon is 10 cm. What is the area (cm^2) of the hexagon?

SSC CHSL 02/06/2022 (Evening)

- (a) $156\sqrt{3}$ (b) $144\sqrt{3}$ (c) $148\sqrt{3}$ (d) $150\sqrt{3}$

Q.542. The area of a square is $9m^2 + 12m + 4$. Find the measure of the side of the square.

SSC CHSL 02/06/2022 (Evening)

- (a) $3m + 2$ (b) $3m + 4$
(c) $3m^2 + 2$ (d) $3m^2 + 2m$

Q.543. If a solid piece of iron in the form of a cuboid, of dimensions $49\text{ cm} \times 33\text{ cm} \times 24\text{ cm}$, is moulded to form a solid sphere, then the radius of the sphere is

SSC CHSL 03/06/2022 (Morning)

- (a) 16 cm (b) 10 cm
(c) 18 cm (d) 21 cm

Q.544. The area of a square is $4x^2 - 12x + 9$, which of the following will be the side of the square?

SSC CHSL 03/06/2022 (Afternoon)

- (a) $(2x + 3)$ (b) $(2x - 3)$
(c) $(3x - 2)$ (d) $(3x + 2)$

Q.545. If the areas of three adjacent faces of a cuboidal box are 729 cm^2 , 529 cm^2 and 289 cm^2 , respectively, then find the volume of the box.

SSC CHSL 03/06/2022 (Evening)

- (a) 10557 cm^3 (b) 10560 cm^3
(c) 10555 cm^3 (d) 10551 cm^3

Q.546. The diameter of the base and the slant height of a right circular cone are 10 cm and 16cm, respectively. Find its total surface area (cm^2). [Use $\pi = \frac{22}{7}$]

SSC CHSL 06/06/2022 (Morning)

- (a) $\frac{5720}{7}$ (b) $\frac{720}{7}$ (c) 330 (d) $\frac{1760}{7}$

Q.547. The diagonal of a rhombus is 8 cm and its area is 48 cm^2 , then the length of each side of the rhombus is:

SSC CHSL 06/06/2022 (Afternoon)

- (a) $\sqrt{13}$ (b) $2\sqrt{13}$ (c) $6\sqrt{13}$ (d) $5\sqrt{13}$

Q.548. The total surface area of a solid cylinder whose radius is r cm and height is $\frac{r}{2}$ cm (in cm^2) is:

SSC CHSL 07/06/2022 (Morning)

- (a) $3\pi \frac{r^2}{2}$ (b) $4\pi r^2$ (c) $3\pi r^2$ (d) $\pi \frac{r^3}{2}$

Q.549. The base of a hemispherical bowl has a circumference of 132 cm. Find the capacity of the bowl [Use $\pi = \frac{22}{7}$]

SSC CHSL 07/06/2022 (Afternoon)

- (a) 18240 cm^3 (b) 19404 cm^3
(c) 19708 cm^3 (d) 16540 cm^3

Q.550. Find the volume of the cylinder circumscribing a sphere with a diameter of 14 cm. [Use $\pi = \frac{22}{7}$]

SSC CHSL 07/06/2022 (Evening)

- (a) 2156 cm^3 (b) 2158 cm^3
(c) 2155 cm^3 (d) 2157 cm^3

Q.551. The surface area of a spherical volleyball is $100\pi \text{ cm}^2$. Find the volume of air enclosed in the ball.

SSC CHSL 08/06/2022 (Morning)

- (a) $\frac{400}{3} \pi$ (b) $\frac{350}{3} \pi$
(c) $\frac{500}{3} \pi$ (d) $\frac{200}{3} \pi$

Q.552. The total surface area of a hemisphere is 64.57 cm^2 . Find its curved surface area (cm^2).

SSC CHSL 08/06/2022 (Morning)

- (a) 64.57 (b) 53.44 (c) 43.04 (d) 33.04

Q.553. Find the length of diagonal (m) of a cuboid 13 m long, 7 m broad and 3 m high.

SSC CHSL 08/06/2022 (Evening)

- (a) $\sqrt{207}$ (b) $\sqrt{232}$ (c) $\sqrt{327}$ (d) $\sqrt{227}$

Q.554. The perimeter of an equilateral triangle is 75 cm. Find its area.

SSC CHSL 08/06/2022 (Evening)

- (a) $\frac{625\sqrt{2}}{4} \text{ cm}^2$ (b) $\frac{625\sqrt{3}}{4} \text{ cm}^2$
(c) $\frac{625\sqrt{3}}{3} \text{ cm}^2$ (d) $\frac{125\sqrt{3}}{4} \text{ cm}^2$

Q.555. If the curved surface area of a closed cylinder is 225 cm^2 and the area of its base is 169 cm^2 , then find the total surface area (cm^2) of the cylinder.

SSC CHSL 09/06/2022 (Morning)

- (a) 394 (b) 343 (c) 563 (d) 195

Q.556. The radius of a roller is 14 cm and its length 20cm. It takes 235 complete revolutions to move once over to level a playground. Find the area of the playground. (Use $\pi = \frac{22}{7}$)

SSC CHSL 09/06/2022 (Afternoon)

- (a) 4136 cm^2 (b) $4136 \times 10^3 \text{ cm}^2$
(c) 41360 cm^2 (d) $4136 \times 10^2 \text{ cm}^2$

Q.557. The length of the hypotenuse of a right-angled triangle is 37cm and the length of one of its other two sides is 12cm. What is the area of the triangle in cm^2 ?

SSC CHSL 09/06/2022 (Evening)

- (a) 222 (b) 204 (c) 216 (d) 210

Q.558. What is the length of the arc of a circle whose radius is 35 cm and arc subtends an angle of 36° at the centre of the circle? [Use $\pi = \frac{22}{7}$]

SSC CHSL 10/06/2022 (Morning)

- (a) 44 (b) 22 (c) 10 (d) 220

Q.559. The area of quadrilateral is 336 m^2 and the perpendiculars drawn to one diagonal from the opposite vertices are 16 m and 12 m long. Find the length of this diagonal.

SSC CHSL 10/06/2022 (Morning)

- (a) 28 cm (b) 26 cm (c) 21 cm (d) 24 cm

Q.560. How many spherical balls of radius 5 cm can be made by melting a spherical clay ball having a radius of 15 cm?

SSC CHSL 10/06/2022 (Afternoon)

- (a) 18 (b) 3 (c) 9 (d) 27

Q.561. What is the height of a solid right circular cylinder whose radius is 3 cm and total surface area is $60\pi \text{ cm}^2$?

SSC CHSL 10/06/2022 (Evening)

- (a) 3 cm (b) 5 cm (c) 7 cm (d) 9 cm

SSC CGL 2021 Tier - 1

Q.562. A solid cube of side 8 cm is dropped into a rectangular container of length 16cm, breadth 8 cm and height 15cm which is partly filled with water. If the cube is completely submerged, then the rise of water level (in cm) is:

SSC CGL 11/04/2022 (Morning)

- (a) 2 (b) 6 (c) 5 (d) 4

Q.563. A solid cube of side 8 cm is dropped into a rectangular container of length 16cm, breadth 8 cm and height 15cm which is partly filled with water. If the cube is completely submerged, then the rise of water level (in cm) is:

SSC CGL 11/04/2022 (Morning)

- (a) 2 (b) 6 (c) 5 (d) 4

Q.564. The length and the breadth of a rectangle are made to increase and decrease, respectively, by 8% and 10%. What is the percentage increase or decrease in its area?

SSC CGL 11/04/2022 (Afternoon)

- (a) Decrease by 1.8 (b) Increase by 1.8
(c) Increase by 2.8 (d) Decrease by 2.8

Q.565. What is the difference in the volume (in cm^3) of a sphere of radius 7 cm and that of a cone of a radius 7 cm and height of 10 cm? [Use $\pi = \frac{22}{7}$]

SSC CGL 12/04/2022 (Morning)

- (a) 205 (b) 704 (c) 924 (d) 1078

Q.566. If the length of a diagonal of a square is $(a + b)$, then the area of the square is:

SSC CGL 13/04/2022 (Morning)

- (a) $a^2 + b^2$ (b) $\frac{1}{2}(a^2 + b^2) + ab$
(c) $a^2 + b^2 + 2ab$ (d) $\frac{1}{2}(a^2 + b^2)$

Q.567. Let $x \text{ cm}^2$ be the surface area and $y \text{ cm}^3$ be the volume of a sphere such that $y = 14x$. What is the radius (in cm) of the sphere?

SSC CGL 13/04/2022 (Evening)

- (a) 102 (b) 42 (c) 51 (d) 68

Q.568. A hemispherical depression of diameter 4 cm is cut out from each face of a cubical block of sides 10 cm. Find

the surface area of the remaining solid

(in cm^2) (Use $\pi = \frac{22}{7}$)

SSC CGL 18/04/2022 (Morning)

- (a) $900\frac{4}{7}$ (b) $112\frac{4}{7}$
(c) $675\frac{3}{7}$ (d) $713\frac{1}{7}$

Q.569. The area of a cardboard (in cm^2) needed to make a closed box of size $20\text{ cm} \times 10\text{ cm} \times 8\text{ cm}$ will be:

SSC CGL 18/04/2022 (Evening)

- (a) 880 (b) 690 (c) 750 (d) 960

Q.570. From the body of a solid cube of edge 7 cm , a solid sphere is removed. The volume of the remaining solid was found to be $163\frac{1}{3}\text{ cm}^3$. What is the diameter (in cm) of the sphere?

(Use $\pi = \frac{22}{7}$)

SSC CGL 19/04/2022 (Morning)

- (a) 10 (b) 7 (c) 5 (d) 8

Q.571. The length of a wire (in cm) of 0.1 mm radius that can be drawn from melting a solid copper sphere of a diameter 6 cm is:

SSC CGL 19/04/2022 (Evening)

- (a) 250000 (b) 360000
(c) 440000 (d) 810000

Q.572. The inner circumference of a circular path enclosed between two concentric circles is 264 m . The uniform width of the circular path is 3 m . What is the area (in m^2 , to the nearest whole

number) of the path? (Use $\pi = \frac{22}{7}$)

SSC CGL 20/04/2022 (Morning)

- (a) 696 (b) 756 (c) 820 (d) 948

Q.573. The total surface area of a right pyramid, with base as a square of side 8 cm , is 208 cm^2 . What is the slant height (in cm) of the pyramid?

SSC CGL 20/04/2022 (Afternoon)

- (a) 7 (b) 10 (c) 9 (d) 8

Q.574. A 22.5 m high tent is the shape of a frustum of a cone surmounted by a hemisphere. If the diameter of the upper and the lower circular end of the frustum are 21 m and 39 m respectively, then find the areas of the cloth (in m^2) used to make the tent (ignoring the wastage).

(Use $\pi = \frac{22}{7}$)

SSC CGL 21/04/2022 (Evening)

- (a) $787\frac{2}{7}$ (b) $2107\frac{2}{7}$
(c) $1635\frac{6}{7}$ (d) $2800\frac{2}{7}$

SSC CGL 2020 Tier - 2

Q.575. A solid metallic cuboid of dimensions $12\text{ cm} \times 54\text{ cm} \times 72\text{ cm}$ is melted and converted into 8 cubes of the same size. What is the sum of the lateral surface areas (in cm^2) of 2 such cubes?

SSC CGL Tier II (29/01/2022)

- (a) 2592 (b) 2268 (c) 3888 (d) 1944

Q.576. The radius of a spherical balloon is inflated from 3.5 cm to 4.9 cm by pushing air into it. What is the percentage increase in the volume of the original balloon?

SSC CGL Tier II (29/01/2022)

- (a) 74.4% (b) 174.4% (c) 73.6% (d) 173.6%

Q.577. The radius of the base of a cylindrical tank is 4 m . If three times the sum of the areas of its two circular faces is twice the area of its curved surface, then the capacity (in kilolitres) of the tank is:

SSC CGL Tier II (29/01/2022)

- (a) 54π (b) 108π (c) 96π (d) 144π

Q.578. The slant height and radius of a right circular cone are in the ratio $29 : 20$.

If its volume is $4838.4\pi\text{ cm}^3$, then its radius is:

SSC CGL Tier II (29/01/2022)

- (a) 20 cm (b) 28 cm (c) 24 cm (d) 25 cm

Q.579. The base of a right prism is a triangle with sides 16 cm , 30 cm and 34 cm . Its height is 32 cm . The lateral surface area (in cm^2) and the volume (in cm^3) are, respectively:

SSC CGL Tier II (29/01/2022)

- (a) 2560 and 7680 (b) 2688 and 7680
(c) 2624 and 7040 (d) 2560 and 6400

Q.580. The areas of three adjacent faces of a cuboidal solid block of wax are 216 cm^2 , 96 cm^2 and 144 cm^2 . It is melted and 8 cubes of the same size are formed from it. What is the lateral surface area (in cm^2) of 3 such cubes?

SSC CGL Tier II (03/02/2022)

- (a) 648 (b) 432 (c) 576 (d) 288

Q.581. A well with inner radius 3 m , is dug 6 m deep. The soil taken out of it has been spread evenly all around it to a width of 2 m to form an embankment. The height (in m) of the embankment is:

SSC CGL Tier II (03/02/2022)

- (a) $4\frac{1}{2}$ (b) $4\frac{1}{4}$ (c) $3\frac{1}{4}$ (d) $3\frac{3}{8}$

Q.582. A solid metallic sphere of radius 4 cm is melted and recast into spheres of 2 cm each. What is the ratio of the surface

area of the original sphere to the sum of surface areas of the spheres, so formed?

SSC CGL Tier II (03/02/2022)

- (a) $2 : 1$ (b) $2 : 3$ (c) $1 : 2$ (d) $1 : 4$

Q.583. A cylindrical tube, open at both ends, is made of a metal sheet which is 0.5 cm thick. Its outer radius is 4 cm and length is 2 m . How much metal (in cm^3) has been used in making the tube?

SSC CGL Tier II (03/02/2022)

- (a) 800π (b) 450π (c) 750π (d) 550π

Q.584. The base of a right prism is a triangle whose sides are 8 cm , 15 cm and 17 cm , and its lateral surface area is 480 cm^2 . What is the volume (in cm^3) of the prism?

SSC CGL Tier II (03/02/2022)

- (a) 540 (b) 600 (c) 720 (d) 640

Q.585. The curved surface area of a right circular cone is 2310 cm^2 and its radius is 21 cm . If its radius is increased by 100% and height is reduced by 50% , then its capacity (in litres) will be correct to one decimal place) (Use $\pi = \frac{22}{7}$)

SSC CGL Tier II (03/02/2022)

- (a) 27.8 (b) 28.2 (c) 26.7 (d) 25.9

SSC MTS 2020 Tier - 1

Q.586. The area of a square is S and that of the square formed by joining the mid points of the given square is A . Then the value of $\frac{A}{S}$ is equal to:

SSC MTS 05/10/2021 (Morning)

- (a) 0.25 (b) 0.5 (c) 0.125 (d) 1.25

Q.587. A container in the shape of a right circular cone, whose radius and depth are equal, gets completely filled by 128000 spherical droplets, each of diameter 2 mm . What is the radius (in cm) of the container?

SSC MTS 05/10/2021 (Morning)

- (a) 6 (b) 4 (c) 8 (d) 2

Q.588. The total surface area of a solid right circular cylinder of height 13 cm , is 880 cm^2 . Its volume (in cm^3) is $11k$. The value of k is: (Use $\pi = \frac{22}{7}$)

SSC MTS 05/10/2021 (Afternoon)

- (a) 208 (b) 182 (c) 104 (d) 91

Q.589. The area of a parallelogram ABCD is 300 cm^2 . The distance between AB and CD is 20 cm and the distance between BC and AD is 30 cm . What is the perimeter (in cm) of the parallelogram?

SSC MTS 05/10/2021 (Evening)

(a) 60 (b) 50 (c) 40 (d) 100

Q.590. The radii of the circular ends of a frustum of a cone are 20 cm and 13 cm and its height is 12 cm. What is the capacity (in litres) of the frustum (correct to one decimal place)? (Use $\pi = \frac{22}{7}$)

SSC MTS 06/10/2021 (Morning)

(a) 10.4 (b) 11.2 (c) 10.8 (d) 11.4

Q.591. If the width of the path around a square field is 4.5 m and the area of the path is 252 m^2 , then the length of the side of the field is:

SSC MTS 06/10/2021 (Morning)

(a) 9.5 m (b) 9 m (c) 8 m (d) 8.5 m

Q.592. A solid metal cube of edge 12 cm is immersed completely in water in a cylindrical vessel whose radius is 20 cm and height is 32 cm, and the water in the vessel is up to a height of 15 cm. The height (in cm) by which the water will rise in the vessel is (correct to one decimal place):

SSC MTS 06/10/2021 (Evening)

(a) 0.9 (b) 1.4 (c) 1.2 (d) 1.1

Q.593. From each of the four corners of a rectangular sheet of dimensions 48 cm $\times$ 27 cm, a square of side 3.5 cm is cut off and a box is made. The volume of the box is:

SSC MTS 07/10/2021 (Morning)

(a) 2880 (b) 2780 (c) 2870 (d) 2860

Q.594. A path 6m wide runs around and outside of a rectangular plot of length 10m and breadth 8m. Find the area (in m^2) of path?

SSC MTS 07/10/2021 (Afternoon)

(a) 440 (b) 600 (c) 80 (d) 360

Q.595. The internal dimensions of a rectangular closed wooden box are 30 cm, 18 cm and 23 cm. What space (in cm^3) will it occupy, if the wood used is 1 cm thick?

SSC MTS 08/10/2021 (Morning)

(a) 12420 (b) 16000
(c) 14136 (d) 9408

Q.596. A solid metallic cylinder with a base radius of 3 cm and a height of 120 cm is melted and recast into a sphere of radius r cm. What is the value of r if there is a 10% metal loss in the conversion?

SSC MTS 08/10/2021 (Afternoon)

(a) 9 (b) 8 (c) 13.5 (d) 12

Q.597. The difference between the areas of two concentric circles is 88 cm^2 . If the radius of the inner circle is 6 cm then the area (in cm^2) of the larger circle is

closest to: (Use $\pi = \frac{22}{7}$)

SSC MTS 08/10/2021 (Afternoon)

(a) 196 (b) 201 (c) 197 (d) 198

Q.598. A rectangular plot is 420 m long and 84 m wide. It is surrounded by semi-circular flower beds along all its sides. The area of the entire plot (in hectares) is x . The value of x (correct to one decimal place) is: (Use $\pi = \frac{22}{7}$)

SSC MTS 08/10/2021 (Evening)

(a) 19.1 (b) 18.7 (c) 17.9 (d) 19.4

Q.599. From a wooden cubical block of side 10 cm, a sphere of radius 4.2 cm is carved out. How much wood is wasted in the process (Use $\pi = \frac{22}{7}$)

SSC MTS 11/10/2021 (Morning)

(a) 476.2 cm^3 (b) 689.536 cm^3
(c) 310.464 cm^3 (d) 523.8 cm^3

Q.600. A typist uses a paper of size 32 cm $\times$ 20 cm. He leaves a margin of 2 cm each on all the sides. If he leaves a margin of 1 cm only on all the sides, then what is the percentage of the increase in the area available for typing (correct to 2 decimal places)?

SSC MTS 11/10/2021 (Morning)

(a) 20.54% (b) 82.69%
(c) 17.04% (d) 79.46%

Q.601. A rectangular water reservoir contains 126 kilolitres of water. The depth of the reservoir is 3.5 m and its length measures 8m. What is the width of the reservoir?

SSC MTS 11/10/2021 (Afternoon)

(a) 4.5 m (b) 4.2 m (c) 4 m (d) 5 m

Q.602. A hall is 18 m long and 12 m broad. If the sum of the areas of the floor and the ceiling is equal to the sum of the areas of the four walls, the volume (in m^3) of the hall is:

SSC MTS 12/10/2021 (Morning)

(a) 1175.5 (b) 1876.2
(c) 1555.2 (d) 1376.4

Q.603. A wooden piece in the shape of a cuboid is divided into five equal parts by cutting it with four cuts perpendicular to its length, and it turns out that each piece is a cube of volume 27 cm^3 . What is the total surface area (in cm^2) of the cuboid?

SSC MTS 12/10/2021 (Afternoon)

(a) 150 (b) 174 (c) 216 (d) 198

Q.604. The material of a sphere of radius r is melted and recast into a hollow cylindrical shell of thickness a and outer radius b . What is its length assuming that

no material is lost in recasting?

SSC MTS 12/10/2021 (Afternoon)

(a) $\frac{4r^3}{3a(2a-b)}$ (b) $\frac{4r^3}{3a(b+a)(b-a)}$
(c) $\frac{4r^3}{3a(b-a)}$ (d) $\frac{4r^3}{3a(2b-a)}$

Q.605. The radius of the base of a cone and also that of a hemisphere each is 14 cm. Also, the volumes of these two solids are equal. The curved surface area of the cone is: (Take $\pi = \frac{22}{7}$)

SSC MTS 12/10/2021 (Evening)

(a) $614\sqrt{5}$ (b) $626\sqrt{5}$ (c) $616\sqrt{5}$ (d) $516\sqrt{5}$

Q.606. A boy walked along two adjacent sides of a rectangular field. If he had walked along the diagonal, then he would have saved a distance equal to one-fourth of the larger side. The ratio of the larger to the smaller side is:

SSC MTS 13/10/2021 (Morning)

(a) 7 : 24 (b) 11 : 18 (c) 24 : 7 (d) 18 : 11

Q.607. The length of a rectangle is three-fifth of the radius of a circle. The radius of the circle is equal to the side of a square whose area is 6400 m^2 . Find the area (in m^2) of the rectangle, if the breadth is 20 m is:

SSC MTS 13/10/2021 (Morning)

(a) 1200 (b) 480 (c) 960 (d) 1000

Q.608. Kamala has three cylindrical containers, each with radius 20 cm that she wants to fill with grain. If her bag of grain contains 100 kg of grain, and she fills all three containers to a height of 70 cm, how much grain will be left in the bag. If 1 kg grain occupies 3000 cm^3 space? (Use $\pi = \frac{22}{7}$)

SSC MTS 13/10/2021 (Afternoon)

(a) 88 kg (b) 50 kg (c) 58 kg (d) 12 kg

Q.609. The radius of a spherical balloon increases from 6 cm to 10 cm when more air is pumped into it. The ratio of the surface area of the original balloon and the inflated balloon is:

SSC MTS 13/10/2021 (Afternoon)

(a) 4 : 5 (b) 27 : 125 (c) 3 : 5 (d) 9 : 25

Q.610. The radius of a cylinder is reduced to 50% of its actual radius. If its volume remains the same as before, then its height becomes k times the original height. The value of k is:

SSC MTS 14/10/2021 (Evening)

(a) 2 (b) 16 (c) 8 (d) 4

Q.611. The length and breadth of a rectangle are 35 cm and 12 cm,

respectively. What will be the circumference of the circle that inscribes this rectangle ? ($\pi = \frac{22}{7}$)

SSC MTS 14/10/2021 (Evening)

- (a) $115\frac{3}{7}$ cm (b) $117\frac{1}{7}$ cm
(c) $114\frac{4}{7}$ cm (d) $116\frac{2}{7}$ cm

Q.612. Four solid cubes, each of volume 1728 cm^3 , are kept in two rows having two cubes in each row. They form a rectangular solid with a square base. The total surface area (in cm^2) of the resulting solid is:

SSC MTS 18/10/2021 (Afternoon)

- (a) 1440 (b) 2304 (c) 576 (d) 1152

Q.613. A right circular cone is inscribed in a cube of side 14 cm occupying the maximum space possible. What is the ratio of the volume of the cube to the volume of the cone ? ($\pi = \frac{22}{7}$)

SSC MTS 18/10/2021 (Afternoon)

- (a) 81 : 22 (b) 11 : 42
(c) 42 : 11 (d) 22 : 81

Q.614. Three circles, each having radius equal to 4 cm, are drawn with the vertices of an equilateral triangle as the centres. If the length of each side of the triangle is equal to 8 cm, then what is the area (in cm^2) of the portion of the triangle that is NOT covered by the sectors of the circles ?

SSC MTS 18/10/2021 (Evening)

- (a) $16(2\sqrt{3} - \pi)$ (b) $8(2\sqrt{3} - \pi)$
(c) $8(2\sqrt{3} - \frac{\pi}{2})$ (d) $16(2\sqrt{3} - \frac{\pi}{2})$

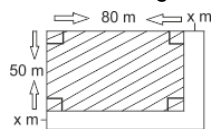
Q.615. Three cubes each having volume

1728 cm^3 are placed one above the other. What is the total surface area of the resulting solid?

SSC MTS 20/10/2021 (Morning)

- (a) 2304 (b) 2592 (c) 2160 (d) 2016

Q.616. In the given figure, the area of the unshaded region is 35% of the area of the shaded region. What is the value of x



SSC MTS 20/10/2021 (Morning)

- (a) 20 (b) 15 (c) 10 (d) 5

Q.617. The sum of the height and radius of the base of a solid right circular cylinder is 46 cm. If the total surface area of the solid cylinder is 6072 cm^2 , what is the volume (in cm^3) of the cylinder ?

SSC MTS 20/10/2021 (Evening)

- (a) 11025π (b) 3030π
(c) 11020π (d) 3036π

Q.618. A solid right circular cone is cut into two parts by a plane parallel to its base and at a point one-third vertically below its vertex. What is the ratio of the volume of the smaller cone to that of the whole cone ?

SSC MTS 20/10/2021 (Evening)

- (a) $\frac{1}{9}$ (b) $\frac{1}{8}$ (c) $\frac{1}{27}$ (d) $\frac{1}{3}$

Q.619. A horse racecourse is in the form of an annular ring whose outer and inner circumferences are 748 m and 396 m, respectively. The width (in m) of the racecourse is: ($\pi = \frac{22}{7}$)

SSC MTS 22/10/2021 (Morning)

- (a) 176 (b) 88 (c) 56 (d) 28

Q.620. The diameter of an iron ball used for shot-put is 21 cm. It is melted and then a cylinder of height 3.5 cm is made. The curved surface area of the cylinder will be

SSC MTS 22/10/2021 (Afternoon)

- (a) 462 (b) 464 (c) 362 (d) 460

Q.621. A hall is 18 m long and 12 m broad. If the area of the floor is equal to the sum of the areas of the four walls, the volume (in m^3) of the hall is:

SSC MTS 22/10/2021 (Evening)

- (a) 876.2 (b) 777.6 (c) 576.4 (d) 675.5

Q.622. The area of a triangular field with sides 60 m, 175 m and 185 m is equal to that of a rectangular park whose sides are in the ratio of 21 : 10. The perimeter (in m) of the park is:

SSC MTS 26/10/2021 (Afternoon)

- (a) 341 (b) 403 (c) 372 (d) 310

Q.623. A circular wire of diameter 56 cm is folded in the shape of a rectangle whose sides are in the ratio 7 : 4. The area enclosed by the rectangle is:

SSC MTS 26/10/2021 (Evening)

- (a) 1842 (b) 1792 (c) 1684 (d) 1782

Q.624. A solid metallic toy has a hemispherical base with radius 3.5 cm surmounted by a cone. If the height of the cone is the same as the radius of its base, then the volume of metal used in cm^3

SSC MTS 27/10/2021 (Morning)

- (a) $89\frac{2}{3}$ (b) $144\frac{3}{4}$ (c) $134\frac{3}{4}$ (d) $179\frac{2}{3}$

Q.625. A bucket in the shape of the frustum of a cone has its top and bottom radii as 20 cm and 10 cm, respectively. The depth of the bucket is 24 cm. The

capacity of the bucket (in cm^3) is :

$$(\pi = \frac{22}{7})$$

SSC MTS 02/11/2021 (Morning)

- (a) 8800 cm^3 (b) 13200 cm^3
(c) 17000 cm^3 (d) 17600 cm^3

Q.626. A rectangle of dimension 10 cm and 5 cm is placed adjacent to another rectangle of the same size to draw an L shape figure. Find the perimeter of the shape so formed.

SSC MTS 02/11/2021 (Morning)

- (a) 100 (b) 40 (c) 50 (d) 60

Q.627. The total surface area of a hemisphere is nearly equal to that of an equilateral triangle. The side of the triangle is how many times (approximately) of the radius of the hemisphere?

SSC MTS 02/11/2021 (Evening)

- (a) $(\frac{8\pi}{\sqrt{3}})^{0.5}$ (b) $(\frac{4\pi}{3})^{0.5}$
(c) $(2\pi\sqrt{3})^{0.5}$ (d) $(4\pi\sqrt{3})^{0.5}$

Q.628. When a rectangle is divided into three equal parts, each of them turns out to be a square of 16 cm^2 . What is the perimeter (in cm) of the rectangle?

SSC MTS 02/11/2021 (Evening)

- (a) 64 (b) 16 (c) 32 (d) 48

Q.629. The area (in cm^2) of a sector of a circle of radius 2 cm is $\frac{4\pi}{5}$. What is the

central angle (in degrees) of the sector?

SSC MTS 02/11/2021 (Evening)

- (a) 60 (b) 75 (c) 36 (d) 72

SSC CGL 2020 Tier - 1

Q.630. A heap of wheat is in the form of a cone whose base diameter is 8.4 m and height is 1.75 m. The heap is to be covered by canvas. What is the area (in m^2) of the canvas required ? ($\pi = \frac{22}{7}$)

SSC CGL 16/08/2021 (Afternoon)

- (a) 60.06 (b) 115.05 (c) 63.6 (d) 115.5

Q.631. The area of the square shaped field is 1764 m^2 . The breadth of a rectangular park is $\frac{1}{3}$ rd the side of the square field and its length is two times its breadth. What is the cost (in Rs.) of leveling the park at Rs.15 per m^2 ?

SSC CGL 16/08/2021 (Evening)

- (a) 4200 (b) 4290 (c) 5880 (d) 4320

Q.632. The volume of a wall whose height is 10 times its width and whose length is 8 times its height, is 51.2 m^3 . What is the

cost (in Rs.) of painting the wall on one side at the rate of Rs.100/m² ?

SSC CGL 20/08/2021 (Afternoon)

(a) 12750 (b) 12500 (c) 12800 (d) 12250

Q.633. A square has the perimeter equal to the circumference of a circle having radius 7 cm. What is the ratio of the area of the circle to the area of the square?

SSC CGL 23/08/2021 (Afternoon)

(a) 7 : 2 (b) 14 : 11 (c) 7 : 11 (d) 121 : 44

Q.634. The area of the circular path enclosed by two concentric circles is 3080 m². If the difference between the radius of the outer edge and that of the inner edge of the circular path is 10m, what is the sum (in m) of the two radii?

SSC CGL 23/08/2021 (Afternoon)

(a) 70 (b) 112 (c) 98 (d) 84

SSC CHSL 2020 Tier - 1

Q.635. One diagonal of a rhombus is $8\sqrt{3}$ cm. If the other diagonal is equal to its side, then the area of rhombus is:

SSC CHSL 12/04/2021 (Morning)

(a) $12\sqrt{3}$ (b) $24\sqrt{3}$ (c) $16\sqrt{3}$ (d) $32\sqrt{3}$

Q.636. The sum of the radius of the base and the height of a closed solid cylinder is 12.5 cm. If the total surface area of the cylinder is 275 cm^2 , then its radius is:

SSC CHSL 12/04/2021 (Evening)

(a) 3 cm (b) 3.5 cm (c) 5 cm (d) 7 cm

Q.637. What is the perimeter of a right-angled triangle whose sides (making 90°) are $20\sqrt{3}$ cm and $15\sqrt{3}$ cm ?

SSC CHSL 13/04/2021 (Morning)

(a) $60\sqrt{3}$ (b) $55\sqrt{3}$ (c) $65\sqrt{3}$ (d) $50\sqrt{3}$

Q.638. The perimeter of an isosceles triangle is 91 cm and its base is $1\frac{1}{4}$ times each of its equal sides. What is the length of its base ? (in cm)

SSC CHSL 16/04/2021 (Morning)

(a) 35 (b) 20 (c) 25 (d) 28

Q.639. The lengths of three sides of a triangle are in the ratio 3:4:5. Among three sides, the difference between the largest side and the smallest side of this triangle is 3.6 cm. The area of this triangle is:

SSC CHSL 16/04/2021 (Evening)

(a) 19.44 (b) 32.4 (c) 21.75 (d) 15.64

Q.640. One - quarter of a circular pizza of diameter 28 cm was removed from the whole pizza. What is the perimeter (in cm) of the remaining pizza? ($\pi = \frac{22}{7}$)

SSC CHSL 04/08/2021 (Morning)

(a) 88 (b) 94 (c) 80 (d) 66

Q.641. The inner and outer radii of two concentric circles are 6.7 cm and 9.5 cm respectively. What is the difference between their circumferences (in cm)?

$$\left(\pi = \frac{22}{7}\right)$$

SSC CHSL 04/08/2021 (Morning)

(a) 20.5 (b) 10.4 (c) 6.5 (d) 17.6

Q.642. From a rectangle sheet of dimensions 32cm × 18 cm, a square of side 3 cm is cut from the four corners of the sheet and a box is made. The volume of the box is:

SSC CHSL 05/08/2021 (Afternoon)

(a) 946 (b) 936 (c) 1305 (d) 1300

Q.643. How many spherical bullets, each bullet being 7 cm in diameter, can be made out of a cube of lead whose edge

measures 77 cm? ($\pi = \frac{22}{7}$)

SSC CHSL 05/08/2021 (Evening)

(a) 1452 (b) 4521 (c) 2451 (d) 2541

Q.644. The base of a triangle is equal to the perimeter of a square whose diagonal is $7\sqrt{2}$ cm and its height is equal to the side of a square whose area is 169 cm². The area (in cm²) of the triangle is :

SSC CHSL 06/08/2021 (Afternoon)

(a) 182 (b) 175 (c) 156 (d) 130

Q.645. How many bricks each measuring $64\text{ cm} \times 11.25\text{ cm} \times 6\text{ cm}$, will be needed to build a wall measuring $8\text{ m} \times 3\text{ m} \times 22.5\text{ m}$?

SSC CHSL 11/08/2021 (Morning)

(a) 200000 (b) 250000

(c) 67500 (d) 125000

Q.646. The perimeter of a rectangle is 86 cm. The measures of its area and breadth are in the ratio of 9 : 1, respectively. The breadth of the rectangle is:

SSC CHSL 11/08/2021 (Evening)

(a) 32 cm (b) 34 cm (c) 36 cm (d) 30 cm

Q.647. The square of the diagonal of a cube is 2175 cm^2 . What is the total surface area (in cm²) of the cube?

SSC CHSL 12/08/2021 (Afternoon)

(a) 4305 (b) 4272 (c) 4530 (d) 4350

SSC CPO 2020 Tier - 1

Q.648. The sides of a triangle are 24cm, 26cm and 10cm. At each of its vertices, circles of radius 4.2 cm are drawn. What is the area (in cm²) of the triangle, excluding the portion covered by the

sectors of the circle ?

SSC CPO 23/11/2020 (Morning)

(a) 27.72 (b) 120 (c) 105.86 (d) 92.28

Q.649. A 9 cm solid metallic cube and a solid metallic cuboid having dimensions 5 cm, 13cm, 31cm are melted and recast into a single cube. What is the total surface area (in cm²) of the new cube?

SSC CPO 23/11/2020 (Evening)

(a) 2744 (b) 1362 (c) 865 (d) 1176

Q.650. A rectangular lawn whose length is twice of its breadth is extended by having four semi-circular portions on its sides. What is the total cost (in Rs.) of levelling the entire lawn at the rate of Rs.100 per square metre if the smaller side of the rectangular lawn is 12 m :

SSC CPO 23/11/2020 (Evening)

(a) 85,320

(b) 86,540

(c) 78,650

(d) 97,625

Q.651. A 9cm solid metallic cube and a solid metallic cuboid having dimensions 5 cm, 13cm, 31cm are melted and recast into a single cube. How much (in Rs.) is the cost to polish the new cube at the rate of Rs.10 per cm²

SSC CPO 24/11/2020 (Morning)

(a) 13,620 (b) 11,760 (c) 8,650 (d) 27,440

Q.652. A hexagon is $1944\sqrt{3}\text{ m}^2$. What will be the cost (in Rs) of fencing it at the rate of Rs.11.50 per metre?

SSC CPO 24/11/2020 (Morning)

(a) 2,256 (b) 2,785 (c) 3,200 (d) 2,484

Q.653. The base of a triangle is equal to the perimeter of a square whose diagonal is $6\sqrt{2}$ cm, and its height is equal to the side of a square whose area is 144 cm^2 .

The area of the triangle (in m²)

SSC CPO 24/11/2020 (Evening)

(a) 144 (b) 216 (c) 288 (d) 72

Q.654. The perimeter of a square is half the perimeter of a rectangle. The perimeter of the square is 40m. If its breadth is two-thirds of its length, then what is the area of the rectangle ?

SSC CPO 25/11/2020 (Morning)

(a) 321 (b) 196 (c) 400 (d) 384

SSC CGL 2019 Tier - 2

Q.655. The areas of three adjacent faces of a cuboidal tank are 3 m^2 , 12 m^2 and 16 m^2 . The capacity of the tank in litres is :

SSC CGL Tier II (15/11/2020)

(a) 48000

(b) 24000

(c) 72000

(d) 36000

Q.656. The base of a right pyramid is a square of a side 10cm. If its height is 10 cm, then the area (in cm^2) of its lateral surface is:

SSC CGL Tier II (15/11/2020)

- (a) 100 (b) $100(\sqrt{5} + 1)$
(c) $50\sqrt{5}$ (d) $100\sqrt{5}$

Q.657. The height of a cylinder is 30 cm and the diameter of its base is 10 cm. Two identical conical holes each of radius 5 cm and height of 12 cm are drilled out. what is the surface area (in cm^2) of the remaining solid?

SSC CGL Tier II 15/11/2020

- (a) 230π (b) 430π (c) 330π (d) 120π

Q.658. The base of a right prism is a regular hexagon of side 5 cm. If its height is $12\sqrt{3}$ cm, then its volume (in cm^3)

SSC CGL Tier II (15/11/2020)

- (a) 900 (b) 1800 (c) 1350 (d) 675

Q.659. A spherical metallic shell with 6 cm external radius weighs 6688 g, What is the thickness of the shell if the density of metal is 10.5 g per cm^3 $\pi = \frac{22}{7}$

SSC CGL Tier II (15/11/2020)

- (a) 2 cm (b) 3 cm (c) $2\frac{1}{2}$ cm (d) 4 cm

Q.660. The sum of the radii of spheres A and B is 14 cm, the radius of A is larger than that of B. The difference between their surface area is 112π . What is the ratio of volumes of A and B?

SSC CGL Tier II (15/11/2020)

- (a) 64 : 27 (b) 8 : 1 (c) 125 : 64 (d) 27 : 8

Q.661. If the radius of a sphere is increased by 2.5 decimeter (dm), then its surface area increases by $110dm^2$. What is the volume (in dm^3) of the sphere

SSC CGL Tier II (16/11/2020)

- (a) $\frac{3}{7}$ (b) $\frac{11}{21}$ (c) $\frac{13}{21}$ (d) $\frac{4}{7}$

Q.662. A cylindrical roller made up of iron is 1.2m long. Its internal radius is 24 cm and thickness of the iron sheet used in making the roller is 15 cm. What is the weight (in kg) of the roller, if $1cm^3$ of the iron has 8 g mass?

SSC CGL Tier II (16/11/2020)

- (a) 892.8π (b) 907.2π
(c) 846.72π (d) 845.75π

Q.663. From a solid cylindrical wooden block of height 18cm and radius 7.5cm, a conical cavity of same radius and same height is taken out. What is total surface

area (in cm^2) of the remaining solid?

SSC CGL Tier II (16/11/2020)

- (a) 326.25π (b) 416.25π
(c) 472.5π (d) 270π

Q.664. A hemispherical tank full of water is emptied by a pipe at the rate of 7.7 litres per seconds. How much time (in hours) will it take to empty $\frac{2}{3}$ part of the

tank if the internal radius of the tank is 10.5 m?

SSC CGL Tier II (16/11/2020)

- (a) $\frac{185}{6}$ (b) $\frac{175}{3}$ (c) $\frac{185}{3}$ (d) $\frac{175}{2}$

Q.665. A solid metallic cuboid of dimensions 18 cm $\times$ 36cm $\times$ 72cm is melted and recast into 8 cubes of the same volume. What is the ratio of the total surface area of the cuboid to the sum of the lateral surface area of all 8 cubes?

SSC CGL Tier II (16/11/2020)

- (a) 2 : 3 (b) 7 : 8 (c) 4 : 7 (d) 7 : 12

Q.666. The base of a right prism is a square having side of 15 cm. If its height is 8 cm, then find the total surface area.

SSC CGL Tier II (18/11/2020)

- (a) $920 cm^2$ (b) $930 cm^2$
(c) $900 cm^2$ (d) $940 cm^2$

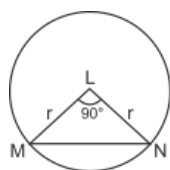
Q.667. If the perimeter of an isosceles right triangle is $8(\sqrt{2} + 1)$ cm, then the length of the hypotenuse of the triangle is:

SSC CGL Tier II (18/11/2020)

- (a) 10 cm (b) 8 cm (c) 24 cm (d) 12 cm

SSC CHSL 2019 Tier - 1

Q.668. In the figure, L is the centre of the circle, and ML is the perpendicular to LN. If the area of the triangle MLN is 36, then the area of the circle is:



SSC CHSL 12/10/2020 (Afternoon)

- (a) 70π (b) 72π (c) 66π (d) 68π

Q.669. A cylindrical vessel with radius 6 cm and height 5 cm is to be made by melting a number of spherical metal balls of diameter 2 cm. The minimum number of balls needed is:

SSC CHSL 13/10/2020 (Morning)

- (a) 125 (b) 135 (c) 115 (d) 105

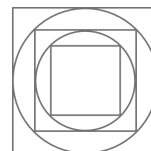
Q.670. The length and breadth of a cuboid are increased by 10% and 20%,

respectively, and its height is decreased by 20%. The percentage increase in the volume of the cuboid is:

SSC CHSL 14/10/2020 (Morning)

- (a) $5\frac{4}{5}\%$ (b) $5\frac{1}{5}\%$ (c) $5\frac{2}{5}\%$ (d) $5\frac{3}{5}\%$

Q.671. In the given figure, the ratio of the area of the largest square to that of the smallest square is:



SSC CHSL 14/10/2020 (Afternoon)

- (a) 4 : 1 (b) $\sqrt{2} : 1$ (c) 3 : 1 (d) 2 : 1

Q.672. Water flows into a tank 180 m $\times$ 140 m through a rectangular pipe of 1.2m $\times$ 0.75m at a rate of 15 km/h. In what time will the water rise by 4 m?

SSC CHSL 14/10/2020 (Afternoon)

- (a) 6 hours 42 minutes
(b) 7 hours 28 minutes
(c) 5 hours 46 minutes
(d) 8 hours 12 minutes

Q.673. The area of a rectangle is $a^2 - b^2$ and its length is $a + b$, what will be its breadth?

SSC CHSL 15/10/2020 (Morning)

- (a) 2ab (b) $a + b$ (c) ab (d) $a - b$

Q.674. The area of the largest triangle that can be inscribed in a semicircle of radius 4 cm in square centimetres is

SSC CHSL 15/10/2020 (Evening)

- (a) 16 (b) 14 (c) 12 (d) 18

Q.675. The area of a square park is $16x^2 + 8x + 1$. What is the length of the park?

SSC CHSL 16/10/2020 (Afternoon)

- (a) $(4x - 1)$ units (b) $(4x + 1)$ units
(c) $(4x + 1)^2$ units (d) $4x$ units

Q.676. A triangle PQR is a right angled triangle at Q. E and F are the mid points of QR and PR respectively. What will be the ratio of the area of quadrilateral PQEF to the area of triangle PQR.

SSC CHSL 19/10/2020 (Morning)

- (a) $\frac{2}{3}$ (b) $\frac{3}{4}$ (c) $\frac{3}{2}$ (d) $\frac{4}{3}$

Q.677. Three sides of a triangle measure 6 cm, 10 cm and x cm. The minimum integral value of x is:

SSC CHSL 19/10/2020 (Evening)

- (a) 2 (b) 1 (c) 3 (d) 5

Q.678. ABC is a right angled triangle, right angled at A. A circle is inscribed in it. The lengths of two sides containing the right angle are 48 cm and 14 cm. The radius of the inscribed circle is:

SSC CHSL 20/10/2020 (Morning)

(a) 4 cm (b) 6 cm (c) 8 cm (d) 5 cm

Q.679. Two concentric circles form a ring. The inner and outer circumference of the ring are 22 cm and 44 cm respectively. The width of the ring is:

SSC CHSL 20/10/2020 (Afternoon)

(a) 3.5 cm (b) 1.5 cm (c) 2.5 cm (d) 3 cm

Q.680. A, B and C are three points on the circle. If $AB = AC = 7\sqrt{2}$ cm and $\angle BAC = 90^\circ$, then the radius is equal to:

SSC CHSL 21/10/2020 (Afternoon)

(a) 14 (b) 7 (c) $7\sqrt{2}$ (d) 6

Q.681. In an isosceles triangle ABC, $AB = AC$ and AD is perpendicular to BC at D. If $AD = 8$ cm and perimeter of $\triangle ABC$ is 64 cm, then the area of $\triangle ABC$

SSC CHSL 26/10/2020 (Evening)

(a) 130 (b) 124 (c) 120 (d) 125

Q.682. If M is the mid point of the side BC of $\triangle ABC$, and the area of $\triangle ABM$ is 18 cm^2 , then the area of $\triangle ABC$ is:

SSC CHSL 17/03/2020 (Morning)

(a) 30 (b) 34 (c) 36 (d) 32

Q.683. A 5 cm long perpendicular is drawn from the centre of a circle to a 24 cm long chord. Find the diameter of the circle.

SSC CHSL 17/03/2020 (Morning)

(a) 32 cm (b) 13 cm (c) 30 cm (d) 26 cm

Q.684. The length of a rectangular plot is five times its breadth. If the area of the rectangular plot is 2000 m^2 , then what is the breadth of the rectangular plot?

SSC CHSL 18/03/2020 (Evening)

(a) 40 m (b) 10 m (c) 20 m (d) 30 m

Q.685. Two small circular grounds of diameters 42 m and 26 m are to be replaced by a bigger circular ground. What would be the radius of the new ground if the new ground has the same area as the two small grounds?

SSC CHSL 18/03/2020 (Evening)

(a) 24.69 (b) 23 (c) 25 (d) 25.01

Q.686. In an isosceles right-angled triangle, the perimeter is 30 m. Find its area (In m^2 Approximate.)

SSC CHSL 18/03/2020 (Evening)

(a) 38.63 (b) 39.60 (c) 37.86 (d) 40

Q.687. The radius a circular cone is R and its height is H. the volume of cone is:

SSC CHSL 19/03/2020 (Morning)

(a) $\frac{\pi R^2 H}{3}$ (b) $\frac{2}{3\pi R^2 H}$

(c) $\pi R^2 H$ (d) $\pi R^2 H$

Q.688. The length of tangents drawn from an external point to the circle are:

SSC CHSL 19/03/2020 (Morning)

(a) not equal (b) perpendicular
(c) parallel (d) equal

SSC CGL 2019 Tier - 1

Q.689. A race track is in the shape of a ring whose inner and outer circumferences are 440m and 506m, respectively. What is the cost of levelling

the track at $\text{₹}6/\text{m}^2$? ($\pi = \frac{22}{7}$)

SSC CGL 03/03/2020 (Afternoon)

(a) $\text{₹}18,966$ (b) $\text{₹}24,832$

(c) $\text{₹}19,866$ (d) $\text{₹}29,799$

Q.690. A cylindrical vessel of radius 30cm and height 42 cm is full of water. Its contents are emptied into a rectangular tub of length 75cm and breadth 44cm. The height(in cm) to which the water rises in the tub is:

($\pi = \frac{22}{7}$)

SSC CGL 04/03/2020 (Afternoon)

(a) 36 (b) 30 (c) 40 (d) 45

Q.691. The diagonal of a square A is (a+b) units. What is the area (in square units) of the square drawn on the diagonal of square B whose area is twice the area of square A?

SSC CGL 05/03/2020 (Morning)

(a) $(a + b)^2$ (b) $2(a + b)^2$

(c) $4(a + b)^2$ (d) $8(a + b)^2$

Q.692. A circular disc of area $0.64\pi \text{ m}^2$ rolls down a length of 1.408 km. The number of revolutions it make is:

($\pi = \frac{22}{7}$)

SSC CGL 06/03/2020 (Morning)

(a) 280 (b) 360 (c) 140 (d) 180

Q.693. If the perimeter of a certain rectangle is 50 units and its area is 150 sq. units, then how many units is the length of its shorter side?

SSC CGL 07/03/2020 (Afternoon)

(a) 9 (b) 12 (c) 10 (d) 15

Q.694. The volumes of sphere A and B are in the ratio 125 : 64. If the sum of radii of A and B is 36 cm, then the surface area (in cm^2) of A is:

SSC CGL 09/03/2020 (Afternoon)

(a) 512π (b) 1600π (c) 800π (d) 1024π

SSC CPO 2019 Tier - 1

Q.695. The height of a right circular cone is 35 cm and the area of its curved

surface is four times the area of its base.

What is the volume of the cone (in m^3 and correct up to 3 decimal places)

SSC CPO 09/12/2019 (Morning)

(a) 3.316 (b) 3.384 (c) 2.994 (d) 2.625

Q.696. 60 discs each of diameter 21 cm and thickness $\frac{1}{3}$ cm are stacked one

above the other to form a right circular cylinder. What is its volume in m^3 if ($\pi = \frac{22}{7}$)?

SSC CPO 09/12/2019 (Morning)

(a) 4.25×10^{-2} (b) 8.05×10^{-3}

(c) 7.62×10^{-2} (d) 6.93×10^{-3}

Q.697. A cylinder 84 cm long is made of steel. Its external and internal diameters are 10 cm and 8 cm respectively. What is the volume of the steel in the cylinder (in 10^{-3} m^3 and correct upto three decimal places)?

SSC CPO 09/12/2019 (Morning)

(a) 2.112 (b) 2.376 (c) 4.752 (d) 9.504

Q.698. A field is in the shape of a trapezium whose parallel sides are 200m and 400m long, whereas each of the other two sides is 260m long. what is the area (in m^2) of the field?

SSC CPO 11/12/2019 (Morning)

(a) 48000 (b) 52000

(c) 72000 (d) 60000

Q.699. A solid metallic sphere of radius x cm is melted and then drawn into 126 cones each of radius 3.5 cm and height 3 cm. There is no wastage of material in this process. What is the value of x?

SSC CPO 11/12/2019 (Morning)

(a) 10.5 (b) 7 (c) 3.5 (d) 21

Q.700. In $\triangle ABC$, $\angle A = 90^\circ$, $AB = 16$ cm and $AC = 12$ cm. D is the midpoint of AC and $DE \perp CB$ at E. What is the area (in cm^2) of $\triangle CDE$?

SSC CPO 11/12/2019 (Evening)

(a) 8.64 (b) 7.68 (c) 5.76 (d) 6.25

Q.701. A right circular cone of largest volume is cut from a solid wooden hemisphere. The Volume of the remaining portion is what percentage of the volume of the original hemisphere?

SSC CPO 12/12/2019 (Morning)

(a) 50% (b) 33.33% (c) 75% (d) 66.66%

Q.702. A conical vessel whose internal base radius is 18 cm and height 60 cm is full of a liquid. The entire liquid of the vessel is emptied into a cylindrical vessel with internal radius 15 cm. The height (in cm) to which the liquid rises in the cylindrical vessel is:

SSC CPO 12/12/2019 (Morning)
(a) 30.2 (b) 28.8 (c) 27 (d) 24

Q.703. From a point P which is at a distance of 10 cm from the centre O of a circle of radius 6 cm, a pair of tangents PQ and PR to the circle at point Q and R respectively, are drawn. Then the area (in sq cm) of the quadrilateral PQOR is equal to :

SSC CPO 12/12/2019 (Evening)
(a) 30 (b) 40 (c) 24 (d) 48

Q.704. A cuboidal tank has 25000 litres of water. If the depth of the cuboid is $\frac{1}{5}$

of its length and breadth is $\frac{1}{8}$ of its length, then the length of the tank is:

SSC CPO 12/12/2019 (Evening)
(a) 12 m (b) 10 m (c) 15 m (d) 18 m

Q.705. A rectangular sheet of paper which is 88 cm long and 11 cm wide is rolled to form a cylinder of height equal to its width of the paper. What is the volume of the cylinder so formed?

SSC CPO 12/12/2019 (Evening)
(a) 7676 (b) 6776 (c) 6546 (d) 6786

Q.706. The diameter of a cycle wheel is 126 cm. A cyclist takes 16.5 minutes to reach the destination at a speed of 72 km/hr. How many revolutions will the wheel make during the journey?

SSC CPO 12/12/2019 (Evening)
(a) 5000 (b) 5200 (c) 4000 (d) 4500

Q.707. A right circular cylinder of maximum possible size is cut out from a solid wooden cube. The remaining material of the cube is what percentage of the original cube? (take $\pi = 3.14$)

SSC CPO 13/12/2019 (Morning)
(a) 22.4 (b) 21.5 (c) 22.8 (d) 21.8

Q.708. The ratio of volumes of two right circular cylinders A and B is x:y and the ratio of their heights is a:b. What is the ratio of their radii?

SSC CPO 13/12/2019 (Evening)

(a) $\sqrt{\frac{xb}{ya}}$ (b) $\frac{xb}{ya}$ (c) $\sqrt{\frac{xa}{yb}}$ (d) $\frac{yb}{xa}$

Q.709. There is a circular park of diameter 'd'. A circular path of uniform width 'x' metre is made outside the park. The area of the circular path outside the park is _____.

SSC CPO 13/12/2019 (Evening)
(a) $\pi x (x + d)$ (b) $\pi x (x + 2d)$
(c) $\pi x (x + \frac{d}{2})$ (d) $\pi x (2x + d)$

Q.710. When the radius of a sphere is increased by 5 cm, its surface area

increases by 704cm^2 . The diameter of the original sphere is _____. (take $\pi = \frac{22}{7}$)

SSC CPO 13/12/2019 (Evening)
(a) 8.2 (b) 5.2 (c) 6.8 (d) 6.2

SSC CGL 2018 Tier - 2

Q.711. A right circular cylinder of maximum volume is cut out from a solid wooden cube. The material left is what percent of the volume (nearest to an integer) of the original cube?

SSC CGL Tier II (11/09/2019)
(a) 19 (b) 28 (c) 23 (d) 21

Q.712. If the radius of the base of a cone is doubled, and the volume of the new cone is three times the volume of the original cone, then what will be the ratio of the height of the original cone to that of the new cone?

SSC CGL Tier II (11/09/2019)
(a) 1 : 3 (b) 4 : 3 (c) 2 : 9 (d) 9 : 4

Q.713. Three solid metallic spheres whose radii are 1cm, x cm and 8 cm, are melted and recast into a single solid sphere of diameter 18 cm. The surface area (in cm^2) of the sphere with radius x cm is :

SSC CGL Tier II (11/09/2019)
(a) 144π (b) 72π (c) 64π (d) 100π

Q.714. The area of the base of a right circular cone is 400π and its height is 15 cm. The curved surface area of the cone (in cm^2) is :

SSC CGL Tier II (12/09/2019)
(a) 480π (b) 500π (c) 450π (d) 560π

Q.715. A cylindrical vessel of radius 3.5 m is full of water. If 15400 litres of water is taken out from it, then the drop in the water level in the vessel will be :

SSC CGL Tier II (12/09/2019)
(a) 72 cm (b) 40 cm (c) 35 cm (d) 60 cm

Q.716. The base of a right pyramid is an equilateral triangle with side 8 cm, and the height of pyramid is $24\sqrt{3}$ cm. The volume (in cm^3) of the pyramid is :

SSC CGL Tier II (12/09/2019)
(a) 1152 (b) 480 (c) 576 (d) 384

Q.717. If the diameter of the base of a right circular cylinder is reduced by $33\frac{1}{3}\%$ and its height is doubled, then the volume of the cylinder will :

SSC CGL Tier II (12/09/2019)

(a) increase by $1\frac{1}{9}\%$
(b) remain unchanged

(c) increase by $11\frac{1}{9}\%$

(d) decrease by $11\frac{1}{9}\%$

Q.718. A solid cube is cut into three cuboids of same volumes. What is the ratio of the surface area of the cube to the sum of the surface areas of any two of the cuboids so formed?

SSC CGL Tier II (12/09/2019)
(a) 9 : 10 (b) 27 : 16 (c) 27 : 10 (d) 9 : 8

Q.719. A field roller, in the shape of a cylinder, has a diameter of 1 m and length of $1\frac{1}{4}$ m. If the speed at which

the roller rolls is 14 revolutions per minute, then the maximum area (in m^2) that it can roll in 1 hour is : (take $\pi = \frac{22}{7}$)

SSC CGL Tier II (12/09/2019)
(a) 3960 (b) 3600 (c) 3300 (d) 3560

Q.720. A solid cylinder of base radius 12cm and height 15 cm is melted and recast into n toys each in the shape of a right circular cone of height 9 cm mounted on a hemisphere of radius 3 cm. The value of n is :

SSC CGL Tier II (13/09/2019)
(a) 27 (b) 64 (c) 48 (d) 54

Q.721. The base of a right pyramid is an equilateral triangle with area $16\sqrt{3}\text{cm}^2$. If the area of one of its lateral faces is 30cm^2 , then its height (in cm) is:

SSC CGL Tier II (13/09/2019)

(a) $\sqrt{\frac{739}{12}}$ (b) $\sqrt{\frac{209}{12}}$

(c) $\sqrt{\frac{611}{12}}$ (d) $\sqrt{\frac{643}{12}}$

Q.722. A sphere of maximum volume is cut out from a solid hemisphere. What is the ratio of the volume of the sphere to that of the remaining solid?

SSC CGL Tier II (13/09/2019)
(a) 1 : 4 (b) 1 : 2 (c) 1 : 3 (d) 1 : 1

Q.723. A 15m deep well with radius 2.8 m is dug and the earth taken out from it is spread evenly to form a platform of breadth 8m and height 1.5m. What will be the length of the platform?

SSC CGL Tier II (13/09/2019)
(a) 28.4 (b) 28.8 (c) 30.2 (d) 30.8

Q.724. The internal diameter of a hollow hemispherical vessel is 24 cm. It is made of a steel sheet which is 0.5 cm thick. What is the total surface area (in cm^2) of the vessel?

SSC CGL Tier II (13/09/2019)
(a) 612.75 π (b) 468.75 π

(c) 600.2π (d) 600.5π

SSC MTS 2019 Tier - 1

Q.725. There is a path of equal width of 3.5 m along with a building measuring 25 m in length and 15 m in breadth. Find the flooring cost of this path at the rate of Rs. 25.50 per sq. m.

SSC MTS 09/08/2019 (Evening)

(a) Rs. 8389.50 (b) Rs.4186.50

(c) Rs. 8146.50 (d) Rs. 9149.50

SSC CHSL 2018 Tier - 1

Q.726. The volume of a right circular cone is 924 cubic cm. If its height is 18 cm, then the area of its base (In square cm) is:

SSC CHSL 02/07/2019 (Morning)

(a) 154 (b) 132 (c) 176 (d) 198

Q.727. What is the area of a rhombus (in cm^2) whose side is 10 cm and the smallest diagonal is 12 cm?

SSC CHSL 09/07/2019 (Afternoon)

(a) 120 (b) 192 (c) 96 (d) 50

SSC CGL 2018 Tier - 1

Q.728. The radii of two circular faces of the frustum of a cone of height 21 cm are 3 cm and 2 cm respectively. What is the volume of the frustum of the cone in cubic cm ? (Take $\pi = \frac{22}{7}$)

SSC CGL 11/06/2019 (Morning)

(a) 154 (b) 286 (c) 345 (d) 418

SSC CPO 2018 Tier - 1

Q.729. How much volume will the wood need to make a closed box of 2.5 cm thickness with outer measurements 90 cm × 75 cm × 50 cm ?

SSC CPO 14/03/2019 (Evening)

(a) 46720 cm^3 (b) 69750 cm^3 (c) 49050 cm^3 (d) 36170 cm^3

Q.730. The perimeter of a square is equal to the perimeter of a rectangle of length 16cm and breadth 14 cm. Find the circumference of a semicircle whose diameter is equal to the side of the square.

SSC CPO 16/03/2019 (Evening)

(a) 38.57 (b) 21.57 (c) 23.57 (d) 25.57

Answer Key :-

1.(b)	2.(a)	3.(c)	4.(c)
5.(b)	6.(d)	7.(a)	8.(d)

9.(a)	10.(b)	11.(a)	12.(a)
13.(c)	14.(c)	15.(b)	16.(b)
17.(c)	18.(d)	19.(b)	20.(c)
21.(a)	22.(b)	23.(c)	24.(d)
25.(a)	26.(b)	27.(d)	28.(a)
29.(a)	30.(d)	31.(d)	32.(b)
33.(c)	34.(c)	35.(a)	36.(d)
37.(c)	38.(a)	39.(c)	40.(b)
41.(d)	42.(c)	43.(d)	44.(c)
45.(b)	46.(c)	47.(c)	48.(d)
49.(d)	50.(b)	51.(d)	52.(b)
53.(d)	54.(c)	55.(d)	56.(a)
57.(c)	58.(d)	59.(b)	60.(d)
61.(d)	62.(c)	63.(a)	64.(a)
65.(d)	66.(c)	67.(b)	68.(b)
69.(a)	70.(b)	71.(c)	72.(c)
73.(a)	74.(c)	75.(d)	76.(d)
77.(a)	78.(a)	79.(d)	80.(a)
81.(a)	82.(a)	83.(d)	84.(a)
85.(d)	86.(a)	87.(c)	88.(a)
89.(b)	90.(b)	91.(b)	92.(a)
93.(b)	94.(b)	95.(b)	96.(d)
97.(d)	98.(c)	99.(a)	100.(c)
101.(b)	102.(c)	103.(a)	104.(d)
105.(b)	106.(d)	107.(a)	108.(c)
109.(c)	110.(a)	111.(a)	112.(c)
113.(a)	114.(d)	115.(d)	116.(d)
117.(d)	118.(d)	119.(d)	120.(d)
121.(c)	122.(c)	123.(b)	124.(c)
125.(a)	126.(b)	127.(b)	128.(c)
129.(c)	130.(c)	131.(b)	132.(c)
133.(c)	134.(c)	135.(a)	136.(b)
137.(b)	138.(b)	139.(c)	140.(b)
141.(b)	142.(d)	143.(c)	144.(a)
145.(c)	146.(c)	147.(a)	148.(d)
149.(d)	150.(a)	151.(c)	152.(b)
153.(a)	154.(b)	155.(a)	156.(d)
157.(b)	158.(c)	159.(a)	160.(c)
161.(c)	162.(d)	163.(a)	164.(a)
165.(a)	166.(c)	167.(d)	168.(d)
169.(d)	170.(d)	171.(a)	172.(c)
173.(c)	174.(c)	175.(d)	176.(c)
177.(a)	178.(c)	179.(b)	180.(a)
181.(d)	182.(b)	183.(b)	184.(a)
185.(d)	186.(a)	187.(c)	188.(a)
189.(a)	190.(b)	191.(a)	192.(a)

193.(a)	194.(d)	195.(a)	196.(c)
197.(d)	198.(a)	199.(b)	200.(b)
201.(a)	202.(a)	203.(c)	204.(a)
205.(d)	206.(a)	207.(a)	208.(a)
209.(d)	210.(b)	211.(a)	212.(d)
213.(b)	214.(d)	215.(d)	216.(b)
217.(b)	218.(d)	219.(b)	220.(a)
221.(d)	222.(d)	223.(a)	224.(b)
225.(c)	226.(d)	227.(d)	228.(a)
229.(c)	230.(d)	231.(a)	232.(d)
233.(c)	234.(c)	235.(c)	236.(c)
237.(b)	238.(c)	239.(a)	240.(d)
241.(c)	242.(c)	243.(c)	244.(d)
245.(a)	246.(c)	247.(b)	248.(c)
249.(a)	250.(a)	251.(a)	252.(a)
253.(c)	254.(c)	255.(a)	256.(c)
257.(b)	258.(d)	259.(b)	260.(c)
261.(d)	262.(c)	263.(a)	264.(a)
265.(b)	266.(d)	267.(b)	268.(b)
269.(a)	270.(c)	271.(d)	272.(a)
273.(d)	274.(b)	275.(d)	276.(d)
277.(c)	278.(b)	279.(b)	280.(d)
281.(b)	282.(b)	283.(b)	284.(c)
285.(a)	286.(c)	287.(a)	288.(c)
289.(a)	290.(a)	291.(b)	292.(b)
293.(b)	294.(b)	295.(a)	296.(c)
297.(b)	298.(b)	299.(a)	300.(d)
301.(c)	302.(c)	303.(b)	304.(a)
305.(c)	306.(c)	307.(d)	308.(b)
309.(c)	310.(c)	311.(c)	312.(a)
313.(a)	314.(c)	315.(b)	316.(c)
317.(a)	318.(c)	319.(d)	320.(b)
321.(b)	322.(a)	323.(d)	324.(d)
325.(d)	326.(d)	327.(c)	328.(c)
329.(a)	330.(b)	331.(c)	332.(c)
333.(c)	334.(c)	335.(a)	336.(b)
337.(d)	338.(d)	339.(d)	340.(b)
341.(a)	342.(d)	343.(c)	344.(d)
345.(c)	346.(a)	347.(b)	348.(b)
349.(d)	350.(d)	351.(a)	352.(a)
353.(c)	354.(a)	355.(b)	356.(a)
357.(d)	358.(d)	359.(b)	360.(b)
361.(a)	362.(a)	363.(c)	364.(a)
365.(a)	366.(b)	367.(c)	368.(c)
369.(b)	370.(d)	371.(b)	372.(c)
373.(a)	374.(b)	375.(d)	376.(d)

377.(a)	378.(d)	379.(d)	380.(b)
381.(d)	382.(c)	383.(a)	384.(c)
385.(c)	386.(a)	387.(a)	388.(a)
389.(a)	390.(d)	391.(a)	392.(a)
393.(c)	394.(c)	395.(a)	396.(a)
397.(b)	398.(c)	399.(b)	400.(c)
401.(c)	402.(b)	403.(b)	404.(c)
405.(c)	406.(d)	407.(a)	408.(c)
409.(c)	410.(d)	411.(a)	412.(c)
413.(a)	414.(b)	415.(d)	416.(a)
417.(a)	418.(a)	419.(d)	420.(a)
421.(d)	422.(b)	423.(d)	424.(d)
425.(d)	426.(a)	427.(a)	428.(b)
429.(b)	430.(c)	431.(c)	432.(d)
433.(b)	434.(a)	435.(b)	436.(d)
437.(c)	438.(a)	439.(c)	440.(a)
441.(c)	442.(b)	443.(d)	444.(a)
445.(d)	446.(b)	447.(d)	448.(a)
449.(b)	450.(b)	451.(c)	452.(b)
453.(a)	454.(b)	455.(b)	456.(c)
457.(c)	458.(d)	459.(a)	460.(c)
461.(a)	462.(d)	463.(d)	464.(d)
465.(d)	466.(c)	467.(c)	468.(d)
469.(d)	470.(b)	471.(c)	472.(b)
473.(c)	474.(b)	475.(d)	476.(c)
477.(c)	478.(c)	479.(c)	480.(d)
481.(a)	482.(d)	483.(d)	484.(c)
485.(b)	486.(b)	487.(d)	488.(b)
489.(b)	490.(a)	491.(c)	492.(c)
493.(a)	494.(d)	495.(a)	496.(a)
497.(d)	498.(d)	499.(c)	500.(a)
501.(c)	502.(a)	503.(d)	504.(b)
505.(b)	506.(a)	507.(d)	508.(c)
509.(c)	510.(c)	511.(a)	512.(d)
513.(c)	514.(c)	515.(d)	516.(c)
517.(b)	518.(d)	519.(a)	520.(c)
521.(d)	522.(d)	523.(a)	524.(b)
525.(c)	526.(b)	527.(a)	528.(c)
529.(c)	530.(a)	531.(c)	532.(c)
533.(c)	534.(b)	535.(c)	536.(b)
537.(a)	538.(a)	539.(b)	540.(a)
541.(d)	542.(a)	543.(d)	544.(b)
545.(a)	546.(c)	547.(b)	548.(c)
549.(b)	550.(a)	551.(c)	552.(c)
553.(d)	554.(b)	555.(c)	556.(d)
557.(d)	558.(b)	559.(d)	560.(d)

561.(c)	562.(d)	563.(d)	564.(d)
565.(c)	566.(b)	567.(b)	568.(c)
569.(a)	570.(b)	571.(b)	572.(c)
573.(c)	574.(b)	575.(a)	576.(b)
577.(c)	578.(c)	579.(a)	580.(b)
581.(d)	582.(c)	583.(c)	584.(c)
585.(d)	586.(b)	587.(c)	588.(b)
589.(b)	590.(a)	591.(a)	592.(b)
593.(c)	594.(d)	595.(b)	596.(a)
597.(b)	598.(c)	599.(b)	600.(a)
601.(a)	602.(c)	603.(d)	604.(d)
605.(c)	606.(c)	607.(c)	608.(d)
609.(d)	610.(d)	611.(d)	612.(b)
613.(c)	614.(b)	615.(d)	616.(c)
617.(a)	618.(c)	619.(c)	620.(a)
621.(b)	622.(d)	623.(b)	624.(c)
625.(d)	626.(c)	627.(d)	628.(c)
629.(d)	630.(a)	631.(c)	632.(c)
633.(b)	634.(c)	635.(d)	636.(b)
637.(a)	638.(a)	639.(a)	640.(b)
641.(d)	642.(b)	643.(d)	644.(a)
645.(d)	646.(b)	647.(d)	648.(d)
649.(d)	650.(a)	651.(b)	652.(d)
653.(a)	654.(d)	655.(b)	656.(d)
657.(b)	658.(c)	659.(a)	660.(a)
661.(b)	662.(b)	663.(c)	664.(b)
665.(b)	666.(b)	667.(b)	668.(b)
669.(b)	670.(d)	671.(a)	672.(b)
673.(d)	674.(a)	675.(b)	676.(b)
677.(d)	678.(b)	679.(a)	680.(b)
681.(c)	682.(c)	683.(d)	684.(c)
685.(a)	686.(a)	687.(a)	688.(d)
689.(d)	690.(a)	691.(b)	692.(a)
693.(c)	694.(b)	695.(c)	696.(d)
697.(b)	698.(c)	699.(a)	700.(a)
701.(a)	702.(b)	703.(d)	704.(b)
705.(b)	706.(a)	707.(b)	708.(a)
709.(a)	710.(d)	711.(d)	712.(b)
713.(a)	714.(b)	715.(b)	716.(d)
717.(d)	718.(a)	719.(c)	720.(c)
721.(c)	722.(c)	723.(d)	724.(a)
725.(a)	726.(a)	727.(c)	728.(d)
729.(b)	730.(a)		

Solutions:-**Sol.1.(b)** Lateral surface area = 120 cm^2

Lateral surface area of prism = perimeter of base $\times$ height

$$120 = 3 \times 4 \times \text{height}$$

$$\text{Height} = 10 \text{ cm}$$

Now,

volume of prism = area of base $\times$ height

$$= \frac{\sqrt{3}}{4} \times 4 \times 4 \times 10 = 40\sqrt{3} \text{ cm}^3$$

Sol.2.(a) According to the question,

Let the number of ice pieces be n

Volume of spherical ice pieces $\times n$

= volume of cylindrical container

$$\frac{4}{3} \times \frac{22}{7} \times 3^3 \times n = \frac{22}{7} \times 9^2 \times 40$$

$$n = 90$$

Sol.3.(c) According to the question,

Volume of hemispherical container

= volume of cylindrical bottles

Let the number of bottles = x

$$\frac{2}{3} \pi r^3 = \pi r^2 h \times x$$

$$\frac{2}{3} \times 18 \times 18 \times 18 = 3 \times 3 \times 6 \times x$$

$$(\text{Number of bottles}) x = 72$$

Sol.4.(c) H.C.F of 90 and 126 = 18

Largest possible side of the tiles

$$= 18 \text{ feet}$$

$$\text{Area of tiles} = 18 \times 18 = 324 \text{ feet}^2$$

Sol.5.(b)

$$\text{Volume of frustum} = \frac{\pi H}{3} (R^2 + r^2 + Rr)$$

$$= \frac{22 \times 45}{7 \times 3} (28^2 + 7^2 + 28 \times 7)$$

$$= \frac{22 \times 45}{7 \times 3} \times 1029 = 48510$$

Now,

Volume of cylinder = Volume of frustum

$$\frac{22}{7} \times 3x \times 3x \times 5x = 48510$$

$$x^3 = 343 \Rightarrow x = 7$$

So, curved surface area of cylinder

$$= 2\pi rh$$

$$= 2 \times \frac{22}{7} \times 21 \times 35 = 4620 \text{ cm}^2$$

Sol.6.(d) Area of square = Base²

Area of rhombus = Base² $\times \sin \theta$ (θ = angle between the sides)

The ratio of area of the square to the

$$\text{area of rhombus} = \frac{1}{\sin 45} = \sqrt{2} : 1$$

Sol.7.(a) Volume of the wood Required =

(Total volume – inner volume)

$$(10 \times 8 \times 6) - [(10 - 1) \times (8 - 1) \times (6 - 1)] = 480 - 315 = 165 \text{ cm}^3$$

Sol.8.(d) According to the question,

Weight $\propto$ (density $\times$ volume)

$$\text{Volume} = k \times \frac{\text{weight}}{\text{density}}$$

Ratio - 1st cube : 2nd cube

Density - 27 : 16
Weight - 4 : 1

Volume - $\frac{4}{27} : \frac{1}{16} \rightarrow 64 : 27$

Side = $\sqrt[3]{64} : \sqrt[3]{27} \rightarrow 4 : 3$

Sol.9.(a) According to the question,
weight = radii² × thickness

Ratio of weight = 4 : 1

Ratio of thickness = 16 : 9

Ratio of their radii = $\sqrt{\frac{4}{16}} : \sqrt{\frac{1}{9}}$
= $\frac{2}{4} : \frac{1}{3} = 3 : 2$

Sol.10.(b) Length and the breadth of the rectangular field along with the path
= 45 + 5 = 50, 35 + 5 = 40
Area of the path = (50 × 40) - (45 × 35)
= 425 m²

Sol.11.(a)

Circumference of dome (2πr) = 154 cm

$r = 154 \times \frac{7}{44} = 24.5$ cm

Curved surface area of hemispherical

dome = $2 \times \frac{22}{7} \times 24.5 \times 24.5$

Required cost = $\frac{2 \times 22 \times 24.5 \times 24.5}{7 \times 100} \times 4$
= $\frac{2 \times 22 \times 3.5 \times 24.5}{25} = 150.92$ ₹

Sol.12.(a) Area of sector = $\frac{\theta}{360^\circ} \times \pi r^2$

$616 \text{ cm}^2 = \frac{10^\circ}{360^\circ} \times \frac{22}{7} \times r^2$

$r^2 = \frac{616 \times 36 \times 7}{22} = 28 \times 36 \times 7$

$r = 84$ cm

Sol.13.(c) Volume of cone

= $\frac{1}{3} \times \frac{22}{7} \times \frac{7}{4} \times \frac{7}{4} \times \frac{7}{2}$

= $\frac{11 \times 7 \times 7}{3 \times 16} = \frac{539}{48} = 11.23 \text{ cm}^3$

Sol.14.(c)

Total surface area of cylinder = 2πr(h + r)

$6336 = 2 \times \frac{22}{7} \times r \times 42$

$r = \frac{6336}{22 \times 12} = 24$ m

Curved surface area of cylinder = 2πrh

= $2 \times \frac{22}{7} \times 24 \times 18 = 2715.43 \text{ m}^2$

Sol.15.(b)

Let the side of cubes is 3x, 4x, 5x

Side of newly formed cube = $\frac{\text{diagonal}}{\sqrt{3}}$

Side of newly formed cube = $\frac{18\sqrt{3}}{\sqrt{3}} = 18$

Equating the volume of cubes :-

$(3x)^3 + (4x)^3 + (5x)^3 = 18 \times 18 \times 18$

$x^3 = \frac{18 \times 18 \times 18}{216} = 27 \Rightarrow x = 3$

Sides are :- 9, 12 and 15

Sol.16.(b)

Perimeter of equilateral angle triangle = 3a

Area of equilateral angle triangle = $\frac{\sqrt{3}}{4} a^2$

According to question, $a = \sqrt[4]{\frac{16}{3}}$

Therefore, required area = $\frac{\sqrt{3}}{4} \times (\sqrt[4]{\frac{16}{3}})^2$
= $\frac{\sqrt{3}}{4} \times \frac{4}{\sqrt{3}} = 1 \text{ unit}^2$

Sol.17.(c)

Curved surface area of hemisphere = $2\pi r^2$

Cost for green colour = $3 \times (2 \times 3.14 \times 4 \times 4) = 301.44$ Rs.

Cost for red colour

= $3 \times (2 \times 3.14 \times 4 \times 4) = 301.44$ Rs.

Area of two bases = $\pi r^2 + \pi r^2 = 2\pi r^2$

= $(2 \times 3.14 \times 4 \times 4) = 100.48$

The cost of painting with blue

= $2 \times 100.48 = 200.96$ Rs.

Total cost = $301.44 + 301.44 + 200.96$

= 803.84 Rs.

Sol.18.(d)

Total surface area of cube = $(6a^2)$

= $6 \times 5^2 = 150 \text{ cm}^2$

Increased area = $6 \times 5 \text{ cm}^2 - 6 \times 1 \text{ cm}^2$

= 24 cm^2

Total increased surface area = $150 + 24$

= 174 cm^2

Sol.19.(b)

Volume of Prism = Area of Base × Height

Volume of Pyramid

= $\frac{1}{3} \times \text{Area of Base} \times \text{Height}$

Required ratio

⇒ Volume of Prism : Volume of Pyramid
= 3 : 1

Sol.20.(c) ATQ, CSA of cone = total area of rectangular cloth

$\frac{22}{7} \times \frac{14}{2} \times 9 = 2 \times (\text{length of cloth})$

⇒ $198 = 2 \times l \Rightarrow l = \frac{198}{2} = 99$ m

Sol.21.(a) Angle covered in 45 minutes

by minute hand = $\frac{360}{60} \times 45 = 270^\circ$

Required area = $\frac{270^\circ}{360^\circ} \times \frac{22}{7} \times 20 \times 20$

= $\frac{6600}{7} \text{ cm}^2$

Sol.22.(b) External radius (R) = 10 cm

Internal radius (r) = 9 cm and

length (h) = 12 cm

Total surface area of hollow cylinder pipe
= $2\pi(R + r)\{h + R - r\}$

= $2 \times \frac{22}{7} \times 19 \times 13$
= $\frac{44 \times 247}{7} = 1552.57 \text{ cm}^2$

Sol.23.(c) Let the length of rectangle is 'l' and breadth of rectangle is 'b'
Area of rectangle = length × breadth = l × b

(Area of rectangle) = (lb)e.q.(1)

According to question,

(Area of resulting square)

= (l - 4) × (b + 2)e.q.(2)

Side of square are equal, so

l - 4 = b + 2 ⇒ l - b = 6e.q.(3)

Equating e.q.(1) and e.q.(2), we get

(lb) = lb + 2l - 4b - 8

l - 2b = 4e.q.(4)

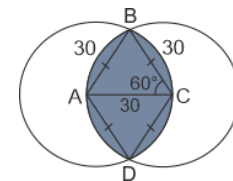
From e.q.(3) and e.q.(4), we get

b = 2, l = 8

So, Perimeter of the original rectangle

= 2(8 + 2) = 20 cm

Sol.24.(d)



Area of sector ACB = $\frac{60^\circ}{360^\circ} \times \pi r^2 = \frac{900\pi}{6}$

Area of sector BAC = $\frac{60^\circ}{360^\circ} \times \pi r^2 = \frac{900\pi}{6}$

Area of equilateral triangle ABC

= $\frac{\sqrt{3}}{4} r^2 = \frac{900\sqrt{3}}{4}$

Upper shaded region = area of sector ACB + area of sector BAC - area of equilateral ΔABC

Area of total shaded region

= 2 × area of upper half shaded region

Required shaded area

= $2 \times \left\{ \frac{900\pi}{6} + \frac{900\pi}{6} - \frac{900\sqrt{3}}{4} \right\}$

= $2 \times \left\{ \frac{1800\pi}{6} - \frac{900\sqrt{3}}{4} \right\}$

= $(600\pi - 450\sqrt{3}) \text{ cm}^2$

Short Trick:-

Area of shaded region = $r^2 \left(\frac{2}{3} \pi - \frac{\sqrt{3}}{2} \right)$

Area = $900 \left(\frac{2}{3} \pi - \frac{\sqrt{3}}{2} \right)$

= $(600\pi - 450\sqrt{3}) \text{ cm}^2$

Sol.25.(a) Given, area of the trapezium
= 1785 sq. feet.

Height of the trapezium = 35 feet, length of one parallel side = 42 feet.

Let the length of the other parallel side be x feet.

Area of trapezium

= $\frac{1}{2} \times (\text{sum of parallel sides}) \times \text{height}$

$$= \frac{1}{2} \times (x + 42) \times 35 = 1785$$

$$= 35x + 1470 = 3570$$

$$= 35x = 3570 - 1470$$

$$= 35x = 2100 \Rightarrow x = 60$$

Hence,

the length of the other side is 60 feet.

Sol.26.(b) Let side of the rhombus be 'a'

$$\text{Perimeter of rhombus} = 4a = 4P \Rightarrow a = P$$

As we know that,

$$D_1^2 + D_2^2 = 4a^2, \quad D_1 + D_2 = L$$

$$\text{Then, } D_1 D_2 = \frac{L^2 - 4a^2}{2}$$

$$\text{Area of Rhombus} = \frac{1}{2} \times D_1 \times D_2$$

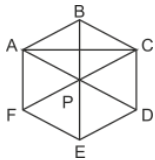
$$= \frac{1}{2} \times \frac{L^2 - 4a^2}{2} = \frac{1}{4} \times (L^2 - 4P^2)$$

Sol.27.(d) Volume of water flowing through the canal in 30 min

$$= \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times (18 \times \frac{5}{18}) \times (30 \times 60)$$

$$= 3,46,500 \text{ m}^3$$

Sol.28.(a) According to the question,



As ABCDEF is a regular hexagon,

$$\text{Area of } \square ABCP = 2 \times \frac{\sqrt{3}}{4} \times (\text{side})^2$$

$$= 2 \times \frac{\sqrt{3}}{4} \times 36^2 = 648\sqrt{3} \text{ cm}^2$$

$$\text{So, area of } \triangle ABC = \frac{1}{2} \times \text{Area of } \square ABCP$$

$$= \frac{1}{2} \times 648\sqrt{3} = 324\sqrt{3} \text{ cm}^2$$

Sol.29.(a)

$$\text{Volume of frustum} = \frac{1}{3} \pi h (R^2 + r^2 + Rr)$$

$$= \frac{1}{3} \pi \times 6 (5^2 + 3^2 + 15)$$

$$= 2\pi (25 + 9 + 15) = 98\pi \text{ cm}^3$$

Sol.30. (d) Let $a = b = 2$ units

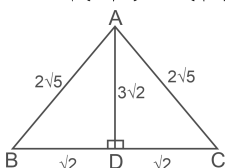
$$BC = \sqrt{a^2 + b^2} = 2\sqrt{2} \text{ units}$$

$$AB = \sqrt{(2a)^2 + b^2} = 2\sqrt{5} \text{ units}$$

$$AC = \sqrt{a^2 + (2b)^2} = 2\sqrt{5} \text{ units}$$

It becomes an isosceles triangle as below.

$$AD = \sqrt{(2\sqrt{5})^2 - (\sqrt{2})^2} = 3\sqrt{2} \text{ units}$$

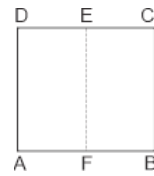


Area of triangle

$$= \frac{1}{2} \times 2\sqrt{2} \times 3\sqrt{2} = 6 \text{ sq units}$$

$$\text{In option (d), } \frac{3}{2} ab = \frac{3}{2} \times 2 \times 2 = 6$$

Sol.31.(d)



Let the side of square = a

$$AB = BC = CD = DA = EF = a$$

$$\text{Perimeter of rectangle ADEF} + \text{Perimeter of rectangle BCEF} = 39 + 39 = 78$$

$$AB + BC + CD + DA + EF + EF = 78$$

$$6a = 78 \text{ cm} \Rightarrow a = 13 \text{ cm}$$

$$\text{Perimeter of Square} = 4a = 52 \text{ cm}$$

Sol.32.(b) Circumference of a circle = $2\pi r$

$$2\pi r = a\pi \Rightarrow 2r = a$$

$$\text{Area} = b\pi \Rightarrow \pi r^2 = b\pi \Rightarrow b = r^2$$

$$a : b = 2r : r^2 \text{ or } 2 : r$$

$$4 : 5 = 2 : r \Rightarrow r = 2.5 \text{ cm}$$

Sol.33.(c) 65 m, 72 m and 97 m are triplets of a right - angled triangle.

So area of triangular park

$$= \frac{1}{2} \times 65 \times 72 = 2340$$

Let sides of the rectangle be $5x$ and $13x$.

$$\text{Area of rectangle} = 65x^2$$

$$\text{ATQ, } 65x^2 = 2340 \Rightarrow x = 6$$

$$\text{Perimeter of rectangle} = 2(5x + 13x) = 36x = 216 \text{ m}$$

Sol.34.(c) Radius of bicycle = 0.42 m

Distance travelled in 25 seconds

$$= 2\pi r \times 40 \Rightarrow 2 \times \frac{22}{7} \times 0.42 \times 40$$

$$= 105.6 \text{ m}$$

$$\text{Speed of bicycle (in m/s)} = \frac{105.6}{25}$$

$$= 4.224$$

$$\text{Speed of bicycle (in kmph)} = 4.224 \times \frac{18}{5}$$

$$= 15.2$$

Sol.35.(a) Area of the square = $a^2 = 169$.

$$\text{So, } a = 13, \text{ Perimeter} = 4a = 4 \times 13 = 52$$

$$\text{Circumference of the circle} = 2\pi r = 52$$

$$r = \frac{7}{44} \times 52 = \frac{91}{11}$$

$$\text{Area} = \pi r^2$$

$$= \frac{22}{7} \times \frac{91}{11} \times \frac{91}{11}$$

$$= \frac{2366}{11} = 215.09 \approx 215 \text{ cm}^2$$

Sol.36.(d) Slant height of the cone

$$= \text{radius of the sector And } \frac{\theta}{360} \times$$

$$\text{perimeter of the circle}$$

$$= \text{perimeter of the base of the cone}$$

$$\text{Let the radius of cone} = r$$

$$\frac{120}{360} \times 2 \times \pi \times 10.5 = 2 \times \pi \times r$$

$$\Rightarrow r = 3.5$$

$$\text{Height of the cone} = \sqrt{l^2 - r^2}$$

$$= \sqrt{(10.5)^2 - (3.5)^2} = 7\sqrt{2}$$

Volume of the cone

$$= \frac{1}{3} \times \pi \times 3.5 \times 3.5 \times 7\sqrt{2}$$

$$= \frac{343\sqrt{2}}{12} \pi \text{ cm}^3$$

Sol.37.(c) 20, 21 and 29 are triplet pairs, clearly a given triangle is a right angle triangle.

$$\text{Area of the triangle} = \frac{1}{2} \times 20 \times 21$$

$$= 210 \text{ cm}^2$$

$$\text{Volume of the prism} = \text{Base Area} \times \text{Height}$$

$$\Rightarrow 7560 = 210 \times \text{Height} \Rightarrow \text{Height} = 36$$

Lateral surface area of the prism

$$= \text{Base perimeter} \times \text{Height}$$

$$= (20 + 21 + 29) \times 36 = 2520 \text{ cm}^2$$

Sol.38.(a) Let the original radius be r .

According to the question

$$4\pi(r + 4)^2 = 4\pi r^2 + 464\pi$$

$$\Rightarrow r^2 + 16 + 8r = r^2 + 116 \Rightarrow r = 12.5$$

$$\text{Volume of the sphere} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \times 12.5 \times 12.5 \times 12.5 \times \pi$$

$$= \frac{15625}{6} \pi \text{ cm}^3$$

Sol.39.(c)

Volume of prism = base area $\times$ height

$$1731.6 = \frac{1}{2} \times (15 + 11) \times 9 \times \text{height}$$

$$\Rightarrow \text{height} = 14.8$$

Sol.40.(b) Let n be the number of turns.

Total length of the ribbon = number of turns $\times$ circumference of the base of

$$\text{cylinder } l = n \times c \Rightarrow n = \frac{l}{c}$$

Sol.41.(d) We know that

$$\text{Volume of cone} = \frac{1}{3} \pi R^2 h$$

$$\text{Volume of sphere} = \frac{4}{3} \pi r^3$$

Given, $R = 2r$

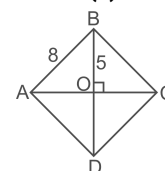
According to the question

$$\frac{1}{3} \pi R^2 h = \frac{4}{3} \pi r^3 \Rightarrow \frac{1}{3} \pi (2r)^2 h = \frac{4}{3} \pi r^3$$

$$r = h \Rightarrow R : h$$

$$\Rightarrow 2r : r = 2 : 1$$

Sol.42.(c)



$$BO = \frac{BD}{2} = \frac{10}{2} = 5$$

$$AO = \frac{AC}{2} = \frac{2\sqrt{x}}{2} = \sqrt{x}$$

$$\text{Now, } AB^2 = AO^2 + BO^2$$

$$8^2 = (\sqrt{x})^2 + 5^2 \Rightarrow x = 39$$

$$\sqrt{x+10} \Rightarrow \sqrt{39+10} = 7 \text{ cm}$$

Sol.43.(d) Perimeter of rhombus

$$= 2\sqrt{d_1^2 + d_2^2} = 2\sqrt{14^2 + 48^2} = 100 \text{ cm}$$

Sol.44.(c)

Circumference of the circle(C) = $2\pi r$

$$r = \frac{C}{2\pi}$$

$$\text{Area of the sector (A)} = \frac{60}{360} \times \pi \times r^2$$

$$= \frac{1}{6} \times \pi \times \left(\frac{C}{2\pi}\right)^2 \Rightarrow \frac{C^2}{24\pi}$$

Sol.45.(b) Let the length of the perpendicular and base of the right angle triangle is a cm.

$$\Rightarrow \text{Hypotenuse of the triangle} = a\sqrt{2}$$

Area of Δ formed on hypotenuse(H)

$$= \frac{\sqrt{3}}{4} \times (a\sqrt{2})^2 = \frac{\sqrt{3}a^2}{2}$$

Area of Δ formed on perpendicular(A)

$$= \frac{\sqrt{3}a^2}{4} \Rightarrow \frac{A}{H} = \frac{\frac{\sqrt{3}a^2}{4}}{\frac{\sqrt{3}a^2}{2}} = \frac{1}{2}$$

Sol.46.(c) Since it is an equilateral triangle. Each side will be formed of 4 sticks and its length will be 4 units.

$$\text{Area of the equilateral triangle} = \frac{\sqrt{3}}{4} a^2$$

$$= \frac{\sqrt{3}}{4} \times 4 \times 4 = 4\sqrt{3}$$

Sol.47.(c) % age change in the area

$$= (-11) + 11 + \frac{(-11) \times (11)}{100} = -1.21\%$$

Here, - ve sign indicates that the area is decreasing.

Sol.48.(d) %age increase in area

$$= 22 + 22 + \frac{22 \times 22}{100} = 48.84\%$$

Sol.49.(d) Let AB = a cm

$$AC = a + 10 \Rightarrow BC = 2a - 2$$

According to the question

$$a + a + 10 + 2a - 2 = 32 \Rightarrow 4a = 24$$

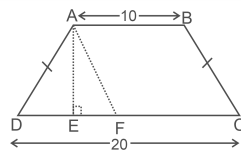
$$a = AB = 6 \text{ cm}$$

$$AC = 6 + 10 = 16 \text{ cm}$$

$$BC = 2(6) - 2 = 10 \text{ cm}$$

Clearly AB is the smallest side.

Sol.50.(b)



Let ABCD be the given trapezium. And AE be its height.

$$\text{Area of trapezium} = \frac{1}{2} \times (CD + AB) \times AE$$

$$180 = \frac{1}{2} \times (20 + 10) \times AE$$

$$AE = 12 \text{ cm}$$

$$\Rightarrow AB = FC = 10 \text{ cm and } AF = BC$$

$$\Rightarrow DF = DC - FC = 20 - 10 = 10 \text{ cm}$$

$$\text{In } \Delta ADF \Rightarrow AF = AD \text{(AD = BC, given)}$$

$$DE = EF = \frac{DF}{2} = 5 \text{ cm}$$

.....(Perpendicular drawn on the base of an isosceles triangle)

$$AF^2 = AE^2 + EF^2, AF = \sqrt{AE^2 + EF^2}$$

$$= \sqrt{12^2 + 5^2} = 13$$

$$\text{Hence } AD = BC = 13 \text{ cm}$$

Sol.51.(d)

Let the breadth of the rectangle = b

Length of the rectangle (l) = b + 20

Perimeter of the rectangle = $2(b + b + 20)$

$$= 4(b + 10)$$

According to the question

$$4(b + 10) \times 53 = 21200$$

$$b + 10 = 100 \Rightarrow b = 90 \text{ m}$$

$$l = 90 + 20 = 110 \text{ m}$$

$$\text{Area of the rectangle} = 90 \times 110 = 9900 \text{ m}^2$$

Sol.52.(b) Let the sides of the rectangle are 7x and 4x.

$$\text{Perimeter of rectangle} = 2(7x + 4x) = 22x$$

$$\text{Perimeter of circle} = 2\pi r \Rightarrow 2 \times \frac{22}{7} \times 31.5$$

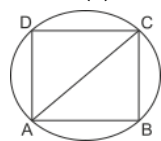
According to the question

$$22x = 2 \times \frac{22}{7} \times 31.5 \Rightarrow x = 9$$

$$\text{Smaller side of rectangle} = 4x$$

$$= 4 \times 9 = 36 \text{ cm}$$

Sol.53.(d)



Let the sides be 4 units and 3 units.

According to the question

$$2(4 \text{ unit} + 3 \text{ unit}) = 56$$

$$1 \text{ unit} = 4 \text{ cm, } 4 \text{ unit} = 16 \text{ cm, } 3 \text{ unit} = 12 \text{ cm}$$

Diagonal of Rectangle = Diameter of circle

$$= \sqrt{AB^2 + BC^2} = \sqrt{16^2 + 12^2} = 20 \text{ cm}$$

$$\text{Radius of the circle} = \frac{20}{2} = 10 \text{ cm}$$

$$\text{Area of circle} = \pi r^2 = \pi 10^2 = 100\pi$$

Sol.54.(c)

Curved Surface area of cylinder = $2\pi rh$

$$2 \times \frac{22}{7} \times 7 \times h = 440, h = 10 \text{ cm}$$

$$\text{Volume of cylinder} = \pi r^2 h$$

$$= \frac{22}{7} \times 7^2 \times 10 = 1540 \text{ cm}^3$$

Sol.55.(d)

Curved Surface area of cylinder = $2\pi rh$

$$\text{Volume of cylinder} = \pi r^2 h$$

According to the question

$$2\pi rh = 132 \text{(1), } \pi r^2 h = 528 \text{(2)}$$

Dividing eq 2 by eq 1, $r = 8 \text{ cm}$

Put $r = 8$ in eq (1)

$$2\pi rh = 132$$

$$2 \times \frac{22}{7} \times 8 \times h = 132 \Rightarrow h = 2\frac{5}{8} \text{ m}$$

Sol.56.(a) Area of Hexagon = $6 \times \frac{\sqrt{3}}{4} a^2$

$$\text{Here, } 6 \times \frac{\sqrt{3}}{4} a^2 = 2400\sqrt{3},$$

$$a^2 = 1600, a = 40$$

Perimeter of the hexagonal field = $6a$

$$6 \times 40 = 240 \text{ metre}$$

$$\text{Required cost of fencing} = 240 \times 18.5 = \text{Rs. 4440}$$

Sol.57.(c) One of the sides of newly formed cuboid = $6 \times 2 = 12 \text{ cm}$

Other two sides will be 2 cm.

Surface area of cuboid = $2(lb + bh + hl)$

$$= 2(12 \times 2 + 2 \times 2 + 2 \times 12) = 104 \text{ cm}^2$$

Sol.58.(d) $40\% = \frac{2}{5}$

Old : New

$$\text{Radius} \rightarrow 5 : 3$$

$$\text{Volume}(\propto r^3) \rightarrow 125 : 27$$

Let the volumes are 125 unit and 27 unit

$$\text{Required \%} = \frac{125 - 27}{125} \times 100 = 78.4\%$$

Sol.59.(b) Area of triangular field with

$$\text{side } x \text{ m} = \frac{\sqrt{3}}{4} x^2$$

Area of other triangular field

$$= \sqrt{s(s-a)(s-b)(s-c)}$$

$$\text{Where } s = \frac{a+b+c}{2}$$

$$\Rightarrow \frac{50+70+80}{2} = 100 \text{ m}$$

Area of field =

$$\sqrt{100(100-50)(100-70)(100-80)}$$

$$= \sqrt{100 \times 50 \times 30 \times 20} = 1000\sqrt{3} \text{ m}^2$$

$$\text{According to question, } \frac{\sqrt{3}}{4} x^2 = 1000\sqrt{3}$$

$$x^2 = 4000 \Rightarrow x = 63.25 \text{ m}^2$$

Sol.60.(d)

$$\text{Area of Sector} = \frac{\theta}{360^\circ} \times \pi r^2$$

$$= \frac{150}{360} \times \frac{22}{7} \times 21^2 = 577.5 \text{ cm}^2$$

Sol.61.(d) Area of circle is $\propto r^2$.

⇒ Area of three circles will be $4^2, 5^2, 7^2$

Desired Ratio = $(5^2 - 4^2) : (7^2 - 5^2)$

$$= 9 : 24 = 3 : 8$$

Sol.62.(c)

Volume of a cylinder = $\pi(R^2 - r^2)h$

Where R = external radius

r = internal radius

h = height or length of the cylinder

$$\text{Volume} = \frac{22}{7} \times (2.5^2 - 1.5^2) \times 756$$

$$= 22 \times (2.5 + 1.5)(2.5 - 1.5) \times 108$$

$$= 9504 \text{ cm}^3$$

Total Weight of pipe = Volume of pipe × density of metal

$$= 9504 \times 7.5 = 71280 \text{ g or } 71.28 \text{ kg}$$

Sol.63.(a)

Volume of a cylinder = $\pi(R^2 - r^2)h$

Where R = external radius

r = internal radius

h = height or length of the cylinder

$$7480 = \frac{22}{7} \times (9^2 - r^2) \times 140$$

$$17 = 81 - r^2 \Rightarrow r^2 = 64 \Rightarrow r = 8 \text{ cm}$$

$$\text{Thickness} = 9 - 8 = 1 \text{ cm}$$

Sol.64.(a) Volume of cuboid = $\sqrt{a \times b \times c}$

Here a, b and c are the area of three

$$\text{faces} = \sqrt{32 \times 24 \times 48} = 192 \text{ cm}^3$$

Sol.65.(d) We know that

$$\text{Inradius} = \frac{\text{area}}{\text{semiperimeter}}$$

$$3 = \frac{15}{\text{semiperimeter}} \Rightarrow \text{Semiperimeter} = 5$$

$$\text{Perimeter} = 2 \times 5 = 10 \text{ cm}$$

Sol.66.(c) Required area of sheet = Total surface area of rectangular tank - Area of circular opening

$$= 2 \times (80 + 48 + 60) - \pi \times 1^2$$

$$= 376 - 3.1 = 372.9 \text{ m}^2$$

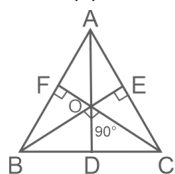
Sol.67.(b) Let side of bigger cube = R

$$R^3 = 13824 \Rightarrow R = \sqrt[3]{13824} = 24$$

Let side of smaller cubes = r

$$13824 = 8 \times r^3 \Rightarrow 1728 = r^3 \Rightarrow 12 = r$$

$$\text{Required ratio} = (6 \times 24^2) : (3 \times 6 \times 12^2) = 4 : 3$$

Sol.68.(b)

$$\Delta OBC, OB = OC = x \Rightarrow BC = 2$$

$$x^2 + x^2 = 2^2 \Rightarrow x^2 = 2 \Rightarrow x = \sqrt{2}$$

$$\text{In } \Delta OBD, BD = 1 \Rightarrow OB = \sqrt{2}$$

$$OD = \sqrt{(\sqrt{2})^2 - 1^2} = 1$$

$$\text{Now, } AO = 2 \times OD \Rightarrow 2 \times 1 = 2$$

$$AD = AO + OD = 3$$

$$\text{Area of } \Delta ABC = \frac{1}{2} \times 3 \times 2 = 3 \text{ cm}^2$$

Sol.69.(a)

Curved surface area of the cylinder

$$= 2 \times \frac{22}{7} \times 7 \times \frac{20}{100} = \frac{44}{5} \text{ m}^2$$

$$\text{Total cost} = \frac{44}{5} \times 20 = 176 \text{ rs}$$

Sol.70.(b) Volume of cubical box

$$= 30 \times 30 \times 30 = 27000 \text{ cm}^3$$

Required rise in water level

$$= \frac{27000}{60 \times 30} = 15 \text{ cm}$$

Sol.71.(c) Old : New

$$\text{Length} \rightarrow 10 : 13$$

$$\text{Breadth} \rightarrow 5 : 4$$

$$\text{Area} \rightarrow 50 : 52$$

$$52 - 50 = 2 \text{ unit} = 100 \Rightarrow 1 \text{ unit} = 50$$

$$\text{New area} = 52 \times 50 = 2600 \text{ cm}^2$$

Sol.72.(c)

$$\text{Required number of buckets} = \frac{12 \times 13.5}{9} = 18$$

Sol.73.(a)

$$\text{Required Cost} = \frac{11 \times 6 \times 100}{60} \times 112.50 = 12,375$$

Sol.74.(c) Remaining area = $119 \times 18 -$

$$17 \times 6 = 2040$$

$$\text{Volume of sand} = 17 \times 6 \times 3 = 306$$

Required height

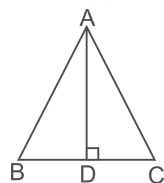
$$= \frac{306 \times 100 \times 100 \times 100}{2040 \times 100 \times 100} = 15 \text{ cm}$$

Sol.75.(d) Total area to be painted

$$= 40 \times 30 + 2(40 \times 2.2 + 30 \times 2.2) = 1508$$

$$\text{Total cost} = 1508 \times 25 = 37700$$

Sol.76.(d) Let the given triangle is ΔABC



Here,

AD is perpendicular drawn to the side BC

Let other two sides ($AB = AC$) = S

According to question

$$6 + 2S = 16 \Rightarrow S = 5$$

$$BD = \frac{6}{2} = 3, AD = \sqrt{5^2 - 3^2} = 4$$

$$\text{Area of } \Delta ABC = \frac{1}{2} \times AD \times BC$$

$$= \frac{1}{2} \times 4 \times 6 = 12 \text{ sq. cm}$$

Sol.77.(a) Inner surface area of all walls

$$= 2(l + b)h = 2(7 + 5)3.5 = 84 \text{ m}^2$$

Sol.78.(a) Since, the height of the triangle formed will be half of the diagonal of the square.

Therefore,

$$\text{Height} = \frac{3\sqrt{2}}{2} = \frac{3}{\sqrt{2}} \text{ m}$$

Sol.79.(d) Surface area of cube = $6 \times a^2 = 1176 \Rightarrow a = 14$

$$\text{Volume of cube} = a^3 = 14^3 = 2744 \text{ cm}^3$$

Sol.80.(a)

No. of boxes that can be painted

$$= \frac{11.28 \times 10000}{2 \times (30 \times 25 + 25 \times 12 + 12 \times 30)} = 40$$

Sol.81.(a) ATQ,

$$\frac{3\sqrt{3}}{2} \times a^2 = 108\sqrt{3} \Rightarrow a = 6\sqrt{2}$$

$$\therefore \text{Perimeter} = 6 \times 6\sqrt{2} = 36\sqrt{2} \text{ cm}$$

Sol.82.(a) Volume = $(12 - 4) \times (12 - 4) \times 2 = 8 \times 8 \times 2 = 128 \text{ cu. cm}$

Sol.83.(d) Given, in a right angle triangle Hypotenuse = 25, Base = 24

Then, Height = 7

$$\therefore \text{Area} = \frac{1}{2} \times 24 \times 7 = 84 \text{ cm}^2$$

Sol.84.(a)

$$\text{Longest pole} = \sqrt{16^2 + 8^2 + 11^2} = 21 \text{ m}$$

$$\text{Sol.85.(d)} \quad \frac{\sqrt{3}}{2} a = 10\sqrt{3} \Rightarrow a = 20$$

$$\text{Area} = \frac{\sqrt{3}}{4} a^2$$

$$= \frac{\sqrt{3}}{4} \times 20 \times 20 = 100\sqrt{3} \text{ cm}^2$$

Sol.86.(a)

$$\text{Rise in level} = \frac{15 \times 15 \times 15}{62 \times 45} = 1.21 \text{ cm}$$

Sol.87.(c)

$$\text{Required time} = \frac{66 \times 35 \times 3}{\frac{22}{7} \times \frac{35}{200} \times \frac{35}{200} \times 8}$$

$$= 9000 \text{ sec} = 2.5 \text{ h}$$

Sol.88.(a)

In the given solid figures. Cuboid has the maximum no. of vertex i.e. 8.

Sol.89.(b)

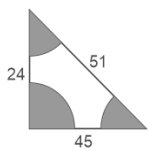
$$\text{Area} = 20 \times (7.5 + 8.6) = 322 \text{ cm}^2$$

Sol.90.(b) Since the chessboard is composed of 64 squares of side 2cm and the border is of thickness 2 cm.

Therefore, the length of each side of chessboard = $2 \times 8 + 4 = 20$

So, the perimeter = $4 \times 20 = 80 \text{ cm}$

Sol.91.(b) Given triangle is a right angled triangle:



$$\text{Area of triangle} = \frac{1}{2} \times 24 \times 45 = 540$$

$$\begin{aligned} \text{Area of all sectors of the circle covering} \\ \text{the triangle} &= \frac{180}{360} \times \frac{22}{7} \times 10.5 \times 10.5 \\ &= 173.25 \end{aligned}$$

$$\begin{aligned} \text{Therefore, Area of triangle excluding the} \\ \text{portion covered} &= 540 - 173.25 \\ &= 366.75 \text{ sq. cm} \end{aligned}$$

Sol.92.(a)

	Old : New
Radius →	2 ² : 5 ²
Height →	5 : 4

$$\text{Volume} \rightarrow \frac{20}{100}$$

$$\text{Required \%} = \frac{100 - 20}{20} \times 100 = 400\%$$

Sol.93.(b)

Before : After

$$\text{Ratio of radius} = 10 : 13$$

$$\therefore \text{Ratio of volume} = (10)^3 : (13)^3$$

$$= 1000 : 2197$$

$$\begin{aligned} \text{\% increase in volume} \\ &= \frac{2197 - 1000}{1000} \times 100 = 119.7\% \end{aligned}$$

Sol.94.(b)

$$\text{Diameter} = 21 \text{ cm, Radius} = \frac{21}{2} \text{ cm}$$

$$\begin{aligned} \text{Volume of hemispherical bowl} &= \frac{2}{3} \pi r^3 \\ &= \frac{2}{3} \times \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2} \times \frac{21}{2} = 2425.5 \text{ cm}^3 \end{aligned}$$

Sol.95.(b)

Before : After

$$\text{Radius} \rightarrow 4^2 : 5^2$$

$$\text{Height} \rightarrow 4 : 5$$

$$\text{Volume} \rightarrow 64 : 125$$

$$\text{\% increase in volume}$$

$$= \frac{125 - 64}{64} \times 100 = 95.31\%$$

Sol.96.(d)

$$\text{Volume of cube} = a^3$$

$$a^3 = 512 \text{ cubic unit}$$

$$\therefore \text{side of cube (a)} = 8 \text{ unit}$$

$$\text{Now, surface area of cube} = 6a^2$$

$$= 6 \times 8 \times 8 = 384 \text{ square unit}$$

Sol.97.(d)

$$\text{Let the total number of packed box} = x$$

$$\therefore \text{volume of big box} = \text{volume of small box} \times x$$

$$160 \times 90 \times 60 = 6 \times 5 \times 4 \times x$$

$$\therefore x = 7200$$

$$\text{Hence no. of packed box} = 7200$$

Sol.98.(c)

$$\text{Let the length} = 6x, \text{width} = 3x, \text{height} = 2x$$

$$\text{ATQ,}$$

$$2(LB + BH + HL) = 1800$$

$$2(6x \times 3x + 3x \times 2x + 2x \times 6x) = 1800$$

$$2 \times 36x^2 = 1800 \Rightarrow x^2 = 25 \Rightarrow x = 5$$

$$\text{Now, volume of cuboid} = L \times B \times H = 6x \times$$

$$= 3x \times 2x = 36x^3 = 36 \times 5^3 = 4500$$

Sol.99.(a) Let the no of small cubes be n
ATQ,

$$\text{Volume of solid cube} = \text{Volume of new cube formed}$$

$$48 \times 48 \times 48 = n \times 64$$

$$n = \frac{48 \times 48 \times 48}{64} = 1728$$

Sol.100.(c) Volume of cylinder = $\pi r^2 h$

$$\text{Radius} \rightarrow 4^2 : 5^2$$

$$\text{Height} \rightarrow 5 : 2$$

$$\text{Ratio of Volume} \rightarrow 80 : 50 = 8 : 5$$

Sol.101.(b) T.S.A. of cylinder = $2\pi r(h + r)$

$$\frac{18,425}{25} = 2 \times \frac{22}{7} \times r(30 + r)$$

$$\frac{737 \times 7}{11} = 4 \times r(30 + r)$$

$$469 = 4r^2 + 120r$$

$$4r^2 + 120r - 469 = 0$$

$$4r^2 + 134r - 14r - 469 = 0$$

$$2r(2r + 67) - 7(2r + 67) = 0$$

$$(2r + 67)(2r - 7) = 0$$

$$r = -\frac{67}{2} \text{ or } r = \frac{7}{2} \text{ m}$$

$$\text{Radius cannot be negative}$$

$$\text{therefore, radius (r)} = \frac{7}{2}$$

$$\text{Volume of cylinder} = \pi r^2 h$$

$$= \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 30$$

$$\text{Volume of cylinder} = 1155 \text{ m}^3$$

Sol.102.(c) According to the question,

$$\text{T.S.A of hemisphere} - \text{C.S.A of hemisphere}$$

$$3\pi r^2 - 2\pi r^2 = \pi r^2 = 4\pi \text{ cm}^2$$

Sol.103.(a) According to the question,

$$\text{C.S.A of cone} = \pi r l$$

$$\text{Required cost} = \frac{22}{7} \times 21 \times 25 \times 1.5$$

$$= 2475 \text{ Rs}$$

Sol.104.(d) According to the question,

$$\text{Volume of cone} = \frac{1}{3} \pi r^2 h$$

$$\frac{1}{3} \times \pi \times 7^2 \times h = 196\pi$$

$$h = 12 \text{ cm}$$

Sol.105.(b)

$$\text{Let the number of hemisphere} = n$$

$$\text{Volume of sphere} = n \times \text{volume of hemisphere}$$

$$\frac{4}{3} \pi \times 12 \times 12 \times 12 = n \times \frac{2}{3} \pi \times 4 \times 4 \times 4$$

$$n = 54$$

Sol.106.(d) Given, Perimeter of rhombus = 52 cm and diagonal (D1) = 10 cm

$$\text{Then, side of the rhombus} = 13 \text{ cm}$$

$$\text{Now, } D1^2 + D2^2 = 4\text{side}^2$$

$$D2^2 = 676 - 100$$

$$D2 = 24 \text{ cm}$$

$$\text{Now, area of rhombus}$$

$$= \frac{1}{2} \times D1 \times D2 = \frac{1}{2} \times 10 \times 24 = 120 \text{ cm}^2$$

Sol.107.(a)

$$\text{Given, radius of hemisphere} = 7 \text{ cm}$$

$$\text{Curved surface area} = 2\pi r^2$$

$$= 2 \times \frac{22}{7} \times 7 \times 7 = 308 \text{ cm}^2$$

$$\text{Total surface area} = 3\pi r^2$$

$$= 3 \times \frac{22}{7} \times 7 \times 7 = 462 \text{ cm}^2$$

Sol.108.(c)

$$\text{Given, } r_1 = 7 \text{ cm and } r_2 = 4 \text{ cm}$$

$$\text{Ratio of the surface area of hemisphere}$$

$$= (r_1)^2 : (r_2)^2 = 49 : 16$$

Sol.109.(c) side of square base = b

$$\text{slant height} = s$$

$$\text{height of pyramid} = h$$

$$\begin{aligned} \text{base area of pyramid (b)}^2 &= \frac{3}{8} \times 1536 \\ &= 576 \text{ m}^2 \end{aligned}$$

$$\text{side of square base (b)} = 24 \text{ m}$$

$$\text{T.S.A. of square based pyramid}$$

$$= (b)^2 + 2 \times b \times s$$

$$1536 = 576 + 2 \times 24 \times s$$

$$\text{Slant height (s)} = \frac{1536 - 576}{48}$$

$$= \frac{960}{48} = 20 \text{ m}$$

$$\text{Height of pyramid (h)} = \sqrt{s^2 - \left(\frac{b}{2}\right)^2}$$

$$= \sqrt{400 - 144} = \sqrt{256} = 16 \text{ m}$$

$$\text{Volume of pyramid} = (b)^2 \times \frac{h}{3}$$

$$= 24 \times 24 \times \frac{16}{3} = 3072 \text{ m}^3$$

Sol.110.(a) According to the question,

$$\text{Let the length and the breadth be } 6x, 5x.$$

$$\text{Area of rectangle} = \text{length} \times \text{breadth}$$

$$480 = 6x \times 5x \Rightarrow x = 4$$

$$\text{So the length will be (6x)} = 6 \times 4 = 24 \text{ m}$$

Sol.111.(a)

$$\text{Volume of sphere} = \frac{4}{3} \times \pi \times (\text{radius})^3$$

$$\text{when diameter is reduced to half then new radius will also reduced to half of}$$

the former sphere

Therefore

Volume of new sphere

$$= \frac{4}{3} \times \pi \times \left(\frac{\text{radius}}{2}\right)^3$$

$$= \frac{1}{8} \times \frac{4}{3} \times \pi \times (\text{radius})^3$$

= reduced by $\frac{1}{8}$ of the former volume

Sol.112.(c) According to the question,

$$\text{Volume of cone} = \frac{1}{3} \pi r^2 h$$

Volume of 1st cone : volume of 2nd cone

$$\frac{1}{3} \pi \times 5^2 \times 7 : \frac{1}{3} \pi \times \left(\frac{21}{2}\right)^2 \times 5$$

$$25 \times 7 : \frac{441}{4} \times 5 \Rightarrow 20 : 63$$

Sol.113.(a) Volume of cone = $\frac{1}{3} \pi r^2 h$

Ratio - initial : final

Radius - 10^2 : 11^2

Height - 10 : 11

Final ratio - 1000 : 1331

Hence, required % = $\frac{331}{1000} \times 100 = 33.1\%$

Sol.114.(d)

Diagonal of rectangle = $\sqrt{4^2 + 3^2} = 5$

According to the question,

Diagonal of rectangle = diameter of hemisphere = 5

So, radius = 2.5 cm

Now, total surface area of hemisphere

$$= 3\pi \times 2.5 \times 2.5 = \frac{75\pi}{4}$$

Sol.115.(d) According to the question,

Volume of cylindrical rod = volume of cylinder (wire)

$$\pi \times 1^2 \times 45 = \pi \times r^2 \times 500$$

$$r^2 = 0.09, r = 0.3 \text{ cm} = 3 \text{ mm}$$

Diameter of the wire = $2r = 6 \text{ mm}$

Sol.116.(d) Let the height and radius of the right circular cone = $12x$ and $5x$

$$\text{Volume of a cone} = \frac{1}{3} \pi r^2 h$$

$$314 = \frac{1}{3} \times 3.14 \times 25x^2 \times 12x$$

$$x^3 = \frac{314 \times 100 \times 3}{314 \times 25 \times 12} = 1 \Rightarrow x = 1$$

Hence, Slant height = $\sqrt{(12x)^2 + (5x)^2}$

$$= \sqrt{144 + 25} = 13 \text{ cm}$$

Sol.117.(d)

Given, diagonal of square = $24\sqrt{2}$

Then, side of the square = 24 cm

Lateral surface area of square pyramid

= $2 \times \text{side} \times \text{slant height}$

$$= 2 \times 24 \times l = 624 \Rightarrow l = 13 \text{ cm}$$

$$\text{Now, height of pyramid} = \sqrt{13^2 - \left(\frac{24}{2}\right)^2}$$

$$= 5 \text{ cm}$$

So, volume of pyramid

$$= \frac{1}{3} \times \text{side}^2 \times \text{height}$$

$$\frac{1}{3} \times 24^2 \times 5 = 960 \text{ cm}^3$$

Sol.118.(d) According to the question,

Volume of cylinder = Volume of sphere

$$\pi r^2 h = \frac{4}{3} \pi r^3 \Rightarrow 12 \times 12 \times h$$

$$= \frac{4}{3} \times 9 \times 9 \times 9 \Rightarrow h = \frac{4 \times 9 \times 9 \times 9}{3 \times 12 \times 12}$$

$$h = 6.75 \text{ cm}$$

Sol.119.(d) According to the question,

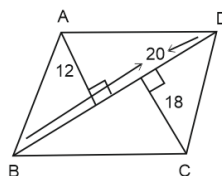
Surface area of sphere = surface area of hemisphere

$$4\pi R^2 = 3\pi r^2 \Rightarrow \frac{R}{r} = \sqrt{\frac{3}{4}}$$

$$\text{Ratio of their volume} = \frac{4}{3} \pi R^3 : \frac{2}{3} \pi r^3$$

$$= 4 \times 3\sqrt{3} : 2 \times 4\sqrt{4} = 3\sqrt{3} : 4$$

Sol.120.(d)



According to the question,

Area of quadrilateral ABCD

$$= \frac{1}{2} \times 20 \times 12 + \frac{1}{2} \times 20 \times 18 = 300 \text{ m}^2$$

Sol.121.(c)

Ratio - ball P : ball Q

Diameter - 8 : 2

Radius - 4 : 1

$$\text{Required ratio} = 4^3 : 1^3 = 64 : 1$$

Sol.122.(c) Given, Slant height (l) = 29 cm

Height (h) = 20 cm

$$\text{Radius } (r) = \sqrt{(29)^2 - (20)^2}$$

$$= \sqrt{841 - 400} = \sqrt{441} = 21 \text{ cm}$$

According to question,

$$\frac{\text{T.S.A. of cone}}{\text{Volume of cone}} = \frac{\pi r(l+r)}{\frac{1}{3} \pi r^2 h} = \frac{(l+r)}{\frac{1}{3} r h}$$

$$\frac{\text{T.S.A. of cone}}{\text{Volume of cone}} = \frac{(29+21)}{\frac{1}{3} \times 21 \times 20} = \frac{50}{140} = 5 : 14$$

Sol.123.(b) Given, Radius (R) = 3 cm

Radius of each hemispherical ball after recasting (r) = 1.5 cm

$$\text{Number of recasted balls} = \frac{\frac{4}{3} \pi R^3}{\frac{2}{3} \pi r^3}$$

$$= \frac{2 \times (3)^3}{(1.5)^3} = 2 \times \left(\frac{3}{1.5}\right)^3 = 2 \times 8 = 16$$

Sol.124.(c) According to the question,

Volume of hollow cylinder = $\pi(R^2 - r^2)h$

$$= \frac{22}{7} \times (10+7)(10-7) \times 49 = 7854 \text{ cm}^3$$

Sol.125.(a) According to the question,

Let the number of balls be n

Volume of small balls $\times n$ = volume of solid metallic sphere

$$\frac{4}{3} \times \pi \times 2 \times 2 \times 2 \times n = \frac{4}{3} \times \pi \times 16 \times 16 \times 16$$

$$(\text{number of balls}) n = 512$$

Sol.126.(b)

Volume of sphere = volume of wire

$$\frac{4}{3} \pi r^3 = \pi r^2 h$$

$$\frac{4}{3} \times 3 \times 3 \times 3 = 0.2 \times 0.2 \times h$$

$$h = 900 \text{ cm or } 9 \text{ m}$$

Sol.127.(b)

$$\text{Area of trapezium} = \frac{1}{2} \times (a + b) \times$$

Distance between parallel sides

$$1720 = \frac{1}{2} \times (48 + b) \times 40$$

$$86 = 48 + b \Rightarrow b = 38$$

Sol.128.(c) Number of small cubes n

$$= \frac{8 \times 6 \times 6}{0.5 \times 0.5 \times 0.5} = 2304$$

Sol.129.(c) Let radius and height of the cone = $3x$ and $4x$

$$\text{Slant height } (l) = \sqrt{(3x)^2 + (4x)^2}$$

$$= \sqrt{25x^2} = 5x$$

According to the question,

$$\text{Volume of the cone} = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \times 3.14 \times 9x^2 \times 4x = 301.44$$

$$= 12x^3 \times 3.14 = 301.44$$

$$= x^3 = \frac{301.44}{12 \times 3.14} = 8 \Rightarrow x = 2$$

$$\text{Slant height } (5x) = 5 \times 2 = 10 \text{ cm}$$

Sol.130.(c) According to the question,

$$1 \text{ liter} = 1000 \text{ cm}^3$$

$$50 \text{ liter} = 50,000 \text{ cm}^3$$

$$50 \text{ liter} = 50,000 \times \frac{1}{1000000} = 0.05 \text{ m}^3$$

$$\text{Required time} = \frac{120}{0.05}$$

$$= 2400 \text{ min. or } 40 \text{ hours}$$

Sol.131.(b) According to the question,

Volume of solid lead balls = Volume of solid sphere

Number of solid lead balls

$$= \frac{\frac{4}{3} \times \pi \times 7 \times 7 \times 7}{\frac{4}{3} \times \pi \times 1 \times 1 \times 1} = 343$$

Sol.132.(c)

Total surface area of cone = $\pi r(l + r)$

Pythagorean triplets 28, 45, 53

$r = 28$ cm, $l = 53$ cm

Total surface area of cone

$$= \frac{22}{7} \times 28 \times (53 + 28)$$

$$88 \times 81 = 7128 \text{ cm}^2$$

Sol.133.(c) Volume of 1st cone = volume of 2nd cone

$$\frac{1}{3} \pi r^2 h = \frac{1}{3} \pi r_1^2 h_1$$

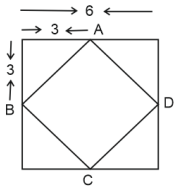
$$4 \times 4 \times 9.8 = 3.5 \times 3.5 \times h_1$$

$$h_1 = 12.8 \text{ cm}$$

Sol.134.(c) according to the question,

Area of the square = $side^2$

$$36 = side^2, \text{ side} = 6$$



$$\text{Length of AB} = \sqrt{3^2 + 3^2} = \sqrt{18}$$

$$\text{Area of square ABCD} = (\sqrt{18})^2$$

$$= 18 \text{ square units}$$

Sol.135.(a) Both the median and the altitude are the same in equilateral triangle.

(height) Median of equilateral triangle

$$= \frac{\sqrt{3}}{2} a$$

$$= \frac{\sqrt{3}}{2} a = 6\sqrt{3} \Rightarrow a = 12 \text{ cm}$$

$$\text{So, area of equilateral triangle} = \frac{\sqrt{3}}{4} a^2$$

$$= \frac{\sqrt{3}}{4} \times 12 \times 12 = 36\sqrt{3} \text{ cm}^2$$

Sol.136.(b) Length of longest pole

$$(\text{diagonal}) = \sqrt{l^2 + b^2 + h^2}$$

$$= \sqrt{576 + 324 + 256} = \sqrt{1156} = 34 \text{ m}$$

Sol.137.(b)

$$\text{Required radius} = \sqrt[3]{6^3 + 8^3 + 10^3}$$

$$= \sqrt[3]{1728} = 12 \text{ cm}$$

Sol.138.(b) Volume of cuboid When areas of three adjacent faces of a

cuboid are given = $\sqrt{pqr} \text{ unit}^3$

Sol.139.(c)

Curved surface area of cone = $\pi r l$

$$\pi r l = 396$$

$$\text{slant height}(l) = \frac{396 \times 7}{10.5 \times 22} = 12 \text{ cm}$$

Sol.140.(b) Volume of the cuboid

$$= 12.2 \times 16 \times 8.5 = 1659.2 \text{ cm}^3$$

Sol.141.(b)

Total surface area of cylinder = $2\pi r(h + r)$

$$2\pi r(h + r) = 64\pi$$

$$2\pi r \times 2r = 64\pi \text{ (given } r = h)$$

$$4\pi r^2 = 64\pi$$

$$r = 4, (h = 4)$$

$$\text{Now, } R = 2, H = 8$$

Total surface area of cylinder

$$= 2\pi \times 2(8 + 2) = 40\pi$$

Sol.142.(d)

$$\text{Area of triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times 7 \times 82 = 287 \text{ cm}^2$$

Sol.143.(c) According to the question,

Volume of hollow metal sphere = volume of solid cone

$$\frac{4}{3} \times \pi \times (6^3 - 5^3) = \frac{1}{3} \times \pi \times (5.2)^2 \times h$$

$$h = \frac{4 \times 91}{5.2 \times 5.2} \approx 13.5$$

Sol.144.(a) Area of rhombus

$$= \frac{1}{2} \times d_1 \times d_2$$

$$= \frac{1}{2} \times 7.5 \times 5.6 = 21 \text{ cm}^2$$

Sol.145.(c)

Let the number of melted cone is n

According to the question,

$$n \times \frac{1}{3} \pi \times 4 \times 4 \times 6$$

$$= \frac{4}{3} \pi \times 12 \times 12 \times 12$$

$$n = \frac{12 \times 12 \times 12}{4 \times 6} = 72$$

Sol.146.(c) According to the question,

$$\text{Volume of the sphere} = \frac{4}{3} \pi r^3$$

$$\frac{2673}{7} = \frac{4}{3} \times \frac{22}{7} \times r^3$$

$$r^3 = 91.125, r = 4.5$$

So the diameter of the sphere

$$= 4.5 \times 2 = 9 \text{ m}$$

Sol.147.(a) According to the question,

Volume of hollow sphere = volume of cone

$$\frac{4}{3} \pi (R^3 - r^3) = \frac{1}{3} \pi r^2 h$$

$$4(3^3 - 2^3) = 4^2 h \Rightarrow 4 \times 19 = 16h$$

$$(\text{height}) h = 4 \frac{3}{4} \text{ cm}$$

Sol.148.(d)

$$\text{Total area of ground} (\pi r^2) = 15 \times 5 = 75 \text{ m}^2$$

$$\text{Volume of the cone} = \frac{1}{3} \pi r^2 h$$

According to the question,

$$\frac{1}{3} \pi r^2 h = 30 \times 15 \Rightarrow \frac{1}{3} \times 75 \times h = 30 \times 15$$

$$\text{Height of cone } (h) = 18 \text{ m}$$

Sol.149.(d) 1 cm = 10 mm

Volume of sphere = Volume of wire(cylinder)

According to the question,

$$\frac{4}{3} \pi \times 1 \times 1 \times 1 = \pi \times 0.2 \times 0.2 \times h$$

$$\text{Length of wire } (h) = \frac{100}{3} \text{ cm} = 33 \frac{1}{3} \text{ cm}$$

Sol.150.(a) Total surface area of cube

$$= 6 \times 3.5 \times 3.5 = 73.5 \text{ cm}^2$$

Sol.151.(c) Radius of cone = 35cm

Curved surface area of cone = $\pi r l$

$$= \frac{22}{7} \times 35 \times l = 4070$$

$$\text{Slant height } (l) = \frac{4070}{22 \times 5} = 37 \text{ cm}$$

$$\text{Height of cone } (h) = \sqrt{(l)^2 - (r)^2}$$

$$= \sqrt{(37)^2 - (35)^2} = \sqrt{(12)^2} = 12 \text{ cm}$$

So, volume of cone

$$= \frac{1}{3} \times \frac{22}{7} \times 35 \times 35 \times 12 = 15,400 \text{ cm}^3$$

Sol.152.(b) diagonal of cube = $\sqrt{3}a$

According to the question,

$$6\sqrt{3} = \sqrt{3}a \Rightarrow a = 6$$

Total surface area of cube = $6a^2$

$$= 6 \times (6)^2 \Rightarrow 6 \times 36 = 216 \text{ cm}^2$$

Sol.153.(a)

Volume of cuboid = Volume of cone

$$11 \times 8 \times 6 = \frac{1}{3} \times \frac{22}{7} \times r^2 \times 14$$

$$r^2 = 36 \Rightarrow r = 6 \text{ cm}$$

Sol.154.(b)

Let length, breadth and height of cuboid be $6x$, $4x$ and $3x$ respectively

Total surface area of the cuboid

$$= 2(lb + bh + hl)$$

$$2(6x \times 4x + 4x \times 3x + 3x \times 6x) = 972$$

$$108x^2 = 972, x = 3$$

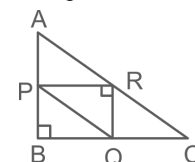
then, length, breadth and height = 18 cm, 12 cm and 9 cm

Diagonal of the cuboid

$$= \sqrt{(18)^2 + (12)^2 + (9)^2}$$

$$= \sqrt{549} = 3\sqrt{61} \text{ cm}$$

Sol.155.(a) (5, 12, 13) is a triplate so the triangle will be a right angle triangle.



triangle made by the mid point of the sides ABC triangle will be also right angle triangle

And the area of ΔPRQ will be $\frac{1}{4}$ of the area of ΔABC .

$$\text{So the area of } \Delta PRQ = \frac{1}{4} \times \frac{1}{2} \times 5 \times 12 = 7\frac{1}{2} \text{ cm}^2$$

Sol.156.(d) C.S.A of cone = πrl

T.S.A of cone = $\pi r(l + r)$

slant height (l) = $\sqrt{h^2 + r^2}$

$$l = \sqrt{20^2 + 15^2} = 25 \text{ cm}$$

C.S.A of cone = $\pi \times 15 \times 25 = 375\pi \text{ cm}^2$

T.S.A of cone = $\pi \times 15(25 + 15) = 600\pi \text{ cm}^2$

Sol.157.(b) circumference of base of cone ($2\pi r$) = $42\pi \Rightarrow r = 21$

Volume of the cone = $\frac{1}{3} \times \pi r^2 h$

$$= \frac{1}{3} \times \frac{22}{7} \times 21 \times 21 \times 24$$

$$= 22 \times 21 \times 24 \Rightarrow 11088 \text{ cm}^3$$

Sol.158.(c) Surface area of spherical

water tank = $4\pi r^2$

$$= 4 \times \frac{22}{7} \times 3.5 \times 3.5 \Rightarrow 88 \times 5 \times 3.5$$

$$= 44 \times 3.5 = 154 \text{ m}^2$$

Required cost = $154 \times 13 = 2002 \text{ Rs.}$

Sol.159.(a)

Pythagoras triplets :- (20, 21, 29)

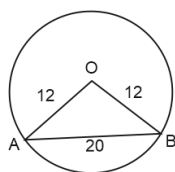
Radius of cone = 20 cm

Volume of cone = $\frac{1}{3} \times \pi r^2 h$

$$= \frac{1}{3} \times \frac{22}{7} \times 20 \times 20 \times 21$$

$$= 400 \times 22 = 8800 \text{ cm}^3$$

Sol.160.(c)



$$\text{Area} = \frac{b}{4} \times \sqrt{4a^2 - b^2}$$

$$= \frac{20}{4} \times \sqrt{4 \times 12^2 - 20^2}$$

$$= 5 \times \sqrt{4 \times 144 - 400}$$

$$= 5 \times \sqrt{176} = 20\sqrt{11} \text{ cm}^2$$

Sol.161.(c)

$$\text{Area} = \frac{\sqrt{3}}{4} \times 22 \times 22 \Rightarrow 121\sqrt{3} \text{ cm}^2$$

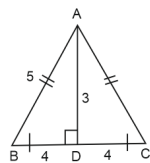
Sol.162.(d)

Perimeter of parallelogram = $2(l + b)$

$$= 48 \text{ cm} \Rightarrow b = 24 - 8 = 16 \text{ cm}$$

$$\text{Area} = \text{base} \times \text{height} = 16 \times 6 = 96 \text{ cm}^2$$

Sol.163.(a)



Pythagoras triplets : (3, 4, 5)

AD = 3 cm

$$\text{Area of triangle} = \frac{1}{2} \times 8 \times 3 = 12 \text{ cm}^2$$

Sol.164.(a) Radius of sphere = 3.5 cm

$$\text{Volume} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times 3.5 \times 3.5 \times 3.5$$

$$= \frac{4}{3} \times 22 \times 0.5 \times 3.5 \times 3.5$$

$$= \frac{44}{3} \times 12.25 = 179.67 \text{ cm}^3$$

Sol.165.(a)

Required area = $2(l + b) \times h + lb$

$$= 2(7 + 5) \times 3 + 7 \times 5 = 107 \text{ m}^2$$

$$\text{Total cost} = 107 \times 16 = ₹1,712$$

Sol.166.(c)

$$\text{Required difference} = (6a^2 - 4a^2) = 2a^2$$

$$= 2 \times 12 \times 12 = 288 \text{ cm}^2$$

Sol.167.(d) 0.2 cm refers to 1 km

1cm refers to 5 km

1cm² refers to 25 km²

$$\text{Now, } 22500 \text{ km}^2 = \frac{1}{25} \times 22500 = 900 \text{ cm}^2$$

Sol.168.(d)

$$\text{Total volume} = (15)^3 + (18)^3 + (25)^3$$

Side of newly formed cube

$$= \sqrt[3]{(15)^3 + (18)^3 + (25)^3}$$

$$= \sqrt[3]{3375 + 5832 + 15625}$$

$$= \sqrt[3]{24832} = \sqrt[3]{64 \times 388} = 4\sqrt[3]{388}$$

Sol.169.(d)

$$\frac{4}{3} \times \pi \times (9)^3 = x \times \frac{4}{3} \times \pi \times (3)^3$$

$$\Rightarrow x = 27 \text{ laddoos}$$

Sol.170.(d) 1m³ = 1000 litre

Volume of hemisphere

$$= \frac{2}{3} \times \frac{22}{7} \times 2.1 \times 2.1 \times 2.1$$

$$= 19.404 \text{ m}^3 = 19404 \text{ litre}$$

Sol.171.(a)

$$\text{Side of square} = \frac{172}{4} = 43 \text{ cm}$$

$$\text{Area of triangle} = \frac{43 \times 43}{2} = 924.5 \text{ cm}^2$$

Sol.172.(c) Required area of sector

$$= \frac{22.5^\circ}{360^\circ} \times \frac{22}{7} \times 21 \times 21$$

$$= \frac{1}{16} \times 66 \times 21 = 86.625 \text{ cm}^2$$

Sol.173.(c) Volume of cuboid = Volume of solid sphere

$$88 \times 21 \times 21 = \frac{4}{3} \times \frac{22}{7} \times r^3$$

$$r^3 = 21 \times 21 \times 21 \Rightarrow r = 21 \text{ cm}$$

Sol.174.(c)

Volume of sphere = $n \times$ Volume of cylinder

$$\frac{4}{3} \times \pi \times 9 \times 9 \times 9 = n \times \pi \times 3^2 \times 2$$

$$n = 2 \times 9 \times 3 = 54$$

Sol.175.(d) 1m³ = 1000 litre

Volume of cuboid = $12 \times 9 \times 7.5$

$$= 810 \text{ m}^3 = 810000 \text{ litre}$$

Sol.176.(c)

Let the no. of solid sphere made = x

$$x = \frac{\frac{1}{3} \times \pi \times 12 \times 12 \times 24}{\frac{4}{3} \times \pi \times 2 \times 2 \times 2}$$

$$= \frac{12 \times 12 \times 3}{4} = 108$$

Sol.177.(a)

Third side = $20 - (5 + 6) = 9 \text{ meter}$

Semiperimeter = 10

$$\text{Area} = \sqrt{10 \times (10 - 5)(10 - 6)(10 - 9)}$$

$$= \sqrt{10 \times (5) \times (4) \times (1)} = 10\sqrt{2} \text{ m}^2$$

Sol.178.(c) Required no. of cube

$$= \frac{20 \times 16 \times 8}{4 \times 4 \times 4} = 5 \times 4 \times 2 = 40 \text{ cubes}$$

Sol.179.(b) Height of the box

$$= \frac{1875}{25 \times 15} = \frac{75}{15} = 5 \text{ cm}$$

Sol.180.(a) Volume of sphere = $\frac{4}{3} \times \pi r^3$

$$= \frac{4}{3} \times \frac{22}{7} \times 7 \times 7 \times 7$$

$$= \frac{88 \times 49}{3} = \frac{4312}{3} = 1437\frac{1}{3} \text{ cm}^3$$

Sol.181.(d) Length of arc = $\frac{\theta}{360^\circ} \times 2\pi r$

$$= 20.6 - 2 \times 7.4 = 5.8 \text{ cm}$$

$$\text{Area of sector} = \frac{\theta}{360^\circ} \times \pi r^2$$

$$= \left(\frac{\theta}{360^\circ} \times 2\pi r\right) \times \frac{r}{2} = 5.8 \times \frac{7.4}{2}$$

$$= 21.46 \text{ cm}^2$$

Sol.182.(b) Total surface area of solid

hemisphere = $3\pi r^2$

$$= 3 \times \frac{22}{7} \times 7 \times 7 = 66 \times 7 = 462 \text{ cm}^2$$

Sol.183.(b)

$$\text{Length of arc} = 15.5 - (2 \times 5) = 5.5 \text{ cm}$$

$$\text{Area of sector} = \text{length of arc} \times \frac{\text{radius}}{2}$$

$$= 5.5 \times \frac{5}{2} = 13.75 \text{ cm}^2$$

Sol.184.(a)

$$\text{Side of the square} = \sqrt{30.25} = 5.5 \text{ cm}$$

$$\text{Circumference of circle} = 4 \times 5.5 = 22 \text{ cm}$$

$$\text{Radius of circle} = 22 \times \frac{7}{2 \times 22} = 3.5 \text{ cm}$$

$$\text{Area of circle} = \frac{22}{7} \times 3.5 \times 3.5 = 38.5 \text{ cm}^2$$

Sol.185.(d) Required number of cones

$$= \frac{\text{volume of sphere}}{\text{volume of cone}}$$

$$= \frac{\frac{4}{3} \pi \times 21^3}{\frac{1}{3} \pi \times 7 \times 7 \times 3} = 252$$

Sol.186.(a) T.S.A of cone = $\pi r(l + r)$

$$= \frac{22}{7} \times 21(28 + 21) = 3234 \text{ cm}^2$$

Sol.187.(c) 8, 15, 17 is a triplate

So the triangle will be a right angle triangle.

$$\text{Area of triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times 8 \times 15 = 60 \text{ cm}^2$$

Sol.188.(a) Total surface area of cylinder

$$= 2\pi r(r + h)$$

$$= 2 \times \frac{22}{7} \times 6 \times (6 + 8) = 22 \times 24$$

$$= 528 \text{ cm}^2$$

Sol.189.(a)

$$\text{Surface area of the sphere} = 4\pi r^2$$

$$= 4 \times \frac{22}{7} \times 3.5 \times 3.5 = 154 \text{ cm}^2$$

Sol.190.(b) According to the question,

$$\text{C.S.A of the cone} = \pi r l$$

$$990 = \frac{22}{7} \times 21 \times l$$

$$(\text{Slant height}) l = 15 \text{ cm}$$

Sol.191.(a) Given:- $r = 0.42 \text{ m}$, $h = 5 \text{ m}$

$$\text{Curved surface area of the pillar} = 2\pi r h$$

$$= 2 \times \frac{22}{7} \times 0.42 \times 5$$

$$\text{Cost of painting the curved surface area}$$

$$\text{of the pillar} = 2 \times \frac{22}{7} \times 0.42 \times 5 \times 15$$

$$= ₹198$$

Sol.192.(a) Inner radius of hemispherical

$$\text{bowl} = 4 \text{ cm}$$

$$\text{So, outer radius} = (4 + 1) = 5 \text{ cm}$$

$$\text{Volume of hemispherical bowl}$$

$$= \frac{2}{3} \times \pi \times (5^3 - 4^3) = \frac{122}{3} \pi = 40\frac{2}{3} \pi$$

Sol.193.(a) Volume of cylinder = $\pi r^2 h$

According to the question,

$$\pi \times 5^2 \times 52 = \pi \times r^2 \times 100 \Rightarrow r^2 = 13$$

$$\text{Radius of the wire } (r) = \sqrt{13}$$

Sol.194.(d)

$$\text{Volume of sphere} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times 4.2 \times 4.2 \times 4.2$$

$$= 88 \times 0.2 \times 17.64 = 310.464 \text{ cm}^3$$

Sol.195.(a)

$$\text{Total surface area of cube } (6a^2) = 24$$

$$\Rightarrow \text{side of cube } (a) = 2$$

$$\text{Volume of cube } (a^3) = 8 \text{ cu.units}$$

Sol.196.(c)

$$\text{Volume of the cylindrical flask} = \pi r^2 h$$

$$= \frac{22}{7} \times 6 \times 6 \times 21 = 2376 \text{ cm}^3$$

Sol.197.(d)

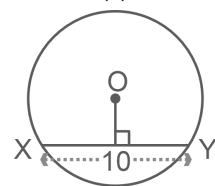
$$\text{Slant height of the cone} = \sqrt{r^2 + h^2}$$

$$\text{Height} = \text{radius} = r$$

$$5 = \sqrt{r^2 + r^2} = 2r^2 = 25 \Rightarrow r = \frac{5}{\sqrt{2}}$$

$$\text{Volume of the cone} = \frac{1}{3} \pi r^2 h$$

$$= \frac{\pi \times 25}{3 \times 2} \times \frac{5}{\sqrt{2}} = \frac{125\pi}{6\sqrt{2}} \text{ cm}^3$$

Sol.198.(a)

$$\text{Area of the circle} = \pi r^2$$

$$616 = \frac{22}{7} \times r^2 \Rightarrow r = 14 \text{ cm}$$

$$\text{Required distance} = \sqrt{14^2 - 5^2} = \sqrt{171} \text{ cm}$$

Sol.199.(b) Area of the triangles

$$= \frac{1}{2} \times \text{base} \times \text{height}$$

According to the question,

$$\frac{5}{3} = \frac{5 \times \text{base}_1}{7 \times \text{base}_2} \Rightarrow \text{Required ratio} = 7 : 3$$

Sol.200.(b)

$$\text{Area of equilateral triangle} = \frac{\sqrt{3}}{4} a^2$$

According to question,

$$\text{Net \% increase in area} = 34 + 34 + \frac{34 \times 34}{100}$$

$$= 79.56 \%$$

Sol.201.(a) Angle Subtended by arc

$$(\text{in radian}) = \frac{\text{Length of arc}}{\text{Radius}}$$

According to question,

$$\frac{\pi}{10} = \frac{23.1}{\text{Radius}}$$

$$\Rightarrow \text{radius} = \frac{231 \times 7}{22} = 73.5 \text{ cm}$$

$$\text{Required area of Circle} = \pi r^2$$

$$= \frac{22}{7} \times 73.5 \times 73.5 = 16978.5 \text{ cm}^2$$

Sol.202.(a)

$$\text{Perimeter of one face of cube} = 4a$$

According to the question,

$$4a = 40 \Rightarrow a = 10 \text{ cm}$$

$$\text{Volume of cube } (a^3) = 10^3 = 1000 \text{ cm}^3$$

Sol.203.(c)

$$\text{Area of the sector} = \frac{\theta}{360^\circ} \times \pi r^2$$

$$= \frac{60}{360} \times \frac{22}{7} \times 5 \times 5 = 13.1 \text{ cm}^2$$

$$\text{Sol.204.(a)} \text{ Area of sector} = \frac{\theta}{360} \times \pi r^2$$

$$= \frac{60}{360} \times 3.14 \times 10 \times 10 = 52.33 \text{ cm}^2$$

Sol.205.(d) According to the question,

$$\text{Volume of the ball} = \frac{4}{3} \pi r^3$$

$$1 \text{ big ball volume} = 3 \text{ small balls volume}$$

$$\frac{4}{3} \pi \times \left(\frac{3}{2}\right)^3 = \frac{4}{3} \pi \left\{ \left(\frac{3}{4}\right)^3 + (1)^3 + (r_3)^3 \right\}$$

$$\left(\frac{27}{8}\right) = \left\{ \frac{27}{64} + 1 + (r_3)^3 \right\}$$

$$(r_3)^3 = \frac{27}{8} - \frac{91}{64} = \frac{125}{64}$$

$$r_3 = \frac{5}{4} = 1.25$$

$$\text{Diameter of the 3rd ball} = 1.25 \times 2 = 2.5 \text{ cm}$$

Sol.206.(a)

$$\text{Area of the sector} = \left(\frac{\theta}{360}\right) \pi r^2$$

$$32\pi = \left(\frac{\theta}{360}\right) \pi \times 12 \times 12$$

$$\theta = \frac{32 \times 360}{12 \times 12} = 80^\circ$$

$$\text{Length of the arc} = 2\pi r \left(\frac{\theta}{360}\right)$$

$$= 2\pi \times 12 \left(\frac{80}{360}\right) = \frac{16}{3} \pi \text{ cm}$$

Sol.207.(a) Area of isosceles triangle

$$= \frac{b}{4} \times \sqrt{4a^2 - b^2}$$

where, b is an unequal side and a is equal sides.

$$30\sqrt{11} \text{ cm}^2 = \frac{5x}{4} \times \sqrt{4 \times 9x^2 - 25x^2}$$

$$30\sqrt{11} \text{ cm}^2 = \frac{5 \times x^2}{4} \times \sqrt{11}$$

$$30 \text{ cm}^2 = \frac{5 \times x^2}{4} \Rightarrow x = \sqrt{24} = 2\sqrt{6}$$

$$\text{Length of unequal side } (5x) = 10\sqrt{6} \text{ cm}$$

Sol.208.(a)

$$\text{Slant height of canvas } (l) = \sqrt{6^2 + 8^2} = 10$$

$$\text{Canvas needed } (\pi r l)$$

$$= \frac{22}{7} \times 6 \times 10 = 188.57 \text{ units}$$

Sol.209.(d) Area of Sector = $\frac{\theta}{360^\circ} \times \pi r^2$

According to question,

$$1848 = \frac{270^\circ}{360^\circ} \times \frac{22}{7} \times r^2$$

$$\text{Radius of circle}(r) = 28 \text{ m}$$

Sol.210.(b)

$$\text{Volume of conical tent} = \frac{1}{3} \pi r^2 h$$

$$\text{Volume} = \frac{1}{3} \times \frac{22}{7} \times 7 \times 7 \times 24$$

$$= 22 \times 7 \times 8 = 22 \times 56 = 1232 \text{ m}^3$$

Let, n number of full bags of rice emptied in space = $1232 \text{ m}^3 = n \times 2 \text{ m}^3$

$$n = \frac{1232}{2} = 616 \text{ bags}$$

Sol.211.(a) Given:- Diameter = 7cm

$$\text{So, radius} = \frac{7}{2} \text{ cm}$$

$$\pi^c = 180^\circ \Rightarrow 1^\circ = \left(\frac{22}{7 \times 180} \right)^\circ$$

$$\theta(\text{in radian}) = \frac{\text{length of arc}}{\text{radius of circle}}$$

$$108 \times \frac{22}{7 \times 180} = \frac{\text{length of arc} \times 2}{7}$$

$$\text{Length of arc} = \frac{108 \times 22 \times 7}{7 \times 180 \times 2}$$

$$= \frac{108 \times 11}{180} = \frac{12 \times 11}{20} = \frac{132}{20} = 6.6 \text{ cm}$$

Sol.212.(d) Lateral surface area of the cylinder = $2\pi rh$

According to the question,

$$2 \times \frac{22}{7} \times r \times 48 = 3168$$

$$r = \frac{3168 \times 7}{2 \times 22 \times 48} = \frac{21}{2} \text{ cm}$$

Now, volume of the cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2} \times 48 = 16632 \text{ cm}^3$$

Sol.213.(b)

Radius of lorry = 182cm = 1.82 metre

Speed of lorry = 66 km/h

$$= \frac{66 \times 1000}{60} = 1100 \text{ m/min}$$

No. of revolution per minute

$$= \frac{\text{distance covered in one minute}}{\text{circumference of wheel}}$$

$$= \frac{1100}{2 \times \frac{22}{7} \times 1.82} = \frac{1100 \times 7}{2 \times 22 \times 1.82}$$

$$= 96.15 \approx 96 \text{ revolution}$$

Sol.214.(d) Surface area of sphere = $4\pi r^2$

According to the question,

$$4\pi r^2 = 9\pi \Rightarrow r = \frac{3}{2}$$

diameter (2r) = 3 cm

Sol.215.(d) Surface area of Cube = $6a^2$

According to question,

$$5046 \text{ cm}^2 = 6a^2 \Rightarrow a = 29 \text{ cm}$$

$$\text{Volume of Cube } (a^3) = 24389 \text{ cm}^3$$

Sol.216.(b)

Curved surface area of cone = $\pi r l$

Area of base = πr^2

According to question, $\pi r l = 2 \times \pi r^2$

$$= \pi r \times 6\sqrt{3} = 2 \times \pi r^2 \Rightarrow r = 3\sqrt{3} \text{ cm}$$

By using pythagoras theorem,

$$\text{Height of cone} = \sqrt{(6\sqrt{3})^2 - (3\sqrt{3})^2} = 9 \text{ cm}$$

Sol.217.(b) Total surface area of cuboid

$$= 2(lb + bh + lh)$$

According to question,

Total surface area of cuboid

$$= 2[5 \times 7 + 7 \times 11 + 11 \times 5] = 334 \text{ cm}^2$$

Sol.218.(d) Total surface area of the cone = $\pi r(r + l)$

According to the question,

$$\pi r(r + l) = \frac{22}{7} \times 6 \times (6 + 10) = 301.44 \text{ cm}^2$$

Sol.219.(b)

$$2(L + B) = 68 \Rightarrow (L + B) = 34 \text{e.q. (1)}$$

$$(L - B) = 2 \text{e.q. (2)}$$

On solving e.q. (1) and e.q. (2), we get the length of the rectangle (L) = 18

Sol.220.(a)

Initial Ratio = Length : Breadth : Height

$$= 1 : 2 : 3$$

Initial volume = $1 \times 2 \times 3 = 6 \text{ unit}^3$

Final Ratio = Length : Breadth : Height

$$= 1 \times 300\% : 2 \times 400\% : 3 \times 400\%$$

$$= 3 : 8 : 12$$

Final volume = $3 \times 8 \times 12 = 288 \text{ unit}^3$

$$\text{Required increase in volume} = \frac{288 - 6}{6}$$

$$= 47 \text{ times}$$

Sol.221.(d) According to the question,

$$l \times b \times h = 2 \times a^3$$

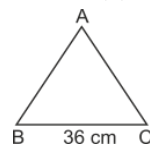
$$= 8 \times 8 \times 16 = 2 \times a^3$$

$$a^3 = 512 \Rightarrow a = 8 \text{ cm}$$

Now, total surface area of the cube = $6a^2$

$$= 6 \times 8 \times 8 = 384 \text{ m}^2$$

Sol.222.(d)



Now, semi-perimeter of the triangle

$$= \frac{100}{2} = 50 \text{ cm}$$

Sol.223.(a)

Equating the volume of sphere

$$= \frac{4}{3} \pi (R)^3 = \frac{4}{3} \pi \{(1)^3 + (6)^3 + (8)^3\}$$

$$(R)^3 = \{1 + 216 + 512\}$$

$$(R)^3 = \{729\} \Rightarrow R = 9 \text{ cm}$$

Sol.224.(b) Area of sector = $\frac{\theta}{360^\circ} \times \pi r^2$

$$112 = \frac{\theta}{360^\circ} \times \frac{22}{7} \times 28 \times 28$$

$$\theta = \frac{360}{22} \text{ (in degree)}$$

$$\theta = \frac{360}{22} \times \frac{22}{7 \times 180} \text{ (in radian)}$$

By Using formula :-

$$\theta(\text{in radian}) = \frac{\text{length of arc}}{\text{radius}}$$

$$\text{Length of arc} = \frac{360}{22} \times \frac{22}{7 \times 180} \times 28 = 8 \text{ cm}$$

Sol.225.(c)

$$\text{Volume of wall} = 14 \times 0.125 \times 5 = 8.75 \text{ m}^3$$

$$\text{Required no. of bricks} = \frac{8.75}{0.0014}$$

$$= 6250 \text{ bricks}$$

Sol.226.(d) Length of arc = $\frac{\theta}{360} \times 2\pi r$

$$19.25 = \frac{18}{360} \times 2 \times \frac{22}{7} \times r$$

$$19.25 = \frac{1}{10} \times \frac{22}{7} \times r$$

$$r = \frac{19.25 \times 10 \times 7}{22} = 61.25$$

hence, area of the circle = πr^2

$$= \frac{22}{7} \times 61.25 \times 61.25 = 11,790.625 \text{ cm}^2$$

Sol.227.(d)

Inner radius of the bowl = 5 cm

Outer radius of the bowl = 5.25 cm

Required Volume of the hemisphere

$$= \frac{2}{3} \times \pi (r_2^3 - r_1^3)$$

$$= \frac{2}{3} \times \frac{22}{7} \times \{(5.25)^3 - (5)^3\}$$

$$= \frac{2}{3} \times \frac{22}{7} \times 19.70 = 41.28 \text{ cm}^3$$

Sol.228.(a) Let the inner radius of pipe be 'r' and outer radius of pipe be 'R'.

$$\text{Volume of pipe} = \pi R^2 h \Rightarrow \frac{88}{7} = \frac{22}{7} \times r^2 \times 1$$

Inner radius = 2 m and outer radius = 4m

Therefore, Required quantity of material

$$= \pi R^2 h - \pi r^2 h$$

$$= \frac{22}{7} \times 4 \times 4 \times 1 - \frac{88}{7} = 37.7 \text{ m}^3$$

Sol.229.(c) Let the length and width of rectangle be 3x and 2x.

According to question,

$$3x \times 2x = 150 \Rightarrow x = 5$$

Require perimeter = 2(length + width)

$$= 2[(3 \times 5) + (2 \times 5)] = 50 \text{ cm}$$

Sol.230.(d) Circumradius of equilateral

$$\text{triangle} = \frac{a}{\sqrt{3}} = \frac{12}{\sqrt{3}} = 4\sqrt{3} \text{ cm}$$

Short trick :-

Ratio $\Rightarrow$ side : circumradius = $6 : 2\sqrt{3}$

Here,

$$6 \text{ units} = 12 \text{ cm} \Rightarrow 2\sqrt{3} \text{ units} = 4\sqrt{3} \text{ cm}$$

Sol.231.(a) Given, length = 10m,

breadth = 8m, height = 6m

Area of four walls = $2(l+b) \times h$

$$= 2(10+8) \times 6 = 216 \text{ m}^2$$

Area of the ceiling = length $\times$ breadth

$$= 10 \times 8 = 80 \text{ m}^2$$

Cost of white washing the walls and

$$\text{ceiling} = (216 + 80) \times 7.5 = \text{Rs. } 2,220$$

Sol.232.(d)

$$\text{Volume of the sphere} = \frac{4}{3} \times \pi r^3$$

According to the question,

Volume of the sphere

$$= \frac{4}{3} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times \frac{7}{2} = 179.67 \text{ cm}^3$$

Sol.233.(c)

$$\text{Area of equilateral triangle} = \frac{\sqrt{3}}{4} a^2$$

According to question,

$$\frac{\sqrt{3}}{4} a^2 = \frac{1.732}{4} \times 12 \times 12 = 62.35 \text{ cm}^2$$

Sol.234.(c) Volume of Sphere = $\frac{4}{3} \pi r^3$

Let inner radius be r and outer radius be R .

According to question,

$$\text{Required material} = \frac{4}{3} \pi R^3 - \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \pi (3^3 - 2^3) = \frac{76}{3} \pi$$

Sol.235.(c)

Equating the volumes of the cubes :-

$$12 \times 12 \times 12 = 6 \times 6 \times 6 + 8 \times 8 \times 8 + x^3$$

$$1728 = 216 + 512 + x^3$$

$$x^3 = 1728 - 728 = 1000 \Rightarrow x = 10 \text{ cm}$$

Sol.236.(c)

Pythagoras triplets :- 150, 80, 170

Length of its diagonal = 170 cm

$$\text{Perimeter} = 2(150 + 80) = 460$$

$$\text{Now, } \frac{460}{170} = 2.7 \text{ times}$$

Sol.237.(b)

$$\text{Required area} = 2(bh + hl) + lb$$

$$= 2(20 \times 15 + 15 \times 10) + 10 \times 20 = 1100 \text{ m}^2$$

So, cost of whitewashing the walls of the

inside of the hall and ceiling =

$$1100 \times 10.20 = \text{₹}11,220$$

Sol.238.(c) Given, radius of the circle = 8

cm, length of the arc = 12 cm

$$\text{Area of sector} = \frac{1}{2} lr$$

$$= \frac{1}{2} \times 12 \times 8 = 48 \text{ cm}^2$$

Sol.239.(a) Given, height of the cylinder

$$= 12 \text{ cm, C.S.A} = 2640$$

Curved surface area of cylinder = $2\pi rh$

$$= 2 \times \frac{22}{7} \times r \times 12 = 2640$$

$$r = \frac{2640 \times 7}{44 \times 12} \Rightarrow r = 35 \text{ cm}$$

Now, volume of the cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times 35 \times 35 \times 12 = 46200 \text{ cm}^3$$

Sol.240.(d)

Length of diagonal of cube = $\text{side}\sqrt{3}$

$$= 7\sqrt{3} \text{ cm} \Rightarrow \text{Side of cube} = 7 \text{ cm}$$

$$\text{Volume of cube (side}^3) = 7^3 = 343 \text{ cm}^3$$

Sol.241.(c) Total surface area of cuboid

= Area of square

$$2(lb + bh + hl) = \text{side}^2$$

$$2(4 \times 6 + 6 \times 8 + 8 \times 4) = \text{side}^2$$

$$2 \times 104 = \text{side}^2$$

$$\text{side} = \sqrt{208} = 14.42 \text{ units}$$

Sol.242.(c)

Curved surface area of the cone = πrl

$$= \frac{22}{7} \times 21 \times 60 = 3960 \text{ cm}^2$$

Sol.243.(c) Given, radius = 35 cm

Surface area of sphere = $4\pi r^2$

$$= 4 \times \frac{22}{7} \times 35 \times 35 = 15400 \text{ cm}^2$$

Sol.244.(d)

$$\text{Area of the sector} = \frac{\theta}{360} \times \pi r^2$$

$$= \frac{56}{360} \times \frac{22}{7} \times r^2 = 110$$

$$r^2 = 225 \Rightarrow r = 15$$

Hence, the radius of the circle = 15 cm

Sol.245.(a) $1 \text{ m}^3 = 1000 \text{ liters}$

Volume of the tank = lbh

$$= 50 \times 25 \times 10 = 12500 \text{ m}^3$$

the capacity of the petrol tank

$$= 12500 \times 1000 = 1,25,00,000 \text{ liters.}$$

Sol.246.(c)

$$\text{Area of equilateral triangle} = \frac{\sqrt{3}}{4} a^2$$

According to question,

$$\frac{\sqrt{3}}{4} a^2 = \frac{\sqrt{3}}{4} \times 16 \times 16 = 64\sqrt{3} \text{ cm}^2$$

Sol.247.(b)

Total surface area of cube = $6a^2$

According to question,

$$6a^2 = 6 \times 12 \times 12 = 864 \text{ cm}^2$$

Sol.248.(c) Volume of cube = a^3

According to question,

$$\text{Side of cube (a)} = \sqrt[3]{4096} = 16 \text{ cm}$$

Sol.249.(a) $l = r\theta$

$$5r = r\theta \Rightarrow \theta = 5^\circ = 5 \times \frac{180^\circ}{\pi}$$

$$\text{Area of sector AOB} = \frac{\theta}{360^\circ} \times \pi r^2$$

$$= \frac{5 \times 180^\circ}{360^\circ \times \pi} \times \pi \times r^2 = 2.5r^2$$

Sol.250.(a) CSA of hemispherical bowl

inside as well as outside = $4\pi r^2$

$$= 4 \times \frac{22}{7} \times 7^2 = 616 \text{ cm}^2$$

Cost of painting the hemispherical bowl

$$= 616 \times \frac{40}{20} = \text{₹}1232$$

Sol.251.(a) Area of the floor = $L \times B$

Area of the flat roof = $L \times B$

Areas of four walls = $2(L + B)H$

ATQ, $2LB = 2(L + B)H$

$$21 \times 14 = (21 + 14)H = 35H$$

$$H = \frac{21 \times 14}{35} = \frac{42}{5} = 8.4 \text{ m}$$

Sol.252.(a)

$$\text{Perimeter of arc} = \frac{\theta}{360^\circ} \times 2\pi r + 2r$$

$$= \frac{45}{360^\circ} \times 8 \times \frac{22}{7} + 8$$

$$= \frac{22 + 56}{7} = \frac{78}{7} \text{ cm}$$

Sol.253.(c) CSA of cylinder = $2\pi rh$

$$1408 = 2 \times \frac{22}{7} \times r \times 56 \Rightarrow 1408 = 352 \times r$$

$$r = \frac{1408}{352} = 4 \text{ cm or } 0.04 \text{ m}$$

$$\text{Diameter of the base of cylinder} = 0.04 \times 2 = 0.08 \text{ m}$$

Sol.254.(c) Volume of cube = $(\text{Side})^3$

$$\text{Side} = \sqrt[3]{12167} = 23 \text{ cm}$$

Surface area of cube = $6 \times (\text{Side})^2$

$$= 6 \times 529 = 3174 \text{ cm}^2$$

Sol.255.(a) Diagonal of cube = $\text{side}\sqrt{3}$

$$10 = \text{side}\sqrt{3} \Rightarrow \text{Side} = \frac{10}{\sqrt{3}} = 5.77 \text{ m}$$

Sol.256.(c) Volume of cylinder = area of base $\times$ height

$$\text{Height} = \frac{1650 \times 7}{550} = 21 \text{ m}$$

Sol.257.(b)

$$\text{Volume of hemisphere} = \frac{2}{3} \pi r^3$$

Let the radius of the hemisphere be R and radius of sphere be r .

According to question,

$$\frac{\frac{2}{3} \pi R^3}{\frac{4}{3} \pi r^3} = 4 \Rightarrow \frac{7 \times 7 \times 7}{2r^3} = 4$$

$$r^3 = \frac{7 \times 7 \times 7}{8}$$

$$r \text{ (radius of sphere)} = 3.5 \text{ cm}$$

Sol.258.(d)

$$\begin{aligned}\text{Volume of hemisphere} &= \frac{2}{3}\pi r^3 \\ &= \frac{2}{3} \times \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2} \times \frac{21}{2} \\ &= \frac{22 \times 21 \times 21}{4} = 2425.5 \text{ cm}^3\end{aligned}$$

$$\text{Sol.259.(b)} \quad S = \frac{a+b+c}{2}$$

Area of triangle

$$= \sqrt{S(S-a)(S-b)(S-c)}$$

$$\text{According to question, } S = \frac{3+5+6}{2} = 7$$

Area of triangle

$$\begin{aligned}&= \sqrt{7(7-3)(7-5)(7-6)} \\ &= \sqrt{7 \times 4 \times 2 \times 1} = \sqrt{56} = 2\sqrt{14} \text{ cm}^2\end{aligned}$$

Sol.260.(c)

$$\text{Diagonal of Cuboid} = \sqrt{l^2 + b^2 + h^2}$$

According to question,

$$\begin{aligned}\text{Diagonal of Cuboid} &= \sqrt{9^2 + 12^2 + 20^2} \\ &= \sqrt{625} = 25 \text{ cm}\end{aligned}$$

Sol.261.(d)

$$\text{Diagonal of cuboid} = \sqrt{l^2 + b^2 + h^2}$$

$$= \sqrt{144 + 100 + 64} = \sqrt{308} = 17.54 \text{ m}$$

Sol.262.(c) Equating the volumes :-

Volume of hollow hemisphere = Volume of cylinder

$$\frac{2}{3}\pi(R^3 - r^3) = \pi \times 7 \times 7 \times H$$

$$\frac{2}{3}(5^3 - 3^3) = 7 \times 7 \times H$$

$$H = \frac{2 \times (125 - 27)}{7 \times 7 \times 3} = \frac{4}{3} = 1.33 \text{ cm}$$

Sol.263.(a) Curved surface area of hemisphere = $2\pi r^2$

$$693 \text{ cm}^2 = 2 \times \frac{22}{7} \times r^2$$

$$r = \sqrt{\frac{693 \times 7}{44}} = \sqrt{\frac{63 \times 7}{4}}$$

$$= \frac{21}{2} = 10.5 \text{ cm}$$

Sol.264.(a)

$$\text{Area of triangle} = \frac{1}{2} \times \text{base} \times \text{Height}$$

As we know, the base and height of an isosceles right angle triangle are equal.

According to question,

$$200 = \frac{1}{2} \times a \times a \Rightarrow a = 20 \text{ cm}$$

By using pythagoras theorem,

$$\text{Hypotenuse} = 20\sqrt{2} \text{ cm}$$

$$\text{Perimeter of isosceles right-angle triangle} = 20\sqrt{2} + 40 = 68.3 \text{ cm}$$

Sol.265.(b) Surface area of sphere = $4\pi r^2$

According to question,

$$616 = 4 \times \frac{22}{7} \times r^2 \Rightarrow r^2 = 49 \Rightarrow r = 7 \text{ cm}$$

Now, total surface area of hemisphere

$$= 3\pi r^2 = 3 \times \frac{22}{7} \times 7 \times 7 = 462 \text{ cm}^2$$

Sol.266.(d)

Volume of sphere = volume of cone

$$\frac{4}{3}\pi r^3 = \frac{1}{3}\pi R^2 h$$

$$4 \times 21 \times 21 \times 21 = R^2 \times 84 \Rightarrow R = 21 \text{ cm}$$

Sol.267.(b)

$$\frac{\pi r^2 h}{2\pi r h} = \frac{1848}{528} \Rightarrow r = \frac{2 \times 7}{2} = 7 \text{ m}$$

$$\text{Now, } 2\pi r h = 528 \text{ m}^2$$

$$h = \frac{528 \times 7}{2 \times 22 \times 7} = 12 \text{ m}$$

Required ratio = 7 : 12

Sol.268.(b)

Pythagoras Triplets :- (20, 21, 29)

So, here slant height (l) = 29 cm

Total surface area of cone = $\pi r(l + r)$

$$= \pi \times 20(29 + 20) = 980\pi \text{ cm}^2$$

Sol.269.(a) Total surface area of Cuboid

$$= 2(lb + bh + lh)$$

$$= 2\{(48 \times 27) + (27 \times 30) + (30 \times 48)\}$$

$$= 2\{1296 + 810 + 1440\} = 7092 \text{ sq. inch}$$

$$= \frac{7092}{12 \times 12} = 49.25 \text{ sq. foot}$$

Sol.270.(c)Let, the breadth of rectangle be x metre, then length be $3x$ metre.

$$\text{Area} = x \times 3x = 3x^2$$

$$3x^2 = 2700 \text{ m}^2 \Rightarrow \text{breadth } (x) = 30 \text{ metre}$$

Sol.271.(d) Surface area of sphere = $4\pi r^2$

According to the question,

$$\frac{4\pi r_1^2}{4\pi r_2^2} = \frac{81}{25} \Rightarrow r_1 : r_2 = 9 : 5$$

Sol.272.(a) Area of square ($side^2$) = 5929

$$\text{Side} = 77$$

Perimeter of square = perimeter of circle

$$4 \times 77 = 2\pi r \Rightarrow 2 \times \frac{22}{7} \times r = 308 \Rightarrow r = 49$$

$$\text{Area of the circle} = (\pi r^2)$$

$$= \frac{22}{7} \times 49 \times 49 = 7546 \text{ cm}^2$$

Sol.273.(d)Perimeter of the semi-circle = $\pi r + 2r$

$$3.6 = r\left(\frac{22}{7} + 2\right)$$

$$3.6 = r\left(\frac{36}{7}\right) \Rightarrow r = \frac{7}{10}$$

$$\text{Area of semi-circle} = \frac{1}{2}\pi r^2$$

$$\frac{1}{2} \times \frac{22}{7} \times \frac{7}{10} \times \frac{7}{10} = 0.77 \text{ m}^2$$

Sol.274.(b) Length of longest diagonal in

$$\begin{aligned}\text{the cuboid} &= \sqrt{(13)^2 + (11)^2 + (20)^2} \\ &= \sqrt{690} = 26.27 \text{ cm}\end{aligned}$$

Sol.275.(d) Let, inner radius and outer radius be r_2 and r_1 respectively.

According to the question,

$$\frac{2\pi r_1}{2\pi r_2} = \frac{17}{15} \Rightarrow \frac{r_1}{r_2} = \frac{17}{15} \Rightarrow r_1 = \frac{17}{15} r_2$$

$$\text{Now, } r_1 - r_2 = 40 \Rightarrow \frac{17}{15} r_2 - r_2 = 40$$

$$r_2 = 300 \text{ and } r_1 = 340$$

Radius of inner circle (r_2) = 300 m**Sol.276.(d)** Height of cylinder = 3.5 cm

Radius = 3 × height = 10.5 cm

Volume of cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times 10.5 \times 10.5 \times 3.5 = 1212.75 \text{ cm}^3$$

Sol.277.(c) Area of the square = ($side^2$)

$$529 = (side^2) \Rightarrow \text{Side} = 23$$

$$\text{Diagonal of the square} = side\sqrt{2} = 23\sqrt{2} \text{ cm}$$

Sol.278.(b) Area of the trapezium

$$= \frac{1}{2} \times \text{sum of parallel side} \times \text{height}$$

$$= \frac{1}{2} \times (15 + 20) \times 22 = 385 \text{ metre}^2$$

Sol.279.(b) Volume of hemisphere = $\frac{2}{3}\pi r^3$

$$= \frac{2}{3} \times \pi \times r^3 = 486\pi \Rightarrow r = 9 \text{ cm}$$

Sol.280.(d) Area of trapezium

$$= \frac{1}{2} \times \text{sum of parallel sides} \times \text{height}$$

$$= \frac{1}{2} \times (6.32 + 7.68) \times 5.50$$

$$= \frac{1}{2} \times (14) \times 5.50 = 38.5$$

$$\text{Required amount} = 38.5 \times 1200 = ₹ 46200$$

Sol.281.(b) Let length of cuboid be l , breadth be b and height be h

According to the question,

Breadth of cuboid = 3cm

$$l = 6b = 6 \times 3 = 18 \text{ cm}$$

$$\text{And, } h = 4b = 4 \times 3 = 12 \text{ cm}$$

Hence,

total surface area of cuboid

$$= 2(lb + bh + hl)$$

$$= 2(18 \times 3 + 3 \times 12 + 12 \times 18) = 612 \text{ cm}^2$$

Sol.282.(b) Ratio of the radius of cylinder and cone = 3 : 1

$$\text{Volume of cylinder} = \pi r^2 h$$

$$\text{Volume of cone} = \frac{1}{3}\pi r^2 h$$

According to the question,

$$\pi(3r)^2 h : \frac{1}{3}\pi(r)^2 h = 9 : \frac{1}{3} = 27 : 1$$

Sol.283.(b) Base of triangle = 34cm

Height of triangle = 22cm

$$\text{Area of triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$\text{Area of triangle} = \frac{1}{2} \times 34 \times 22 = 374 \text{ cm}^2$$

Sol.284.(c) Area of scalene triangle

$$= \sqrt{s(s-a)(s-b)(s-c)}$$

$$\text{Semiperimeter } (s) = \frac{11+11+4}{2} = 13$$

$$\sqrt{13(13-11)(13-11)(13-4)}$$

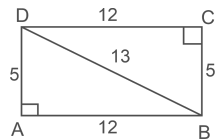
$$= \sqrt{13 \times 2 \times 2 \times 9} = 6\sqrt{13} \text{ unit}$$

According to the question,

$$6\sqrt{13} \text{ unit} = 2\sqrt{117} \text{ cm}^2 \Rightarrow 1 \text{ unit} = 1$$

Required side = 11 cm

Sol.285.(a) Triplate = 5, 12, 13



So, the area of the parallelogram

$$= 2 \left(\frac{1}{2} \times 12 \times 5 \right) = 60 \text{ cm}^2$$

Sol.286.(c)

Diameter = 12mm, Radius = 6mm

Area of the circle = πr^2

$$= \pi(6)^2 = 36\pi \text{ mm}^2$$

Sol.287.(a) Formula used :- Total surface area of cylinder(T.S.A) = $2\pi r(r+h)$

$$\text{T.S.A of cylinder} = 2 \times \frac{22}{7} \times 21(21+16)$$

$$= 4884 \text{ cm}^2 \text{ After bisecting the cylinder from its height}$$

T.S.A of both cylinder

$$= 2 \times \left[2 \times \frac{22}{7} \times 21(21+8) \right] = 7656 \text{ cm}^2$$

Increased in T.S.A of cylinder

$$= 7656 - 4884 = 2772 \text{ cm}^2$$

$$\text{Required \%} = \frac{2772}{4884} \times 100 = 56.76$$

Sol.288.(c) Formula used:-

curved surface area of cylinder = $2\pi rh$

According to the question,

$$\text{Curved surface area} = 2 \times 3.14 \times 20 \times 34 = 4270.4 \text{ cm}^2$$

Sol.289.(a) Volume of cuboid = $l \times b \times h$

Volume of cube = side^3

Volume of cuboid = volume of cube

$$8 \times 27 \times 64 = \text{side}^3$$

$$\text{Side} = 2 \times 3 \times 4 = 24 \text{ cm}$$

$$\text{TSA of cuboid} = 2(lb + bh + hl)$$

$$\text{TSA of cube} = 6\text{side}^2$$

Required ratio

$$= (8 \times 27 + 27 \times 64 + 64 \times 8) : 6 \times 24^2$$

$$= (216 + 1728 + 512) : 6 \times 576$$

$$= 2 \times 2456 : 6 \times 576 = 307 : 216$$

Sol.290.(a) Circumference of circle = $2\pi r$

$$2 \times \frac{22}{7} \times r = 44 \times 4 \Rightarrow r = 28 \text{ cm}$$

Sol.291.(b)

$$\text{Volume of hemisphere} = \frac{2}{3} \pi r^3$$

$$= \frac{2}{3} \times \frac{22}{7} \times 4.2 \times 4.2 \times 4.2 = 155.232 \text{ m}^3$$

Now,

$$\frac{1}{21} \text{th of volume} = \frac{155.232}{21} = 7.3 \text{ m}^3$$

Sol.292.(b) After joining two cubes each of edge 9 cm to form cuboid

Length of cuboid(l) = 18cm,

width (b) = 9cm and height (h) = 9 cm

Then, Total surface area of cuboid

$$= 2(lb + bh + hl)$$

$$= 2(18 \times 9 + 9 \times 9 + 9 \times 18) = 810 \text{ cm}^2$$

Sol.293.(b)

$$\text{Area of triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times 120 \times 35 = 2100 \text{ cm}^2$$

Sol.294.(b) CSA of cone = $\pi r l$

$$\text{Slant height of the cone } (l) = \sqrt{r^2 + h^2}$$

$$l = \sqrt{8^2 + 15^2} = \sqrt{64 + 225} = 17$$

$$\text{Required cost} = \frac{\pi \times 8 \times 17 \times 54}{\pi} = ₹ 7344$$

Sol.295.(a) For largest possible sphere, diameter = side of cube

$$\text{Radius of sphere} = \frac{6}{2} = 3 \text{ cm}$$

$$\text{Volume of the sphere} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times 3 \times 3 \times 3 = \frac{792}{7} \text{ cm}^3$$

Sol.296.(c)

Formula used :- area of rhombus

$$= \frac{1}{2} \times \text{first diagonal} \times \text{second diagonal}$$

According to the question,

$$\text{Area of rhombus} = \frac{1}{2} \times 16 \times 20 = 160 \text{ cm}^2$$

Sol.297.(b) Volume of cube = side^3

Volume of 2 small cube = volume of 1 big cube

$$7^3 + 11^3 = \text{volume of 1 big cube}$$

$$\text{volume of 1 big cube} = 343 + 1331$$

$$= 1674 \text{ cm}^3$$

Sol.298.(b)

Area of rectangle = length $\times$ breadth

$$= 45 \times 55 = 2475 \text{ cm}^2$$

Sol.299.(a)

$$\text{Area of rectangle} = 58 \times 30 = 1740 \text{ m}^2$$

$$\text{Required cost} = 1740 \times 20 = ₹ 34800$$

Sol.300.(d)

$$\text{Volume of cube } (a)^3 = 5832 \text{ (cm)}^3$$

$$\text{Side}(a) = 18 \text{ cm}$$

$$\text{Lateral surface area} = 4(a)^2$$

$$= 4 \times 18 \times 18 = 1296 \text{ cm}^2$$

Sol.301.(c) According to the question,

Length of cuboid (l) = 23cm, breadth(b)

= 18cm and height(h) = 35 cm

Total surface area of cuboid

$$= 2(lb + bh + hl)$$

$$= 2(23 \times 18 + 18 \times 35 + 35 \times 23)$$

$$= 2(1849) = 3698 \text{ cm}^2$$

Sol.302.(c) Volume of the cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times \frac{126}{10} \times \frac{126}{10} \times 400 \text{ cm}^3$$

$$= 199584 \text{ cm}^3 \Rightarrow 1 \text{ cm}^3 = 2 \text{ grams}$$

$$199584 \text{ cm}^3 = \frac{2}{1} \times 199584 \times \frac{1}{1000}$$

$$= 399.168 \text{ kg}$$

Sol.303.(b) Here, (20, 48, 52) is the pythagorean triplet

$$\text{Hence, Area of triangle} = \frac{1}{2} \times 20 \times 48$$

$$= 480 \text{ cm}^2$$

Sol.304.(a) Area of kite

$$= \frac{1}{2} \times \text{first diagonal} \times \text{second diagonal}$$

$$= \frac{1}{2} \times 22 \times 6 = 66 \text{ cm}^2$$

Sol.305.(c) According to the question,

$$\text{Base} = 56 \times \frac{1}{4} = 14 \text{ cm}$$

$$\text{Area of the triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times 14 \times 56 = 392 \text{ cm}^2$$

Sol.306.(c)

Length with the path = $150 + 20 = 170 \text{ m}$

Breadth with the path = $25 + 20 = 45 \text{ m}$

Area of path = $(170 \times 45) - (150 \times 25)$

$$= 3900 \text{ metre}^2$$

Sol.307.(d)

Outer radius of spherical shell (R) = 13cm

Inner radius (r) = $13 - 3 = 10 \text{ cm}$

$$\text{Volume of the spherical sell} = \frac{4}{3} \pi (R^3 - r^3)$$

$$\text{Required volume} = \frac{4}{3} \times \frac{22}{7} \times (13^3 - 10^3)$$

$$= \frac{4}{3} \times \frac{22}{7} \times 1197 = 5016 \text{ cm}^3$$

Sol.308.(b)

$$\text{Diagonal of square } (\sqrt{2} a) = 5\sqrt{2}$$

$$\text{Side}(a) = 5 \text{ cm}$$

$$\text{Area of square } (a)^2 = 5 \times 5 = 25 \text{ cm}^2$$

Sol.309.(c) According to question,

$$\text{Volume of sphere} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times (65)^3 = 1150809.52$$

$$= 1.15 \times 10^6 \text{ cm}^3$$

Sol.310.(c)

Diameter = 12 cm So, radius = 6 cm

According to question,

$$\text{Volume of Cone} = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \times \frac{22}{7} \times 6 \times 6 \times 21.7 = 818.4 \text{ cm}^3$$

Sol.311.(c)

Perimeter of the picture = $2(l + b)$

$$= 2(160 + 40) = 400 \text{ cm}$$

Ratio of width and perimeter

$$= 40 : 400 = 1 : 10$$

Sol.312.(a)

$$\frac{\text{C.S.A of cone}}{\text{base area of cone}} = \frac{\pi r l}{\pi r^2} = \frac{l}{r}$$

According to the question,

$$\frac{l}{r} = \frac{3}{1}, \frac{l}{\frac{1}{2}} = \frac{3}{1} \Rightarrow l = \frac{3}{2}$$

T.S.A of cone = $\pi r(l + r)$

$$= \pi \times \frac{1}{2} \left(\frac{3}{2} + \frac{1}{2} \right) = \pi \text{ cm}^2$$

Sol.313.(a) According to the question,

circumference of a circle = $2\pi r$

$$= 2 \times \frac{22}{7} \times r = 220 \Rightarrow r = 35 \text{ cm}$$

Area of the circle = πr^2

$$= \frac{22}{7} \times 35 \times 35 = 3850 \text{ cm}^2$$

Sol.314.(c) Let radius of sphere = R

According to the question, $R = \frac{3}{4} r$

Ratio of volumes

$$= \frac{4}{3} \times \pi \times \left(\frac{3}{4} r \right)^3 : \frac{2}{3} \times \pi \times (r)^3$$

$$\Rightarrow 2 \times \frac{27}{64} : 1 = 27 : 32$$

Sol.315.(b) According to the question,

$$720 \times h = 28800$$

$$\text{Depth of the water } (h) = \frac{28800}{720}$$

= 40 meters.

Sol.316.(c) Circumference of wheel = $2\pi r$

$$= 2 \times \frac{22}{7} \times 21 = 132 \text{ cm}$$

According to question,

$$\text{Required Distance} = 132 \times 20 = 2640 \text{ cm}$$

$$\text{Sol.317.(a)} \text{ Volume of cone} = \frac{1}{3} \pi r^2 h$$

According to question,

$$\text{Volume of cone} = \frac{22}{7} \times 14 \times 14 \times \frac{12}{3}$$

$$= 2464 \text{ cm}^3$$

Sol.318.(c)

From the given sides of triangle

$$= (6, 8, 10)$$

$$6^2 + 8^2 = 10^2 \Rightarrow 100 = 100$$

So, the given triangle is a right angled triangle.

Therefore, Area of triangle

$$= \frac{1}{2} \times \text{base} \times \text{height} = \frac{6 \times 8}{2} = 24 \text{ cm}^2$$

Sol.319.(d) According to the question,

$$\frac{\text{volume of 1st cylinder}}{\text{volume of 2nd cylinder}} = \frac{\pi r_1^2 h_1}{\pi r_2^2 h_2}$$

$$\frac{6}{5} = \frac{r_1^2 \times 5}{r_2^2 \times 24} \Rightarrow \frac{144}{25} = \frac{r_1^2}{r_2^2}$$

$$\frac{r_1}{r_2} = \sqrt{\frac{144}{25}} = \frac{12}{5}$$

Ratio of radii = 12 : 5

Sol.320.(b) According to the question,

circumference of the base of the cylinder

$$= 2\pi r$$

$$22 = 2 \times \frac{22}{7} \times r \Rightarrow r = 3.5 \text{ cm}$$

$$\text{Height} = 3.5 \times 4 = 14 \text{ cm}$$

$$\text{C.S.A of cylinder} = 2\pi r h$$

$$2 \times \frac{22}{7} \times 3.5 \times 14 = 308 \text{ cm}^2$$

Sol.321.(b) According to the question,

Volume of sphere = volume of cylinder

$$\frac{4}{3} \pi r^3 = \pi R^2 h$$

$$\frac{4}{3} \times 18 \times 18 \times 18 = 4 \times 4 \times h$$

$$(\text{Height})h = 486 \text{ cm}$$

Sol.322.(a) According to the question,

$$\text{Volume of sphere} = \frac{4}{3} \pi r^3$$

total Number of spheres

$$= \frac{\text{volume of big sphere}}{\text{volume of small spheres}}$$

$$= \frac{r^3}{r_2^3} \Rightarrow \frac{28 \times 28 \times 28}{3.5 \times 3.5 \times 3.5} = 512$$

Sol.323.(d)

$$\text{Volume of Hemisphere} = \frac{2}{3} \pi r^3$$

According to question,

$$\text{Volume of Hemisphere} = \frac{2}{3} \times \frac{22}{7} \times 5^3$$

$$= 261.9 \text{ cm}^3$$

Sol.324.(d) Volume of Cylinder = $\pi r^2 h$

According to question,

$$\text{Required ratio} = \frac{\pi r_1^2 h_1}{\pi r_2^2 h_2}$$

$$= \frac{\pi \times \frac{6}{2} \times \frac{6}{2} \times 7}{\pi \times \frac{7}{2} \times \frac{7}{2} \times 9} = \frac{4}{7}$$

Sol.325.(d) According to the question,

$$2\pi r h = \frac{2}{5} \times 2\pi r(r + h)$$

$$3h = 2r \Rightarrow \frac{r}{h} = \frac{3}{2}$$

Hence, ratio of diameter to height

$$= 3 \times 2 : 2 = 3 : 1$$

Sol.326.(d)

Circumference of the circle ($2\pi r$) = 13.2 cm

$$\text{Radius of the circle } (r) = \frac{13.2 \times 7}{22 \times 2} = 2.1 \text{ cm}$$

Sol.327.(c)

circumferences of circles = $2\pi r$

Let the radius of larger circle be R .

According to question,

$$\left(2 \times \frac{22}{7} \times R \right) - \left(2 \times \frac{22}{7} \times 1.4 \right) = 4.4$$

$$\left(2 \times \frac{22}{7} \times R \right) = 13.2 \Rightarrow R = 2.1 \text{ cm}$$

Sol.328.(c) According to question,

$$\text{Slant height } (l) = \sqrt{(24)^2 + (18)^2} = 30 \text{ cm}$$

curved surface area of the cone = $\pi r l$

$$= \frac{22}{7} \times 18 \times 30 = 1697.14 \text{ cm}^2$$

Sol.329.(a) According to the question,

$$9 \times \pi r^2 = 3 \times \text{side}^2 \Rightarrow \frac{r^2}{\text{side}^2} = \frac{1}{3 \times \pi}$$

$$\frac{r}{\text{side}} = \sqrt{\frac{1}{3 \times \pi}} = \frac{1}{\sqrt{3\pi}}$$

$$\text{Diameter of the circle} = 2r = 2 \times 1 = 2$$

Diagonal of the square = $\sqrt{2}$ side

$$= \sqrt{2} \times \sqrt{3\pi}$$

$$\text{Required ratio} = 2 : \sqrt{2} \times \sqrt{3\pi} = \sqrt{2} : \sqrt{3\pi}$$

Sol.330.(b)

$$\text{Volume of hemispherical bowl} = \frac{2}{3} \pi r^3$$

$$\frac{2}{3} \times \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2} \times \frac{21}{2} = 2425.5 \text{ m}^3$$

$$1 \text{ m}^3 = 1000 \text{ liter}$$

$$2425.5 \text{ m}^3 = 2425500 \text{ liter}$$

Sol.331.(c) Total Surface area of Cylinder

$$= 2\pi r(r + h) = 2 \times \frac{22}{7} \times 28(28 + 24)$$

$$= 9152$$

Total Surface area of bisected cylinder

$$= 2\{2\pi r(r + \frac{h}{2})\}$$

$$= 2 \times 2 \times \frac{22}{7} \times 28(28 + \frac{24}{2}) = 14080$$

$$\text{Required\%} = \frac{14080 - 9152}{9152} \times 100 = 53.85\%$$

Sol.332.(c)

Curved surface area of Cylinder = $2\pi r h$

$$330 = 2 \times \frac{22}{7} \times r \times 15$$

$$r = \frac{330 \times 7}{2 \times 15 \times 22} = 3.5 \text{ cm}$$

Required Diameter = 7 cm

Sol.333.(c) According to question,
Length, Breadth and Height of room
= 640 cm, 480 cm and 320 cm
HCF(640, 480, 320) = 160 cm
So, Required length of tape = 1 m 60 cm

Sol.334.(c) According to the question,

$$\text{Volume of hemisphere} = \frac{2}{3}\pi r^3$$

$$\text{Volume of cone} = \frac{1}{3}\pi r^2 h$$

$$N \text{ number of cone} = \frac{\text{Volume of hemisphere}}{\text{volume of cone}}$$

$$= \frac{\frac{2}{3}\pi r^3}{\frac{1}{3}\pi r^2 h} \Rightarrow \frac{2 \times 8 \times 8 \times 8}{4 \times 4 \times 2} = 32$$

Sol.335.(a) Length of cuboid = 80,
breadth of cuboid = 64 and
Height of cuboid = 50
Now, cubes of side 8 cm should be placed.

So, 2 cm of cuboid height should be vacant. No. of cube should be placed

$$= \frac{80 \times 64 \times 48}{8 \times 8 \times 8} = 480$$

$$\text{Sol.336.(b)} \text{ Area of sector} = \frac{\theta}{360} \times \pi r^2$$

$$= \frac{60}{360} \times \frac{22}{7} \times 7 \times 7 = 25.66 \text{ cm}^2$$

Sol.337.(d) According to the question,

$$\frac{2\pi R}{2\pi r} = \frac{11}{7} \Rightarrow \frac{R}{r} = \frac{11}{7}$$

Now, width of path (11 - 7) unit

= 20 meter $\Rightarrow$ 1 unit = 5 meter

Radius of inner circle (r) = 7 $\times$ 5 = 35 meter

$$\text{Sol.338.(d)} \quad S = \frac{a+b+c}{2} \text{ and Area of}$$

$$\text{triangle} = \sqrt{S(S-a)(S-b)(S-c)}$$

According to question,

$$S = \frac{7+4+5}{2} = 8$$

$$\text{Area of triangle} = \sqrt{8(8-7)(8-4)(8-5)}$$

$$= \sqrt{8 \times 1 \times 4 \times 3} = \sqrt{96} = 4\sqrt{6} \text{ cm}^2$$

Sol.339.(d)

circumference of base of a cylinder = $2\pi r$

$$2\pi r = 28\pi = r = 14 \text{ cm}$$

Curved surface area of cylinder = $2\pi r h$

$$= 2 \times \pi \times 14 \times 17 = 476\pi \text{ cm}^2$$

Sol.340.(b) C.S.A of cone = $\pi r l$

$$= 3.14 \times 12 \times 14 = 527.52 \text{ cm}^2$$

Sol.341.(a) According to the question,

$$\frac{4}{3}\pi R^3 = n \times \frac{4}{3}\pi r^3$$

$$\text{Number of balls be made (n)} = \frac{R^3}{r^3}$$

$$= \frac{66 \times 66 \times 66}{33 \times 33 \times 33} = 8$$

$$\text{Sol.342.(d)} \text{ Area of square} = \frac{(\text{diagonal})^2}{2}$$

$$= \frac{(18.4)^2}{2} = \frac{338.56}{2} = 169.28 \text{ cm}^2$$

Sol.343.(c) Volume of Cylinder = $\pi r^2 h$
According to question,

$$5000 = 3.14 \times 10 \times 10 \times h \Rightarrow h = \frac{50}{3.14} \text{ m}$$

Now,

Curved Surface Area of Cylinder = $2\pi r h$

$$= 2 \times 3.14 \times 10 \times \frac{50}{3.14} = 1000 \text{ m}^2$$

Sol.344.(d) Volume of cylinder = $\pi r^2 h$

$$\text{Volume of cone} = \frac{1}{3}\pi r^2 h$$

Let the Radius of cylinder = x

And the radius of cone = $2x$

According to the question,

$$\frac{1}{3}\pi r^2 h : \pi r^2 h = \frac{1}{3} \times 2x \times 2x : x \times x$$

Required ratio = 4 : 3

Sol.345.(c) According to the question,

$$l - b = 40$$

Now, perimeter of rectangle $2(l + b) = 96$

$$l + b = 48$$

Then $l = 44$ and $b = 4$

$$\text{Area of rectangle} = 44 \times 4 = 176 \text{ m}^2$$

Sol.346.(a)

$$\text{Breadth of the floor} = \frac{600}{30} = 20 \text{ feet}$$

Sol.347.(b)

$$\text{Area of Trapezium} = \frac{(a+b)}{2} \times h$$

According to question,

Area of Trapezium

$$= \frac{7.58 + 6.42}{2} \times 6.50 = 45.5 \text{ square yard}$$

Required price = $45.5 \times 2000 = \text{Rs. } 91000$

Sol.348.(b)

$$\text{Area of base of the 2nd cylinder} = \pi r_2^2$$

$$\pi r_2^2 = 154 \Rightarrow \frac{22}{7} \times r_2^2 = 154 \Rightarrow r_2 = 7 \text{ cm}$$

$$\text{Volume of cylinder} = \pi r^2 h$$

$$\frac{(\text{Volume of cylinder})_1}{(\text{Volume of cylinder})_2} = \frac{r_1^2 \times 13}{7 \times 7 \times 9} = \frac{13}{1}$$

$$r_1 = 21 \text{ cm}$$

Sol.349.(d) Curve surface of cone = $\pi r l$

$$= 3.14 \times 5 \times 30 \Rightarrow 471 \text{ cm}^2$$

Sol.350.(d) Formula used :-

$$\text{Area of trapezium} = \frac{1}{2} \times \text{sum of parallel}$$

side $\times$ height

$$= \frac{1}{2} \times (25 + 30) \times 20 = 550 \text{ m}^2$$

Sol.351.(a) According to question,

$$\pi r^2 H = \pi \left(\frac{88r}{100}\right)^2 h$$

$$r \times r \times H = \frac{22r}{25} \times \frac{22r}{25} \times h \Rightarrow \frac{H}{h} = \frac{484}{625}$$

$$\text{Required \%} = \frac{(625 - 484)}{484} \times 100 = 29.13\%$$

Sol.352.(a) Volume of Cylinder = $\pi r^2 h$

Let the radii of two cylinders be r and R

And their height be h and H .

According to question,

$$\frac{\pi r^2 5}{\pi R^2 7} = \frac{7}{5} \Rightarrow \frac{r^2}{R^2} = \frac{49}{25} \Rightarrow \frac{r}{R} = \frac{7}{5}$$

Sol.353.(c) According to the question,

Height = $7 \times 3 = 21 \text{ cm}$

CSA of cylinder = $2\pi r h$

$$= 2 \times \frac{22}{7} \times 7 \times 21 = 924 \text{ cm}^2$$

Sol.354.(a) Semiperimeter of triangle (s)

$$= \frac{3x + 3x + 4x}{2} = 5x$$

Now, area of triangle

$$= \sqrt{5x(5x - 3x)(5x - 3x)(5x - 4x)}$$

$$= \sqrt{5x(2x)(2x)(x)} = 2\sqrt{5} x^2$$

According to the question,

$$2\sqrt{5} x^2 = 32\sqrt{5} \Rightarrow x = 4 \text{ cm}$$

Length of equal sides (3x) = 12 cm

Sol.355.(b) Area of outer square park

$$= 18 \times 18 = 324 \text{ cm}^2$$

Area of inner square park

$$= (18 - 1 - 1)^2 = 256 \text{ cm}^2$$

$$\text{Required area} = 324 - 256 = 68 \text{ cm}^2$$

Sol.356.(a)

$$\text{Area of triangle} = \frac{1}{2} \times \text{base} \times \text{Height}$$

According to question,

$$\text{Area of triangle} = \frac{1}{2} \times 7 \times 12 = 42 \text{ cm}^2$$

Sol.357.(d) According to the question,

Length of the cuboid = $14 + 14 = 28 \text{ cm}$

Breadth and height of the the cuboid = 14 cm

Volume of the cuboid = $l \times b \times h$

$$= 28 \times 14 \times 14 = 5488 \text{ cm}^3$$

Sol.358.(d) Length = 4cm

ATQ

Breadth = 16cm, Height = 8cm

Lateral surface area of cuboid

$$= 2(\text{length} + \text{breadth})\text{height}$$

$$= 2(4 + 16) \times 8 = 2(20) \times 8 = 320 \text{ cm}^2$$

Sol.359.(b)

Let length of longer Diagonal be 10 unit

Then, the length of the smaller diagonal is 7 units.

$$\text{Area of Rhombus} = \frac{\text{product of Diagonals}}{2}$$

$$= \frac{7 \times 10}{2} = 35$$

square of the length of larger diagonal = 100

According to question,

$$\text{Required Ratio} = \frac{35}{100} = 7 : 20$$

Sol.360.(b) According to question

$$\begin{array}{lcl} & \text{initial} : & \text{final} \\ \text{Radius} \rightarrow & (25)^2 : & (21)^2 \\ \text{Height} \rightarrow & h_1 : & h_2 \end{array}$$

$$\text{Volume} \rightarrow 625 h_1 : 441 h_2$$

According to question ,
volume remains same

Volume is directly proportional to height-
Increase in height:-

$$\frac{625 - 441}{441} \times 100 = 41.72\%$$

Sol.361.(a) According to the question,

$$\frac{\pi r_1^2 \times 3}{\pi r_2^2 \times 4} = \frac{27}{25} \Rightarrow \frac{r_1^2}{r_2^2} = \frac{36}{25} \Rightarrow \frac{r_1}{r_2} = \frac{6}{5}$$

$$\text{Area of base of second cylinder} = \left(\frac{22}{7}\right)$$

$$\times 25x^2 = 3850 \text{ cm}^2 \Rightarrow x^2 = 49 \Rightarrow x = 7 \text{ cm}$$

$$\text{So, radius of first cylinder} = 6 \times 7 = 42 \text{ cm}$$

Sol.362.(a) According to the question,

$$\text{Height of the cuboid} = 16 \times \frac{1}{4} = 4 \text{ cm}$$

CSA of cuboid

$$= 2(\text{length} + \text{breadth}) \times \text{height}$$

$$= 2(16) \times 4 = 128 \text{ cm}^2$$

Sol.363.(c) Volume of the spheres = $\frac{4}{3}\pi r^3$

$$\text{Required ratio} = \frac{4}{3} \times \pi \times 1^3 : \frac{4}{3} \times \pi \times 5^3$$

$$= 1 : 125$$

Sol.364.(a) According to question,

Length, breadth and height of Cuboid
= 10 cm, 5 cm and 20 cm respectively

lateral surface area of the cuboid

$$= 2(\text{length} + \text{breadth}) \times \text{height}$$

$$= 2(10 + 5) \times 20 = 600 \text{ cm}^2$$

Sol.365.(a) Total surface area of cuboid

$$= 2(lb + bh + hl)$$

$$= 2(4 \times 6 + 6 \times 10 + 10 \times 4) = 248 \text{ cm}^2$$

Sol.366.(b) Length and breadth of

garden with path = 228m, 78 m

According to the question,

$$\text{Area of path} = (228 \times 78) - (220 \times 70)$$

$$= 2384 \text{ metre}^2$$

Sol.367.(c) According to the question,

$$2\pi r \times 11 = 968$$

Diameter of cylinder (2r)

$$= \frac{968 \times 7}{22 \times 11} = 28 \text{ cm}$$

Sol.368.(c) According to the question,

Breadth of the cuboid (b) = 12 cm

$$\text{Length of the cuboid (l)} = 12 \times \frac{1}{2} = 6 \text{ cm}$$

$$\text{Height of the cuboid (h)} = 12 \times 3 = 36 \text{ cm}$$

$$\text{Total surface area of cuboid} = 2(lb + bh + hl)$$

$$= 2(6 \times 12 + 12 \times 36 + 36 \times 6)$$

$$= 2(72 + 432 + 216) = 1440 \text{ cm}^2$$

Sol.369.(b) Perimeter of rectangle

$$= 2(\text{length} + \text{breadth})$$

$$56 = 2(l + b) \Rightarrow (l + b) = 28 \text{ m}$$

$$\text{Area of rectangle} = l \times b$$

$$l \times b = 192$$

$$(l + b) = 28 \text{ m (squaring both sides)}$$

$$l^2 + b^2 + 2 \times 192 = 784 \Rightarrow l^2 + b^2 = 400$$

$$\text{Diagonal of the rectangle} = \sqrt{l^2 + b^2}$$

$$\text{Diagonal of the rectangle} = \sqrt{400} = 20 \text{ meter}$$

Sol.370.(d)

$$\text{Surface area of sphere} = 4\pi r^2$$

$$\text{Surface area of spherical ball}$$

$$= 4 \times \frac{22}{7} \times 1.1 \times 1.1 = 15.2 \text{ cm}^2$$

Sol.371.(b) According to the question,

$$\text{Area of circle} = \pi r^2 \Rightarrow \pi \times r^2 = 1560$$

$$\text{total surface area of the hemisphere}$$

$$= 3\pi r^2 \Rightarrow 3 \times 1560 = 4680 \text{ m}^2$$

Sol.372.(c) According to the question,

$$\text{Height of the cuboid} = 40 \times \frac{1}{5} = 8$$

$$\text{Lateral Surface area of cuboid}$$

$$= 2(l + b)h \Rightarrow 2(40) \times 8 = 640 \text{ cm}^2$$

Sol.373.(a)

$$\text{Semiperimeter (s)} = \frac{13 + 14 + 15}{2}$$

$$= 21 \text{ inches}$$

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\text{Area} = \sqrt{21(8)(7)(6)} = 84 \text{ square inches}$$

Sol.374.(b) According to the question,

$$\text{Circumference of the base of the cone}$$

$$(2\pi r) = 88 \text{ cm}$$

$$\text{Then, } r = 88 \times \frac{7}{22} \times \frac{1}{2} = 14 \text{ cm}$$

$$\text{Slant height of the cone (l)}$$

$$= \sqrt{(r)^2 + (h)^2} = \sqrt{(14)^2 + (28)^2}$$

$$= \sqrt{980} = 14\sqrt{5}$$

$$\text{Therefore, curved surface area of the}$$

$$\text{cone } (\pi rl) = \frac{22}{7} \times 14 \times 14\sqrt{5} = 616\sqrt{5} \text{ cm}^2$$

Sol.375.(d) According to the question,

$$\pi r_1^2 h_1 : \pi r_2^2 h_2 = 6 : 1 \Rightarrow \pi r_1^2 h : \pi r_2^2 6h = 6 : 1$$

$$r_1^2 : r_2^2 = 36 : 1 \Rightarrow r_1 : r_2 = 6 : 1$$

Sol.376.(d)

$$\text{Volume of the sphere} = \frac{4}{3}\pi r^3$$

After decreasing the radius of sphere by

10% then volume of the sphere

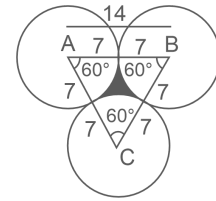
$$= \frac{4}{3}\pi \left(\frac{9}{10}r\right)^3 = \frac{4}{3}\pi \left(\frac{729}{1000}r^3\right)$$

Therefore, decrement in the volume

$$\frac{r^3 - \left(\frac{729}{1000}r^3\right)}{r^3} \times 100$$

$$= \frac{271r^3}{1000r^3} \times 100 \Rightarrow 27.1\%$$

Sol.377.(a)



Side of the equilateral triangle (2r) = 14 cm

So, area of the equilateral triangle =

$$\frac{\sqrt{3}}{4} \times 14 \times 14 = 49\sqrt{3} \text{ cm}^2$$

Now, the area of the each sector

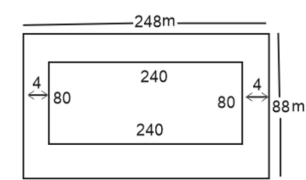
$$= 3 \times \frac{60^\circ}{360^\circ} \times \pi r^2$$

$$= 3 \times \frac{1}{6} \times \frac{22}{7} \times 7 \times 7 = 77 \text{ cm}^2$$

Therefore, area of the portion enclosed
by three circle = (area of equilateral
triangle - area of all three sector)

$$\Rightarrow (49\sqrt{3} - 77) \text{ cm}^2$$

Sol.378.(d)



$$\text{Area of the path} = (248 \times 88 - 240 \times 80)$$

$$= (21824 - 19200) \text{ m}^2 = 2624 \text{ m}^2$$

Sol.379.(d) According to the question,

$$l + b + h = 20 \text{ cm}$$

$$\text{Total Surface area of the cuboid}$$

$$= 2(lb + bh + hl) = 256 \text{ cm}^2$$

$$\text{Now, } (l + b + h)^2 = (l)^2 + (b)^2 + (h)^2 + 2(lb + bh + hl)$$

$$\Rightarrow (l)^2 + (b)^2 + (h)^2 = 400 - 256$$

$$= 144 \text{ cm}^2$$

Maximum Length of stick that can be
placed inside the cuboid box =

$$\sqrt{(l)^2 + (b)^2 + (h)^2}$$

$$\sqrt{144} = 12 \text{ cm}$$

Sol.380.(b) Let side of third triangle (z) = a

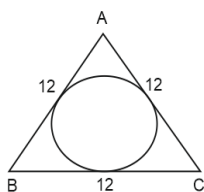
According to the question,

$$\text{Area of the triangle} \rightarrow X + Y = Z$$

$$\frac{\sqrt{3}}{4} \times (6)^2 + \frac{\sqrt{3}}{4} \times (8)^2 = \frac{\sqrt{3}}{4} \times (a)^2$$

$$(a)^2 = (6)^2 + (8)^2 \Rightarrow a = 10 \text{ cm}$$

Sol.381.(d)



$$\text{Inradius} = \frac{\text{side}}{2\sqrt{3}}$$

Side of the equilateral triangle

$$= \text{inradius} \times 2\sqrt{3}$$

$$\text{inradius} = \frac{12}{2\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 2\sqrt{3}$$

$$\text{Area of the equilateral triangle} = \frac{\sqrt{3}}{4} \times a^2$$

$$= \frac{\sqrt{3}}{4} \times 144 = 36 \times 1.732 = 62.35 \text{ sq. cm}$$

$$\text{Area of the incircle} = \pi r^2$$

$$= \frac{22}{7} \times (2\sqrt{3})^2 = 37.71 \text{ sq. cm}$$

Therefore, area of the triangle which is not in the incircle (shaded region)

$$= (62.35 - 37.71) = 24.64 \text{ sq. cm}$$

Sol.382.(c)

Total surface area of the cuboidal box

$$= 2(lb + bh + hl)$$

$$= 2(18 \times 12 + 12 \times 10 + 10 \times 18)$$

$$= 2(516) \Rightarrow 1032 \text{ cm}^2$$

$$\text{Sol.383.(a) Area of the rhombus} = \frac{1}{2}$$

$$\times \text{first diagonal} \times \text{second diagonal}$$

$$= \frac{1}{2} \times 8 \times 13 = 52 \text{ cm}^2$$

Sol.384.(c)

$$\text{Volume of the sphere} = \frac{4}{3} \pi (R)^3$$

$$\text{Volume of the hemisphere} = \frac{2}{3} \pi (r)^3$$

According to the question,

$$\text{Required ratio} \rightarrow \frac{4}{3} \pi (R)^3 : \frac{2}{3} \pi (r)^3$$

$$\frac{4}{3} \pi (3r)^3 : \frac{2}{3} \pi (r)^3 \Rightarrow 54 : 1$$

Sol.385.(c) Let, the radius of the 1st cylinder be 'R' and radius of the smaller cylinder be 'r'.

Ratio $\rightarrow$ 1st cylinder : 2nd cylinder

$$\text{Volume} \rightarrow 32 : 9$$

$$\text{Height} \rightarrow 8 : 9$$

According to the question,

Area of base of 2nd Cylinder (πr^2)

$$= 616 \text{ cm}^2 \Rightarrow r = \sqrt{616 \times \frac{7}{22}} = 14 \text{ cm.}$$

Now,

$$\frac{\text{Volume of 1st Cylinder} (\pi R^2 H)}{\text{Volume of 2nd Cylinder} (\pi r^2 h)} = \frac{32}{9}$$

$$\frac{\pi r^2 (8)}{\pi \times 196 \times (9)} = \frac{32}{9}$$

$$r^2 = 784 \Rightarrow r = 28 \text{ cm}$$

$$\text{Sol.386.(a) } \frac{2(l+b)}{2} = 30 \text{ cm}$$

$$(l+b) = 30 \text{ cm}$$

According to question;

$$l = (6+x) \text{ and } b = x$$

$$(6+x) + x = 30 \text{ cm} \Rightarrow x = 12$$

$$l = (6+12) = 18 \text{ cm and } b = 12 \text{ cm}$$

$$\text{Area of rectangle} = l \times b = 18 \times 12 = 216 \text{ cm}^2$$

Alternate method :

Formula : Perimeter of rectangle

$$= 2(l+b) \Rightarrow \frac{2(l+b)}{2} = 30 \text{ cm}$$

$$(l+b) = 30 \text{ cm} \dots \text{eq(1)}$$

$$(l-b) = 6 \text{ cm} \dots \text{eq(2)}$$

Solving the eq(1) and eq(2), we get,

$$l = 18 \text{ cm, and } b = 12 \text{ cm,}$$

$$\text{Area of rectangle} = l \times b = 18 \times 12$$

$$= 216 \text{ cm}^2$$

Sol.387.(a) 24 solid hemisphere volume is equal = volume of cylinder

$$24 \times \frac{2}{3} \times \frac{22}{7} \times r^3 = \frac{22}{7} \times 12 \times 12 \times 24$$

$$r = 6$$

Sol.388.(a) Area of triangle

$$= \sqrt{s(s-a)(s-b)(s-c)}$$

$$S = \frac{a+b+c}{2} = \frac{7+9+8}{2} = \frac{24}{2} = 12$$

$$\Rightarrow \sqrt{12(12-7)(12-9)(12-8)}$$

$$\Rightarrow \sqrt{12 \times 5 \times 3 \times 4} \rightarrow 12\sqrt{5} \text{ Sq. cm}$$

$$\text{Sol.389.(a) } (a \times b \times c)^2 = 1024 \text{ units}$$

$$a \times b \times c = 32$$

$$\text{Area of its circumcircle} = 16\pi$$

$$\pi R^2 = 16\pi \Rightarrow R = 4$$

$$\text{circumradius}(R) = \frac{abc}{4 \times \text{area of } \Delta}$$

$$4 = \frac{32}{4 \times \text{area of } \Delta}$$

$$\text{Area of given triangle} = 2 \text{ sq. units}$$

Sol.390.(d)

Let the length of the rectangle = 5x

Breadth = 3x

$$\text{Area of rectangle} = \text{length} \times \text{breadth}$$

$$60 = 5x \times 3x \Rightarrow x = 2$$

Now,

$$\text{Length} = 5x \rightarrow 10 \text{ m, Breadth} = 3x \rightarrow 6 \text{ m}$$

$$\text{Difference between length and breadth}$$

$$\Rightarrow 10 - 6 \rightarrow 4 \text{ m}$$

Sol.391.(a) Volume of Brick

$$= 25 \text{ cm} \times 12.5 \text{ cm} \times 16 \text{ cm} = 5000 \text{ cm}^3$$

$$\text{Volume of Wall} = 800 \text{ cm} \times 500 \text{ cm} \times$$

$$22.5 \text{ cm} = 90,00,000 \text{ cm}^3$$

$$\text{Bricks required} = \frac{90,00,000 \text{ cm}^3}{5000 \text{ cm}^3} = 1,800$$

Sol.392.(a) Total surface area of cylinder

$$2\pi r(h+r) = 2 \times \frac{22}{7} \times 7(5+7) = 528 \text{ cm}^2$$

When Cylinder is bisected from its height

$$\text{, then Increased area} \rightarrow 2 \times (\pi r^2)$$

$$= 2 \times \frac{22}{7} \times 7 \times 7 = 308 \text{ cm}^2$$

Required % increase

$$= \frac{308}{528} \times 100 \approx 58.33\%$$

Alternate method :

After cutting the cylinder in two identical cylinder the total increase in the surface area will be $2\pi r^2$

% increase in the surface area

$$= \frac{2\pi r^2}{2\pi r(h+r)} \times 100 \Rightarrow \frac{r}{(r+h)} \times 100$$

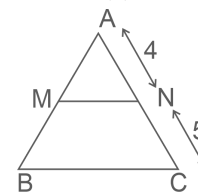
$$= \frac{7}{(5+7)} \times 100 = 58.33\%$$

Sol.393.(c) Volume of cube = side^3

$$1728 = \text{side}^3 \Rightarrow \text{side} = 12 \text{ cm}$$

$$\text{T.S.A} = 6 \times \text{side}^2 \Rightarrow 6 \times 12 \times 12 = 864 \text{ cm}^2$$

Sol.394.(c)



$$\text{Area of } \Delta AMN = (4)^2 \Rightarrow 16 \text{ units}$$

$$\text{Area of } \Delta ABC = (9)^2 \Rightarrow 81 \text{ units}$$

$$\text{Area of quadrilateral MBCN} = \text{Area of } \Delta ABC - \text{Area of } \Delta AMN$$

$$81 - 16 \Rightarrow 65 \text{ units}$$

$$65 \text{ units} = 130 \text{ cm}^2$$

$$\text{Area of } \Delta AMN (16 \text{ units}) = 32 \text{ cm}^2$$

Sol.395.(a)

Circumference of circle = $2\pi r$ (Small circle)

$$2\pi r = 264 \text{ m} \Rightarrow r = 42 \text{ m}$$

$$(\text{Big circle}) \Rightarrow 2\pi R = 308 \text{ m, } R = 49 \text{ m}$$

$$\text{Area of circle} = \pi r^2$$

$$\text{Difference between bigger and smaller circle's area} = \pi(49^2 - 42^2) \rightarrow 2002 \text{ m}^2$$

Sol.396.(a)

$$\text{Circumference of circle} = 2\pi r \Rightarrow 22 = 2\pi r$$

$$r = 3.5, \text{ Area of circle} = \pi r^2$$

$$\frac{22}{7} \times 3.5 \times 3.5 \Rightarrow 38.5 \text{ cm}^2$$

Sol.397.(b) Formula:- Area of the

$$\text{trapezium} = \frac{1}{2} \times \text{sum of parallel sides}$$

$$\times (\text{distance between parallel sides})$$

then, distance between parallel sides

$$= \frac{2 \times 875}{21 + 14} \Rightarrow 50 \text{ cm}$$

Sol.398.(c) Ratio of their heights $\rightarrow 2 : 3$

Ratio of their radius $\rightarrow 6 : 5$

$$\frac{\text{Volume}_1}{\text{Volume}_2} = \frac{\pi \times 6 \times 6 \times 2}{\pi \times 5 \times 5 \times 3} = \frac{24}{25}$$

Sol.399.(b) Length + breadth = 4 cm

Length = 3 cm, Breadth = 1 cm

$$\text{Height} = \frac{4}{2} = 2 \text{ cm}$$

$$\begin{aligned} \text{T.S.A cuboid} &= 2(\text{length} \times \text{breadth} + \text{breadth} \times \text{height} + \text{height} \times \text{length}) \\ &= 2(3 \times 1 + 1 \times 2 + 2 \times 3) = 22 \text{ cm}^2 \end{aligned}$$

Sol.400.(c) Area of square = side^2

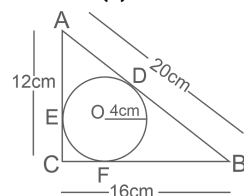
$$1156 = \text{side}^2, \text{side} = 34 \text{ cm}$$

$$\begin{aligned} \text{So the length of rectangle} &= 34 \times 2 \\ &= 68 \text{ cm} \end{aligned}$$

$$\text{Breadth of rectangle} = 34 - 14 = 20 \text{ cm}$$

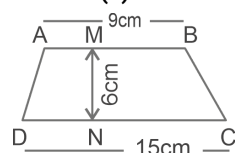
$$\begin{aligned} \text{Ratio of length and breadth of rectangle} \\ &= 68 : 20 \Rightarrow 17 : 5 \end{aligned}$$

Sol.401.(c)



$$\text{Area of } \triangle ABC = \frac{16 \times 12}{2} = 96 \text{ cm}^2$$

Sol.402.(b)



$$\text{Area of Trapezium} = \frac{1}{2} \times (AB + CD) \times MN$$

$$= \frac{1}{2} \times (9 + 15) \times 6 = 72 \text{ cm}^2$$

$$\begin{aligned} \text{Total cost of copper plating} \\ &= 72 \times 3 = \text{Rs. } 216 \end{aligned}$$

Sol.403.(b) According to the question, Semiperimeter of the $\triangle ABC$,

$$s = \frac{22}{2} = 11 \text{ cm}$$

Then, Area of triangle $\triangle ABC$

$$\Rightarrow \sqrt{11(7)(3)(1)} = \sqrt{231} \text{ cm}^2$$

$$\therefore \triangle ABC \sim \triangle PQR$$

$$\text{Therefore, } \frac{\text{ar}(\triangle PQR)}{\text{ar}(\triangle ABC)} = \left(\frac{QR}{BC}\right)^2$$

$$= \frac{\text{ar}(\triangle PQR)}{\sqrt{231}} = \frac{256}{64}$$

$$= \text{ar}(\triangle PQR) = 4\sqrt{231} \text{ cm}^2$$

Sol.404.(c) Volume of cube = side^3

Volume of three small cube = volume of bigger cube

Total Volume of Ist, IInd and IIIrd cube

$$= 6^3 + 8^3 + 10^3 = 1728$$

$$1728 = \text{volume of bigger cube}$$

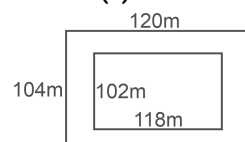
$$a^3 = 1728 \text{ Side of bigger cube} = 12 \text{ cm}$$

Sol.405.(c) Length of each side of the cube = 21 cm

$$\text{Radius of sphere} = \frac{21}{2} \text{ cm}$$

$$\begin{aligned} \text{Volume of sphere} &= \frac{4}{3} \pi r^3 \\ &= \frac{4}{3} \times \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2} \times \frac{21}{2} \\ &= 4851 \text{ cm}^3 \end{aligned}$$

Sol.406.(d)



Required perimeter for marking

$$= 2(l + b) = 2(102 + 118) = 440 \text{ metre}$$

$$\text{Total price} = 440 \times 2.5 = 1100 \text{ Rs.}$$

Sol.407.(a)

Let the number of bottles be n .

$$\text{Volume of hemispherical bowl} = \frac{2}{3} \pi r^3$$

$$\text{Volume of cylinder} = \pi r^2 h$$

Number of Bottles (n)

$$= \frac{\text{Volume of Hemispherical bowl}}{\text{Volume of Cylinder}}$$

$$(n) = \frac{2 \times 18 \times 18 \times 18}{3 \times 3 \times 3 \times 6} = 72.$$

Sol.408.(c)

$$\begin{aligned} \text{TSA of Pyramid} &= \text{Area of Base} + \left(\frac{1}{2} \times \right. \\ &\quad \left. \text{Perimeter of Base} \times \text{Slant height}\right) \end{aligned}$$

$$\text{Slant Height } (l) = \sqrt{r^2 + h^2}$$

$$l = \sqrt{9 + 16} = \sqrt{25} = 5 \text{ cm}$$

$$\text{TSA of Pyramid} = 64 + \left(\frac{1}{2} \times 32 \times 5\right)$$

$$= 144 \text{ cm}^2.$$

Sol.409.(c) Longest diagonal in a room

$$= \sqrt{l^2 + b^2 + h^2}$$

According to Question,

$$\text{Breadth} = \frac{\text{Length}}{2} \text{ and}$$

$$\text{Breadth} = 2 \times \text{height}$$

$$\text{Now, Diagonal} = \sqrt{(2b)^2 + (b)^2 + \left(\frac{b}{2}\right)^2}$$

$$= \sqrt{\frac{16b^2 + 4b^2 + b^2}{4}} = \frac{\sqrt{21} b}{2}$$

$$\text{Sol.410.(d)} \text{ LSA} = \frac{1}{2} \times \text{Perimeter of Base}$$

$\times$ Slant height

$$\text{TSA of pyramid} = \text{LSA} + \text{Area of Base}$$

$$(\text{Edge})^2 = (\text{Slant height})^2 + \left(\frac{\text{Side}}{2}\right)^2$$

$$(24)^2 = (\text{Slant height})^2 + \left(\frac{8}{2}\right)^2$$

$$\Rightarrow \text{Slant height} = 4\sqrt{35} \text{ cm}$$

$$\text{LSA} = 48\sqrt{35} \text{ cm}$$

$$\text{TSA} = 16\sqrt{3} + 48\sqrt{35} \text{ cm}^2$$

Sol.411.(a)

$$\text{C.S.A of Cylinder } (2\pi rh) = 126 \pi$$

$$2 \times \pi \times r \times 14 = 126 \pi \Rightarrow r = \frac{9}{2} \text{ cm}$$

$$\begin{aligned} \text{Volume of cylinder } (\pi r^2 h) \\ &= \pi \times \frac{9}{2} \times \frac{9}{2} \times 14 = \frac{567}{2} \pi \text{ cm}^3 \\ &= 283\frac{1}{2} \pi \text{ cm}^3 \end{aligned}$$

Sol.412.(c) Let the parallel sides of trapezium be a and b cm.

$$\begin{aligned} \text{Area of Trapezium} &= \frac{1}{2} \times \text{Sum of two} \\ &\quad \text{parallel sides} \times \text{height} \end{aligned}$$

$$1380 = \frac{1}{2} \times (a + b) \times 60$$

$$a + b = 46 \text{ and } a - b = 12$$

On Solving these equations, we get

$$a = 29 \text{ cm and } b = 17 \text{ cm.}$$

Sol.413.(a)

Area of rectangle = Length $\times$ Breadth

Area of rectangle is directly proportional to product of length and breadth of rectangle.

$$\% \text{ change in area} = 8 + 5 + \frac{8 \times 5}{100} = 13.4\%$$

Sol.414.(b) Volume of rectangular leads = $n \times$ volume of spherical lead

$$L \times B \times H = n \times \frac{4}{3} \times \pi \times r^3$$

$$88 \times 63 \times 42 = n \times \frac{4}{3} \times \frac{22}{7} \times 4.2^3$$

$$n = \frac{63 \times 10 \times 21}{4.2 \times 4.2} = 750$$

Sol.415.(d) If any 3D figure cast into another 3D figure always equate their Volume.

Volume of sphere = Volume of Cone

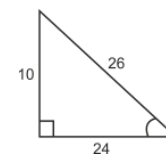
$$\frac{4}{3} \pi r_1^3 = \frac{1}{3} \pi r_2^2 h$$

$$\frac{4}{3} \times (13)^3 = \frac{1}{3} \times \left(\frac{13}{2}\right)^2 h$$

$$h = \frac{4 \times 13 \times 13 \times 13 \times 3 \times 2 \times 2}{3 \times 13 \times 13} = 208 \text{ cm}$$

Sol.416.(a)

Ratio of sides of triangle = $10x : 24x : 26x$



$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height} = 480 \text{ cm}^2$$

$$= \frac{1}{2} \times 24x \times 10x = 480 \text{ cm}^2$$

$$x^2 = 4 \text{ cm} \Rightarrow x = 2 \text{ cm}$$

$$\text{Perimeter} = 20 + 48 + 52 = 120 \text{ cm}$$

Sol.417.(a)

Statement 1 :- volume of cuboid

$$= \text{Length} \times \text{Breadth} \times \text{height}$$

$$= 6 \times 7 \times 8 = 336 \text{ cm}^3$$

Statement 2 :- volume of cube

$$= (\text{Side})^3 = (7)^3 = 343 \text{ cm}^3$$

Clearly, the cube has more volume

Sol.418.(a) Curved surface area of hemisphere = $2\pi r^2$

$$C.S.A. = 2 \times \frac{22}{7} \times 42 \times 42 = 11088 \text{ cm}^2$$

$$\text{Cost of tin painting} = 11088 \text{ cm}^2 \times \frac{21}{100} = 2328.48 \text{ Rs.}$$

Sol.419.(d) Area of equilateral Triangle

$$= \frac{\sqrt{3}}{4} \times (\text{side})^2 = \frac{\sqrt{3}}{4} \times (24)^2$$

$$= 144\sqrt{3} \text{ cm}^2$$

Sol.420.(a)

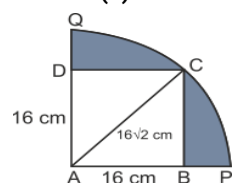
Height of the cylinder = 3.5 cm

Then radius of the cylinder = 17.5 cm

Volume of cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times (17.5)^2 \times 3.5 = 3368.75 \text{ cm}^3$$

Sol.421.(d)



$$\text{Area of square} = (\text{Side})^2 = 16^2 = 256 \text{ cm}^2$$

$$\text{Diagonal of square} = \sqrt{2} \times \text{Side} = 16\sqrt{2} \text{ cm}$$

Here, diagonal of square is the radius for the quadrant,

$$\text{So, area of the quadrant} = \frac{1}{4} \times \pi r^2$$

$$= \frac{1}{4} \times 3.14 \times 16\sqrt{2} \times 16\sqrt{2} = 401.92 \text{ cm}^2$$

Now, Area of shaded portion

$$= \text{Area of quadrant} - \text{area of square}$$

$$= 401.92 \text{ cm}^2 - 256 \text{ cm}^2 = 145.92 \text{ cm}^2$$

$$\text{Sol.422.(b)} \text{ Volume of sphere} = \frac{4}{3} \times \pi r^3$$

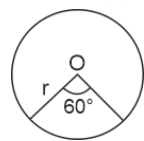
$$\text{Now, } 24416.64 = \frac{4}{3} \times 3.14 \times r^3$$

$$r^3 = \frac{7776 \times 3}{4} = 5832 \Rightarrow r = 18 \text{ cm}$$

$$\text{Now, surface area of sphere} = 4\pi r^2$$

$$= 4 \times 3.14 \times 18 \times 18 = 4069.44 \text{ cm}^2$$

Sol.423.(d)



$$\text{Area of sector} = \frac{\theta}{360} \times \pi r^2$$

$$66 \text{ cm}^2 = \frac{60}{360} \times \frac{22}{7} \times r^2$$

$$r^2 = \frac{66 \times 6 \times 7}{22} = 3 \times 6 \times 7 \Rightarrow r = 3\sqrt{14} \text{ cm}$$

Sol.424.(d) Height = 2.1 cm

According to question, radius = 6.3 cm

Volume of cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times 6.3 \times 6.3 \times 2.1 = 261.95 \text{ cm}^3$$

Sol.425.(d) If any solid object cast into another solid object always equates their Volume.

$$\text{Volume of sphere} = \text{Volume of cylinder}$$

$$\frac{4}{3} \pi r_1^3 = \pi r_2^2 h$$

$$\frac{4}{3} \times \frac{22}{7} \times 9^3 = \frac{22}{7} \times 2^2 \times h$$

$$h = \frac{729}{3} = 243 \text{ cm}$$

Sol.426.(a) Side of given cube = 5.6 cm.

For largest sphere to be inscribed in the cube,

diameter of sphere should be equal to edge length of cube,

$$\Rightarrow 2 \times \text{radius} = 5.6 \text{ cm} \Rightarrow \text{radius (r)} = 2.8$$

$$\text{Volume of the sphere} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \pi (2.8)^3 = 91.98 \text{ cm}^3$$

Sol.427.(a) Volume of hollow sphere

$$= \frac{4}{3} \pi (R^3 - r^3) = \frac{4}{3} \pi (4^3 - 1^3) = 84\pi \text{ cm}^3$$

Sol.428.(b) Diagonal of cube = $\text{side} \times \sqrt{3}$

Side of cube = 6 cm.

Three cube of equal volume are joined end to end, then a cuboid is formed, having length=18 cm, breadth = 6 cm and height = 6 cm

Now, Surface area of cuboid

$$= 2(lb + bh + hl)$$

Surface area of cuboid

$$= 2(108 + 36 + 108) = 504 \text{ cm}^2$$

Sol.429.(b) When a 3D figure is cast into another 3D figure, then their volume is equal. According to the Question,

$$\frac{C.S.A. \text{ of cylinder}}{T.S.A. \text{ of cylinder}} = \frac{2}{5}$$

$$\frac{2\pi rh}{2\pi r(h+r)} = \frac{2}{5}$$

$$\frac{h}{(h+r)} = \frac{2}{5} \Rightarrow r = 3x$$

Let height is 2x then radius be 3x

Again, Volume of solid hemisphere

$$= \text{volume of cylinder} = \frac{2}{3} \pi R^3 = \pi r^2 h$$

$$\frac{2}{3} \times 21 \times 21 \times 21 = 9x^2 \times 2x$$

$$x^3 = \frac{2 \times 7 \times 21 \times 21}{9 \times 2} \Rightarrow x = 7,$$

$$\text{then radius} = 3x = 3 \times 7 = 21 \text{ cm}$$

$$\text{Sol.430.(c)} \text{ Area of sector} = \frac{\theta}{360} \times \pi r^2$$

$$\text{Area} = \frac{90^\circ}{360^\circ} \times 3.14 \times 10^2 = 78.5 \text{ m}^2$$

Sol.431.(c) Circumference of circle = $2\pi r$

Ratio of circumference of circle is

directly proportional to ratio of radius of

circle. $R : r = 5 : 4$

Width of path

$$= (\text{outer radius} - \text{inner radius}) = 1 \text{ unit}$$

According to question, 1 unit = 50 m

Then, Inner radius (r) = 200 m

Sol.432.(d)

$$\text{Diagonal of the first square} = 8\sqrt{2}$$

$$\text{Side of the first square} = \frac{8\sqrt{2}}{\sqrt{2}} = 8 \text{ cm}$$

$$\text{Area of the square} = 8 \times 8 = 64 \text{ cm}^2$$

$$\text{Area of the new square} = 64 \times 3 = 192 \text{ cm}^2$$

$$\text{Side of the new square} = \sqrt{192} = 8\sqrt{3} \text{ cm}$$

Diagonal of the new square

$$= 8\sqrt{3} \times \sqrt{2} = 8\sqrt{6} \text{ cm}$$

Sol.433.(b) Given, inner radius = 6 cm, thickness = 0.37 cm

Outer radius = 6 + 0.37 = 6.37 cm

Outside curved surface area = $2\pi R^2$

$$= 2 \times \frac{22}{7} \times 6.37 \times 6.37 = 255.0548 \text{ cm}^2$$

Sol.434.(a)

Old : New

Length $\rightarrow 5 : 6$

Breadth $\rightarrow 4 : 5$

Height $\rightarrow 10 : 7$

Volume $\rightarrow 20 : 21$

Percentage increase in volume

$$= \frac{1}{20} \times 100 = 5\%$$

Sol.435.(b) Given, height of cone = 8 m and slant height = 10 m

$$\text{Radius} = \sqrt{10^2 - 8^2} = 6 \text{ m}$$

C.S.A of cone = $\pi R l$

$$= \pi \times 6 \times 10 = 60\pi \text{ m}^2$$

Sol.436.(d) Given, diameter of the rod

= 4 cm, radius = 2 cm and height = 12 cm

Volume of cylinder = $\pi R^2 h$

$$= 2 \times 2 \times 12 \times \pi = 48\pi$$

Now, new length = 12m = 1200 cm

$$\pi r^2 \times 1200 = 48\pi$$

$$r = \sqrt{\frac{48}{1200}} = 0.2 \text{ cm}$$

So, the thickness of the wire is 0.4 cm

Sol.437.(c) Volume of cone

$$= \frac{1}{3} \pi \times \left(\frac{42}{2}\right)^2 \times 42 = 6174\pi$$

Volume of sphere

$$= \frac{4}{3} \pi \times (24)^3 = 18,432\pi$$

The amount of wood wasted

$$= 18,432\pi - 6174\pi = 12,258\pi$$

Required % of wood wasted

$$= \frac{12,258\pi}{18,432\pi} \times 100 = 66.50\%$$

Sol.438.(a) diameter the largest sphere that can be taken out from cube will be 2.8cm

$$\begin{aligned}\text{Volume of sphere} &= \frac{4}{3} \times \frac{22}{7} \times \left(\frac{2.8}{2}\right)^3 \\ &= \frac{34.496}{3} \approx 11.50 \text{ cm}^3\end{aligned}$$

Sol.439.(c) CSA of cone = $\pi r l$

$$594 = \frac{22}{7} \times 21 \times l$$

$$594 = 66 \times l \Rightarrow l = \frac{594}{66} = 9 \text{ cm}$$

Sol.440.(a)

Let number of solid sphere be 'n'

$$n \times \frac{4}{3} \pi \times \left(\frac{6}{2}\right)^3 = \pi \times \left(\frac{4}{2}\right)^2 \times 90$$

$$n \times \frac{4}{3} \pi \times 27 = \pi \times 4 \times 90$$

$$n \times 36\pi = 360\pi \Rightarrow n = \frac{360}{36} = 10$$

Sol.441.(c) It is given that height = 14 cm

So radius = $4 \times 14 = 56 \text{ cm}$

Volume of cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times 56 \times 56 \times 14 = 1,37,984 \text{ cm}^3$$

Sol.442.(b) Volume of a frustum

$$= \frac{1}{3} \times \pi \times h(R^2 + r^2 + R \times r)$$

$$= \frac{1}{3} \times \frac{22}{7} \times 7(5^2 + 3^2 + 5 \times 3)$$

$$= \frac{22}{3}(25 + 9 + 15)$$

$$= \frac{22 \times 49}{3} = \frac{1078}{3} = 359.3 \text{ cm}^3$$

Sol.443.(d)

Let a be the edge of the cubical room.

ATQ, Area of the floor = $a^2 = 64 \Rightarrow a = 8 \text{ m}$

So,

length of the longest rod = $\sqrt{3} \times 8 = 8\sqrt{3} \text{ m}$

Sol.444.(a)

Let the height of water raised be h

ATQ,

(Volume of conical = (Volume of cylindrical vessel))

$$\frac{1}{3} \pi \times 20 \times 20 \times 27 = \pi \times 15 \times 15 \times h$$

$$3600 = 225h \Rightarrow h = \frac{3600}{225} = 16 \text{ cm}$$

Sol.445.(d) CSA of cylinder = $2\pi RH$

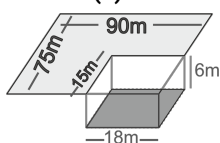
Volume of cylinder = $\pi R^2 H$

ATQ,

$$2 \times 2\pi RH = \pi R^2 H, \Rightarrow R = 4$$

So, radius of cylinder = 4 unit

Sol.446.(b)



Let the rise in the level of the field be h

The area of field = $90 \times 75 = 6750 \text{ m}^2$

Area of pit = $18 \times 15 = 270 \text{ m}^2$

Area of the remaining part of the field

$$= 6750 - 270 = 6480 \text{ m}^2$$

ATQ, Volume of earth taken out = Volume

of earth spread over remaining part of field $270 \times 6 = 6480 \times h \Rightarrow 1620 = 6480 h$

$$h = \frac{1620}{6480} = \frac{1}{4} \text{ m} = 25 \text{ cm}$$

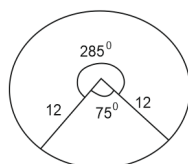
Sol.447.(d) Area of sector OAB

$$= \frac{\text{length of arc} \times \text{radius}}{2} = \frac{2r \times r}{2} = r^2$$

Sol.448.(a) TSA of hemisphere = $3\pi r^2$

$$= 3\pi \times \frac{25}{\pi} = 75 \text{ unit}^2$$

Sol.449.(b)



Perimeter of major sector

$$= \frac{285^\circ}{360^\circ} \times 2\pi \times 12 + 12 \times 2 = (19\pi + 24)\text{m}$$

Sol.450.(b)

Let the no of sphere formed be N

Volume of frustum = N $\times$ Volume of sphere

$$\frac{1}{3} \pi(R^2 + r^2 + Rr)h_1 = N \times \frac{4}{3} \pi \times R_1^3$$

$$(25^2 + 20^2 + 500)18 = 4N$$

$$(625 + 400 + 500)18 = 4N$$

$$1525 \times 18 = 4N \Rightarrow N = \frac{1525 \times 18}{4} = 6862$$

Sol.451.(c) C.S.A of hemisphere = $2\pi R^2$

75% of CSA of hemisphere = $\frac{3}{4} \times 2 \times$

$$\frac{22}{7} \times 21 \times 21 = 2079 \approx 2080 \text{ cm}^2$$

Sol.452.(b) Given, radius of the cylinder

= 7 cm, height = 8 cm

TSA of cylinder = $2\pi r(r + h)$

$$= 2 \times \frac{22}{7} \times 7 \times (7 + 8) = 44 \times 15 = 660 \text{ cm}^2$$

Sol.453.(a) TSA of hemisphere = $3\pi r^2$,

$$462 = 3 \times \frac{22}{7} \times r^2$$

$$r^2 = \frac{462 \times 7}{66} = 49 \Rightarrow r = 7 \text{ cm}$$

Volume of hemisphere = Volume of cone

$$\frac{2}{3} \pi \times 7^3 = \frac{1}{3} \pi \times 7^2 \times h$$

$$686 = 49h \Rightarrow h = \frac{686}{49} = 14 \text{ cm}$$

Sol.454.(b) Let the radius of each of the smaller semicircle be r

Then, the diameter of larger semicircle = 6r

The area of larger semicircle

$$= \frac{\pi}{2} \times \left(\frac{6r}{2}\right)^2 = 4.5\pi r^2$$

The area of unshaded region

$$= 3 \times \frac{1}{2} \pi \times r^2 = 1.5\pi r^2$$

The required ratio = $1.5\pi r^2 : \pi r^2(4.5 - 1.5)$

$$= 1.5\pi r^2 : 3\pi r^2 = 1 : 2$$

Sol.455.(b) Length of the longest pole

$$= \sqrt{60^2 + 30^2 + 20^2}$$

$$= \sqrt{3600 + 900 + 400} = \sqrt{4900} = 70 \text{ feet}$$

Sol.456.(c) Let the breadth of wall be x

Then, height of wall = 6x

And length of wall = 3x

ATQ,

$$3x \times x \times 6x = 23958 \Rightarrow 18x^3 = 23958$$

$$x^3 = \frac{23958}{18} = 1331 \Rightarrow x = 11 \text{ cm}$$

Sol.457.(c)

CSA of cylinder = $2\pi rh = 2200$ ----- (1)

Circumference at the base of cylinder = 2

$$\pi r = 110$$
 ----- (2)

Dividing eqn(1) and (2) we have :

$$h = \frac{2200}{110} = 20 \text{ cm}$$

Sol.458.(d)

Let the breadth of a cold storage be x

Then, length of the cold storage = 5x

The area of four walls = $2(l+b) \times h$

$$720 = 2(5x + x) \times 8$$

$$45 = 6x \Rightarrow x = \frac{45}{6} = 7.5 \text{ m}$$

Now,

$$2lb = 2 \times 5x \times x = 10x^2 = 10 \times 7.5^2 = 562.5$$

TSA of cold storage = $2(l + b) \times h + 2lb$

$$= 720 + 562.5 = 1282.5 \text{ m}^2$$

Sol.459.(a) ATQ, r = 6 cm, h = 8 cm

Slant height of cone

$$= \sqrt{6^2 + 8^2} = \sqrt{100} = 10 \text{ cm}$$

Total CSA of cones

$$= 3 \times \pi \times 6 \times 10 = 180\pi \text{ cm}^2$$

Sol.460.(c) ATQ,

$$\frac{2\pi R_1}{2\pi R_2} = \frac{5}{3} \Rightarrow \frac{R_1}{R_2} = \frac{5}{3}$$

$$\text{So, } \frac{\text{volume of cone 1}}{\text{volume of cone 2}} = \frac{\frac{1}{3}\pi R_1^2 h_1}{\frac{1}{3}\pi R_2^2 h_2}$$

$$= \frac{5^2 \times 1}{3^2 \times 5} = \frac{25}{45} = \frac{5}{9}$$

Sol.461.(a) The area of sector

$$= \frac{\text{length of arc} \times \text{radius}}{2} \Rightarrow 128 \times 2 = 64r$$

$$256 = 64r$$

$$\Rightarrow r = \frac{256}{64} = 4 \text{ cm}$$

Sol.462.(d)

Volume of cone-shaped storage

$$= \frac{1}{3} \times 3.14 \times 7^2 \times 9 = 461.58 \text{ cubic feet}$$

Sol.463.(d) TSA of hall to be painted

$$= 2(l + b) \times h + l \times b$$

$$= 2(10 + 5) \times 6 + 10 \times 5 = 2 \times 15 \times 6 + 50$$

$$= 180 + 50 = 230 \text{ m}^2$$

Sol.464.(d) Volume of hemispherical bowl = volume of cylindrical bottles

$$\frac{2}{3} \pi R^3 = n \times \pi r^2 h$$

Required number of bottles

$$= \frac{2 \times 6 \times 6 \times 6}{3 \times 1 \times 1 \times 4} = 36 \text{ bottles}$$

Sol.465.(d) Given, total surface area of hemisphere = 462 cm²

$$3 \pi R^2 = 462 \Rightarrow R = \sqrt{\frac{462 \times 7}{3 \times 22}} = 7$$

So, the diameter of the hemisphere is 14 cm

Sol.466.(c)

Given, s - a = 18, s - b = 17 and s - c = 25

$$S = \frac{a + b + c}{2} \Rightarrow a + b + c = 2s$$

Now, s - a + s - b + s - c = 60

$$3s - (a + b + c) = 60$$

$$3s - 2s = 60 \Rightarrow s = 60$$

So, area of the triangle

$$= \sqrt{s(s - a)(s - b)(s - c)}$$

$$= \sqrt{60 \times 18 \times 17 \times 25} = 30\sqrt{510} \text{ cm}^2$$

Sol.467.(c)

CSA + TSA of cylinder = 2068 cm²

$$2\pi rh + (2\pi rh + 2\pi r^2) = 2068$$

$$2\pi r(2h + r) = 2068$$

$$2 \times \frac{22}{7} \times 7(2h + 7) = 2068$$

$$2h + 7 = \frac{2068}{44} = 47$$

$$2h = 40 \Rightarrow h = 20 \text{ cm}$$

So, the volume of cylinder

$$= \frac{22}{7} \times 7^2 \times 20 = 3080 \text{ cm}^3$$

Sol.468.(d) Slant height l = 20 cm

According to the question,

Area of base of cone = 616 cm²

$$\pi r^2 = 616 \Rightarrow \frac{22}{7} \times r^2 = 616$$

$$r^2 = 196 \Rightarrow r = 14 \text{ cm}$$

So, CSA of cone = $\pi r l$

$$= \frac{22}{7} \times 14 \times 20 = 880 \text{ cm}^2$$

Sol.469.(d) Given that, r = 14 cm

According to the question,

Height h = r + 6 = 14 + 6 = 20 cm

So, volume of cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times 14^2 \times 20 = 12320 \text{ cm}^3$$

Sol.470.(b) Let the radius of a small solid

sphere be r.

According to the question,

Volume of large solid sphere

= 8 × Volume of Small solid sphere

$$\frac{4}{3} \pi R^3 = 8 \times \frac{4}{3} \pi r^3 \Rightarrow R^3 = 8 \times r^3$$

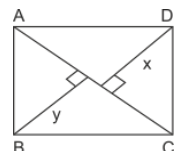
$$14^3 = 8 \times r^3 \Rightarrow r^3 = \frac{14^3}{8}$$

$$r = \frac{14}{2} = 7 \text{ cm}$$

So, TSA of all 8 small solid spheres

$$= 8 \times 4\pi r^2 = 8 \times 4 \times \frac{22}{7} \times 7^2 = 4928 \text{ cm}^2$$

Sol.471.(c) Let the length of the perpendicular drawn from point D to AC and point B to AC be x and y respectively. According to the question,



So, area of quadrilateral

= Area of $\triangle ABC$ + Area of $\triangle ADC$

$$= \frac{1}{2} \times AC \times y + \frac{1}{2} \times AC \times x$$

$$= \frac{1}{2} \times AC \times (x + y)$$

$$= \frac{1}{2} \times 24 \times (10) = 120 \text{ cm}^2$$

Sol.472.(b)

Volume of prism = Area of base × height

$$\left(\frac{1}{2} \times 9 \times 12\right) \times h = 648$$

$$54h = 648 \Rightarrow h = \frac{648}{54} = 12 \text{ cm}$$

So, the height of the prism = 12 cm.

Sol.473.(c) Let the radius of the inner and outer circle be r and R respectively.

According to the question,

Difference between the areas of two concentric circles = 264 cm²

$$\pi R^2 - \pi r^2 = 264 \Rightarrow \pi(R^2 - r^2) = 264$$

$$\frac{22}{7} \times (R^2 - r^2) = 264$$

$$R^2 - r^2 = 84 \text{ cm}^2$$

So, option (c) is the correct answer.

Sol.474.(b) According to the question,

Volume of cone = 400 cm³

Volume of cone with radius r and height

$$h = \frac{1}{3} \pi r^2 h = 400$$

$$\frac{1}{3} \times \frac{22}{7} \times r^2 h = 400 \Rightarrow r^2 h = \frac{4200}{11}$$

Volume of cone with radius 2r and height

$$h = \frac{1}{3} \pi (2r)^2 h = \frac{4}{3} \pi r^2 h$$

$$= \frac{4}{3} \times \frac{22}{7} \times \frac{4200}{11} = 1600 \text{ cm}^3$$

Sol.475.(d)

Given, radius of the cone = 4 cm,

C.S.A of cone = $\pi r l$

$$\pi r l = 20\pi \Rightarrow l = \frac{20}{4} = 5 \text{ cm}$$

Now, height of the cone = $\sqrt{(l)^2 - (r)^2}$

$$= \sqrt{25 - 16} = 3 \text{ cm}$$

Volume of the conical vessel = $\frac{1}{3} \pi r^2 h$

$$= \frac{1}{3} \times \pi \times 4 \times 4 \times 3 = 16\pi \text{ cm}^3$$

Sol.476.(c) Area of rhombus = $\frac{1}{2} \times d_1 \times d_2$

$$= \frac{1}{2} \times 6 \times 8 = 24 \text{ cm}^2$$

Sol.477.(c) Volume of a cube = a³

Where a = side of a cube

$$\frac{\text{Side of smaller cube}}{\text{Side of larger cube}} = \sqrt[3]{\frac{\text{Volume of smaller cube}}{\text{Volume of larger cube}}}$$

$$\frac{\text{Side of smaller cube}}{\text{Side of larger cube}} = \sqrt[3]{\frac{8}{27}} = \frac{2}{3}$$

Then, side of smaller cube = 2x

and side of larger cube = 3x

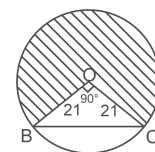
According to question,

$$27x^3 - 8x^3 = 513 \text{ cm}^3 \Rightarrow 19x^3 = 513$$

$$x^3 = 27 \Rightarrow x = 3$$

So, Side of smaller cube = 2 × 3 = 6 cm

Sol.478.(c)



Area of minor sector(OBC)

$$= \frac{90}{360} \times \frac{22}{7} \times 21 \times 21 = 346.5 \text{ cm}^2$$

Area of $\triangle BOC$

$$= \frac{1}{2} \times 21 \times 21 \times \sin 90^\circ = 220.5 \text{ cm}^2$$

Area of minor segment

$$= 346.5 - 220.5 = 126 \text{ cm}^2$$

Now, area of major segment = area of circle - area of minor segment

$$= \frac{22}{7} \times 21^2 - 126 = 1386 - 126 = 1260 \text{ cm}^2$$

Short Trick:- Area of major segment :- area of major sector + area of triangle

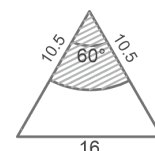
$$\text{Required area :- } \frac{270}{360} \times \frac{22}{7} \times 21 \times 21 +$$

$$\frac{1}{2} \times 21 \times 21 \times \sin 90^\circ$$

$$\text{Required area :- } \frac{21}{2} (99 + 21)$$

$$= 60 \times 21 = 1260 \text{ cm}^2$$

Sol.479.(c) As it is given that plot is in the shape of an equilateral triangle then, the grazed area is part of circle of radius 10.5m. so, angle is 60°



Hence, required area

$$= \frac{60^\circ}{360^\circ} \times \frac{22}{7} \times 10.5 \times 10.5$$

$$= \frac{2425.5}{42} = 57.75 m^2$$

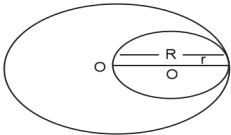
Sol.480.(d) Let the radius and height of the cone be r and h respectively.

ATQ,

$$\frac{\text{New volume}}{\text{Original volume}} = \frac{\frac{1}{3}\pi \times 4r^2 \times 2h}{\frac{1}{3}\pi \times r^2 \times h} = 8$$

So, the volume of the cone becomes 8 times its previous volume.

Sol.481.(a)



Area of the smaller circle = 192

$$\pi r^2 = 192 \Rightarrow \pi \times \left(\frac{R}{2}\right)^2 = 192$$

$$\pi R^2 = 192 \times 4 = 768$$

So, area of the bigger circle = $768 cm^2$

Sol.482.(d) Let the length of the rectangle be L cm and the side of the square be a cm.

Area of square = $400 cm^2$

$$a^2 = 400 \Rightarrow a = 20 \text{ cm}$$

ATQ, Perimeter of square = perimeter of rectangle

$$4a = 2(L + 12) \Rightarrow 4 \times 20 = 2(L + 12)$$

$$40 = L + 12 \Rightarrow L = 40 - 12 = 28$$

So, the area of rectangle

$$= 28 \times 12 = 336 cm^2$$

Sol.483.(d) ATQ,

TSA of Sphere = TSA of cube

$$4\pi r^2 = 6s^2 \Rightarrow 4 \times \frac{22}{7} \times r^2 = 6s^2$$

$$\frac{r^2}{s^2} = \frac{6 \times 7}{4 \times 22} = \frac{42}{88} = \frac{21}{44}$$

So, we can see that $r < s$

Sol.484.(c) The ratio of length, breadth and height of the wall = $3 : 1 : 6$,

Let the length, breadth and height of the wall be $3x$, x and $6x$ respectively

ATQ,

The volume of wall = 23958

$$6x \times 3x \times x = 23958$$

$$18x^3 = 23958 \Rightarrow x^3 = \frac{23958}{18} = 1331$$

$$x = \sqrt[3]{1331} = 11 \text{ cm}$$

Sol.485.(b) Area of trapezium

$$= \frac{1}{2} \times (\text{sum of parallel side}) \times \text{height}$$

$$= \frac{1}{2} \times 12 \times (15 + 20) = 6 \times 35 = 210 cm^2$$

Sol.486.(b)

Diameter of a wire = 1 mm = 0.1 cm

$$\text{Radius of a wire} = \frac{0.1}{2} = 0.05 \text{ cm}$$

Let the no of gold chain required be n
ATQ,

Volume of wire $\times n$ = Volume of gold

$$\frac{22}{7} \times 0.05 \times 0.05 \times 56 \times n = 110$$

$$n = \frac{110 \times 7}{22 \times 0.05 \times 0.05 \times 56} = 250$$

Sol.487.(d) The outer dimensions of a closed metal box is 15 cm, 18 cm, 8 cm

Thickness of the metal = 1.5 cm

So, the inner dimension of the metal box-

$$= 15 - 1.5 \times 2 = 15 - 3 = 12 \text{ cm},$$

$$= 18 - 1.5 \times 2 = 18 - 3 = 15 \text{ cm},$$

$$= 8 - 1.5 \times 2 = 8 - 3 = 5 \text{ cm}.$$

So, the required volume of the metal

$$= 15 \times 18 \times 8 - (12 \times 15 \times 5)$$

$$= 2160 - 900 = 1260 cm^3$$

Sol.488.(b)

Perimeter of an equilateral triangle = $3a$

Length of equilateral triangle = a

ATQ,

$$3a - a = 120 \Rightarrow 2a = 120 \Rightarrow a = 60 \text{ cm}$$

So, the area of equilateral triangle -

$$= \frac{\sqrt{3}}{4} \times 60 \times 60 = 900\sqrt{3} cm^2$$

Sol.489.(b) It is given that,

$$L + B = 37 - (1), \quad L - B = 13 - (2)$$

Adding eqn. (1) and (2), we have :

$$L = \frac{37 + 13}{2} = \frac{50}{2} = 25 \text{ cm}, \quad B = 12 \text{ cm}$$

ATQ, Volume of cuboid = $25 \times 12 \times H$

$$3300 = 300H \Rightarrow H = \frac{3300}{300} = 11 \text{ cm}$$

Sol.490.(a) Area of the rectangle -

$$= 40 \times 35 = 1400 cm^2$$

We can cut the biggest square by taking side of breadth of rectangle

Breadth of rectangle = 35 cm

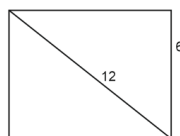
Then area of the square -

$$= 35 \times 35 = 1225 cm^2$$

Area of remaining part-

$$= 1400 - 1225 = 175 cm^2$$

Sol.491.(c) In rectangle



$$\text{Another side} = \sqrt{12^2 - 6^2} = \sqrt{144 - 36}$$

$$= \sqrt{108} = 6\sqrt{3}$$

$$\text{Area of rectangle} = l \times b = 6\sqrt{3} \times 6 = 36\sqrt{3}$$

Sol.492.(c) Sphere is melted and recast into pyramid

Then the volume of sphere = volume of pyramid

$$\frac{4}{3}\pi r^3 = \frac{1}{3} \times (\text{area of base}) \times \text{height}$$

Let side of pyramid having square base = a

$$4 \times \frac{22}{7} \times 21^3 = a^2 \times 66$$

{ area of square = (side)² }

$$a^2 = 4 \times \frac{22}{66 \times 7} \times 21 \times 21 \times 21$$

$$a^2 = 4 \times 21 \times 21$$

$$\text{Therefore, } a = \sqrt{21 \times 21 \times 4} = 42 \text{ cm}$$

Sol.493.(a) According to the question,

The total surface area of cube having 8 cm side = area of a square + $128 cm^2$

Let side of square is, X

$$6 \times 8 \times 8 = X^2 + 128$$

$$X^2 + 128 = 384$$

$$X^2 = 256 \Rightarrow X = \sqrt{256} \Rightarrow 16 \text{ cm}$$

Sol.494.(d) According to the question, height of cylinder = 4 m

$$\text{Radius of cone and cylinder}(r) = \frac{10.5}{2} \text{ m}$$

And slant height of cone (l) = 10 m

Canvas used for all tents = $3861 cm^2$

Let number of tents = N ,

$$= N \times 3861$$

$$= N \times [\pi r(l + 2h)] = 3861$$

$$= N \times \left[\frac{22}{7} \times \frac{10.5}{2} (10 + 8)\right] = 3861$$

$$= N \times \frac{22}{7} \times \frac{10.5}{2} \times 18 = 3861 \Rightarrow N = 13$$

Sol.495.(a)

Area of a rectangle = $l \times b = 480$

Perimeter of rectangle = $2(l + b) = 92$

$$l + b = 46 = 30 + 16$$

$$l \times b = 480 = 30 \times 16$$

So clearly length is 30 cm and breadth is- 16 cm

Now diagonal of a rectangle

$$= \sqrt{l^2 + b^2} = \sqrt{30^2 + 16^2}$$

$$= \sqrt{1156} = 34 \text{ cm}$$

Sol.496.(a) According to question,

Circumference of circle = perimeter of square,

$$\Rightarrow 2\pi r = 4a \Rightarrow r = \frac{2a}{\pi}$$

Area of circle = πr^2

$$= \pi \times \left(\frac{2a}{\pi}\right)^2 = \frac{4a^2}{\pi} = \frac{14}{11} \times a^2$$

Area of square = a^2

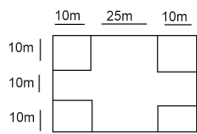
Hence, we can see that the area of a circle is more than the area of a square.

Sol.497.(d) If length is $2x$ then breadth is x

Perimeter of rectangle = $2(l + b) = 54$

$$2(2x + x) = 54 \Rightarrow x = 9$$

$$\text{Length} = 2x = 2 \times 9 = 18 \text{ cm}$$

Sol.498.(d)

When the 4 square shaped cuts are made in rectangular sheet then
Length = 25m, Breadth = 10m,
Height = 10m

$$\text{Volume} = l \times b \times h = 25m \times 10m \times 10m = 2500 m^3$$

Sol.499.(c)

$$\text{Total surface area of hemisphere} = 3\pi R^2$$

$$R = \sqrt{\frac{16632}{3\pi}} = 42 \text{ cm}, D = 2R = 84 \text{ cm}$$

$$\text{Radius} + \text{Diameter} = 42 + 84 = 126 \text{ cm}$$

$$\text{Sol.500.(a) Side of the cube} = \frac{50\sqrt{3}}{\sqrt{3}} = 50$$

$$\text{Total surface area of cube} = 6 \text{ side}^2 = 6 \times 50 \times 50 = 15000 \text{ cm}^2$$

Sol.501.(c) Let the smaller diagonal be x and larger diagonal be $= \frac{150}{100}x \Rightarrow \frac{3}{2}x$

$$432 = \frac{1}{2} \times \frac{3}{2}x^2$$

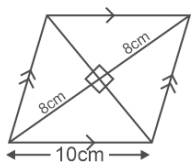
$$x = \sqrt{\frac{432 \times 4}{3}} = 24 \text{ cm}$$

So, $d_1 = 24 \text{ cm}$ and $d_2 = 36 \text{ cm}$
Side of the rhombus

$$= \sqrt{12^2 + 18^2} = 6\sqrt{13}$$

$$\text{Sol.502.(a) } 4\pi R^2 = 2464$$

$$R = \sqrt{\frac{2464 \times 7}{4 \times 22}} = 14 \text{ cm}$$

Sol.503.(d)

$$D_2 = 2 \times 6 = 12 \text{ cm}$$

$$\text{Area of rhombus} = \frac{1}{2} \times 16 \times 12 = 96 \text{ cm}^2$$

Sol.504.(b) T.S.A of cube (6side^2) = 150
side = 5 cm



Now,

$$\text{T.S.A of cuboid} = 2(10 \times 5 + 5 \times 5 + 10 \times 5) = 2(50 + 25 + 25) = 250 \text{ cm}^2$$

Sol.505.(b) According to the question,
Radius of circle = 84cm

Circumference of circle = perimeter of an equilateral triangle.

$$2\pi r = 3 \times a \quad (\text{let side of triangle} = a)$$

$$2 \times \frac{22}{7} \times 84 = 3 \times a$$

$$3 \times a = 2 \times 22 \times 12$$

$$a = 2 \times 22 \times 4 = 176 \text{ cm}$$

Sol.506.(a) Height of a cylinder = 10 cm

According to the question,
radius = $10 - 3 = 7 \text{ cm}$

$$\text{So, volume of cylinder} = \pi r^2 h$$

$$\frac{22}{7} \times 7 \times 7 \times 10 \Rightarrow 22 \times 7 \times 10 = 1540 \text{ cm}^3$$

Sol.507.(d) Area of large field = area of three smaller field

$$\Rightarrow \pi R^2 = \pi 8^2 + \pi 9^2 + \pi 12^2$$

$$\Rightarrow \pi R^2 = \pi(8^2 + 9^2 + 12^2)$$

$$\Rightarrow R^2 = 64 + 81 + 144$$

$$\Rightarrow \text{Radius (R)} = \sqrt{289} \Rightarrow 17 \text{ m}$$

Sol.508.(c)

$$\text{Radius of common base } r = 15 \text{ cm}$$

$$\text{Volume of toy} = \frac{2}{3} \pi r^3 + \frac{1}{3} \pi r^2 h$$

$$\Rightarrow 2850 \pi = \frac{1}{3} \pi r^2 (2r + h)$$

$$\Rightarrow \frac{1}{3} \times 15 \times 15 \times (30 + h) = 2850$$

$$\Rightarrow (30 + h) = 38 \Rightarrow h = 8$$

Hence, by Pythagoras

$$L(\text{Slant Height}) = \sqrt{15^2 + 8^2} = 17 \text{ cm}$$

Therefore,

$$\text{Surface area of toy} = 2\pi r^2 + \pi r l$$

$$\Rightarrow \pi r (2r + l) \Rightarrow \pi \times 15 \times (30 + 17) = 705\pi \text{ cm}^2$$

Sol.509.(c) According to the question,

$$\text{Side of the large cube} = 12 \text{ cm}$$

$$\text{Radius of small spheres} = 3 \text{ cm}$$

$$\text{So} \Rightarrow 12 \times 12 \times 12$$

$$= N \times \frac{4}{3} \pi r^3 + (N + 2) 4 \times 4 \times 6.5$$

$$= N \times \frac{4}{3} \pi r^3 + (N + 2) 4 \times 4 \times 6.5 = 1728$$

$$= N \times \frac{4}{3} \times \frac{22}{7} \times 3 \times 3 \times 3 + 104N + 208 = 1728$$

$$N \left[\frac{792}{7} + 104 \right] = 1728 - 208$$

$$N \left[\frac{792 + 728}{7} \right] = 1520$$

$$N \left[\frac{1520}{7} \right] = 1520 \Rightarrow N = 7$$

Sol.510.(c) Area of trapezium

$$= \frac{1}{2} (68 \times 75) = 2550 \text{ cm}^2$$

$$\text{Area of square} = \frac{2550 \times 17}{6} = 7225 \text{ cm}^2$$

$$\text{Side of square} = \sqrt{7225} = 85 \text{ cm}$$

$$\text{Diagonal of square} = \sqrt{2} \times 85 = 119.85 \text{ cm}$$

Sol.511.(a) L = 56 m, B = 34 m, H = 8 m

The length of the longest ladder

$$= \sqrt{56^2 + 34^2 + 8^2} = 66 \text{ m}$$

Sol.512.(d) $D = 2R = 7 \text{ cm}$

$$\text{Hence, } R = 3.5 \text{ cm}$$

$$\text{Total surface area} = 3\pi R^2$$

$$3 \times \frac{22}{7} \times 3.5 \times 3.5 = 115.5 \text{ cm}^2$$

Sol.513.(c) Let the length and width of the rectangle be $3x, 2x$

$$\text{Area of rectangle} = 6x^2$$

According to question,

$$\text{Now, } \left(\frac{125}{100} \times 3x \right) \times 2x - 6x^2 = 24$$

$$\frac{15}{2}x^2 - 6x^2 = 24 \Rightarrow x = \sqrt{\frac{24 \times 2}{3}} = 4$$

$$\text{Width of the rectangle} = 8 \text{ m}$$

Sol.514.(c) ratio of an external angle and an internal angle = 1 : 5

As per question,

$$x + 5x = 180 \Rightarrow x = 30$$

$$\text{Number of sides} = \frac{360}{30} = 12$$

Sol.515.(d) Total cost of painting four walls = $[2(80 + 70) \times 80] \times 0.5 = \text{Rs } 12,000$

Sol.516.(c) Area of a square = 1296

$$a^2 = 1296 \Rightarrow a = 36$$

$$\text{Perimeter of square} = 4a = 4 \times 36 = 144$$

$$\text{Radius} = 36 \times \frac{7}{6} = 42$$

Circumference of circle -

$$= 2 \times \frac{22}{7} \times 42 = 264$$

$$\text{Required ratio} = 144 : 264 = 6 : 11$$

Sol.517.(b) According to question,

$$\frac{2}{3} = \frac{4 \times 3 \times H_1}{5 \times 4 \times H_2} \Rightarrow H_1 : H_2 = 10 : 9$$

Sol.518.(d)

Let breadth of the rectangle = $4x$

Radius of the circle = $5x$

Side of the square = $25x$

$$\text{Area} = 625 \Rightarrow (25x)^2 = 625 \Rightarrow x = 1$$

So,

$$\text{breadth of the rectangle} = 4x = 4$$

$$\text{Length} = 20$$

$$= \text{Area of rectangle} = 4 \times 20 = 80 \text{ cm}^2$$

Sol.519.(a)

Let the diameter of circle = $7x$

Base of triangle = $4x$

$$\text{Area of the triangle} = \frac{1}{2} \times b \times h$$

$$= 56 = \frac{1}{2} \times 4x \times 14 \Rightarrow x = 2$$

$$\text{Diameter} = 7x = 14 \text{ cm} \Rightarrow \text{Radius} = 7 \text{ cm}$$

Circumference of circle -

$$= 2 \times \frac{22}{7} \times 7 = 44 \text{ cm or } 0.44 \text{ m}$$

Sol.520.(c)

Given length of side of square = p

$$\text{So, length of rectangle} = \frac{p}{2}$$

$$\text{Breadth of rectangle} = \frac{p}{6}$$

Now, required area = area of square -

$$\text{area of rectangle} = p^2 - \frac{p^2}{12} = \frac{11p^2}{12}$$

Sol.521.(d) Desired area

$$= \frac{22}{7} \times [(9)^2 - (5)^2] = 176 \text{ m}^2$$

Sol.522.(d)

Since, Pillar is cylindrical in shape.

So, to reach the required answer we have to find the volume of the cylinder.

$$\text{Volume of cylinder (pillar)} = \pi r^2 h$$

$$= \frac{22}{7} \times 11 \times 11 \times 21 = 7986 \text{ m}^3$$

Sol.523.(a) Diagonal of cube = diameter

$$\text{of sphere} = 14\sqrt{3} \text{ cm}$$

$$\text{Radius of sphere} = \frac{14\sqrt{3}}{2} = 7\sqrt{3} \text{ cm}$$

$$\text{Volume of Sphere} = \frac{4}{3} \times \frac{22}{7} \times 7\sqrt{3} \times$$

$$7\sqrt{3} \times 7\sqrt{3} = 4312\sqrt{3} \text{ cm}^3$$

Sol.524.(b) Capacity of the pond

$$= 50 \times 35 \times 2.5 = 4375 \text{ m}^3$$

Sol.525.(c) $1 \text{ m}^3 = 1000 \text{ litre}$

$$\text{Volume of tank} = 1.5 \times 3 \times 3.5 \times 1000 = 15750 \text{ litre.}$$

Sol.526.(b) Number of bricks

$$= \frac{8 \times 100 \times 6 \times 100 \times 22.5}{25 \times 11.25 \times 6} = 6,400$$

Sol.527.(a) Edge of cube = 10 cm

Height of the largest cone = Edge of the cone = 10 cm

$$\text{Radius of the largest cone} = \frac{10}{2} = 5 \text{ cm}$$

Volume of the largest cone

$$= \frac{1}{3} \times \pi \times 5 \times 5 \times 10 = \frac{250\pi}{3} \text{ cm}^3$$

Sol.528.(c) Volume of 15 spherical ball

$$= 15 \times \frac{4}{3} \times \pi \times 5^3 = 2500\pi \text{ cm}^3$$

Since 35% of the mass wasted,

So the volume of sphere

$$= 2500\pi \times \frac{65}{100} = 1625\pi \text{ cm}^3$$

Let the radius of new formed sphere = r

Volume of big Sphere = volume of new

$$\text{formed sphere} \Rightarrow 1625\pi = \frac{4}{3} \times \pi \times r^3$$

$$r^3 = \frac{1625 \times 3}{4} = \frac{4875}{4}$$

$$r = \sqrt[3]{1218.75} = 10.68 \text{ cm}$$

Sol.529.(c)

$$\text{Volume of sphere} = r^3 \left(\frac{4}{3}\pi \text{ is constant}\right)$$

$$\text{Original Volume } 343 = 7^3$$

$$\text{New Volume } 2744 = 14^3$$

$$\text{Required ratio} = 343 : 2744 = 1 : 8$$

Sol.530.(a)

Volume of cylinder = Volume of cone

$$\pi \times 15^2 \times 12 = \frac{1}{3} \times \pi \times 24^2 \times H$$

$$H = \frac{225 \times 36}{576} = \frac{8100}{576} = 14.0625 \text{ cm}$$

$$\text{Sol.531.(c) Radius} = \frac{14}{2} = 7 \text{ cm}$$

$$\text{TSA of cylinder} = 2\pi r(h + r)$$

$$= 2 \times \frac{22}{7} \times 7(25 + 7)$$

$$= 44 \times 32 = 1408 \text{ cm}^2$$

Sol.532.(c) According to the question,

Percentage contribution of hemisphere in

$$\text{total volume of toy} = 100 - (50 + 20)$$

$$= 100 - 70 = 30\%$$

$$\text{So, required ratio} = 20 : 50 : 30 = 2 : 5 : 3$$

Sol.533.(c) Circumference of the base of cylinder = $2\pi r$

$$\Rightarrow 4\pi = 2\pi r \Rightarrow r = 2 \text{ cm}$$

$$\text{CSA of cylinder} = 2\pi rh,$$

$$12\pi = 2 \times \pi \times 2 \times h \Rightarrow h = \frac{12}{4} = 3 \text{ cm}$$

$$\text{Volume of cylinder} = \pi \times 2 \times 2 \times 3 = 12\pi \text{ cm}^3$$

Sol.534.(b) $L + B + H = 23 \text{ cm}$ (Given)

Length of the diagonal of cuboid

$$= \sqrt{L^2 + B^2 + H^2}$$

$$\Rightarrow 5\sqrt{7} = \sqrt{L^2 + B^2 + H^2}$$

$$\Rightarrow 175 = L^2 + B^2 + H^2$$

$$\Rightarrow (L + B + H)^2 = L^2 + B^2 + H^2 + 2(LB + BH + HL)$$

$$\Rightarrow 23^2 = 175 + 2(LB + BH + HL)$$

$$\Rightarrow 529 = 175 + 2(LB + BH + HL)$$

$$\Rightarrow 354 = 2(LB + BH + HL)$$

$$\text{So, TSA of cuboid} = 354 \text{ cm}^2$$

Sol.535.(c)

CSA of cone-shaped temple = πrl

$$= \frac{22}{7} \times 11 \times 10 = \frac{2420}{7} \text{ m}^2$$

Total cost for painting its CSA

$$= 7 \times \frac{2420}{7} = 2420 \text{ Rs.}$$

Sol.536.(b)

Let the base of the triangle = 24cm

$$\text{Area of } \Delta = \frac{1}{2} \times 24 \times \text{height}$$

$$96 = 12 \times \text{height} \Rightarrow H = \frac{96}{12} = 8 \text{ cm}$$

Sol.537.(a) According to the question,

Length = Breadth + 2(Breadth)

$$= 8 + 2(8) = 8 + 16 = 24 \text{ cm}$$

Breadth = 2(Height)

$$8 = 2(\text{Height}), \text{ Height} = 4 \text{ cm}$$

Volume of cuboid = Length $\times$ Breadth $\times$

$$\text{Height} = 24 \times 8 \times 4 = 768 \text{ cm}^3$$

Sol.538.(a) Volume of a cube = a^3

$$64 = a^3 \Rightarrow a = 4 \text{ cm}$$

$$\text{T.S.A of cube} = 6 \times 4 \times 4 = 96 \text{ cm}^2$$

Sol.539.(b)

$$\text{Diagonal of cuboid} = \sqrt{l^2 + b^2 + h^2}$$

$$= \sqrt{20^2 + 5^2 + 4^2} = \sqrt{441} = 21 \text{ m}$$

Sol.540.(a) Volume of Hemisphere

$$= \frac{2}{3} \times \frac{22}{7} \times 7 \times 7 \times 7$$

$$= \frac{2156}{3} = 718\frac{2}{3} \text{ cm}^3$$

Sol.541.(d) Length of each side of hexagon = 10 cm

Area of Hexagon = 6 $\times$ area of equilateral triangle

$$\text{Area of Hexagon} = 6 \times \frac{\sqrt{3}}{4} \times 10 \times 10$$

$$= \frac{600\sqrt{3}}{4} = 150\sqrt{3} \text{ cm}^2$$

Sol.542.(a)

$$\text{Area of square} = 9m^2 + 12m + 4$$

$$= (3m)^2 + 2 \times 3m \times 2 + 2^2 \Rightarrow (3m + 2)^2$$

$$\text{Side of square} = \sqrt{(3m + 2)^2} = 3m + 2$$

Sol.543.(d)

Volume of cuboid = Volume of sphere

$$49 \times 33 \times 24 = \frac{4}{3} \times \frac{22}{7} \times r^3$$

$$r^3 = \frac{49 \times 33 \times 24 \times 21}{4 \times 22} = 27 \times 343$$

$$r = 3 \times 7 = 21 \text{ cm}$$

Sol.544.(b) Area of square = $4x^2 - 12x + 9$

$$= (2x)^2 - 2 \times 2x \times 3 + 3^2 = (2x - 3)^2$$

$$\text{Side of the square} = \sqrt{(2x - 3)^2} = 2x - 3$$

Sol.545.(a) $L \times B = 729 \text{ cm}^2$, $B \times H = 529$

$$\text{cm}^2, H \times L = 289 \text{ cm}^2$$

$$L^2 \times B^2 \times H^2 = 729 \times 529 \times 289$$

$$L \times B \times H = \sqrt{729 \times 529 \times 289}$$

$$= 27 \times 23 \times 17 = 10557 \text{ cm}^3$$

Sol.546.(c) TSA of cone = $\pi r(r + l)$

$$= \frac{22}{7} \times \frac{10}{2} \left(\frac{10}{2} + 16\right)$$

$$= \frac{22}{7} \times 5 \times 21 = 330 \text{ cm}^2$$

Sol.547.(b)

Diagonal of rhombus(D_1) = 8cm

$$\text{Area of rhombus} = \frac{1}{2} \times 8 \times D_2$$

$$48 = 4 \times D_2 \Rightarrow D_2 = \frac{48}{4} = 12 \text{ cm}$$

Now, side of rhombus

$$= \sqrt{\left(\frac{8}{2}\right)^2 + \left(\frac{12}{2}\right)^2} = \sqrt{16 + 36}$$

$$= \sqrt{52} = 2\sqrt{13} \text{ cm}$$

Sol.548.(c) TSA of cylinder = $2\pi r(r + h)$

$$= 2\pi r\left(r + \frac{r}{2}\right) = 2\pi r \times \frac{3r}{2} = 3\pi r^2$$

Sol.549.(b) Circumference of the base of hemispherical bowl = 132 cm

$$2 \times \frac{22}{7} \times r = 132 \Rightarrow \frac{44}{7} \times r = 132$$

$$r = \frac{132 \times 7}{44} = 21 \text{ cm}$$

The capacity of the bowl

$$= \frac{2}{3} \times \frac{22}{7} \times 21 \times 21 \times 21 = 19404 \text{ cm}^3$$

Sol.550.(a)

$$\text{Radius of cylinder} = \frac{14}{2} = 7 \text{ cm}$$

Diameter of sphere = height of cylinder
...(Since cylinder is circumscribing a sphere)

So, volume of cylinder

$$= \frac{22}{7} \times 7^2 \times 14 = 2156 \text{ cm}^3$$

Sol.551.(c) Surface area of spherical volleyball = 100π

$$4\pi r^2 = 100\pi \Rightarrow r^2 = 25 \Rightarrow r = 5$$

Volume of air

$$= \frac{4}{3} \times \pi \times 5 \times 5 \times 5 = \frac{500}{3}\pi$$

Sol.552.(c) The total surface area of a hemisphere = 64.57 cm^2

$$\Rightarrow 3\pi r^2 = 64.57$$

$$\Rightarrow 3 \times \frac{22}{7} \times r^2 = 64.57 \Rightarrow r^2 = 6.848$$

$$\text{Curved surface area} = 2 \times \frac{22}{7} \times 6.848$$

$$= 43.04 \text{ cm}^2$$

Short trick :- CSA of hemisphere =

$$\frac{64.57}{3} \times 2 = 43.04 \text{ cm}^2$$

Sol.553.(d) Length of diagonal of cuboid

$$= \sqrt{13^2 + 7^2 + 3^2} = \sqrt{169 + 49 + 9}$$

$$= \sqrt{227} \text{ m}$$

Sol.554.(b) Perimeter of an equilateral triangle = $3a$

$$75 = 3a \Rightarrow a = \frac{75}{3} = 25 \text{ cm}$$

Area of an equilateral triangle

$$= \frac{\sqrt{3}}{4} \times 25^2 = \frac{625\sqrt{3}}{4} \text{ cm}^2$$

Sol.555.(c) TSA of cylinder

$$= \text{CSA of cylinder} + 2 \times \text{Area of base}$$

$$= 225 + 2 \times 169 = 225 + 338 = 563 \text{ cm}^2$$

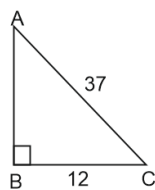
Sol.556.(d) Area of playground = CSA of roller $\times$ Number of complete revolutions

$$= 2\pi r h \times 235$$

$$= 2 \times \frac{22}{7} \times 14 \times 20 \times 235$$

$$= 413600 = 4136 \times 10^2 \text{ cm}^2$$

Sol.557.(d) According to the question,



$$AB^2 = \sqrt{AC^2 - BC^2} = \sqrt{37^2 - 12^2}$$

$$= \sqrt{1369 - 144} = \sqrt{1225} = 35 \text{ cm}$$

$$\text{Area of triangle} = \frac{1}{2} \times AB \times BC$$

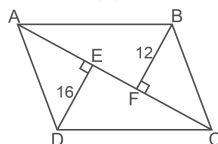
$$= \frac{1}{2} \times 35 \times 12 = 210 \text{ cm}^2$$

Sol.558.(b)

$$\text{Length of the arc} = \frac{\theta}{360} \times 2\pi r$$

$$= \frac{36}{360} \times 2 \times \frac{22}{7} \times 35 = 22 \text{ cm}$$

Sol.559.(d)



Area of ABCD

$$= \text{area of ADC} + \text{area of ABC}$$

$$336 = \frac{1}{2} \times AC \times DE + \frac{1}{2} \times AC \times BF$$

$$336 = \frac{1}{2} \times AC (DE + BF)$$

$$672 = AC(16 + 12)$$

$$672 = AC \times 28 \Rightarrow AC = 24 \text{ cm}$$

Sol.560.(d)

Let the number of spherical balls be n .

According to the question,

Volume of n spherical balls

= Volume of spherical clay ball

$$n \times \frac{4}{3} \pi r^3 = \frac{4}{3} \pi R^3 \Rightarrow n \times r^3 = R^3$$

$$n \times 5^3 = 15^3 \Rightarrow n = 3^3 = 27$$

Sol.561.(c) Total surface area = $2\pi r(h + r)$

$$60\pi = 2 \times \pi \times 3(h + 3)$$

$$10 = h + 3 \Rightarrow h = 10 - 3 = 7 \text{ cm}$$

Sol.562.(d) Let,

the rise of water level = a cm

According to the question,

Volume of cube = Length of rectangle $\times$

Breadth of rectangle $\times a$

$$= 8 \times 8 \times 8 = 16 \times 8 \times a \Rightarrow a = 4 \text{ cm}$$

Sol.563.(d)

Let, the rise of water level = a

According to the question,

Volume of cube = Length of rectangle $\times$

Breadth of rectangle $\times a$

$$8 \times 8 \times 8 = 16 \times 8 \times a \Rightarrow h = 4 \text{ cm}$$

Sol.564.(d)

Percentage change in area of rectangle

$$= 8 - 10 + \frac{8(-10)}{100}$$

$$= -2 + (-0.8) = -2.8\%$$

So there is a decrease of 2.8% in the area of the rectangle.

Sol.565.(c) Required difference = Volume of sphere - Volume of cone

$$= \frac{4}{3} \pi r^3 - \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi r^2 (4r - h)$$

$$= \frac{1}{3} \times \frac{22}{7} \times 7^2 \times (4 \times 7 - 10)$$

$$= \frac{1}{3} \times 154 \times (18) = 924 \text{ cm}^3$$

Sol.566.(b) Length of square

$$= \frac{\text{Diagonal}}{\sqrt{2}} = \frac{(a+b)}{\sqrt{2}}$$

$$\text{So, Area of square} = \left[\frac{(a+b)}{\sqrt{2}} \right]^2$$

$$= \frac{a^2 + b^2 + 2ab}{2} = \frac{1}{2}(a^2 + b^2) + ab$$

Sol.567.(b) According to the question,

$$y = 14x$$

Volume of sphere = $14 \times$ surface area of sphere

$$\frac{4}{3} \pi r^3 = 14 \times 4\pi r^2$$

$$\frac{1}{3} \times r = 14 \Rightarrow r = 42 \text{ cm}$$

Sol.568.(c) Curved surface area of the hemisphere = $2\pi r^2$

Base area of the hemisphere = πr^2

Effective area increase on each face of

the cube = $2\pi r^2 - \pi r^2 = \pi r^2$

Area of each face of the cube

$$= (10)^2 + \pi r^2 = 100 + \frac{22}{7} \times 2^2 = 112 \frac{4}{7} \text{ cm}^2$$

Total surface area of the cube

$$= 6 \times 112 \frac{4}{7} = 675 \frac{3}{7} \text{ cm}^2$$

Sol.569.(a) Here, length (l) = 20 cm, breadth (b) = 10 cm and height (h) = 8 cm

Area of a cardboard needed

$$= 2(lb + bh + hl)$$

$$= 2(20 \times 10 + 10 \times 8 + 8 \times 20)$$

$$= 2(200 + 80 + 160) = 2(440) \Rightarrow 880 \text{ cm}^2$$

Sol.570.(b) Volume of cube = volume of sphere + remaining part volume

$$a^3 = \frac{4}{3} \pi r^3 + 163 \frac{1}{3}$$

$$7^3 = \frac{4}{3} \times \frac{22}{7} \times r^3 + \frac{490}{3}$$

$$343 - \frac{490}{3} = \frac{4}{3} \times \frac{22}{7} \times r^3$$

$$\frac{539}{3} = \frac{4}{3} \times \frac{22}{7} \times r^3$$

$$r^3 = \frac{7 \times 7 \times 7}{2 \times 2 \times 2} \Rightarrow r = \frac{7}{2} \Rightarrow d = 2r = 7 \text{ cm}$$

Sol.571.(b) When we melt and convert any solid then volume of both the solids remains same. $R = 3 \text{ cm} = \text{radius of sphere}$

$$r = 0.1 \text{ mm} = \frac{1}{100} \text{ cm} = \text{radius of wire}$$

$h = \text{length of wire}$

$$\frac{4}{3} \pi R^3 = \pi r^2 h$$

$$\frac{4}{3} \pi \times 3 \times 3 \times 3 = \pi \frac{1}{10000} h$$

$$h = 360000 \text{ cm}$$

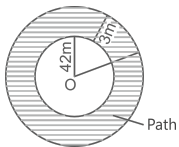
Sol.572.(c) Take inner radius = r

Outer radius = $R = (r + 3)$

According to the question

$$2\pi r = 264 \text{ m} \Rightarrow r = \frac{264 \times 7}{22 \times 2} = 42 \text{ m}$$

$$R = (r + 3) = 42 + 3 = 45 \text{ m}$$

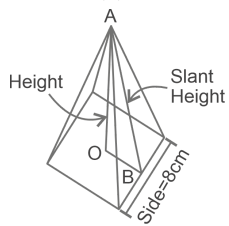


Area of path = $\pi (R^2 - r^2)$

$$\frac{22}{7} (R + r)(R - r)$$

$$= \frac{22}{7} (87 \text{ m} \times 3 \text{ m}) = 820.28 \approx 820 \text{ m}^2$$

Sol.573.(c)



Total surface area of square based pyramid = Surface area of base + $4 \times$ surface area of wall

$$8^2 + 4 \times \frac{1}{2} \times 8 \times AB = 208 \text{ cm}^2$$

$$AB = \text{Slant height} = \frac{208 - 64}{16} = 9 \text{ cm}$$

Sol.574.(b) Upper radius = $21/2 = 10.5$

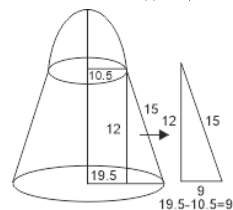
$$\text{Lower radius} = \frac{39}{2} = 19.5$$

Height of frustum = $22.5 - \text{radius of hemisphere}$

$$\text{Height of frustum} = 22.5 - 10.5 = 12$$

Now by pythagoras theorem

$$\text{Slant height (l)} = \sqrt{12^2 + 9^2} = 15$$



Total surface area = surface area of hemisphere + surface area of frustum

$$\text{Total surface area} = 2\pi r^2 + \pi(R + r)l$$

$$= 2 \times \frac{22}{7} \times 10.5 \times 10.5 + \frac{22}{7} \times$$

$$(19.5 + 10.5) \times 15 \\ = \frac{14751}{7} = 2107 \frac{2}{7} \text{ cm}^2$$

Sol.575.(a) Let, side of each cube so formed be a .

Volume of cuboid = number of cubes formed $\times a^3$

$$\text{Volume of cuboid} = 8 \times a^3$$

$$12 \times 54 \times 72 = 8 \times a^3$$

$$a^3 = 5832 \Rightarrow a = 18 \text{ cm}$$

Lateral surface area of 2 such cube

$$= 2 \times (4 \times 18^2) = 8 \times 324 = 2592 \text{ cm}^2$$

Sol.576.(b) Ratio of old radius to new

$$\text{radius} = 3.5 : 4.9 = 5 : 7.$$

$$\text{Old} : \text{New}$$

$$\text{Radius} \quad 5 : 7$$

$$\text{Volume} \quad 5^3 : 7^3$$

$$125 : 343$$

Required percentage increase

$$= \frac{343 - 125}{125} \times 100 = 174.4\%$$

Sol.577.(c) Let the height of the cylindrical tank be " h "

$$\text{Curved surface area} = 2\pi \times 4 \times h = 8\pi h$$

3 times of sum of areas of two circular

$$\text{faces} = 6\pi r^2 = 6\pi \times 4 \times 4 = 96\pi$$

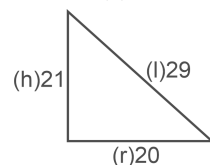
According to the question,

$$96\pi = 2 \times 8\pi h \Rightarrow h = 6$$

Volume of cylindrical tank

$$= \pi r^2 h = \pi \times 4 \times 4 \times 6 = 96\pi.$$

Sol.578.(c)



As height, radius and slant height of a cone form a triplet, we have height $21x$ and radius = $20x$

$$\frac{1}{3} \pi (20x)^2 \times 21x = 4838.4\pi$$

$$x^3 = 1.728 \Rightarrow x = 1.2$$

$$\text{radius} = 20x = 20 \times 1.2 = 24 \text{ cm}$$

Sol.579.(a) 16, 30 and 34 forms a triplet so it is right angled triangle.

$$\text{So, base area} = \frac{1}{2} \times 16 \times 30 = 240 \text{ cm}^2$$

Volume of prism = base area $\times$ height

$$\Rightarrow 240 \times 32 = 7680 \text{ cm}^3$$

Lateral surface area of prism = perimeter of base $\times$ height

$$(16 + 30 + 34) \times 32 = 80 \times 32 = 2560 \text{ cm}^2$$

Sol.580.(b) According to the question,

$$(l \times b) \times (b \times h) \times (h \times l)$$

$$= 216 \times 96 \times 144$$

$$(lbh)^2 = 216 \times 96 \times 144$$

$$lbh = \sqrt{216 \times 96 \times 144} = 1728 \text{ cm}^3$$

Let the edge of each cube = x

So,

volume of cuboid = 8 (volume of cube)

$$lbh = 8(x)^3$$

$$1728 = 8(x)^3 \Rightarrow (x)^3 = 216 \Rightarrow x = 6 \text{ cm}$$

Lateral surface area of three cubes

$$= 3 \times 4(x)^2 = 3 \times 4(6)^2 = 432 \text{ cm}^2$$

Sol.581.(d)

Let inner radius = r and outer radius = R

According to the question,

$$R - r = 2 \Rightarrow R - 3 = 2 \Rightarrow \text{So, } R = 5 \text{ m}$$

Volume of the soil taken out = Volume of embankment

$$\pi r^2 h = \pi (R^2 - r^2) H \Rightarrow r^2 h = (R^2 - r^2) H$$

$$(3)^2 \times 6 = (5^2 - 3^2) \times H$$

$$54 = 16 \times H \Rightarrow H = \frac{54}{16} = 3 \frac{3}{8} \text{ m}$$

Sol.582.(c)

Let " n " small spheres have been formed.

$$\frac{4}{3} \pi (4)^3 = n \times \frac{4}{3} \pi (2)^3 \Rightarrow n = \frac{64}{8} = 8.$$

Surface area of original sphere

$$= 4\pi \times 4^2 = 64\pi$$

Surface area of 8 small spheres

$$= 8 \times 4\pi \times 2^2 = 128\pi$$

Required ratio = $1 : 2$.

Sol.583.(c) Volume of the sheet

= volume of cylinder with outer radius - volume of cylinder with inner radius.

$$\text{Inner radius} = 4 \text{ cm} - 0.5 \text{ cm} = 3.5 \text{ cm}$$

$$h = 2 \text{ m} = 200 \text{ cm}.$$

$$\text{Required volume} = \pi \times 200(4^2 - 3.5^2)$$

$$= \pi \times 200(16 - 12.25)$$

$$= \pi \times 200 \times 3.75 = 750\pi$$

Sol.584.(c) Lateral surface area of prism

= Perimeter of base $\times$ Height

$$480 = 40 \times \text{Height} \Rightarrow \text{Height} = 12 \text{ cm}$$

Volume of prism = Area of base $\times$ height

$$= (\frac{1}{2} \times 8 \times 15) \times 12 = 720 \text{ cm}^3$$

Sol.585.(d) According to the question,

$$\text{CSA of cone} = 2310 \text{ cm}^2$$

$$\frac{22}{7} \times 21 \times l = 2310 \Rightarrow l = 35 \text{ cm}$$

$$\text{So, } h = \sqrt{l^2 - r^2} = \sqrt{35^2 - 21^2}$$

$$= \sqrt{1225 - 441} = \sqrt{784} = 28 \text{ cm}$$

Let new height = H and new radius = R

$$\text{So, } H = 28 \times \frac{50}{100} = 14 \text{ cm and } R$$

$$= 21 \times \frac{200}{100} = 42 \text{ cm}$$

$$\text{Capacity of cone} = \frac{1}{3} \pi R^2 H$$

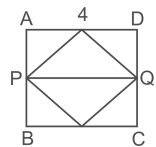
$$= \frac{1}{3} \times \frac{22}{7} \times (42)^2 \times 14 = 25872 \text{ cm}^3$$

So, capacity (in litres)

$$= \frac{25872}{1000} = 25.872 \approx 25.9 \text{ litres}$$

Sol.586.(b) Let the side of square = 4

So area of square $S = 4^2 = 16$



For square formed by joining mid point, diagonal (PQ) = 4

So, area of square which formed

$$= \frac{1}{2} \times (\text{diagonal})^2 = \frac{1}{2} \times (4)^2 = 8$$

$$\text{So, } \frac{A}{S} = \frac{8}{16} = \frac{1}{2} = 0.5$$

Sol.587.(c)

Radius of spherical droplet (in cm) = R

$$= \frac{2}{10 \times 2} = \frac{1}{10} \text{ cm}$$

Radius of cone = Height of cone = r

Volume of 128000 droplets = Volume of cone

$$128000 \times \frac{4}{3} \pi R^3 = \frac{1}{3} \pi r^2 h$$

$$128000 \times 4 \left(\frac{1}{10}\right)^3 = r^2 \times r$$

$$r^3 = 512 \Rightarrow r = 8 \text{ cm}$$

Sol.588.(b) TSA of cylinder = 880 cm^2

$$2\pi r(h + r) = 880$$

$$2 \times \frac{22}{7} \times r(13 + r) = 880$$

$$13r + r^2 = 140 \Rightarrow r^2 + 13r - 140 = 0$$

$$r^2 + 20r - 7r - 140 = 0$$

$$r(r + 20) - 7(r + 20) = 0 \Rightarrow r = 7 \text{ cm}$$

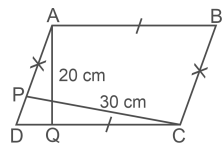
According to question,

Volume of cylinder = $11k$

$$\pi r^2 h = 11k$$

$$= \frac{22}{7} \times (7)^2 \times 13 = 11k \Rightarrow k = 182$$

Sol.589.(b) According to question,



Area of parallelogram = Base $\times$ Altitude

For base AD,

$$300 = AD \times CP \Rightarrow 300 = AD \times 30$$

$$\Rightarrow AD = 10 \text{ cm,}$$

$$BC = AD = 10 \text{ cm}$$

Similarly, for base CD,

$$300 = CD \times AQ \Rightarrow 300 = CD \times 20$$

$$\Rightarrow CD = 15 \text{ cm, } AB = CD = 15 \text{ cm}$$

So, perimeter = $AB + BC + CD + AD$

$$= 10 + 15 + 10 + 15 = 50 \text{ cm}$$

Sol.590.(a) Volume of frustum

$$= \frac{1}{3} \pi h (r_1^2 + r_2^2 + r_1 r_2)$$

$$= \frac{1}{3} \times \frac{22}{7} \times 12 \times (20^2 + 13^2 + 20 \times 13)$$

$$= \frac{1}{3} \times \frac{22}{7} \times 12 \times (400 + 169 + 260)$$

$$= \frac{1}{3} \times \frac{22}{7} \times 12 \times 829 \approx 10422 \text{ cm}^3$$

$$= 10422 \text{ cm}^3 \Rightarrow 10.4 \text{ liters}$$

Sol.591.(a)

Let each side of the square be "a"

Side including the road

$$= a + 2 \times 4.5 = a + 9$$

According to the question,

$$(a + 9)^2 - a^2 = 252$$

$$a^2 + 18a + 81 - a^2 = 252$$

$$18a = 171 \Rightarrow a = \frac{171}{18} = 9.5 \text{ m}$$

Sol.592.(b)

Volume of the cube = base area of the cylinder $\times$ (raised height of water)

$$12 \times 12 \times 12 = \frac{22}{7} \times 20 \times 20 \times h$$

$$h = \frac{12 \times 12 \times 12 \times 7}{22 \times 20 \times 20} \approx 1.4 \text{ cm}$$

Sol.593.(c) For box prepared after

cutting squares from rectangular sheet,

$$\text{Length} = 48 - 2(3.5) = 48 - 7 = 41 \text{ cm}$$

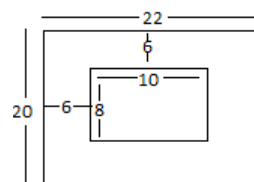
$$\text{Breadth} = 27 - 2(3.5) = 27 - 7 = 20 \text{ cm}$$

Height = 3.5 cm

$$\text{Volume of the box} = 41 \times 20 \times 3.5$$

$$= 2870 \text{ cm}^3$$

Sol.594.(d)



$$\text{Area of the path} = (22 \times 20) - (10 \times 8)$$

$$= 440 - 80 = 360 \text{ m}^2$$

Sol.595.(b) Here the thickness = 1 cm,

so for outer surface : Length

$$= 30 + 2(1) = 32 \text{ cm,}$$

$$\text{Breadth} = 18 + 2(1) = 20 \text{ cm,}$$

$$\text{Height} = 23 + 2(1) = 25 \text{ cm}$$

Space occupied by rectangular box

$$= \text{Length} \times \text{Breadth} \times \text{Height}$$

$$= 32 \times 20 \times 25 = 16000 \text{ cm}^3$$

Sol.596.(a) According to question,

$$\frac{90}{100} \times [\text{Volume of cylinder}] = \text{Volume}$$

$$\text{of sphere} = \frac{90}{100} \times [\pi R^2 h] = \frac{4}{3} \pi r^3$$

$$= \frac{9}{10} \times [3^2 \times 120] = \frac{4}{3} r^3$$

$$= \frac{9}{10} \times [1080] = \frac{4}{3} r^3$$

$$= r^3 = 729 \Rightarrow r = 9 \text{ cm}$$

Sol.597.(b) Difference between areas of

two concentric circles = 88 cm^2

$$\pi R^2 - \pi r^2 = 88 \Rightarrow \pi(R^2 - r^2) = 88$$

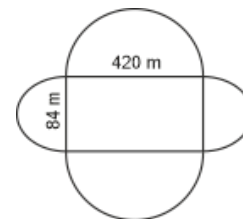
$$\frac{22}{7} (R^2 - r^2) = 88$$

$$R^2 - r^2 = 28 \Rightarrow R^2 = 64$$

Area of largest circle = πR^2

$$= \frac{22}{7} \times 64 = 201.14 \approx 201 \text{ cm}^2$$

Sol.598.(c)



Total area(x) = Area of rectangular plot + Area of two semi-circular of diameter 84 m + Area of two semi-circular of diameter 420 m

$$= 420 \times 84 + 2\left(\frac{\pi r^2}{2}\right) + 2\left(\frac{\pi R^2}{2}\right)$$

$$= 35280 + \pi(r^2 + R^2)$$

$$= 35280 + \frac{22}{7} \left[\left(\frac{84}{2}\right)^2 + \left(\frac{420}{2}\right)^2 \right]$$

$$= 35280 + \frac{22}{7} [(42)^2 + (210)^2]$$

$$= 35280 + \frac{22}{7} [1764 + 44100]$$

$$= 35280 + \frac{22}{7} [45864]$$

$$= 35280 + 144144 = 179424 \text{ m}^2$$

$$1 \text{ hectare} = 10000 \text{ m}^2$$

so, Area of plot = x

$$= \frac{179424}{10000} = 17.9 \text{ hectares}$$

Sol.599.(b)

Let the side of cubical block be x.

Volume of cubical box

$$= x^3 = 10^3 = 1000 \text{ cm}^3$$

$$\text{Volume of sphere} = \frac{4}{3} \pi r^3$$

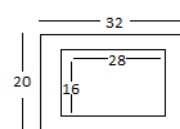
$$= \frac{4}{3} \times \frac{22}{7} \times (4.2)^3 = 310.464 \text{ cm}^3$$

So, wood wasted

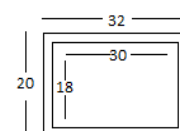
$$= 1000 - 310.464 = 689.536 \text{ cm}^3$$

Sol.600.(a)

(Figure 1)



(figure 2)



When margin of 2 cm is left, area available for typing (figure 1)

$$= (32 - 4) \times (20 - 4) = 28 \times 16 = 448 \text{ cm}^2$$

When margin of 1 cm is left, area available for typing (figure 2)

$$= (32 - 2) \times (20 - 2) = 30 \times 18 = 540 \text{ cm}^2$$

So percentage increase in typing area

$$= \frac{540 - 448}{448} \times 100 = \frac{92}{448} \times 100$$

$$= 20.54\%$$

Sol.601.(a) $1 \text{ m}^3 = 1000 \text{ litres}$, so 126 kilolitres = $126 \times 1000 = 126000 \text{ litres}$ which is equivalent to 126 m^3 .

$$\text{Width of reservoir} = \frac{126}{8 \times 3.5} = 4.5 \text{ m}$$

Sol.602.(c) Area of four walls = Area of floor and ceiling

$$2h(l + b) = 2(l \times b)$$

$$2h(18 + 12) = 2(18 \times 12)$$

$$60h = 432 \Rightarrow h = 7.2 \text{ m}$$

$$\text{Volume of hall} = l \times b \times h$$

$$= 18 \times 12 \times 7.2 = 1555.2 \text{ m}^3$$

Sol.603.(d)

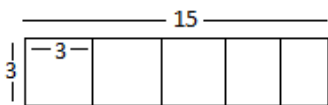
Let the edge of each cube be x .

Volume of each cube so formed

$$= 27 \text{ cm}^3 \Rightarrow x^3 = 27 \Rightarrow x = 3$$

$$\text{So length of cuboid} = 3 \times 5 = 15 \text{ cm,}$$

$$\text{breadth} = 3 \text{ cm, and height} = 3 \text{ cm}$$



So, TSA of cuboid = $2(lb + bh + hl)$

$$= 2(15 \times 3 + 3 \times 3 + 3 \times 15)$$

$$= 2(45 + 9 + 45) = 2(99) = 198 \text{ cm}^2$$

Sol.604.(d) For hollow cylindrical shell:

Outer radius $R = b$, Inner radius $P = b - a$, and $R + P = b + (b - a) = 2b - a$

According to the question,

Volume of sphere = Volume of hollow cylindrical shell

$$\frac{4}{3}\pi r^3 = \pi(R^2 - P^2)H$$

$$\frac{4}{3}r^3 = (R^2 - P^2)H$$

$$\frac{4}{3}r^3 = (R - P)(R + P)H$$

$$\frac{4}{3}r^3 = (a) \times (2b - a) \times H$$

$$H = \frac{\frac{4}{3}r^3}{(a) \times (2b - a)} = \frac{4r^3}{3a(2b - a)}$$

Sol.605.(c)

Volume of cone = Volume of hemisphere

$$\frac{1}{3}\pi r^2 h = \frac{2}{3}\pi r^3, h = 2r$$

$$\text{So, slant height of cone} = \sqrt{r^2 + h^2}$$

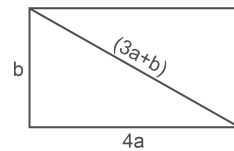
$$= \sqrt{r^2 + (2r)^2} = \sqrt{5r^2} = \sqrt{5}r = 14\sqrt{5} \text{ cm}$$

CSA of cone = $\pi r l$

$$= \frac{22}{7} \times 14 \times 14\sqrt{5} = 616\sqrt{5} \text{ cm}^2$$

Sol.606.(c) Let the larger side of the rectangle be "4a", and other side be "b"

According to the question, length of diagonal = $4a + b - a = 3a + b$



Applying pythagoras theorem we get,

$$b^2 + (4a)^2 = (3a + b)^2$$

$$b^2 + 16a^2 = 9a^2 + b^2 + 6ab$$

$$7a^2 = 6ab \Rightarrow \frac{a}{b} = \frac{6}{7},$$

ratio of larger side : smaller side

$$= 4a : b \Rightarrow 4 \times 6 : 7 = 24 : 7$$

Sol.607.(c) Area of the square = 6400 m^2

$$\text{Side of the square} = \sqrt{6400} = 80 \text{ m}$$

So, radius of circle = 80 m

$$\text{Length of rectangle} = \frac{3}{5} \times 80 = 48 \text{ m}$$

Area of rectangle = length $\times$ breadth

$$= 48 \times 20 = 960 \text{ m}^2$$

Sol.608.(d) Volume of 3 cylinder to be filled with grain

$$= 3 \times \text{volume of each cylinder}$$

$$= 3 \times (\pi r^2 h) = 3 \times \left(\frac{22}{7} \times (20)^2 \times 70\right)$$

$$= 3 \times (88000) = 264000 \text{ cm}^3$$

3000 cm^3 corresponds to 1 kg of grain,

$$\text{so } 264000 \text{ cm}^3 = \frac{264000}{3000} = 88 \text{ kg}$$

$$\text{Grain left in the bag} = 100 - 88 = 12 \text{ kg}$$

Sol.609.(d) Surface area of original balloon : Surface area of inflated balloon

$$= 4\pi r^2 : 4\pi R^2 = r^2 : R^2 = (6)^2 : (10)^2$$

$$= 36 : 100 = 9 : 25$$

Sol.610.(d) Let the original radius $r = 8$,

$$\text{so new radius } R = \frac{r}{2} = \frac{8}{2} = 4$$

Let the height of original cylinder $h = 10$

and new height = H

Volume of original cylinder = Volume of new cylinder

$$\pi r^2 h = \pi R^2 H \Rightarrow r^2 h = R^2 H$$

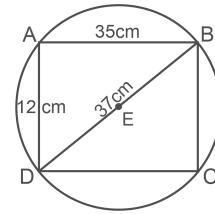
$$8^2 \times 10 = 4^2 \times H$$

$$H = 40 = 4(10) = 4H, \text{ so value of } k = 4$$

Sol.611.(d) $BD = \sqrt{AB^2 + AD^2}$

$$= \sqrt{35^2 + 12^2} = \sqrt{1225 + 144}$$

$$= \sqrt{1369} = 37 \text{ cm}$$

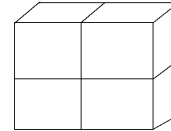


Diagonal of rectangle = Diameter of circle = $BD = 37 \text{ cm}$

Circumference of circle

$$= 2\pi r = 2 \times \frac{22}{7} \times \frac{37}{2} = \frac{814}{7} = 116\frac{2}{7} \text{ cm}$$

Sol.612.(b) Clearly, the following cuboidal figure with dimensions $24 \times 24 \times 12$ will be formed.



Total surface area of a cuboid

$$= 2(lb + bh + lh)$$

$$= 2(24 \times 24 + 24 \times 12 + 24 \times 12)$$

$$= 2(576 + 288 + 288)$$

$$= 2 \times 1152 = 2304 \text{ cm}^2$$

Sol.613.(c)

For maximum volume of the cone we must have the height and diameter of the cone equal to the side of the cube.

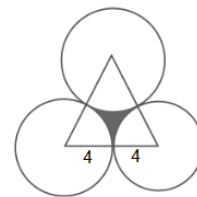
The side of the cube be 14 units, then the height of the cone is 14 units and its radius is 7 units.

Required ratio

$$= 14 \times 14 \times 14 : \frac{1}{3} \times \frac{22}{7} \times 7 \times 7 \times 14$$

$$= 42 : 11$$

Sol.614.(b)



Required area = Area of equilateral triangle - Area of 3 sectors with angle 60°

$$\text{Area of equilateral triangle} = \frac{\sqrt{3}}{4} a^2 = \frac{\sqrt{3}}{4} \times 8 \times 8 = 16\sqrt{3} \text{ cm}^2$$

Area of three sectors with central angle 60° and radius 4 cm

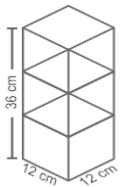
$$= 3 \times \frac{60^\circ}{360^\circ} \times \pi \times 4 \times 4 = 8\pi.$$

$$\text{Required area} = 16\sqrt{3} - 8\pi = 8(2\sqrt{3} - \pi)$$

Sol.615.(d)

$$\text{Side of each cube} = \sqrt{1728} = 12 \text{ cm}$$

By putting three cubes one above the other, the resulting figure will be cuboid with dimension of $36 \text{ cm} \times 12 \text{ cm} \times 12 \text{ cm}$



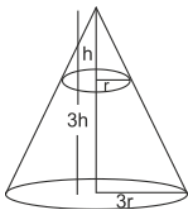
Total surface area of the cuboid
 $= 2 (LB + BH + HL)$
 $= 2(12 \times 12 + 12 \times 36 + 12 \times 36)$
 $= 2 \times 1008 = 2016 \text{ cm}^2$

Sol.616.(c) Area of shaded region

$= 80 \times 50 = 4000 \text{ m}^2$
 So, area of unshaded region
 $= \frac{35}{100} \times 4000 = 1400 \text{ m}^2$
 Total area $= (80 + x) \times (50 + x)$
 $4000 + 1400 = 4000 + 130x + x^2$
 $x^2 + 130x - 1400 = 0$
 $x^2 + 140x - 10x - 1400 = 0$
 $x(x + 140) - 10(x + 140) = 0$
 $x \neq -140$, so $x = 10$

Sol.617.(a) Here, $(r + h) = 46 \text{ cm}$,
 Total surface area of cylinder
 $= 2\pi r(r + h) \Rightarrow 2\pi r(r + h) = 6072$
 $2 \times \frac{22}{7} \times r \times 46 = 6072$
 $r = \frac{6072 \times 7}{2 \times 22 \times 46}$, $r = 21 \text{ cm}$
 $h = 46 - 21 = 25 \text{ cm}$.
 Volume $= \pi r^2 h = \pi \times 21 \times 21 \times 25$
 $= 11025\pi$

Sol.618.(c) The respective ratio of height and radius of the two cones will be 1 : 3.
 Let radius and height of smaller cone be "r" and "h"
 Radius and height of whole cone become 3r and 3h



$$= \frac{\text{volume of smaller cone}}{\text{volume of whole cone}}$$

$$= \frac{\frac{1}{3}\pi r^2 h}{\frac{1}{3}\pi (3r)^2 3h} = \frac{1}{3^2 \times 3} = \frac{1}{27}$$

Sol.619.(c) Let the radius of the outer circumference be 'R' and the radius of the inner circumference be 'r'.
 Now, $2\pi R = 748 \text{ m}$.
 $2 \times \frac{22}{7} \times R = 748$.
 $R = \frac{748 \times 7}{22 \times 2} \Rightarrow R = 119 \text{ m}$.
 Again, $2\pi r = 396 \text{ m}$.

$$2 \times \frac{22}{7} \times r = 396 \Rightarrow r = \frac{396 \times 7}{22 \times 2} = 63 \text{ m}$$

Width of the track $= (R - r)$
 $= (119 - 63) = 56 \text{ m}$.

Sol.620.(a)

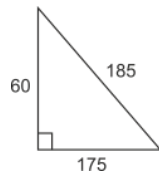
Radius of iron ball $(r) = \frac{21}{2} \text{ cm}$
 Volume of sphere = Volume of cylinder
 $\frac{4}{3}\pi r^3 = \pi R^2 h$
 $\frac{4}{3} \left(\frac{21}{2}\right)^3 = R^2 \times \frac{7}{2}$
 $R^2 = 441 \Rightarrow R = 21 \text{ cm}$
 CSA of cylinder $= 2\pi R h$
 $= 2 \times \frac{22}{7} \times 21 \times 3.5 = 462 \text{ cm}^2$

Sol.621.(b)

Area of the floor $= l \times b = 18 \times 12 = 216$
 Area of four walls
 $= 2 \times (l + b) \times h \Rightarrow 2 \times (18 + 12) \times h$
 Given, $2 \times (30) \times h = 216$
 So, $h = 3.6$
 Now volume of a cuboidal shape
 $= l \times b \times h = 12 \times 18 \times 3.6 = 777.6 \text{ cm}^3$

Sol.622.(d)

Sides of the triangular field are triplets.



Area of Δ = Area of rectangular park

$$\frac{1}{2} \times 60 \times 175 = 21x \times 10x$$

$$x^2 = 25 \Rightarrow x = 5$$

Perimeter of rectangular park
 $= 2 [21x + 10x] = 62x = 62(5) = 310 \text{ m}$

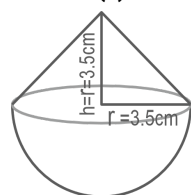
Sol.623.(b) $r = \frac{56}{2} = 28 \text{ cm}$

Length of circular wire
 $= 2\pi r = 2 \times \frac{22}{7} \times 28 = 176 \text{ cm}$

Length of circular wire = Perimeter of rectangle
 $176 = 2(7x + 4x) \Rightarrow 88 = 11x \Rightarrow x = 8$

Area of rectangle $= 7x \times 4x = 28x^2$
 $= 28(8^2) = 28(64) = 1792 \text{ cm}^2$

Sol.624.(c)



Height of cone = radius of base, so $h = r$
 Volume of metal used = Volume of cone
 $+ \text{volume of hemisphere}$

$$= \frac{1}{3} \pi r^2 h + \frac{2}{3} \pi r^3$$

$$= \frac{1}{3} \pi r^3 + \frac{2}{3} \pi r^3$$

$$= \frac{1}{3} \pi r^3 (1 + 2) = \frac{1}{3} \pi r^3 (3)$$

$$= \frac{22}{7} \times \left(\frac{7}{2}\right)^3 = \frac{539}{4} = 134\frac{3}{4} \text{ cm}^3$$

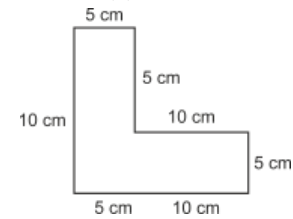
Sol.625.(d) Volume of frustum

$$= \frac{1}{3} \pi h [r_1^2 + r_2^2 + (r_1 \times r_2)]$$

$$= \frac{1}{3} \times \frac{22}{7} \times 24(20^2 + 10^2 + 20 \times 10)$$

$$= \frac{1}{3} \times \frac{22}{7} \times 24 \times 700 = 17600 \text{ cm}^3$$

Sol.626.(c) The given figure is



Total perimeter
 $= (5 + 10 + 5 + 10 + 5 + 10 + 5 + 10) = 50 \text{ cm}$.

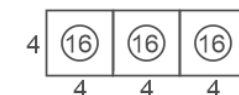
Sol.627.(d) Let the radius of the hemisphere be "r" and the side of the triangle be "a".

$$\text{Now, given } 3\pi r^2 = \frac{\sqrt{3}}{4} a^2$$

$$= \frac{a^2}{r^2} = \frac{3\pi \times 4}{\sqrt{3}} \Rightarrow \frac{a}{r} = (4\pi\sqrt{3})^{0.5}$$

Sol.628.(c)

Each side of the square $= \sqrt{16} = 4 \text{ cm}$



Clearly, the breadth of the rectangle is 4 cm while the length of the rectangle is $3 \times 4 = 12 \text{ cm}$

Perimeter of the rectangle
 $= 2(l + b) = 2(4 + 12) = 32 \text{ cm}$.

Sol.629.(d) Area of sector making angle

$$\theta \text{ at the centre} = \frac{\theta}{360^\circ} \times \pi r^2$$

$$= \frac{\theta}{360^\circ} \times \pi (2)^2 = \frac{4\pi}{5} \Rightarrow \theta = \frac{360^\circ}{5} = 72^\circ$$

Sol.630.(a)

Radius = 4.2m, Height = 1.75m
 As radius and height is in the ratio of 12 : 5 So slant height = 13 unit

Slant height $l = \frac{1.75}{5} \times 13 = 4.55 \text{ m}$

Curved surface area of the canvas

$$= \pi r l = \frac{22}{7} \times 4.2 \times 4.55$$

$$= 13.2 \times 4.55 = 60.06 \text{ m}^2$$

Sol.631.(c)

Side of the square $= \sqrt{1764} = 42 \text{ m}$

Breadth = $\frac{1}{3}$ of 42 = 14 m

Length = 28 m, Area = 28×14

Cost of levelling = $28 \times 14 \times 15 = ₹ 5880$

Sol.632.(c) Let width of the wall = x

Height of the wall = 10x

Length of the wall = 80x

Volume of the wall

$$= 800 x^3 = 51.2 \text{ m}^3 \Rightarrow x = 0.4 \text{ m}$$

Length of the wall = 32 m

Height of the wall = 4 m

Area of the one side of the wall = 128 m^2

Cost of painting = $128 \times 100 = ₹ 12800$

Sol.633.(b)

circumference of the circle = $2\pi r = 44 \text{ cm}$

Side of the square = $a = \frac{44}{4} = 11 \text{ cm}$

Ratio of area = $\pi r^2 : a^2$

$$= \frac{22}{7} \times 7 \times 7 : 11 \times 11 = 14 : 11$$

Sol.634.(c)

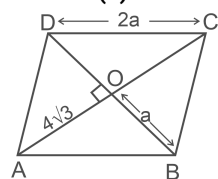
Area of the ring = $\pi(R^2 - r^2) = 3080$

$$(R^2 - r^2) = 3080 \times \frac{7}{22} = 980$$

$$(R + r)(R - r) = 980$$

$$(R + r) \times 10 = 980 \Rightarrow R + r = 98$$

Sol.635.(d)



Let side of the rhombus = 2a

Other diagonal = 2a; OB = OD = a cm

In $\triangle OAB$; $AB = 2a$, $OB = a$, $AO = 4\sqrt{3}$

$$(2a)^2 = a^2 + (4\sqrt{3})^2$$

$$\Rightarrow 4a^2 = a^2 + 48 \Rightarrow a = 4$$

Other diagonal = 2a = 8

Area of the rhombus

$$= \frac{1}{2} \times 8 \times 8\sqrt{3} = 32\sqrt{3}$$

Short trick : Let the other diagonal = a

So, side = a In a rhombus,

$$4a^2 = a_1^2 + a_2^2 \Rightarrow 4a^2 = (8\sqrt{3})^2 + a^2$$

$$a^2 = 64, a = 8$$

Area of the rhombus =

$$\frac{1}{2} \times 8 \times 8\sqrt{3} = 32\sqrt{3} \text{ cm}^2$$

Sol.636.(b) Radius of cylinder = r

Height of cylinder = h $\Rightarrow r + h = 12.5$

Total area of cylinder = 275 cm^2

$$2\pi r(r + h) = 275 \text{ cm}^2$$

$$2\pi r(12.5) = 275 \text{ cm}^2,$$

$$r = \frac{7}{2} = 3.5 \text{ cm}$$

Sol.637.(a) As we know that 15, 20 and

25 are pythagoras triplet so third side is

$$25\sqrt{3}$$

So, total perimeter

$$= 15\sqrt{3} + 20\sqrt{3} + 25\sqrt{3} = 60\sqrt{3}$$

Sol.638.(a) Let the equal side of the

isosceles triangle = 4x and its base = 5x.

$$4x + 4x + 5x = 91 \Rightarrow x = 7 \text{ cm}$$

Length of its base = 35 cm

Sol.639.(a) Let the sides of a triangle are

3x, 4x and 5x respectively.

$$5x - 3x = 3.6 \Rightarrow x = 1.8 \text{ cm}$$

$$\text{Area of triangle} = \frac{1}{2} \times 3x \times 4x$$

$$= 6 \times 1.8 \times 1.8 = 19.44 \text{ cm}^2$$

Sol.640.(b) Diameter = 28 cm

Radius = 14 cm

Perimeter of circular pizza

$$= 2\pi r = 88 \text{ cm}$$

Perimeter of circular part of one - quarter

$$= \frac{1}{4} \times 88 \text{ cm} = 22 \text{ cm}$$

Perimeter of the circular part of

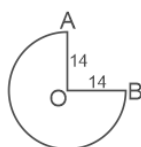
remaining pizza = $(88 - 22) = 66 \text{ cm}$

Perimeter of remaining pizza

$$= \text{Perimeter of the circular part} + 2 \times r$$

$$= 66 + 28 = 94 \text{ cm}$$

Alternate method :



Perimeter of remaining pizza = length of

arc AB + AO + OB

$$= \frac{3}{4} \times 2 \times \frac{22}{7} \times 14 + 14 + 14 = 94 \text{ cm}$$

Sol.641.(d) Inner radii = 6.7 cm

Outer radii = 9.5 cm

Difference between their circumference

$$= 2\pi(9.5 - 6.7) \text{ cm}$$

$$= 2 \times \frac{22}{7} \times 2.8 = 17.6 \text{ cm}$$

Sol.642.(b) After cutting,

Length of Box = $32 - (3 + 3) = 26$

Breadth of Box = $18 - (3 + 3) = 12$

Height = 3

$$\text{Volume of the box} = 26 \times 12 \times 3 = 936 \text{ cm}^3$$

Sol.643.(d)

Volume of n sphere = volume of cube

$$\frac{4}{3} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times \frac{7}{2} \times n$$

$$= 77 \times 77 \times 77 \Rightarrow n = 2541$$

Sol.644.(a) Diagonal of the square = $7\sqrt{2}$

Side of the square = 7 cm

Perimeter = 4a = 28 = base of a triangle

Area of another square = 169

Side of another square = height of the triangle = 13

Required area of triangle

$$= \frac{1}{2} \times 28 \times 13 = 182 \text{ cm}^2$$

Sol.645.(d) No. of bricks

$$= \frac{8 \times 3 \times 22.5 \times 1000000}{64 \times 11.25 \times 6} = 125000$$

Sol.646.(b)

Let the area and breadth of the rectangle are 9x and x respectively.

$$\text{Area} = l \times b \Rightarrow 9x = l \times x \Rightarrow l = 9$$

Perimeter of rectangle = $2(l + b)$

$$86 = 2(l + b) \Rightarrow l + b = 43$$

$$= 9 + b = 43 \Rightarrow b = 34 \text{ cm}$$

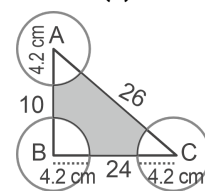
Sol.647.(d) If side of a cube is = a

then its diagonal = $a\sqrt{3}$ and total surface area = $6a^2$

$$(\sqrt{3}a)^2 = 2175 \text{ then } 3a^2 = 2175,$$

$$a^2 = 725 \text{ and } 6a^2 = 6 \times 725 = 4350 \text{ cm}^2$$

Sol.648.(d)



Since we know that (10,24,26) is a pythagorean triplet so we can easily find out the area of the triangle to be

$$= \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times 24 \times 10 = 120 \text{ cm}^2$$

So the area of the triangle will be equal to 120. At each vertices a circle of radius 4.2 cm is drawn so the total area of circle will be = $\frac{1}{2} \times \text{area of a circle (as the sum of angles of a triangle is equal to } 180^\circ)$

$$\text{Area of circle} = \frac{1}{2} \times \pi \times 4.2 \times 4.2 = 27.72$$

Area of triangle excluding the portion covered by circle = $120 - 27.72 = 92.28 \text{ cm}^2$

Sol.649.(d)

Volume of cube = $a^3 = 9^3 = 729 \text{ cm}^3$

Volume of cuboid = $5 \times 13 \times 31 = 2015$

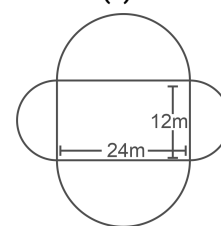
Total volume = $2015 + 729 = 2744 \text{ cm}^3$

Side of resultant cube = 14 cm

Surface area of resultant cube

$$= 6a^2 = 6 \times 14^2 = 1176 \text{ cm}^2$$

Sol.650.(a)



Total area of rectangular lawn

$= 24 \times 12 + \pi R^2 + \pi r^2$ (Where R = radius of bigger semicircle and r = radius of smaller semicircle)

$$\text{Area} = 288 + 3.14 \times (12^2 + 6^2)$$

$$= 288 + 565.2 = 853.2 \text{ m}^2$$

$$\text{Cost} = 853.2 \times 100 = 85320$$

Sol.651.(b)

$$\text{Volume of cube} = a^3 = 9^3 = 729 \text{ cm}^3$$

$$\text{Volume of cuboid} = 5 \times 13 \times 31 = 2015$$

$$\text{Total volume} = 2015 + 729 = 2744 \text{ cm}^3$$

$$\text{Side of a new cube} = 14 \text{ cm}$$

$$\text{Surface area of a new cube}$$

$$= 6a^2 = 6 \times 14^2 = 1176 \text{ cm}^2$$

$$\text{Cost} = 1176 \times 10 = 11,760$$

$$\text{Sol.652.(d) Area of a hexagon} = \frac{3\sqrt{3}}{2} a^2$$

$$= 1944\sqrt{3} = \frac{3\sqrt{3}}{2} a^2 \Rightarrow a = 36$$

$$\text{Perimeter} = 36 \times 6 = 216 \text{ m}$$

$$\text{Cost} = 216 \times 11.50 = \text{Rs. } 2,484$$

$$\text{Sol.653.(a) Diagonal of square} = 6\sqrt{2} \text{ cm}$$

$$\text{Side of square} = 6 \text{ cm}$$

$$\text{perimeter of square} = 24 \text{ cm}$$

$$\text{Base of triangle} = 24 \text{ cm}$$

$$\text{Height of triangle} = \sqrt{144} = 12 \text{ (area of square = side}^2\text{)}$$

$$\text{Area of triangle} = \frac{1}{2} \times 24 \times 12 = 144 \text{ cm}^2$$

$$\text{Sol.654.(d) Perimeter of square} = 40 \text{ m}$$

$$\text{Perimeter of rectangle} = 2 \times 40 = 80 \text{ m}$$

$$\text{Let length of rectangle} = 3x$$

$$\text{Breadth of rectangle} = 2x$$

$$\text{Total perimeter of rectangle}$$

$$= 2(3x + 2x) \Rightarrow 10x = 80 \Rightarrow x = 8$$

$$\text{length of rectangle} = 24 \text{ m}$$

$$\text{Breadth of rectangle} = 16 \text{ m}$$

$$\text{area of rectangle} = 24 \times 16 = 384 \text{ m}^2$$

$$\text{Sol.655.(b) Let height} = h; \text{Length} = l$$

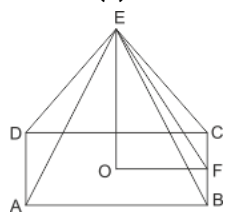
$$\text{Breadth} = b$$

$$lb = 3, bh = 12, hl = 16$$

$$(lbh)^2 = 3 \times 12 \times 16 \Rightarrow lbh = 24 \text{ m}^3$$

$$\text{As } 1 \text{ m}^3 = 1000l$$

$$\text{So total volume} = 24000l$$

Sol.656.(d)

$$\text{Perimeter of base} = 40 \text{ cm}$$

$$\text{Height(OE)} = 10 \text{ cm}$$

$$\text{In right angled triangle OEF}$$

$$OF = 5 \Rightarrow EF^2 = OF^2 + OE^2 \Rightarrow EF = 5\sqrt{5}$$

$$\text{Slant height(EF)} = 5\sqrt{5}$$

$$\text{Lateral surface area}$$

$$= \frac{1}{2} \times \text{perimeter of base} \times \text{Slant height}$$

$$= \frac{1}{2} \times 40 \times 5\sqrt{5} \Rightarrow 100\sqrt{5} \text{ cm}^2$$

Sol.657.(b) Curved surface area of cylinder $= 2\pi rh = 2\pi \times 5 \times 30 = 300\pi$
Slant height of cone

$$= \sqrt{5^2 + 12^2} = 13$$

$$\text{Surface area of cone} = \pi rl$$

$$= \pi \times 5 \times 13 = 65\pi$$

$$\text{Surface area of 2 cone} = 130\pi$$

$$\text{Total surface area} = 430\pi$$

Sol.658.(c) Regular hexagon consists of 6 equilateral triangles.

$$\text{Area of triangle} = \frac{\sqrt{3}}{4} a^2 = \frac{25\sqrt{3}}{4}$$

$$\text{Area of base} = 6 \times \text{Area of triangle} \\ = \frac{150\sqrt{3}}{4}$$

$$\text{Volume of prism} = \text{area of base} \times \text{height}$$

$$= \frac{150\sqrt{3}}{4} \times 12\sqrt{3} = 1350 \text{ cm}^3$$

$$\text{Sol.659.(a) Density} = \frac{\text{mass}}{\text{volume}}$$

$$\Rightarrow 10.5 = \frac{6688}{\text{volume}} \Rightarrow \text{volume} = \frac{6688}{10.5}$$

$$\text{Volume of spherical shell}$$

$$= \frac{4}{3}\pi(R^3 - r^3)$$

$$\frac{6688}{10.5} = \frac{4}{3} \times \frac{22}{7} (216 - r^3)$$

$$\text{Solving the above eq}$$

$$216 - r^3 = 152 \Rightarrow r = 4$$

$$\text{Thickness} = 6 - 4 = 2 \text{ cm}$$

$$\text{Sol.660.(a) Given that } R + r = 14$$

$$4\pi R^2 - 4\pi r^2 = 112\pi$$

$$R^2 - r^2 = 28$$

$$(R + r)(R - r) = 28$$

$$R - r = 2 \Rightarrow R = 8, r = 6$$

$$\text{Volume is in the ratio of the cube of their}$$

$$\text{radius} = 8^3 : 6^3 = 512 : 216 = 64 : 27$$

$$\text{Sol.661.(b) } R = \text{new radius} \Rightarrow R = r + 2.5$$

$$\text{Surface area of the sphere} = 4\pi r^2$$

$$4 \times \pi \times (R^2 - r^2) = 110$$

$$(R^2 - r^2) = \frac{35}{4}$$

$$(R + r)(R - r) = \frac{35}{4} \Rightarrow (R + r) = 3.5$$

$$R = 3 \text{ and } r = 0.5 \Rightarrow \text{Volume} = \frac{4}{3}\pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{11}{21}$$

$$\text{Sol.662.(b) } l = 1.2 \text{ m} = 120 \text{ cm}$$

$$r = 24 \text{ cm}, R = 39 \text{ cm}$$

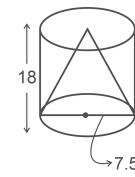
$$\text{Volume} = \pi(R^2 - r^2) h$$

$$= \pi \times 945 \times 120 = 113400\pi \text{ cm}^3$$

$$\text{Mass} = \text{density} \times \text{volume}$$

$$\text{Mass} = (113400\pi \times 8)/1000$$

$$\text{Mass} = 907.2\pi \text{ kg}$$

Sol.663.(c)

$$h = 18; r = 7.5$$

$$l = 19.5 \text{ (by pythagoras theorem)}$$

$$\text{Total surface area} = \text{lateral surface area of cylinder} + \text{lateral surface area of cone} + \text{area of circular side (above)}$$

$$\text{Total surface area} =$$

$$2\pi r h + \pi r l + \pi r^2$$

$$\text{Total surface area} = \pi r(2h + l + r)$$

$$\text{Total surface area}$$

$$= \pi \cdot 7.5(36 + 19.5 + 7.5)$$

$$\text{Total surface area} = \pi \cdot 7.5(63)$$

$$\text{Total surface area} = 472.5\pi \text{ cm}^2$$

$$\text{Sol.664.(b) Volume of the tank}$$

$$= \frac{2}{3} \times \pi \times 10.5^3 \times \frac{2}{3} = 1617\pi \text{ m}^3$$

$$1\text{ m}^3 = 1000l \Rightarrow 1617\text{ m}^3 = 1617000l$$

$$\text{Time required} = \frac{1617000}{7.7 \times 3600} = \frac{175}{3} \text{ hrs}$$

$$\text{Sol.665.(b) Volume of cuboid}$$

$$= 18 \text{ cm} \times 36 \text{ cm} \times 72 \text{ cm} = 46656 \text{ cm}^3$$

$$\text{Volume of 1 cube} = \frac{46656}{8} = 5832 \text{ cm}^3$$

$$\text{Side of cube} = \sqrt[3]{5832} = 18 \text{ cm}$$

$$\text{Lateral surface area of 8 cubes}$$

$$= 8 \times 4 \times 18 \times 18 = 10368 \text{ cm}^2$$

$$\text{Total surface area of the cuboid}$$

$$= 2((18 \times 36) + (36 \times 72) + (72 \times 18))$$

$$= 9072 \text{ cm}^2$$

$$\text{Required ratio} = 9072 : 10368 = 7 : 8$$

$$\text{Sol.666.(b) Total surface area of prism}$$

$$= 2 \times \text{Area of base} + \text{Curved surface area}$$

$$\text{Area of base} = 15^2 = 225 \text{ cm}^2$$

$$\text{Curved surface area} = \text{Perimeter of base}$$

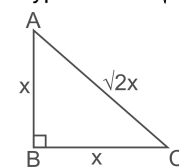
$$\times \text{height} = 4 \times 15 \times 8 = 480 \text{ cm}^2$$

$$\text{Total surface area of prism}$$

$$= 2 \times 225 + 480 = 930 \text{ cm}^2$$

$$\text{Sol.667.(b) Let the length of equal sides (Perpendicular and Base)} = x$$

$$\text{Hypotenuse} = \sqrt{2} x,$$



$$\text{Perimeter} = (2 + \sqrt{2})x$$

$$8(\sqrt{2} + 1) = (2 + \sqrt{2})x$$

$$\text{On solving } x = 4\sqrt{2}$$

$$\text{Hypotenuse} = \sqrt{2} \times 4\sqrt{2} = 8 \text{ cm}$$

Sol.668.(b) Area of the triangle MLN

$$\Rightarrow \frac{1}{2} \times r^2 = 36 \Rightarrow r = 6\sqrt{2}$$

$$\text{Area of circle} = \pi r^2 = \pi (6\sqrt{2})^2 = 72\pi$$

Sol.669.(b) let the number of spherical metal balls = n Volume of n spherical metal balls
= Volume of cylindrical vessel

$$= n \times \frac{4}{3} \pi r^3 = \pi r^2 h$$

$$= n \times \frac{4}{3} \pi (1)^3 = \pi (6)^2 \cdot 5 \Rightarrow n = 135$$

Sol.670.(d) Let each side = 10

$$\text{Volume} = 10 \times 10 \times 10 = 1000$$

New length, breadth and height

$$= 11, 12, 8$$

$$\text{New volume} = 11 \times 12 \times 8 = 1056$$

Required percentage

$$= \frac{56}{1000} \times 100 = 5.6\%$$

Short trick :

Old : New

Length $\rightarrow 10 : 11$ Height $\rightarrow 5 : 4$ Breadth $\rightarrow 5 : 6$ Volume $\rightarrow 250 : 264$

$$\% \text{age increase} = \frac{264 - 250}{250} \times 100 = 5.6\%$$

Sol.671.(a)

Let side of smallest square = 2 unit

Then, diameter of circle in which smallest square is inscribed will equal to diagonal of square = $2\sqrt{2}$ unit and this diameter is equal to the side of the middle square.

Now, diameter of outer circle = 4 unit

So side of the largest square = 4 unit

Therefore, the ratio of the area of the largest square to that of the smallest square is 4 : 1.

Sol.672.(b) Let the pipe take T hours to raise the water level by 4m.

$$180 \text{ m} \times 140 \text{ m} \times 4 \text{ m}$$

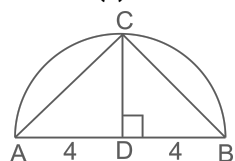
$$= 1.2 \text{ m} \times 0.75 \text{ m} \times 15000 \text{ m} \times T$$

$$T = 7\frac{7}{15} \text{ hours} = 7 \text{ hours } 28 \text{ minutes}$$

Sol.673.(d) Area of rectangle = $a^2 - b^2$

$$(a + b) \times \text{breadth} = (a + b)(a - b)$$

$$\text{breadth} = (a - b)$$

Sol.674.(a)

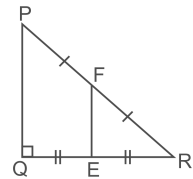
$$AD = DB = CD = 4 \text{ cm}, AB = 8 \text{ cm}$$

$$\text{Area of triangle} = \frac{1}{2} \times 8 \times 4 = 16 \text{ cm}^2$$

Sol.675.(b) Area of a square park

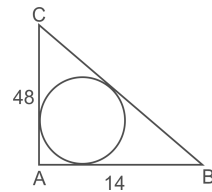
$$= 16x^2 + 8x + 1 \Rightarrow a^2 = (4x + 1)^2$$

$$a = 4x + 1$$

Sol.676.(b)

$$QE : ER = 1 : 1 \Rightarrow QR = 2 \text{ unit}$$

$$\frac{\text{Area of quadrilateral PQEF}}{\text{Area of triangle PQR}} = \frac{2^2 - 1}{2^2} = \frac{3}{4}$$

Sol.677.(d) The minimum integral value of $x = 6 + x > 10$, $x > 4$, $x = 5$ **Sol.678.(b)**

$$BC = \sqrt{48^2 + 14^2} = 50$$

$$\text{Inradius} = \frac{P + B - H}{2}$$

$$\text{Inradius} = \frac{48 + 14 - 50}{2} = 6 \text{ cm}$$

Sol.679.(a)Let the radius of outer circle = R And radius of inner circle = r

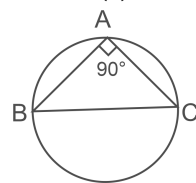
$$2\pi r = 22 \Rightarrow 2 \times \frac{22}{7} \times r = 22$$

$$\Rightarrow r = \frac{7}{2} \text{ cm}$$

$$2\pi R = 44 \Rightarrow 2 \times \frac{22}{7} \times R = 44$$

$$R = 7 \text{ cm}$$

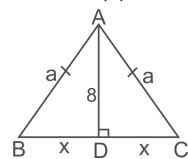
$$\text{Width of ring} = 7 - \frac{7}{2} = 3.5 \text{ cm}$$

Sol.680.(b)

$$AB = AC = 7\sqrt{2}$$

$$BC = \sqrt{(7\sqrt{2})^2 + (7\sqrt{2})^2}; BC = 14$$

$$\text{Radius} = \frac{14}{2} = 7 \text{ cm}$$

Sol.681.(c)Perimeter of $\triangle ABC$

$$= a + a + 2x = 2a + 2x = 64$$

$$\Rightarrow a + x = 32 \dots (i)$$

Using pythagoras theorem in $\triangle ABD$;

$$a^2 = x^2 + 8^2$$

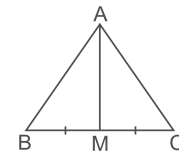
$$a^2 - x^2 = 8^2 \Rightarrow (a - x)(a + x) = 64$$

$$(a - x)(32) = 64 \Rightarrow a - x = 2 \dots (ii)$$

From (i) and (ii) $\therefore a = 17$ and $x = 15$

$$\text{Area of } \triangle ABC = \frac{1}{2} \times BC \times AD$$

$$= \frac{1}{2} \times 30 \times 8 = 120 \text{ cm}^2$$

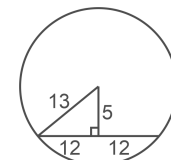
Sol.682.(c)AM is the median of $\triangle ABC$.

$$\text{Area of } \triangle ABM = 18 \text{ cm}^2$$

Therefore, area ($\triangle ABM$) = area ($\triangle ACM$)
(Median divides the triangle into two equal areas).

$$\text{area of } \triangle ABC = 2 \times \text{area of } \triangle ABM$$

$$= 2 \times 18 = 36 \text{ cm}^2$$

Sol.683.(d)

$$\text{Radius} = 13 \text{ cm}$$

$$\text{Diameter} = 2 \times 13 = 26 \text{ cm}$$

Sol.684.(c) In a rectangular plot, Length = 5 $\times$ breadthTherefore, $L : B = 5 : 1$

$$\text{Area of rectangular plot} = 2000 \text{ m}^2$$

$$L \times B = 2000$$

$$5 \times B \times B = 2000 \Rightarrow B = 20 \text{ m}$$

Sol.685.(a) Diameters of two grounds = 42 m and 26 m

Radius of two grounds = 21 m and 13 m

Sum of area of two grounds

$$= \pi \times (21)^2 + \pi \times (13)^2 = \pi \times [441 + 169]$$

$$= \pi \times [610]$$

Let the radius of new ground = R

$$\pi \times (R)^2 = \pi \times [610] \Rightarrow R = 24.69 \text{ m}$$

Sol.686.(a) For an isosceles righttriangle, sides = $a, a, a\sqrt{2}$

$$\text{Perimeter} = a + a + a\sqrt{2} = 30$$

$$2a + \sqrt{2}a = 30$$

$$a(2 + \sqrt{2}) = 30, a = \frac{30}{2 + \sqrt{2}} = 15(2 - \sqrt{2})$$

Area of triangle

$$= \frac{1}{2} \times a^2 = \frac{1}{2} \times 225 \times (2 - \sqrt{2})^2$$

$$= \frac{1}{2} \times 225 \times (4 + 2 - 4\sqrt{2})$$

$$= \frac{1}{2} \times 225 \times (6 - 4\sqrt{2}) = 225 \times (3 - 2\sqrt{2})$$

$$= 225 \times (3 - 2[1.414]) = 225 \times (3 - 2.828)$$

$$= 225 \times (0.172) \approx 38.7 \text{ m}^2$$

Sol.687.(a) Volume of cone = $\frac{1}{3}\pi R^2 H$

Sol.688.(d) The length of tangents drawn from an external point to a circle are equal in length.

Sol.689.(d)

Circumference of a circle = $2\pi R$

Let r_1 and r_2 be the inner and outer radius of the ring respectively. And c_1 and c_2 be inner and outer circumference.

$$2\pi r_1 = 440 \text{ m} \Rightarrow r_1 = 70 \text{ m}$$

$$2\pi r_2 = 506 \text{ m} \Rightarrow r_2 = 80.5 \text{ m}$$

$$\text{Area of the ring} = \pi[(r_2)^2 - (r_1)^2]$$

$$= \frac{22}{7} \times [(80.5)^2 - (70)^2] = 4966.5 \text{ m}^2$$

Cost of levelling the ring

$$= 6 \times 4966.5 \text{ m}^2 = ₹ 29,799$$

Sol.690.(a) Volume of water in cylinder

= Volume of rectangular tub

$$\pi r^2 h = l \times b \times h$$

$$\frac{22}{7} \times 30 \times 30 \times 42 = 75 \times 44 \times h$$

$$H = 36 \text{ cm}$$

Sol.691.(b) Let the side of square A = x

$$\text{Diagonal}_A = \sqrt{2}x = (a + b) \text{ units}$$

$$\text{Side of square B} = \sqrt{2}x$$

$$\text{Area}_B = 2(a + b)^2$$

Sol.692.(a) Area of disc = $0.64 \pi \text{ m}^2$

Thus, radius of disc = 0.8 m

$$\text{Circumference} = 2\pi r = 1.6\pi \text{ m}$$

$$\text{Distance} = 1.408 \text{ km} = 1408 \text{ m}$$

$$\text{Number of revolution} = \frac{1408}{1.6\pi} = 280$$

Sol.693.(c) Let 'l' be the length and 'b' be the breadth of a rectangle.

$$2(l + b) = 50 \Rightarrow l + b = 25 \Rightarrow l \times b = 150$$

$l = 15$ and $b = 10$ satisfy the above equation. $\Rightarrow$ So, Shorter side = 10

Sol.694.(b) Volume $\propto (\text{radius})^3$

$$\frac{\text{Volume}_A}{\text{Volume}_B} = \left(\frac{r_A}{r_B}\right)^3 = \frac{125}{64}, \frac{r_A}{r_B} = \frac{5}{4}$$

$$r_A + r_B = 9x = 36 \Rightarrow x = 4$$

$$r_A = 20 \text{ and } r_B = 16$$

$$\text{Surface area of sphere A} = 4 \times \pi \times r_A^2$$

$$= 4 \times \pi \times 400 = 1600 \pi \text{ cm}^2$$

Sol.695.(c)

Let r be the radius of the cone

Height of the cone = 35 cm

Curved surface area (l is the slant height of the cone) = $4(\text{Area of the base})$

$$\pi r l = 4 \times \pi r^2, l = 4r$$

$$\text{As per formula, } l^2 = h^2 + r^2$$

Putting the value of $l = 4r$ from equation 1 in the above formula

$$16r^2 = h^2 + r^2 \Rightarrow 16r^2 - r^2 = h^2 \Rightarrow 15r^2$$

Put $h=35$ in above equation

$$15 \times r \times r = 35 \times 35 \Rightarrow r^2 = \frac{245}{3}$$

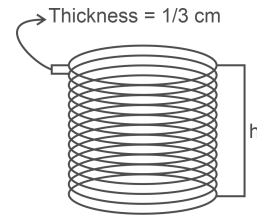
Now, volume of the cone (V)

$$= \frac{1}{3} \times \pi \times r^2 \times h$$

$$= \frac{1}{3} \times \frac{22}{7} \times \frac{245}{3} \times 35 = 2994 \text{ cm}^3$$

$$= 2.994 \text{ m}^3$$

Sol.696.(d)



As per the question

60 discs are stacked in the above manner having thickness $\frac{1}{3} \text{ cm}$

Diameter of each disc = 21 cm

$$\text{Radius} = \frac{21}{2} \text{ cm}$$

Total number of discs $\times$ thickness of one disc = height of the cylinder

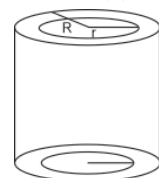
$$\text{Height}(h) = 60 \times \frac{1}{3} = 20 \text{ cm}$$

Thus, Volume of the cylinder is (V)

$$= \pi r^2 h = \frac{22 \times 21 \times 21 \times 20}{7 \times 2 \times 2}$$

$$= 6930 \text{ cm}^3 = 6.93 \times 10^{-3} \text{ m}^3$$

Sol.697.(b)



The volume of steel in the cylinder

= outer volume - inner volume

Outer radius (R) = 5 cm

Inner radius (r) = 4 cm

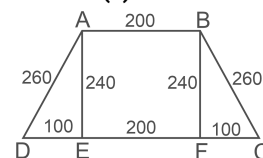
The volume of the steel in cylinder

$$= \pi R^2 h - \pi r^2 h \Rightarrow \pi h(R^2 - r^2)$$

$$= \frac{(22 \times 84 \times 9)}{7} = 216 \times 11$$

$$= 2376 \text{ cm}^3 = 2.376 \times 10^{-3} \text{ m}^3$$

Sol.698.(c)



In $\triangle ADE$, by pythagoras theorem

$$260^2 - 100^2 = AE^2$$

$$360 \times 160 = AE^2$$

$$AE = 6 \times 4 \times 100 \Rightarrow AE = 240$$

$$\text{area of trapezium} = \frac{1}{2} \times (200 + 400) \times 240$$

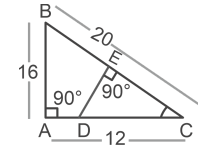
$$= 600 \times 120 = 72000$$

Sol.699.(a) As per question

$$\frac{4}{3} \pi x^3 = 126 \times \frac{1}{3} \times \pi \times \frac{7}{2} \times \frac{7}{2} \times 3$$

$$x = \frac{9 \times 7 \times 7 \times 7 \times 3}{8} \Rightarrow x = 10.5$$

Sol.700.(a)



$\triangle CAB$ & $\triangle CED$ are similar triangles

$$\angle CAB = \angle CED = 90^\circ$$

$$\angle C = \angle C, \angle CBA = \angle CDE$$

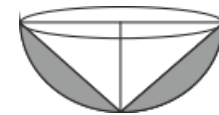
$$\text{Now, } \frac{\text{Area of } \triangle CED}{\text{Area of } \triangle CAB} = \left(\frac{6}{20}\right)^2 = \frac{36}{400}$$

Here, 400 units corresponds to 96 cm^2

$$\text{So } 36 \text{ units corresponds to } \frac{36 \times 96}{400}$$

$$= 8.64 \text{ cm}^2 \Rightarrow \text{area } \triangle DEC = 8.64 \text{ cm}^2$$

Sol.701.(a) Here, the height of the cone will be equal to radius.



Volume of the remaining portion (shaded portion) = volume of hemisphere - volume of cone

$$\frac{2}{3} \pi r^3 - \frac{1}{3} \pi r^3 = \frac{1}{3} \pi r^3$$

$$\text{Required percentage} = \frac{\frac{\pi}{3} \times r^3}{\frac{2}{3} \pi r^3} \times 100$$

$$= 50\%$$

Sol.702.(b) The volume of the cone and cylinder will be same

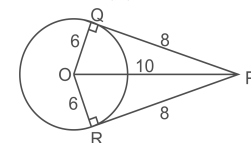
$$\frac{1}{3} \times \pi \times r^2 \times H = \pi r^2 h$$

On substituting the values

$$6 \times 18 \times 60 = 15 \times 15 \times H$$

$$H = \frac{18 \times 8}{5} = \frac{144}{5} = 28.8 \text{ cm}$$

Sol.703.(d)



$$\text{Area of PQOR} = 2 \times \frac{1}{2} \times 6 \times 8$$

$$= 48 \text{ sq. cm}$$

Sol.704.(b) As per question

$$\text{Let, length} = x, \text{height} = \frac{x}{5}, \text{breadth} = \frac{x}{8}$$

We know that

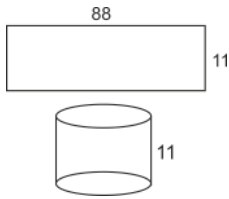
$$1 \text{ m}^3 = 1000 \text{ L} \Rightarrow 25 \text{ m}^3 = 25000 \text{ L}$$

$$\text{Volume} = 25 \text{ m}^3$$

$$\text{Volume of the cuboid} = 25$$

$$x \times \frac{x}{5} \times \frac{x}{8} = 25$$

$$x^3 = 25 \times 5 \times 8 \Rightarrow x = 10\text{m}$$

Sol.705.(b)

Height of the cylinder = 11cm

Thus, $2\pi r$ = length of the rectangle

$$2\pi r = 88 \Rightarrow r = 14$$

Volume of the cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times 14 \times 14 \times 11 = 6776 \text{ cm}^3$$

Sol.706.(a)

Radius of the Cycle wheel = 63cm

Distance = Speed $\times$ Time

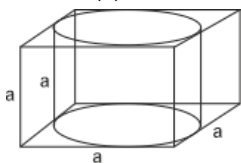
$$= 72 \times \frac{5}{18} \times \frac{33}{2} \times 60$$

$$= 33 \times 600 = 19800 \text{ cm}$$

The circumference of the Cycle wheel

$$= 2 \times \frac{22}{7} \times 63 = 396$$

$$\text{Number of revolutions} = \frac{1980000}{396} = 5000$$

Sol.707.(b)

Diameter of the cylinder = a

$$r = \frac{a}{2} \Rightarrow h = a$$

The percentage of remaining material of the cube to the original cube

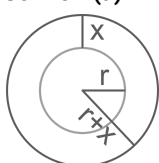
$$= \frac{a^3 - \frac{\pi a^3}{4}}{a^3} \times 100 \Rightarrow \frac{4 - \pi}{4} \times 100$$

$$= \frac{0.86 \times 25}{100} = 21.42 \approx 21.5\%$$

Sol.708.(a) According to question,

$$\frac{\pi r_1^2 h_1}{\pi r_2^2 h_2} = \frac{V_1}{V_2} \Rightarrow \frac{r_1^2 h_1}{r_2^2 h_2} = \frac{V_1}{V_2}$$

$$\frac{r_1^2}{r_2^2} \times \frac{a}{b} = \frac{x}{y} \Rightarrow \frac{r_1}{r_2} = \sqrt{\frac{xb}{ya}}$$

Sol.709.(a)

$$\text{Area of the path} = \pi[r + x]^2 - \pi[r]^2$$

$$= \pi \times (r + x + r) \times (r + x - r) = \pi x(x + d)$$

Sol.710.(d) Initially area of sphere = $4\pi r^2$ Increased area of sphere = $4\pi(r + 5)^2$

$$\text{ATQ, } 4\pi(r + 5)^2 - 4\pi r^2 = 704$$

$$4\pi\{(r + 5)^2 - r^2\} = 704$$

$$4 \times \frac{22}{7} \times (2r + 5) \times 5 = 704$$

$$2r + 5 = 8 \times \frac{7}{5} \Rightarrow 2r = 6.2$$

diameter = 6.2 cm

Sol.711.(d) Let, the side of cube = aVolume of cube = a^3 Now, Radius of the cylinder = $\frac{a}{2}$

Height of the cylinder = a

Volume of the cylinder = $\pi r^2 h$

$$= \pi\left(\frac{a}{2}\right)^2 a = \frac{\pi}{4} a^3$$

$$\text{Required \%} = \frac{a^3 - \frac{\pi}{4} a^3}{a^3} \times 100$$

$$= \frac{a^3(1 - \frac{22}{7 \times 4})}{a^3} \times 100 = \frac{6}{28} \times 100 \approx 21$$

Sol.712.(b) Let original volume = v, original radius = r and original height = h and new height = H

$$v = \frac{1}{3} \pi r^2 h \Rightarrow h = \frac{3v}{\pi r^2}$$

According to the question,

New radius = 2r

$$3v = \frac{1}{3} \pi (2r)^2 H \Rightarrow H = \frac{9v}{4\pi r^2}$$

$$\text{Required ratio} = \frac{3v}{\pi r^2} : \frac{9v}{4\pi r^2} = 4 : 3$$

Sol.713.(a) Volume of the sphere = $\frac{4}{3} \pi r^3$

Here,

$$\frac{4}{3} \pi (9)^3 = \frac{4}{3} \pi (1)^3 + \frac{4}{3} \pi (x)^3 + \frac{4}{3} \pi (8)^3$$

$$\frac{4}{3} \pi (9)^3 = \frac{4}{3} \pi (1^3 + x^3 + 8^3)$$

$$729 = 1 + x^3 + 512 \Rightarrow x = 6$$

Surface area of the sphere = $4\pi r^2$

$$= 4 \times \pi \times 6^2 \Rightarrow 144\pi \text{ cm}^2$$

Sol.714.(b)Area of the base of a cone = πr^2

$$400\pi = \pi r^2 \Rightarrow r = 20$$

Slant height (l) of the cone = $\sqrt{h^2 + r^2}$

$$= \sqrt{15^2 + 20^2} = 25$$

curved surface area of the cone

$$= \pi r l = \pi \times 20 \times 25 = 500\pi \text{ cm}^2$$

Sol.715.(b)We know that, 1000 litres = 1m^3 $\Rightarrow$ Volume of the water taken out

$$= \frac{15400}{1000} = 15.4\text{m}^3$$

Volume of water taken out = $\pi r^2 h$

$$15.4 = \pi \times 3.5 \times 3.5 \times h$$

h = 0.4 m or 40 cm

Sol.716.(d) Volume of the pyramid

$$= \frac{1}{3} \times \text{Base Area} \times \text{Height}$$

Area of an equilateral triangle = $\frac{\sqrt{3}}{4} a^2$

$$= \frac{\sqrt{3}}{4} \times 8 \times 8 = 16\sqrt{3}$$

Volume of the pyramid

$$= \frac{1}{3} \times 16\sqrt{3} \times 24\sqrt{3} = 384 \text{ cm}^3$$

Sol.717.(d) $33\frac{1}{3}\% = \frac{1}{3}$

We know that volume is directly proportional to the product of the square of the radius and the height

Old : New

Radius $\rightarrow$ 3 : 2Height $\rightarrow$ 1 : 2Volume $\rightarrow$ 9 : 8

Decrease in volume = 9 - 8 = 1

$$\text{Required \%age} = \frac{1}{9} \times 100 = 11\frac{1}{9}\%$$

Sol.718.(a) Let the side of cube = a $\Rightarrow$ sides of the cuboid a, a and $\frac{a}{3}$ Total surface area of the cube = $6a^2$

Total surface area of the cuboid

$$= 2\{(a \times a) + (a \times \frac{a}{3}) + (a \times \frac{a}{3})\}$$

$$= \frac{10}{3} a^2$$

$$\text{Required ratio} = 6a^2 : 2 \times \frac{10}{3} a^2 = 9 : 10$$

Sol.719.(c) Area covered in onerevolution = $2\pi r h$

$$= 2 \times \frac{22}{7} \times \frac{1}{2} \times \frac{5}{4} = \frac{55}{14}$$

Total revolution in one hour = 60 $\times$ 14

Area covered in one hour

$$= 60 \times 14 \times \frac{55}{14} = 3300 \text{ m}^2$$

Sol.720.(c)Volume of the cylinder = $\pi R^2 H$

$$\text{Volume of the toy} = \frac{1}{3} \pi r^2 h + \frac{2}{3} \pi r^3$$

According to the question

$$\pi R^2 H = n \times \left(\frac{1}{3} \pi r^2 h + \frac{2}{3} \pi r^3\right)$$

$$\frac{22}{7} \times 12 \times 12 \times 15 = n \times \frac{22}{7} \times$$

$$\left(\frac{1}{3} \times 3 \times 3 \times 9 + \frac{2}{3} \times 3 \times 3 \times 3\right)$$

$$n = 48$$

Sol.721.(c)

$$\text{Area of an equilateral triangle} = \frac{\sqrt{3}}{4} a^2$$

$$16\sqrt{3} = \frac{\sqrt{3}}{4} a^2 \Rightarrow a = 8$$

$$\text{Radius of the pyramid} = \text{inradius of the triangle} = \frac{a}{2\sqrt{3}} = \frac{8}{2\sqrt{3}} = \frac{4}{\sqrt{3}}$$

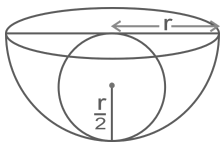
$$\text{And area of its lateral face} = \frac{1}{2} \times \text{base} \times \text{slant height}$$

$$\frac{1}{2} \times 8 \times \text{slant height} = 30$$

$$\Rightarrow \text{slant height} = \frac{15}{2}$$

$$\text{Height of the pyramid} = \sqrt{l^2 - r^2}$$

$$= \sqrt{\left(\frac{15}{2}\right)^2 - \left(\frac{4}{\sqrt{3}}\right)^2} = \sqrt{\frac{611}{12}}$$

Sol.722.(c)

Let the radius of hemisphere = r

$$\Rightarrow \text{radius of the sphere} = \frac{r}{2}$$

$$\text{Volume of hemisphere} = \frac{2}{3} \pi r^3$$

$$\text{Volume of sphere} = \frac{4}{3} \pi \left(\frac{r}{2}\right)^3 = \frac{1}{6} \pi r^3$$

Area of the remaining solid

$$= \frac{2}{3} \pi r^3 - \frac{1}{6} \pi r^3 = \frac{1}{2} \pi r^3$$

$$\text{Required ratio} = \frac{1}{6} \pi r^3 : \frac{1}{2} \pi r^3 = 1 : 3$$

Sol.723.(d)

$$\text{Mud taken out from earth} = \pi r^2 h$$

$$= \frac{22}{7} \times 2.8 \times 2.8 \times 15 = 369.60$$

Let the length of platform = l

$$369.60 = l \times b \times h$$

$$l = \frac{369.60}{8 \times 1.5} = 30.8 \text{ m}$$

Sol.724.(a) Internal radius of hemisphere

$$= \frac{24}{2} = 12 \text{ cm}$$

External radius of the hemisphere

$$= 12 + 0.5 = 12.5 \text{ cm}$$

Total surface area of a hollow hemisphere

$$= 2\pi(R^2 + r^2) + \pi(R^2 - r^2)$$

$$= \frac{22}{7} [2 \times \{(12.5)^2 + (12)^2\} + (12.5)^2 - (12)^2] = 612.75\pi \text{ cm}^2$$

$$\text{Sol.725.(a) Area of the Path} = 2(\text{width of path})\{\text{length} + \text{breadth} + 2(\text{width of path})\} = 2(3.5)\{25 + 15 + 2(3.5)\} = 329 \text{ cm}^2$$

$$\text{Required cost} = 329 \times \frac{51}{2} = \text{Rs. } 8389.50$$

$$\text{Sol.726.(a) Volume of cone} = \frac{1}{3} \pi r^2 h$$

$$\text{Area of its base} = \pi r^2$$

According to the question

$$\frac{1}{3} \pi r^2 h = 924 \Rightarrow \frac{1}{3} \pi r^2 \times 18 = 924$$

$$\pi r^2 \times 6 = 924 \Rightarrow \pi r^2 = 154 \text{ cm}^2$$

Sol.727.(c) We know that

$$a^2 = \left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2$$

Where a = side of the rhombus and d_1, d_2 are the diagonals of the rhombus.

$$10^2 = \left(\frac{12}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2$$

$$\frac{d_2}{2} = \sqrt{10^2 - 6^2} = 8 \Rightarrow d_2 = 16$$

$$\text{Area of the rhombus} = \frac{1}{2} \times d_1 \times d_2$$

$$= \frac{1}{2} \times 12 \times 16 = 96 \text{ cm}$$

Sol.728.(d) Volume of Frustum

$$= \frac{1}{3} \pi (R^2 + r^2 + R \times r)h$$

$$= \frac{1}{3} \times \frac{22}{7} \times (3^2 + 2^2 + 3 \times 2) \times 21 = 418$$

Sol.729.(b) Required volume

$$= (90 \times 75 \times 50) - (85 \times 70 \times 45) = 337500 - 267750 = 69750 \text{ cm}^3$$

Sol.730.(a) Perimeter of square

= Perimeter of rectangle

$$= 2(16 + 14) = 60 \text{ cm}$$

$$\text{Side of square} = \frac{60}{4} = 15 \text{ cm}$$

Perimeter of semicircle

$$= \pi r + 2r = \frac{22}{7} \times \frac{15}{2} + 15$$

$$= 23.58 + 15 = 38.58 \text{ cm}$$